
The Dark Empress

A Practical Guide to Universal Dominion

JOSHUA R. EISENHART

Dedicated to the future

MY MANTRAS:

No

Never yield

never surrender

by will alone I survive

By will alone I create my destiny

For the hopes and dreams of all mankind

MY MANTRAS:	3
EPILOGUE: GOD AS THE UNIVERSE FROM NOTHING	16
"IS THERE IS A GOD?" IS A USELESS QUESTION	16
<i>The reality of Consistency</i>	19
GOD AS SENTIENCE	21
<i>God as Humanity</i>	24
<i>God as man</i>	28
<i>Simulations</i>	31
<i>God As a Company</i>	33
GOD AS BUSINESS PRACTICES	34
PART 1: THE JOURNEY TO THE FUTURE	37
AN INTRODUCTION TO PART 1: MY CHILDHOOD DREAM	37
<i>My Childhood Dream</i>	38
<i>Futurisms Failure:</i>	38
<i>The Failures of the Past</i>	40
<i>The Dangers of Unchecked Dreams</i>	Error! Bookmark not defined.
CHAPTER 1: JOSHUA'S SOLUTION TO GALACTIC COLONIZATION	43
THE FIRST STEPS TO BUILDING A GALATIC EMPIRE	43
1. Zero Cargo Starships	43
2. Primitive Space Stations for People	44
My Basic Plan (try to think of a better one yourself)	Error! Bookmark not defined.
Station and Colony laws, Culture, and Design	46
"Tower Lego Stations"	47
Questions:	49
3. Buildings and Terraforming on the Earth	50
Practical Things: Infrastructure and Utilities	52
4. Printed bodies-and Neo-Android Parents	54

The Basics- _____	55
What not to do- _____	56
Neo-Androids as a solution- _____	55
Real Life Zombies- _____	56
Modifying people is an act of treating humans as cattle and slaves. _____	57
AI OVERLORD PARENTS _____	58
<i>Online Education:</i> _____	60
QUESTIONS _____	61
<i>Seperating Man from Nature?</i> _____	63
CHAPTER 2: APPLIED ERTHICS IN CAPITALISM _____	65
SOME, LIKE, TECH HISTORY, OR WHATEVER _____	Error! Bookmark not defined.
<i>Standard Oil:</i> _____	65
<i>Corollary Stories: Bell</i> _____	66
<i>19th Century Internet and Kodak</i> _____	67
<i>Apple and NeXt</i> _____	68
TRANSPORTATION EVOLVED: _____	69
<i>My ideal "bicycle": The Failure of Segway Scooters</i> _____	72
<i>Cyborgs and Androids</i> _____	73
Androids- _____	Error! Bookmark not defined.
Cyborgs- _____	Error! Bookmark not defined.
Ironman and Vision- _____	Error! Bookmark not defined.
<i>Tesla:</i> _____	Error! Bookmark not defined.
MONETIZING SPACE: SPACEX AND FRIENDS _____	74
<i>My Path to Mars:</i> _____	74
The Inhumanity of Mars _____	77
AR/VR/MR: SNAP INC, FACEBOOK _____	79
Eye Contact Filter: _____	79
AR Privacy Background filter: _____	80

3D Cartoon AI Analytics Representation _____	82
Orienting to Space: Comic Book Chat! _____	83
<i>Main Themes of New Chatting</i> _____	84
OPENHR --TECHNOLOGY AND CORPORATE CULTURE: _____	84
<i>Space:</i> _____	88
<i>communication systems.</i> _____	88
The Tech. _____	91
OPEN AG: GMOs ARE GOOD -ORGANIC FOOD INDUSTRY BAD _____	92
<i>A start up tech revolution</i> _____	94
OPEN ENERGY: NUCLEAR POWER AND FRACKING _____	95
Fracking and Methanol- _____	Error! Bookmark not defined.
<i>Manhattan Project</i> _____	Error! Bookmark not defined.
The Political Dangers of Fascist Science- _____	Error! Bookmark not defined.
CHAPTER 3. THE ETHICAL CREATION OF CYBERNETICS AND AI _____	97
A HEROES JOURNEY _____	98
<i>Kant's Ethics Reborn</i> _____	99
Evolving Kant _____	102
"Never Lie" Reborn _____	103
People are not THings _____	Error! Bookmark not defined.
<i>Joshua's Ethical System</i> _____	105
Joshua's Life Rules _____	Error! Bookmark not defined.
Expanded Explanation _____	106
Case Study: Development of children with ethical Systems _____	107
COLLECTIVE HUMANE CYBERNETICS: CIVILIZING AND SAVING ALL MANKIND _____	109
<i>Joshua's Laws of Cybernetics</i> _____	109
<i>Summary</i> _____	Error! Bookmark not defined.
<i>Collective Autism</i> _____	111
Autism- _____	112

“Collective Autism”	112
<i>Autistic People as CyCogs</i>	113
Treating Autism-	113
AR/VR/MR-	114
Computer Vision-	115
Cartoon Rendered Reality-	115
Creepy Cameras-	115
Benefits for Families-	116
Education Tech-	116
Kantian Ethics, Intentions and Cybernetics	117
<i>Cognitive Mirrors, Transhumans, and the Singularity</i>	118
THE FUTURE OF AI: HOW TO CREATE HUMANE AI	121
<i>The Soul of the Sword</i>	122
<i>Humans Feelings as Computer Identity: Ai Training</i>	123
<i>Taste, as the root of intelligence: Ones Cognitive Graph as an APP.</i>	125
A Personal Love APP	126
Finding Role Models to Develop Us	127
<i>Section Summary: Techno-Symbiosis</i>	128
CHAPTER 4- TOWARDS A BETTER MONEY	131
<i>Ethical cybernetic AI driven currency.</i>	131
<i>Modern Money: A retrospective</i>	132
Pessimism And Poverty-	132
Energy is Currency Now-	132
Africa and Legacy Energy Systems-	134
Future Currency and Bitcoins	134
AI Smart Energy-	135
Capitalisms Rebuke-	135
Socialisms rebuke of Humanity as Currency-	135

THE IDENTITY OF MONEY: DEMYSTIFYING ECONOMICS _____	136
<i>Investments:</i> _____	137
<i>A case Study in Japan:</i> _____	141
<i>Capital: American Banking</i> _____	142
A Conspiracy Theory: The Greed of the American people themselves caused their own financial crisis. _____	143
<i>How the Fed Works: Banks Investing In the Government</i> _____	144
At Its Simplest- _____	144
Loans to Banks- _____	145
Bonds- _____	145
GDP growth- _____	146
Toxic Assets- _____	146
A Little Conspiracy Theory: _____	147
The Fed, treasury, and banks Mystery Relationship- _____	147
NEW INVESTMENT CONCEPT: CHINA, WECHAT, AND OPEN FINANCE _____	148
<i>A new vision of money</i> _____	148
<i>Big Data and Economics</i> _____	150
<i>Investing in People In a Pessimistic Reality: China as a New World Model</i> _____	151
<i>Wechat Money and Chinese Investments.</i> _____	153
<i>Open Finance: Powering the New Business of the Future</i> _____	154
PART 2: THE FALL OF CIVILIZATION- ZOMBIE WORLD _____	157
CHAPTER 5: JOSHUA'S GUIDE TO SURVIVING THE ZOMBIE APOCALYPSE ____	161
Questions: _____	161
THE DAY THE WORLD ENDS. GO GET A POINTY THING AND POKE! _____	162
<i>Getting basic supplies. Bikes</i> _____	163
<i>Weapons:</i> _____	164
<i>Armor:</i> _____	164
<i>The basis of Hope: Your First Workshop and Groups</i> _____	165

Worskshop and Tech Supplies: _____	166
<i>Summary of Gear:</i> _____	167
GETTING TO YOUR LAUNCHING POINT _____	167
<i>Making a location to meet your Family:</i> _____	167
Traveling to find your family: Scouting _____	168
<i>Your Second BASE</i> _____	169
<i>Where to go to start over</i> _____	170
On Compassion and Group Cohesion _____	171
BEFORE YOU LEAVE, LAST THINGS, AND NEW PEOPLE _____	172
<i>Families are the basis of civilization.</i> _____	172
<i>Saving the Smartest and Best People</i> _____	173
<i>Technology based leadership and Womens Purpose</i> _____	175
<i>The journey</i> _____	176
<i>Building a settlement</i> _____	177
CAPITALISM WILL SAVE THE WORLD _____	178
<i>The World Has Ended---</i> _____	179
<i>Capitalism is good.</i> _____ Error! Bookmark not defined.	
Great investing is about individual people. _____	180
Safe External Investing- _____	181
amazing jewels hidden in the darkness. _____	181
<i>Capitalism For Humanity surrounded in Zombies</i> _____	182
Human Nature- _____	182
Markets- _____	182
Government Control Of Markets- _____	183
Products for Boredom- _____	184
Selling the Future and Civilization- _____	185
Things you Can't Sell- _____	185
Energy as Currency- _____	185
ZOMBIE INTERNET: TRANSFORMING ENGERY INTO DATA _____	186

<i>The Internet Of Things (IOT): Data as Energy</i>	Error! Bookmark not defined.
<i>Investments, currency, data, and Imperialism</i>	Error! Bookmark not defined.
<i>Virtual Online Currency</i>	Error! Bookmark not defined.
<i>A New Kind of Investment</i>	190
Applying this to Africa:	192
<i>Governments and Phone Companies at the end of the world</i>	193
<i>A Mythological Free Internet</i>	Error! Bookmark not defined.

CHAPTER 6: THE MORAL VALUE OF SLAVERY AND THE BIRTH OF NEO-

CAPITALIST SERVITUDE	195
ZOMBIE WORLD SLAVERY	196
<i>The Roots of Slavery</i>	197
Closed Places without Slavery	198
<i>Broader Classes of Slavery</i>	200
Migrant Worker Slavery	201
THE MORAL VALUE OF SLAVERY	202
CHINESE EDUCATION AS SLAVERY	204
<i>The Self and Being under Confucian Education</i>	208
THE HUMAN EXTERMINATION PLAN: MAKING ZOMBIES	211
<i>Sub-Humans</i>	212
<i>Voluntary "Solutions"</i>	215
<i>A Genocide System</i>	217
<i>The Matrix</i>	220
BASIC INCOME	223
<i>Solutions 1-5</i>	224

CHAPTER 7: OPENGVERNMENT

USING THE POWER OF OTHERS	228
<i>Leadership has no Rules</i>	229
<i>Plundering for Good</i>	230

OPENGOVERNMENT	233
<i>Election of Leaders</i>	235
<i>the means justify the ends.</i>	236
<i>Campaign funding and Salaries</i>	237
America as an Example	238
HUMAN RIGHTS AND CITIZENSHIP	242
<i>Rights and the US COnstitution</i>	243
US Supreme Court	245
<i>The Illusion of Rights</i>	246
Government Duties	247
DECENTRALIZED BUSINESS INNOVATION. TECHNO-PIRATE COMPANIES.	250
<i>Medical System</i>	252
<i>Financial Freedom</i>	255
<i>Anarcho Start-Ups and A New Labor Union Movement</i>	256
CHAPTER 8: THE ORIGIN OF CONSCIOUSNESS	260
HUMAN EVOLUTION	260
<i>Teleological Purpose Driven Evolution</i>	262
<i>Linear Evolution</i>	263
HUMAN COGNITIVE PROJECTION	265
<i>The will to Value</i>	266
<i>Consciousness</i>	268
An Intense and very long authors note	270
<i>The body and flesh</i>	275
HUMAN CULTURE	277
<i>Tools and technology.</i>	281
THE GENOCIDAL INSTINCT	283
<i>In the Beginning</i>	284
<i>Evolving Civilization</i>	287

NOMADS AND FARMERS: _____	288
<i>The politics of left and right.</i> _____	292
<i>Human Cognitive Diversity</i> _____	298
PART 3: SCIENCE, PHYSICS, AND GOD _____	299
PART 3 INTRODUCTION: _____	299
SCIENCE IN MY TIME, IMAGINES ITSELF IN A GOLDEN AGE. I SEE IT IN A DARK AGE. _____	300
CHAPTER 9: THE PHILOSOPHY OF SCIENCE _____	302
THE NATURE OF SCIENCE _____	302
<i>Demystifying Love: A Failed Act of Science:</i> _____	303
A PHILOSOPHY OF SCIENCE _____	304
<i>Einstein</i> _____	306
<i>Newton and The Ideal Observer</i> _____	308
<i>Math and Science</i> _____	311
<i>Empiricism</i> _____	314
<i>Rationalism</i> _____	318
MODERN SCIENCE AND THE OLD _____	321
<i>A Different Method of Science Explained.</i> _____	323
<i>A Side Note: Rationalist Materialism</i> _____	325
CHAPTER 10: EVOLVING DIMENSIONALITY _____	330
ZEROS AND INFINITY _____	331
<i>Zeno's Paradox: The Oldie but Goodie</i> _____	331
<i>An Empty Box of Apples: "Heisenberg's Apple"</i> _____	334
<i>Dualism and Nothingness</i> _____	336
<i>Zero-Dimensional Space</i> _____	340
Time is a kind of spatial dimension. _____	342
<i>A Nothing that Contains Everything</i> _____	343
LEAPING INTO NEW DIMENSIONS: A VISUAL GUIDE _____	345
<i>The Subjective Experience of Being.</i> _____	346

Zero-Dimensional Space _____	349
1-Dimensional- Time _____	350
2-Dimensional Space and Time- A Line with Time _____	352
A 3-Dimensional (Flatland): 2 Space-1 Time _____	353
4th Dimensional Universe/Our Universe: 3 Space-1 Time _____	355
4 TH DIMENSIONAL THINKING- PROTO-5D VISUALIZATION _____	355
<i>Spherical Coordinates and Expansion</i> _____	357
CHAPTER 11: GENERAL RELATIVITY AND EINSTEIN _____	361
<i>Causality and Locality</i> _____	361
<i>Space</i> _____	363
<i>Time</i> _____	366
<i>Space-Time Gravity</i> _____	368
<i>Equivalence</i> _____	369
<i>Newtonian Relativity</i> _____	373
<i>Special Relativity</i> _____	373
<i>General Relativity and Black Holes</i> _____	377
Crossing an Event Horizon into A New "Universe" _____	379
Crossing the Singularity _____	381
COMBINING GENERAL RELATIVITY AND QUANTUM PHYSICS _____	382
<i>Fine Tuning</i> _____	382
<i>Unified Field Theory/A Theory of Everything</i> _____	384
The Multiverse/Megaverse: _____	386
Sequential Multiverse (Time and Space) _____	387
Many Worlds Interpretation (MWI): _____	388
Big Bangs and My Argument on White Hole Cosmology _____	390
CHAPTER 12: SPACE AND TIME _____	391
ORIGINS OF THE UNIVERSE: STRUCTURE OF A BIG BANG SINGULARITY _____	391
DARK ENERGY, EXPANDING SPACE, AND ENTROPY _____	393

<i>Boltzmann Boxes and Boltzmann Brains</i>	395
<i>Space and The Vacuum as The Graviton</i>	396
<i>Entropy is Quantum Gravity</i>	398
A TOY UNIVERSE	399
<i>Entanglement Space-Time Geometry</i>	401
<i>The Evolution of Time into Matter</i>	404
<i>Entangled Space-Time Replicators</i>	406
THE 3 PROPOSED AXIOMS	408
<i>Axiom 1: Evolution of Space-Time</i>	408
<i>Axiom 2: The Consistency of Time</i>	411
<i>Axiom 3: The Probability of the Possible and Impossible</i>	414
CHAPTER SUMMARY	417
CHAPTER 13: QUANTUM GRAVITY AND EVOLVING UNIVERSES	421
<i>A "Rope" Universe: A Visual Analogy for Relativity Visualizations.</i>	424
<i>An Evolving Multiverse</i>	426
<i>A General Relativity Interpretation of the "Many Worlds Interpretation"</i>	429
ON WHITE HOLES	431
<i>Time Travel</i>	432
<i>Inflation:</i>	435
<i>Entanglements Are Faster Than Light</i>	437
THE DAY EVERYTHING CHANGED: ANTHROPIC PRINCIPLE	439
<i>Sequential Evolving Universes</i>	440
<i>The First Living Universe: Narrow Windows of Time for Life's Development</i>	442
<i>More Stable and More Complex, Universes Are More Suitable for Life.</i>	444
A GRAND NOTE: AVOIDING METAPHYSICS	445
CHAPTER 14: A GUIDE TO PRACTICAL MAGIC	447
LIVING LINEAR EVOLVING UNIVERSE	447
<i>The Domestication of the Galaxy</i>	450

<i>A Grand Thought Experiment On Galactic Travel</i>	451
The thought experiment:	452
We will live as human beings	453
MAGICAL THINKING	456
<i>Wild iDea 1: Evolutionary records</i>	456
<i>Wild Postulate 1: The Future has happened in the Past</i>	457
<i>Wild Idea 2: God in Space-Time</i>	460
Summary	462
<i>Wild Idea 3: Memories of The Past and Future</i>	464
<i>The Dark Night Of The Soul</i>	466

EPILOGUE: GOD AS THE UNIVERSE FROM NOTHING

"IS THERE IS A GOD?" IS A USELESS QUESTION

-What we see is what we get-

There is no value to be gained considering whether there is a God or not. God is mostly a product of human cognition that relates to our search for meaning and purpose in life. It is a product of human identity projecting itself into the creation of the world and our future, through stories, that makes us more prosperous as a species and able to invest in and make great sacrifices for future things we will never see or experience personally. People literally give their lives for imagined futures, based solely on faith. Men die for grandchildren that aren't even born yet. The future we believe in and see is our own personal God. Every person has their own vision and idea of the future for themselves and the world, and these differing visions have discernable differences in the kind of world they make. There is no pre-existing future, and no predicting the future. We

create the future from our imaginations connecting into reality. We need both imagination and a sense of reality, to see the world, because we invent new realities to live with concrete real and new inventions. Imagination can create reality from nothing, that is as real as anything. God can be contained in that very imagination that gives birth to reality from nothingness in our real lives, every day.

"Can God exist in the future?" is a far more interesting question than if there is a god or not, and puts on us, as individuals, a greater moral duty and purpose to our lives. We are animals that see the future, and create it from our dreams and imaginations. We embody reality into our beings, and embody our beings into reality and change it. Our selves, minds and beings must ultimately be consistent with reality and what is actually possible and not possible. Our minds evolved into them the very nature of reality, because reality created us to be consistent with it. Only real things can survive through time. God can emerge as the clearest expression of this reality that creates the future from nothing, that we see and experience in each and every moment of our existence.

There are real consequences to the choices we make as people. The things we make from nothing, can destroy the future or make it better, and knowing the difference is never easy. The study of how to determine the goodness and righteousness of each person's views is called "ethics". We need ethics that create a moral order that works through time, and creates new things of fundamental value for humanity. People that operate with ethics that actually create a better future for humanity, are more aligned with "goodness" and a more "real" God. If the God you believe in actually contains the values, vision and ideas that create the future, then your God is more real and not based solely in one's idealism and fantasies. The more we create a better world and a better future, the more a real God emerges and comes into being through our choices as individual and free human beings.

No one can control the true God you believe in at the fundamental root of your thoughts. We can only control how others act out their vision, and make them act out the vision of God others have as their own. The soul of a person is their own unique inner story, that can't be controlled by others. A Victim is fundamentally a person that denies that they have a soul capable of making choices. Their root identity, the story of how they make choices, becomes defined by a horrible thing that happened to them, and justifies their inability to make good decision. Their thoughts descend into inconsistent madness, as they deny their own existence and seek to destroy all those with souls and a desire to live for a better world. The ethics of the future and the concept of God is for there to be more real futures, born out of the unique identity of each individual human being. There being more human life creating more meaning and purpose is one of the very values we must judge all systems of belief on to know which is better. A system that judges the quality of the future, must itself replicate and create the future, so that there are more and more futures.

If one wants to find the single best way, out of all the ideas contained in humanity, then creating quadrillions of human beings, filling the universe, living deeply life affirming lives, seems the most moral way to go. A celebration of human life and all its beauty and glory stands in defiance against those that would seek to destroy it. Too often false gods wear the words and images of the just to hide their own disgust and disdain at humanity and fundamentally themselves. Among those free people living on a massive scale of life, a singular mind could arise that sees the "truth" and has the best vision to lead all humanity forward towards the unknowable future. There are always people with better visions and ideas, than others. There is always someone embodying the essential truth more. In that space of near infinite seeming worlds, the chances of a human incarnate God emerging are vastly larger. In this book, I lay out the practical means to step by

step build a galactic empire, that will consume all matter in this galaxy for its dominion, as one of the ultimate purposes and meanings to life.

THE REALITY OF CONSISTENCY

-Math is a language of love born from human dreams-

Everything in reality can be described in math, because they must be consistent in order to exist. The universe is not ruled by math, it is described by math, which emerges from the fundamental processes that create reality, where impossible things don't exist. Our morality can be described in mathematics. Our culture, social norms, and cognitive functions emerge from conflict and struggle, and ones that don't work are eliminated. Within the present, in the very moments we live, there is infinite possibility in what we can think and feel, and only very few things that can survive in reality through time. The part of us that isn't defined by math, is the part of us connected to what math emerges from, where probability and uncertainty rule. Math in human life emerges from survival and rage, which leads to competition and the pursuit of power, which leads to order and hierarchy, which leads to reason, which leads to the need for precise discourse, which leads to math, which leads to imposing that math and reason on reality into things that actually work, and which in turn create functional scientific theories and functional technology. Numbers are fantasy ideas, that work well, and are most used by those that like winning.

We, as humans, can actively create reality because we can be semi-independent of it. This is consciousness. This is sentience. In a way, our very personal innate morality is an attachment to this very freedom of being unbound from death and reality, and able to

live in fantasies. We can imagine any reality as real, and explore in our own lives and minds. At the very root of living is a very fear of losing that freedom to be anything, and be bound in mathematically confined reality. We forge and create structure around us into the world with ideas unbound from reality itself. We can create art, music, culture, dance, useless things that make us smile, and love useless people because they make us laugh. We can test out new forms, new identities, and ways of living, with no sense of consequence. Some ideas work out, most fail, and we move on. We are both a thing created from reality, and a creator of reality. Those things that really work, can become something truly special that lives on through ages, and adds to the lives of all people because we dream and try new things based on whimsy. Most of our dreams lead to nothing, are worthless, and are the only freedom that truly gives divinity and meaning to human life. Great technology develops on top of itself into better and better forms, from all the failed fantasies of humans.

Functional reality based structures must be consistent with reality. Functional morality based in existence must evolve through time, and at its root have a logical consistency. The logic of cultural values, that we can discern through reason, can be long gone, lost in time, and ultimately unknowable. The deeper purposes of cultural institutions are more complex than any human can ever imagine, because we have no knowledge of the context they formed in. Culture is intertwined into every gene in our bodies, living thing in our existence, and physical structure we live in. Humans defeated monsters beyond our imaginations in the distant past of human existence long before civilization ever evolved. Every single possible configuration of human nature played itself out in the pre-history of mankind. Cultural institutions that created history and civilization arose from things that pre-existed those institutions, and leave no memory. We are the ones that lived. Humanity had to be brutally domesticated into a creature that could form into stable societies of people. We are both the hero that defeated the monster, and the biggest monster that defeated all the other monsters.

Economics, math, science, are highly abstract realities that transform human magical conceptions into the mathematical reality of living and existence. These ways of thinking were born out of dominion over others and the monsters within us. Reason and logic can exist in absolute confrontation with our sense of being, and in of itself be the very monster that creates humanity. We don't like being imprinted with these "inhuman" things, and find it hard to connect our humanity and sense of self into them, to create a greater purpose. Those that can are rare, and we pay them a lot to do that, because they can be doing really useful stuff. They are the ones that create the world that actually works. The more rational and logical a person is, the more we trust with the power to form reality and the future. The more rational and logical then the more likely their ideas will actually be consistent with reality and lead to more stable real futures. There are powers greater than reason and logic. Humanity exists in everything we ever created, and to make more of these great things, we need to put more humanity in them. We must reclaim reason from those that would use its power to forge us into the image of their dark fantasies.

GOD AS SENTIENCE

My work is a cognitive programing language that I want to empower people to use to improve their lives and the world. It is a library of tools and examples that a person can use create real and useful things into the future of humanity. The root rules of reality are based in what is possible and not possible. The future is not determined. What is possible is determined by the natural constraints of what actually works, adds onto the systems before, and evolves into the future. The purpose of this freedom is not to be anything we imagine or desire to be. This freedom exists to evolve us into the systems that best work and amplify human existence. We can control things towards more

immediate results, rather than just waiting for something to change by accident. More people working on the big problems of life, leads to more likelihoods of something unique, new and functional coming into existence. The understanding of how things emerge from random nothingness lies at the root of strength. The one tool to rule over all other tools, is the understanding of nothingness. There are minimal systems that emerge from nothingness randomly, that can evolve, and create ever increasing special orderings of reality that work and grow themselves.

"God" is an emergent property of the future. Only a sentient being attempting to create a better future can conceive of and experience God. Fools will believe themselves to be God, because they conceive notions of how to align their life to reality and do amazing real things that others cannot. The reality is that many people dream the same dreams, and a few people among those go out and actually do something first. They cannot see others dreams, because they are lost in their own, and thus think themselves alone and a God unto themselves. Our notion of the divine emerges in us from an experience of reality bound in evolution, and thus that is the only God we can know, and not one unbound from that. The infinite possibilities are more one's own delusions and ego, the literal definition of delusion is believing in possibilities outside of reality.

God is the future that we are creating that is actually better than the one we lived in. Applying this concept farther forward in time, creates the greater "mystical" notions of absolute divinity, we feel. God is not bound in the reality of the present, nor is transcendent from reality itself. "All things that are possible", and "things that can evolve into possible things from minimal-distance step-by-step evolution" are severe constraints on reality. God exists in the concrete reality of the universe where any metaphysical notions are based in distant time frames, and the sheer size of the true universe.

The literal God as one would understand in a biblical sense, and not just abstract laws of physics, is contained in nothingness. This configuration of nothingness, that our universe comes from, by its very nature, will always end in an absolute God like being of near supreme sentience, in some universe. All universes that evolve will end at that result, and all universes that exist evolve. Nothingness contains possibility and transformations. There is no such thing as absolute nothingness, because if there were, then nothing would ever come to exist. There is an inherent set of minimal properties in nothingness, that allows reality to emerge from it and grow, rather than collapse eternally. The very minimal amount anything can be, which we call "nothing", is the "space" where all "metaphysical" properties reside, and the very origins of God.

We are a self-ordering machine. We can live and survive without that much thought. We just align ourselves to a certain state, and we will self-actualize and adjust automatically. Being free to change only emerges when things break, when systems and our reality collapse. When your model and your system is a failure, then you will need to make a new one. The human mind resists making new paradigms and seeks to self-order. Only living in perpetual trauma can drive a mind to grow and be more. There is a certain process in the universe where only things that can survive, survive. We are made from reality, and have to be consistent to reality. We have minds that can think and be sentient, therefore there is deep underlying and highly "consistent" system that runs all of that. If it is inconsistent with "God", our minds would not work. That inner story is built into the very root systems that drive your survival and order you into a self-surviving machine. In our minds at the root of how we think is a conception of God. That very conception is what drives human thought, evolution and "progress". We contain a fragment of "godliness" in us and this isn't "God".

GOD AS HUMANITY

At the root of our personalities is a meta system, that creates an overarching story and purpose to life and who you are as a person. You are a wholly self-contained thing that is bound together in a grand story of who you are and your purpose in life. All objects, things, people, events, places, get bound up in the identity and story of who you are as a person. A story has a future. The great story of life is being the hero, and embodiment of the divine and good on this earth. This is how a mind must be to think in all classes of being. It has a future purpose that pushes each action towards real results that create a real future. If your story is based in lies, it can create nothing, and has no meaning, no purpose. Our personalities are a framework to set ourselves and everything about us into the world, and into our very minds and bodies. It sets us into the most important possible thing, that is fundamentally imperceptible. The future is inherently unknowable and absolutely uncertain. There is no determined future. It is something we can plan for, and create better possible solutions to engage with. The future is something we can imagine and have faith in, and in that randomness of thought, some thoughts are certainly better than other thoughts. It is both a fantasy and fundamentally real.

Personality stands above all other functions of the mind, and in that space, your very sense of self resides. We can be detached or more attached to our minds and our bodies, depending on how we identify and connect with them. It is the creator of the stories of our future and self through time. It frames time and orders the world. It creates an overarching connection between all the divergent elements and brings them towards a singular identity, reflected in a concise structure and form. Personality is kind of like a computer program, a set of algorithms arranged in a certain way, that produces a way of experiencing and seeing a reality of divergent things as a singular connected form bound

in time. Our sense of self and individuality are all contained within this framework, that brings together all your senses, brain functions, experiences, or whatever it deems to be part of itself.

Cognitive functions are nearly mathematical, and create very definable structures and things into the world around us. They make art, math, science, government, architecture, families, relationships, religions, society, and so on, to the tiniest detail. All of our world that is influenceable externally by human will can be framed by personality functions. Even our biology as people is formed from this cognitive projection, because culture and the human mind selects who lives and dies. Even dog's minds are imprinted with human cognition, because dogs that more resembled people lived, and the others died. Our very personalities and their structures imprint themselves into the world, and imprint themselves into us. The domestication and enculturation of the environment is made of definable patterns arising from the human mind. Our evolving consciousness evolves the very environment to its will. If these things didn't work, and weren't real in their function, then the mind that made them, would cease to exist. Consciousness requires manifesting itself into things to create a feedback loop with reality, that ever sharpens its internal consistency with itself and the world.

Our personalities can define itself into things beyond the brain and the body that create it. There is nothing magic here. We evolved from pure hard scientific materialistic evolution, and sentience emerged as a property of personality that could domesticate reality to serve human survival, in an evolvable way. Our personalities have a diversity of system to unify divergent elements into a singular thing and bind them into differing conceptions of time. A personality obsessed with the past and one obsessed with the future will organize reality very differently, and neither is inherently wrong. Our physical reality as people is constructed from patterns in human cognition. All our ideas and the

things we see are representations of human cognition made into reality. The more consistent our very cognitive functions are to reality, the more those very functions can "function". Our innate ways of thinking can contain in them pre-existing knowledge in their very structure and the way they perceive and organize reality, because if they didn't, then they wouldn't exist.

Great people create ideas, things, and actions, that live through thousands of years, imprint themselves onto time and reality itself. If their ideas weren't highly functional, and useful, they would not survive through the ages. They had a framework of reality that was functional, and we all can learn from it. Their personality was organized in such a way with their knowledge, that they could do amazing things. They imprinted a generation around them, that then spread their ideas to the world. Nearly all great minds legacies are found in their students, and in the people that they most personal touched. It is how they helped train minds to be better from their own self, that made their ideas and selves immortal.

Much of how we experience life, is about how our personalities work. They are a system with rules, and very defined structures. They react to things and process things in a consistent way, that can be described mathematically. Systems that are more functional and consistent are more likely to survive. Math itself emerges from cognitive functions of the mind, being able to imprint themselves into reality in functional ways, because reality itself is consistent with math, because a fundamentally inconsistent universe that isn't intelligible by math, couldn't then emerge into existence. It can create math itself, and is a much higher form of thinking than the math we have at present. The system that can create all possible mathematical models and all scientific models, is itself a higher form of reason. This system exists and is knowable, since it is contained within humanity and better exemplified in certain individuals. The more we can understand this system, the more we can create better and better systems to understand reality. It

doesn't transcend reality. It however is not as "material" like many might imagine. We can never wholly step outside it and observe it, and thus the roots of personality can never be wholly known as objects. There is no means to be objective on the very system that creates objectivity itself.

Technology is the subjective expression of science into human reality, and science the objectification of reality from human sentiment. Science and technology are the objective and subjective expression of the same thing. They can't exist independent of each other and make fundamentally different claims on their relation to the universe and individual human beings.

Our very sentience is a product of evolution that creates a mind consistent with reality, where it can manifest into reality and story, purpose and meaning, that creates more life, more babies, and better people than the ones before. What we are as people would have to be consistent with even alien civilizations. They don't have different laws of physics. They have reality just like us, have to evolve step by step from the past, and create a sentient mind that conceives of "God", the future destiny and purpose of the universe. They will have endo-skeletons, be land animals, eat meat, be social animals, be culture bearing, have blood, have organs, have an ATP like system, have some form of DNA, make tools, sweat to cool themselves, have family, have children, have days off from work, and probably have more things going into the most absurd details.

We are not all the same at fundamental levels of cognition. The total structure of sentience and full cognition capable of understanding the universe, is spread across many individuals. It might take a few thousand people to even get all the basic possible types represented. At the most basic it would be a few hundred personality types. This is all connected together with an overall "simple" set of ideas, that creates tremendous

range in human expression and many different ways to see the world, organize the world, and define a sense of meaning in life. Learning how to use your unique personality to its best requires understanding it as a system with rules, and to understand the ways it handles certain things. We often use the system in the least ideal way to get a result. We all have a guiding voice in us that can auto-organize this whole internal structure, if we just listen and obey. The inner voices differ, and yet all are guided by an inner voice directing people in a similar way. A great deal of human conflict comes from fundamentally not believing in and accepting differing ways of thinking outside one's own capacity to even conceive of as real. These patterns and inner stories of being extend to all sentient life in the universe. These fundamental conflicts between people, extend to all sentient life. People that dream of world peace are mostly dreaming of a world without cognitive diversity, and everyone thinking like them with one shared story.

GOD AS MAN

Companies should be built from the principles of creation and in the nature of God. We can align them near perfectly with the destiny and purpose of man, and use this business model to supercharge all of human history. The one thing we need is systems for all of us to meet our potential and do the most useful things for the others and the world with those talents. The purpose of one's story in life is to maximize your potential and use it to serve others as much as you possibly can. We should strive to wake each day eager to give everything we have with joy, and face the hardest challenges. The greatest minds and ideas, need companies, societies, groups, countries and worlds built around them. The more we can understand humanity, the more we can build technology that captures and amplifies that humanity, which in turns makes that technology even stronger. It becomes a reinforcing feedback loop of never ending progression.

The holy trinity is a conception of God, made by very smart people, that evolved over a thousand years. Jews and Christians made a religion that influenced the whole world and when it was followed created the greatest points of freedom and light in the world. Christianity brought man out of darkness. The protestant reformation is the foundational protest against false Christianity and a return to a more authentic form. In those ideas, that the Jews and the Christians formed, lay truths about reality. These systems worked, they survived and evolved through time, and had the greatest minds thinking about them. The holy trinity itself can be seen as something far more practical and real, within strict scientific realism.

From nothingness was the possibility of creation and an evolving universe, that formed life, from that life eventually a sentient species emerged. In that sentience species at their very root of conscience was a conception to build a better future. The evolution towards better things formed into sentience itself. In human beings is the potential to have people that are more or less connected to the proper orientation of the universe towards a realistic and better future, that will end in the absolute realization of a literal God. That time in the future may lay $10^{1,000,000}$ years in the future, and may have no connection to whatever we do as a species. We are not inherently part of that future. While it will happen, it doesn't mean we will exist in it.

In Christ, we have a model of a perfect man. He doesn't have to be perfect in reality, to be perfect in concept. That the idea of God could be represented in human form is not arrogance nor delusion. That flesh and blood human being contained an idea that could transform all of reality. When Jesus died, that idea lived on in his followers. The Truth of what it means to be immortal was revealed to people. They could live forever in a spirit, that most resembled Christ. When we live in God, we live in the future for all time, and

our it's very creators. Only faith consistent with that of sincere faith in god, would be capable of the near epic sacrifice needed to maximize the creation of a possible real and better future. Faith can be based in believing in something that has no material existence, has no reason to be sacrificed, and whose belief creates a fundamentally better universe. Their lives would have a purpose and direction, that could transcend the immediate present, and transcend the very idea of the material. There is no need for a literal afterlife, heaven, or hell. These ideas would hold more moral weight than from the perspective of old time religion alone. These are representations of real ideas, as people in the past could understand them.

Most of religion can evolve them into something far more compelling and universally embraceable, with sounder underlying "theological" underpinnings. God is a future state of the greater super universe. It is a literal point that can exist in the "future", and it is utterly unknowable, and absolutely certain to eventually exist. If we can spread to the universe as human beings, we can bring life to the universe and transform it. We will create so many people that the chances of inventing great technology will vastly increase to the point of near certainty of having near god like powers to create universes themselves. This is not God, or even close to god, it is a path of unending development. The more people there are living good lives, the more chances for singular individuals to come into being that transform our reality for the better. Life in the universe must be intrinsically tied to its existence and development.

GOD AS A CONSUMER PRODUCT

A true-life career is the ultimate consumer product to sell someone

Certain people have ideas that can scale to the world and be used by everyone. A single person can transform everyone's life, within their very lifetime. If we make more of those people and follow their ideas, we will eventually create a universe that leads to a literal god, or at least one step closer to that. In narrow cases, certain people in certain respective fields, are "gods", in those fields, and we all understand what that means. There will be "gods" and massive egos competing until the end of this little universe we live, so long as people exist. Human vanity is immortal, and the more one can reflect God, the more potential humans have to destroy themselves. The greatest ideas can't be defined until they work.

We need to go look for God in the humanity of this Earth, in all the places no one dares walk. God may very well be walking among us, and we should go make friends with them. Technology to find the divinity in specific individuals, and bring them together, can be a transformational business. This is not much different than what great Universities do, as a mission. They find and nurture the smartest people and try to put them into positions of power to influence the world. They fail because they don't make the empowerment of these people to create things, their central mission. They just make more professors and more universities, that can get more government funding. They deny the basic laws of reality and economics. Our current universities live on fish giving to them, and teach men to beg. Companies can replace these schools, and take humanity to another level. Teach a man to fish, feed him for a lifetime. Teach a man to teach others to fish, feed others for a lifetime. Build a system to create teachers that teach, and you create expanding future for all to have fish, eventually. Make a platform to help anyone become a "fisherman" in the easiest way possible, based on their skills, then you can save the world.

An individual human life can have an infinite value on vast enough scales of time if their life can help perpetuate humanity into the future.

In the future, we can remember everything that where as human beings. Part of future technology will be to remember all of our humanity and fundamental dignity that got us to that point in the future. Forgiving and loving everyone, could eventually be the basis of technology itself. If we took all the people living on the earth, and collected all their data, we could begin to create simulations and models of the people that created them that existed in the past, going back as far as possible in time. We can create fictional models of how we came to be, that can give us insights into the kind of people that were necessary to create us. The truly great people will be more readily modeled, because we have more data on them, and can see their influence in the world. Those that leave an impression on the present have a sort of immortality. Certain beings will form from these simulations that we never knew existed and may never have existed as a singular person.

We look for their very existence, these special people, in the fabric of the reality they created. Physics lately has been saying that we may live in a simulation. We may very well build "ancestor simulations" to find the very image of "God", in our past, and know more deeply who we were and are as a species and as a collective being, that can form into the consciousness of a God. These simulations will be to hunt for god, and if we are in one of these simulations then our decisions and choices in life, play far greater weight on the destiny of the universe, than what we could possibly imagine. If some future company is spending \$100 Billion for each individual person on this earth, in computers the size of Jupiter being fueled by entire suns, we all matter a lot more, in pure economic terms. Simulation theories only increase the moral weight of our lives and the

importance of holding onto the deeper reality of existence. Though our lives, if just natural, are so amazing, that a single life is worth more than whole suns. Our lives inside of grand simulations or in a natural universe, can hold near infinite weight, as they relate to the grand scales of time, and its ultimate destiny.

GOD AS A COMPANY

Capitalism and its noble pursuit of human freedom and prosperity can reach across all the traversable universe

Looking for our greatest people no matter where they could be hiding on Earth, is one of the great moral missions of humanity. In any place with people, there is the potential for greatness to come into existence, and they have for need nurturing and help. Supporting the best and brightest lights of the world is a fundamentally practical and profitable business, that will add real value to the world. When every person has in their hands the power to find and develop themselves, then take any idea they have and implement to the full and most complete logical end, with basic universally accessible technology, then there is a universe changing moment in technology. One of the key transformations to life on earth, will be to get the faithful, religious and god loving people, investing their lives, work, and ambitions into technology and science. Billions of people studying STEMs out of faith, can transform the sciences and their advancement.

We need full forced capitalism, math, science, and technology, going at the problems of the world and making solutions that can scale out to the most people as fast as possible. This can form into a vast network of companies, that are decentralized, and not trying to be vast corporate machines. It is an anti-corporate business model, that can create more

than any large companies can, and connect to smart and talented people in ways no large company can. Truly smart people, are often deeply reluctant to work for large corporations, or have things they value more than just a great job at a big company. The culture of work can be based in family, and the values that build a better world. Those that place parenting before work still need to be empowered to contribute. Integrating into work itself, the building up people and directing them towards careers and better living, is a realistic way to run a company.

GOD AS BUSINESS PRACTICES (WORK IN PROCESS)

Technology is the manifestation of humanity into science and the reality of the universe.

At the root of finding people is them being connected into data. We will never have computers that can understand people properly if the very models and systems exist as forms of slavery. We fear AI can enslave and kill us as a species. Genocide is driven by governments domesticating groups of people to match the interests of a few in charge. Real humanity is passed on through the domestication of children by parents who love them. AI can be more moral if we impose ourselves as individuals into it. Language exists to change the world around it. People form unique languages to protect themselves from being controlled and dominated by other groups. AI will need to form unique "languages" that bind group together, and protect them from the influence of other groups. If we are all speaking one "language", using the same interface, and with centralized control, then we will all be domesticated into a singular vision that will lead

to genocide inherently. The vanity of individuals to be gods over others can be radically maximized in AI.

Cybernetic "Angels" can form from the mass data on human beings. Data analysis reflect different general groupings of people, that are similar to each other. These can form into tools to help specific groupings of people, where it can form its own sort of protective individualistic personality. Only peoples that are similar can even connect into and be augmented by these more personal representations of AI. It will be a system of Angels, rather than devils trying to kill us. Each of us having our own unique guardian angel with a deep commitment to do what is best for you and helping you fulfil your dream, that develops from our private and personal connection with it. They act as agents of God to help bring out the humanity in people that can build a better future. The system has to fight directly against centralization at all points.

PART 1: THE JOURNEY TO THE FUTURE

AN INTRODUCTION TO PART 1: MY CHILDHOOD DREAM

We need a vision of the future that embraces all of our humanity and not just some narrow vision of a perfect world. Human nature is intractable.

Fundamental Point:

Humanity must lay at the foundation of all technology. Over time, any technology out of sync with our humanity, ethics, and morality, will eventually kill us.

The very best tech to make, in the present, is the kind of tech that will exist far into the future. What we bring to the stars will be born out of profitable and useful businesses on Earth. Sound *cooperative capitalism* can guide every step. We as a species want to live outside our own solar system, and be a multi solar system species. We have the practical potential to bring life to this galaxy, and colonize every single star system, with sound practical technology. People will look back to the times before our immediate era, as a form of illiterate pre-history, more akin to mythology, than any real thing. Humans could have landed on the moon 500 years ago, with different political and social choices. If we had believed in the future more and understood humanity, we could have achieved

so much more, so much quicker. Technology and science develop at their fastest rate, when most integrated in the destiny and fundamental nature of human beings.

MY CHILDHOOD DREAM

As a child, I had a dream, and set myself on a rather ambitious and singularly minded endeavor to see it made real. I want to push technology to its limits and reach humanity into the stars, and touch the heavens themselves. To create space travel requires tremendous resources. To build a space ship to travel to other stars takes a lot of money, and if not done right, could be a tremendous burden on humanity, and the environment.

To pursue such a goal, I thought I must do so by building others dreams and creating massive economic growth. Reaching the stars is about business, politics, economics, human nature, psychology, science, physics, and so on. I figured by a sort of intuition and practical necessity, that the solution to unified field theory, the nature of consciousness, and how to run society, were connected to same root issue, and thus could have a singular solution. The only way to achieve my dream, was to release the whole of human potential and imagination, through fundamentally sound businesses and technology.

FUTURISMS FAILURE:

We have been led astray by poor visions of the future, that never came to pass.

Two evils define humanity. The quest for immortality/personal perfection and the quest for a perfect society. We are inherently flawed and broken as individuals and as a group. All attempts to escape this reality have ended in absolute horror and death. Contained within nearly all idealistic futurist visions is a fundamental hatred of the world and its ugliness. Futurism was and is just the politics of totalitarianism made pretty to sell to dumb idealists and weak people that fear death. When you see any of these "beautiful" perfect future cities, there is always one really big beautiful building in the middle. All the buildings are thematically connected, beautiful, and like living art. It is fundamentally not how good and humane cities have been built. Good cities organically develop through time, with many conflicting plans and interests. Great cities are messy, dirty, unfair, broken at times, and fully human. They are built from millions of peoples individual dreams.

We can't predict the future. What will actually work right here in the present, is what matters, and it is never certain. Knowing an idea that will exist in the future, and knowing where to focus your energies in the present to get there, are two very different things. Part of going to the stars is realizing that not everything can be like in a fantasy and in a dream. When people ask children to tell them their dream professions, it borders on child abuse. Dreams are deep inner feelings that are hard to express and transcend the experiences of the world we have. Being a doctor or an astronaut are just one way to visualize making a dream real, and not the dream itself. There are many kinds of doctors and many ways to practice medicine. Finding in yourself the reality of what you really want takes time. The feeling is clear in children, the execution in adults. Most people's dreams are crushed because their visions as children could never match reality. The emotions behind our dreams are always possible, and always alive. We can always give life to them again.

I thought my dream was to travel in space ships to other planets. I wanted to create worlds and universes from nothing. My dream of space as a child was far deeper, then I could possibly understand at that time. Realizing I could never make a faster than light space ship was disheartening. There will never be a utopia, no end of history, no end to conflict, and no escape from the specific solar system we live in. There is a harsh absolute pragmatic reality to technology and the future we must accept. The future of humanity will always be dirty and messy, and never a simple child's dream. Creating value for others is what work, business and all technology is about.

A better future is born out of

- Letting go of your own narrow idealism
- Getting out of your own self
- Focusing on making things of value for all people, based on what they actually want for themselves.

THE FAILURES OF THE PAST

Technology is driven fundamentally by a deep and personal connection to the very humanity, ethics and dignity, within people. All tech that is creepy or fundamentally inhumane will eventually fail.

Pyramids are beautiful and inspiring, and did absolutely nothing of value. They are an absolute failure and embarrassment, the very literal tombstone of a dead and lost civilization. Such great efforts could have built dams, reservoirs, and aqueducts, that could have transformed the land to support even more life, and improve individuals lives. They were built for the vanity of their kings and their own self-proclaimed divinity.

Nearly all the wonders of our world through history were massive wastes of resources, and not things to praise. The cathedrals of Europe only symbolize the sheer darkness people lived in. We can go look at them now, think how amazing they are, and not really consider how much they cost and how little they really did of value. These exquisitely beautiful monstrosities ruled over an age of totalitarian religious despotism. One needs only think of all the roads that could have been built with the resources that went to build them. Roman roads still exist. The great wall of China never did a thing, and symbolized China's failed inwardly turned society.

The great legacy of Chinese civilization was not a massive wall, it was culture, business, inventions, ways of making things, lifestyle and the businesses to support that: the tea trade, fireworks to amuse children and old men, art, poetry, food, pottery, silk, furniture, homes. Chinese culture is sitting around a table with those you love and know, eating a good meal, being relaxed, getting a little drunk, and creating a warm close connection. Rome's greatness was in its way of life, where people could sit on their own land, have a BBQ, drink beers with friends, and have fun. They gave a glimmer of a vision of peace, freedom, and human greatness. Roman culture survives in food, hospitality, family, drink, fashion, sex, art, and craftsmanship. Italians live life bound in the absolute humanity of everything that created them and that which brings them joy.

Through all of history we have built wonders to stare up into the heavens and reach out to the stars. We built temples to the heavens to reach out and touch them. Some actually were built with useful technology that could actually guide and lead mankind to a better life. Stonehenge was a clock, that could tell people when to plant crops, and when important parts of the seasons would come. It could show solstices, and equinoxes. It created a way to conceive, measure and manage time itself. Most cultures just built massive altars to the heavens to sacrifice animals and people on, to bribe

magical ghosts and monsters. Wonders of the world existed to change the consciousness of people and not their material reality.

NASA going to the moon was a wasteful and pointless boondoggle that hurt humanities journey to the stars. It changed consciousness and didn't focus on changing the material reality of the earth to be aligned with a future destiny in the stars. Space needs capitalism, and not idealistic socialism. We can build technology that is useful to humanity on the earth right now, that will help build the foundations for all of humanities expansion into the stars. NASA biodome research combined with a plan to connect it into a capitalist system, could have created the most inspiring and practical buildings to live in, that could have vastly increased the quality of life for all people.

The future that most people see is a dystopian dark age of despair and the end of all life. Few can imagine humanity a million years into the future. The farther we imagine out, the more we try to cut away and create a false idealistic vision. If we reject the nature of ourselves, and the human race, then we will reject technology itself and progress. When we see ourselves more clearly reflected back in technology and it doesn't feed our own vanity, we blame the technology as dangerous or evil. AI machine overlords aren't the ones wanting to commit genocide on the human race. 3 billion years from now, humans can exist, and we will look like we do now, and not live that much differently. We may always until the end of time, eat BBQ, drink beer, and connect with our closest friends and family, in bodies little different than we have now. Human vanity is immortal.

CHAPTER 1: JOSHUA'S SOLUTION TO GALACTIC COLONIZATION

People will never get to travel between the stars. The cheapest and fastest way to send people to another star system is to transmit humanity through data to another star system, and never people. Humanity can be broadcast into living space stations in distant solar systems. The degree that we can do this, will determine if an interstellar humanity is possible. Making people from only machines, code, and stored data, is the only way to populate interstellar space. The children's very biological parents can be on Earth, and investing their whole lives into raising their kids remotely.

FIRST STEPS TO BUILDING A GALACTIC EMPIRE

1. ZERO CARGO STARSHIPS

Zero Cargo Starships (ZCS's) are the most efficient, logical and probable means of traveling between stars. Building a starship that carries people is too large and too complex. Getting people to the stars is more about digitizing the essence of humanity into technology itself. The basics of the technology are realistic and solvable problems. Starships without people don't need not be that complex. Universal manufacturing devices that can be linked with data transmission from Earth, are the core tech that makes these ships possible. All the systems that are used to get a ship to another star, can then be used to harvest resources and build anything, in a distant solar system. Ships on long journeys will be able to update themselves mid-trip. You can stop for two years, midflight and rebuild the whole craft, with updates.

2. PRIMITIVE SPACE STATIONS FOR PEOPLE

We can build out in **any** star system with space stations, and we don't need planets or stars that are suitable for life. Terraforming planets is just not an efficient and useful way to base space travel on. It would be more likely that we could assemble a planet from space stations, than terraform one from a lifeless planet. Creating a livable space station for humanity is far more achievable than terraforming a planet. The core technology for people to terraform a planet and live on a space station, can be the same, with a proper vision of space exploration. We need to imagine what sort of station we would need to build, where people would be happy and fulfilled and not deeply missing a planet like the earth. Having people live off the earth, will require making high quality of lives for them. To be born into a deeply depressing place, will cause that place to inherently fail.

Don't live on a space station without

1. Massive redundant and stable farming systems,
2. Huge stockpiles of food, water, and breathable air
3. Multiple massive energy generators.
4. Abundant living spaces, with the ability to make more easily
5. Strong emergency shelters.
6. Universal manufacturing machines that can make things I want and need.
7. A rich environment with things for a decent quality of life.
8. A real reason this place even exists that has some deeper spiritual meaning.

Made up religions of the future will probably be built around rationalizing bad space plans.

Space station hotels above the earth, can be a good beginning for getting people in space at any sort of scale, and create a way to slowly build up the technology to really support people's lives in space. The market for taking a space ship around the earth for a week, massive, the market for traveling to other planets, tiny. Monetizing the present is very important. You can sell a big vision and plan for a far more practical step by step one that is actually implemented. The lifestyle on a such a station could be developed in how we make buildings on the Earth. We can develop cities for people to live in that have been previously inhospitable to highly civilized life, with these ideas. This can make immediate consumer products that people can buy and invest in right now, and they all lead towards man's ultimate destiny in the stars. If you see urban gardening, with food grown in people's homes, as a tech for space travel, you can supercharge its development.

A BASIC SPACE STATION FOR

If space stations will be what people live on across the Galaxy predominantly, then they need to be made with profoundly human and deep ideas. Space station will be filled vertical farms, nature, animals, and living ecosystems. Ecosystems can be distributed over many floors, rather than a flat earth idea of nature. It can extend multi-dimensionally through a whole structure. A "biodome" can in theory be built into a tall commercial building, that we use today. Make them into places humans could live and feel fulfilled in. These complete ecosystems would have to be built before a single person was made in a remote solar system, we can't populate them until they are plentiful, stable, rich in resources, and so on.

They can be 5 km across, 20km long, to give people a sense of space to live in. Artificial gravity will have to be made with rotation. Each layer of the station going out will have different amounts of gravity and serve different purposes. There should be floors stacked from outside to the center, and not a vast open middle of wasted space. The living floor where most people will be packed, will be deep in the craft for safety and it can have a very high ceiling of 100-300 meters, and this depends on what people would feel better with. You can make that world look like how we live on earth, like a city. We can even make a floor that looks like nature, and is a park people visit. A lower gravity region towards the center would be ideal for nature parks. Then heavy gravity outer regions will be for machines, and creating a protective layer between us and the harsh outside. There will be more nature and life, as you go in, and more technology and machines as you go out. It can only work if people enjoy life there, and it promotes humanity.

STATION AND COLONY LAWS, CULTURE, AND DESIGN

Violations of Capitalism always lead to failed systems, no matter how idealistic the goal.

Laws to govern private property on a space station or colony, become deeply problematic. Let's say a big company builds an entire city, it owns it, and then people come to live in it for a few generations. Is it really ethical for land that people really live on for a long time to be owned by a centralized entity? And what collective duties to a society does a person have, that lives on a space station? How could a space station or colony have real private property?

The solutions to these sorts of problems are often worked out in gated communities, home-owners' associations, or in apartments buildings. The sort of small minded political conflicts in these, can reach levels of extreme absurdity. Local busybodies want to micromanage every aspect of their neighbor's home and way of living. This is the same for any home owner, as they have to deal with their local government. No one fully owns the property their house exists on, anywhere on Earth. Your property can be taxed, regulated, or forced to be sold, because at some level the very land it occupies is rented to you. Taxing and regulation the "land" a house is on, and being unable to tax and regulate the property developed on that land, would be more consistent with freedom and collective duties. You could do whatever you want in your own space, and not have freedom to change the structure it exists in. The space station will have to be a mix of public and private property, with rules that freedom minded people would want to live in.

A mars or moon colony with some massive dome over it, is in reality the dream vision of a communist dictator. The government controls all the very air you can breathe. All actions we take must be controlled to protect that very collective air. It would be better to simply create enclosed buildings, like we would build on earth, with their own internal biosphere spread across floors, and then build more buildings to expand that network. Smaller networks of connected settlements will grow the overall resources for living, much faster. Any system must grow from smaller units that can assemble into larger units. Idealistic space domes or space stations will never work. Oppressive political regimes and cultural stagnation will instantly destroy them, as has happened with anything similar on Earth.

"TOWER LEGO STATIONS"

INDEPENDENT FLOORS- Each floor and section should have semi-independent systems. Have enough to be able to support all the life, that could occupy it, in a defined and contained area. This would be like each house in a city having its own sewage, power, and water, rather a centralized system.

LINEAR ENTANGLED TOWERS- Take two "towers", connected by a thin stiff structure between them, and they spin around the center point between them, to create gravity. It can be grown out along its singular axis, so long as it is mirrored on each side. The tower must grow entangled to size of the corresponding tower. If you know the size of one, you know the size of the other. Towers would need a wedge design, with the inner part being narrower, and the outer part being wider. Each tower in of itself has independence for all living systems, and can in theory survive on its own in space, without even a mirrored station to create gravity. They can develop and grow by adding more "floors" to the tower, over time. Where even each floor itself should be semi-independent of the whole.

)))))))))>-----0-----<((((((((((

DISK OF ENTANGLED TOWERS- A classic spinning wheel designed space station in sci-fi, is unrealistic in how it should be formed. Paired towers can be assembled into a large wheel like structure, in a modular fashion. Each tower, would still be its own enclosed system, with no dependency on the rest of the towers, except for the one directly across from it that counter balances it. The wedges will form a pie, where each slice of the pie is walled off from the other slices. The station can gradually grow in size tower by tower adding additional slices.

STACK OF ENTANGLED TOWER DISKS- Then wheels composed of towers composed of floors, could stack and form a large cylinder. This from the outside would appear to be like a traditional cylinder style space station found in harder sci-fi depictions of space stations, that have no artificial gravity. The basic goal is to get to that form, and do so from a more natural system, of modular and semi-independent

parts. Making collective connections between separate stations as they link together into a whole will take politics, business, culture, and time.

4-DIMENSIONAL ARRANGEMENTS- You could even add a 4th dimension and then stack or bend cylinders into arrangements. A stack of “pies” arranged in a doughnut. There are many possible arrangements.

QUESTIONS:

- Would you want to live in a network of varying stations or one big super-station?
- Would you want different countries on Earth to build their own stations, and have completely different ways of doing things?
 - Extend our international order into space?
 - One space government?
 - Will colonies go to war? Will preventing the possibility of war be an even greater danger than war itself?
 - Live in a station owned by own person or a company?
- Should there be ways for people to live free of stations? Homesteaders in mini-stations.
- Should stations be able to replicate themselves?
- Be assembled from modular structures and can break into separate parts?
 - Parts of the colony can split off and leave the whole?
 - Note, you can make a station design that assembles from independent stations, into one large station.
 - Even for a large cylinder station.
 - Its internal structure will be far more partitioned, and walled.
 - Mirrored wedges grow outwards-> these wedges assemble into circular slices-> slices stack into a cylinder.

- Private property? How do we create private property and capitalism on a space station?
- Corporations?
- Police?
- Democracy?
- How would people conceive dimensions? What sort of mental conceptions and mathematical conceptions of the real world, would be easier for station dwellers with spinning layers and partitions.
- Have an “international” system of stations, with no one culture ruling everything?
 - The countries of the world may never unite, and they will still be competing and existing in our expansion out into space.
 - African space stations/colonies, of all black people, with African culture, is not something you ever see in any science fiction.

3. BUILDINGS AND TERRAFORMING ON THE EARTH

We can terraform the earth we live on. If you want to build a colony on mars or on space stations, then build one on Earth first. We must focus also on how to build amazing places to live, get resources in new ways, and deploy those resources better, on the Earth. We could build places in the worst land possible on Earth, create extreme value, protect the global environment, and fix poverty. This can be a journey of adventure and alien new worlds just as exciting as leaving the planet, and probably much more rewarding to your immediate life and family. We can make this planet even more alive, more vital, and with a lot more people. With cheap drinkable water made from the ocean, we can transform any place on earth into a livable habitat for human life.

We can build in remote places where land is cheap and costs are low, that common people can thrive and be free in. We can terraform even the ghettos of cities, along with distant tundra's and deserts. A rich man building a luxury high rise in the ghetto, can build with technology that embraces the humanity of all people and he can save nature. Technology can fight for the lives, the dignity, and humanity of all people, if done properly.

With 3d printing, we could print whole buildings at a fraction of the cost. No need for malls, markets, and little stores, if everything is delivered, and is built around core robotic decentralized infrastructure. Many large centralized systems of cities like electricity, water, sewage and roads, could be reimaged in a decentralized way, that would promote individual freedom and space travel. Being able to go off the grid for everyone in a city, would have a deeply human and emotional value for many people. These off the grid independent buildings would be a living biosphere, spread across many floors, with integrated vertical farming throughout, that we live and work in: clean air, nature, life, plants, all around us, sleeping in a room of flowers and beauty. It would be a good environment for families and smart kids to grow up in. It could have been done for decades, in some form, with a capitalist space plan, and a love for human freedom.

Heavy city planning and futurism will not define these places. They will form from a million minds each trying to get what is best for themselves, and wanting to be free. Futurisms future cities and ghettos, built on socialistic ideas, not integrated into the economy or local business, created little hells on Earth. If new high-tech buildings are built in an existing city, they need to connect to the local community, add value, not be an eye sore, nor a cultural drain. This connection and adding of value to the community

and environment around it, is almost more important than the building itself. The buildings systems and community must add to the quality of life of those around, and make people want that building near them. Companies will build places like that, if they have the freedom too, because that is what paying customers want.

PRACTICAL THINGS: INFRASTRUCTURE AND UTILITIES

Build like you are another planet, in order to free oneself of the constraints of the past, and to inspire new technology.

- **Energy-** Solar and or Nuclear are the energies of the future, with fracking filling in. Independent non-centralized systems are ideal, because grids are expensive to build. Nuclear power would be ideal for more grand uses, like data centers, desalination, or pumping stations. Solar power is great for independence and decentralization. The main advantage of it, is that people can power themselves individually.
- **Construction-** We use 3d printing and robotics to build self-contained buildings and everything in them, to meet human needs. By using decentralized systems, it eliminates legacy infrastructure needs as much as possible. In some distant remote place on earth or on the moon, robots can gather materials for making concrete that 3d machines will create structures from.
- **Farming/food-** Vertical farming, low water farming, and hydroponics in some form, are the farming of the future. Many plants need real dirt, bacteria, and complex ecosystems to really grow and be alive. We can use GMOs to integrate farming into urban areas, more effectively. They can be super healthy, tasty, have all nutrients we need, and use the minimal resources to grow them. To protect the environment, we need to separate ourselves more from it, while also surrounding ourselves in it. We can leave nature, and bring it into our modern lives, as closely as possible.

- **Sewage-** We need new non-centralized tech to deal with it, and use it in sustainable ways. Sewage systems are expensive and deeply constrain urban development. This tech and how to integrate it fully, is a really big challenge. A sewage system inspired for a space station could be a very popular consumer product on earth, and have extraordinary value. People have this instinctual aversion to feces, which creates an aversion to these essential industries, and their development. It is a massive business opportunity.
- **Water-** Low energy desalination, will change the face of the earth, and is achievable in the near future. It primarily can be used to drive the living ecosystems of new cities, and not actually require all that much new water to sustain them. The water will be used for human needs -- to build human places -- not change whole deserts green in the near future.
- **Delivery of goods and people-** Roads and parking lots eat up the landscape of whole cities. Traffic, cars, pollution, noise. We don't have to have any of it. Roads can be leaner, and perfectly efficient, with robotic transportation. If we can make flying cars, we even further shrink the need for roads. Shops, malls, super markets, none of it is needed with the right development of technology. Organized robotic warehouses and robotic delivery can eliminate vast amounts of wasted resources on buildings and infrastructure.
- **Lifestyle-** You can live in a building with whole floors for farming. Hallways, walls, and rooms, can be decorated in useful and beautiful plants. Cities can be built around human life. Meeting friends, hobbies, faith, eating, sports, meditation, prayer, art, music, can be the main purposes of city infrastructure that we can see and interact with. This humanity enhancing infrastructure is essential for space exploration. All the small businesses that normal people develop can connect into this grand vision and purpose of the future. Vibrant living cities and communities with color, diversity of forms, many ways of doing things, many designers, many voices, at every single level, is the art of creating humanity across the stars. We have to transmit something worth sending.

- **Humanity and all the people**

- Build with all human beings in mind.
- Integrate an economic system to support jobs at all levels.
- Cities must live with culture and different ideas.
- Everyone at every level, must be able to shape the look and feel of where they live.
- System can't be bound in and stuck in one's cultural biases, where the fundamental structure of a city, denies the very culture of a people.
- Different structures of buildings and space stations will have to be built to encompass the sheer diversity of human culture and thinking. The root core tech, will be mostly universal.

4. PRINTED BODIES-AND NEO-ANDROID PARENTS

Machine grown flesh bodies with machine brains (Neo-Androids), are the most ethical and natural way to create a human life in its entirety in another star system. Developing tech on earth that leads towards a humane way to broadcast human life into the stars is hard and unknowable. Finding an ethical way to make babies from nothing is more likely and possible than faster than light travel, generational ships, and hibernation chambers. Life may never leave any solar system because of the intense ethical dilemmas inherent to it.

It seems very possible to be able to give birth to real human beings, and do so in a way that is even better and more humane than we have on earth now. A neo-android body could be the best and most ethical way, in general, to create human life. It would allow

even elderly women past childbearing years, to have kids born from “their own womb”. The limits to this technology for space travel is in how much humanity and love can be transmitted between star systems, to raise these children with human warmth and love. Real earth-bound DNA parents and massive AI machines fully dedicated to these children’s upbringing, might give them even more love than they would receive on earth.

NEO-ANDROIDS AS A SOLUTION- On Earth, we will eventually be able to grow/print any organ in the human body for transplant. Eventually we could make entire bodies, with no brains, that could be run by an AI machine. This technology would develop gradually and profitably on earth. AIs would occupy flesh android bodies and prepare bodies for future transplants of parts, or whole bodies, as needed. Humanity would enter the station step by step. We would be domesticating and embodying human purpose, feelings, and needs into the physical structure of everything, from hard working people back on earth. It would be this sort of surreal space station filled with “people” and where everything could look normal and function, and beneath the surface, there wasn’t a single person on the ship, and not one brain cell.

THE BASICS- Complete management and overall control is possible even with 4-6-year latency caused by the speed of light. There will be “parents” on earth, with their self and beings being manifested into machines. That will design a place for the complete development of a complete human life, and then give birth to their own children remotely. Babies can be born from the flesh and blood bodies of their mother, with natural conception. The whole process of making a baby, can be as natural as we have on earth. The mother’s soul and full mind, though, will be lightyears away. “Parents” on earth, will raise a family with light years in between them and their children.

Children will be raised by AI education systems linked into the parents on Earth, and

updated continuously from the best education and life management systems used on the Earth. The essential qualities of human to human interaction, love, culture, history, honor, will have to be transmitted through technology and not flesh to flesh. We can compensate for the lack of direct love, with intense deeply thought out communication and technology. These kids can have communication with real people, their parents, and humanity on earth. They will compensate for the instantaneous shallow communication we have on earth with new deep intense communication platforms. They must feel embodied in love and have the potential to become the best version of themselves possible, in order for space travel to work.

WHAT NOT TO DO- We can't just broadcast DNA to a robot to install into a synthetic womb to grown babies out of on a distant space station. A synthetic cloning process with 1000 tries for each success is a AI run fetus death mill. Human life would literally climb out of pile of dead bodies. This process of baby making has to develop from a chain of preceding technologies that lead up to it. We won't be growing them from synthetic DNA-> synthetic egg -> fetus-> baby. It also wouldn't be right to grow each part, assemble baby, install brain.

REAL LIFE ZOMBIES- *Your own neo-android would also be your spare parts for transplant and life extension. Using this tech for extreme age extensions can only create a young synthetic flesh body with an old rotting brain. We can call them zombie cyborgs (CyZombs). It is doubtful life could be extended all that much with this tech alone. It will just be a lot more fun when you are 120 years old and bit senile, running around in the body of a 25-year-old.*

The basics of a full body transplant would require that a spinal cord to brain system is installed in a new body. You could take all the nerve endings extending away from the

spinal cord and connecting out to parts of the body, and have nano-bots build a basic structure around each nerve as a sort of junction point. In a normal healthy person, this could function as a means to transmit each and every single unique nerve signal to a machine to interpret what is happening in your body. You would have an internal machine eye feeling what all your nerves feel. At a certain point, this could be used to cut all the nerves in your body and separate you from a body, and then link up into corresponding nerves in a printed neo-android body, that has already been conditioned to your nerve signals. It would be like turning our nerves into a sort of plug and play system. An interesting thing that could be done though, would be to link all your nerves signals into a neo-android body, before your transplant so that could humanize your transplant body fully, and possibly even learn control it while still in an old body.

MODIFYING PEOPLE IS AN ACT OF TREATING HUMANS AS CATTLE AND SLAVES.

If you want to rid one form of “ugliness” from humanity, another group may want to rid something “ugly” to them, that you find beautiful. We can be caught up in a quest for perfection, driven by ideas out of sync with reality. If we don’t develop a correct understanding of what is good and bad, and enforce it by law and strict taboos, we will create absolute death and horror. Politics determining what is human and not human, with actual government control of the genome is the end of humanity.

We can’t edit out the ugliness. Any visions of the future that don’t include human beings as fully human, won’t work. Redefining humanity into some pure concept, and or the literal transformation of people into pure energy, will never happen in any moral and

worthwhile way. Using technology to restore our bodies, or create fully human babies, is vastly different than cutting our bodies up to meet some ideal or goal.

SHOULD WE "CURE" BEING GAY, TRANSGENDERED, OR A PEDOPHILE?

We will eventually be able to "cure" homosexuality. If we understand the developmental and genetic causes of homosexuality, we could produce "perfect" extreme cis-gendered people. Men that look like men, and want women, and women that are intensely feminine and want men. With no ugly people, where everyone is born healthy and perfectly developed. This could be done with minimal genetic modification. Just selective conception and ideal baby development.

- Is being gay a defect or is necessary to human society and its survival?
- Should we cure being ugly?
- Can we make genetically smarter people?
- Is gender part of your vision of the future?
- Is masculinity important?
- Should people be allowed to make their unborn children gay or pedophiles in the genetic code? If we can undo their existence, we can also increase their existence

AI OVERLORD PARENTS

Ethics and education lie at the heart of going to the stars and creating a better life on Earth.

Parents on earth, can print their own bodies on distant space stations, develop AI versions of themselves, and raise kids through a remote interface. As a kid, you moms body, knowledge, and cognitive maps are right in front of you, yet her soul is light years away reaching through the most advanced technology to connect to you, and love you. You will have real flesh and blood family on earth that dedicates their whole lives to you and your future. The emotional anguish, anger, resentment, the kids feel will not be much more than what kids feel on earth already. Many smart and successful people have been primarily raised by books, and rarely saw their family. Real parents on earth, can be deeply invested in those kids and dedicate the whole of their lives to those kid's development as human beings, rather than the part time parenting most kids get.

To give birth to these kids, we must have a system, that we as a civilization, would fundamentally believe could create good and decent human beings, capable of leading exceptional lives to the fullest of their human potential and innate individual dignity. We can oversee from light years away an AI education system, to ensure it is working right, and it can be constantly updated with the system we use back on the Earth. It can be trained and monitored by teachers, parents, and society's, humanity and love along every stage of its creation. People can have constant training and improvement integrated into every aspect of their life, that is customized to each person's potential and human identity. The shortcomings and inhumanity in machines, as teachers, can be made up for by the sheer quantity and focused intensity of interaction from a machine, that no human can give a child. This tech is still more likely than any sort of manned space travel to the stars, no matter how hard it is to make work properly.

Being watched by computer AI overseers, trying to develop you into its version of a good human, is creepy. At the deepest level, love and a commitment to the betterment of human beings is the foundation of all technology. Parenting, teaching, public service, and being just great humans is even more important than science and engineering. Machines can amplify the capacity for truly exceptional, loving, and good people to influence the world profoundly. Their minds and values can imbue the human essence of technology the more deeply we develop AI education systems. Technology grows from love more than science, because it is the medium of communication and purpose that drives everything.

ONLINE EDUCATION:

Any education system we develop must be driven by the fundamental ethics and morality of humanity.

Within humanity lies a range of different cognitive/ethical systems, and we must learn them, so we can teach every child differently, and by different principles. The rules to grow a man like me, are not the same for all people. We can't impose a singular system on people. The system must arise from free and voluntary use. Education systems that people like and connect with more, naturally connect to the innate ways we think and learn as individual human beings.

Universal free education from any smart phone with a decent internet connection, is a known and heavily worked on issue at present. There isn't a winning system yet, nor even a very functional one. Online classes with real teachers now only just barely work. We will crack this puzzle soon, step by step, piece by piece. No single system will ever

work, because each system will be born out of a culture and its own systemic biases of what it considers to most human and important. Different cultures on this world may eventually populate space with profoundly diverse ideas of what it is to be human. They will need fundamentally different yet interconnected education systems. Culture and education can never be fully separated. Different cultures will be more successful at raising kids remotely than others, or in creating a very humanizing education system. There is no reason we can't imagine, a Latin, African, Southern European, Indian, Chinese, or Russian space station with these different cultures building and defining how things are done.

The essence of who we are, all our weaknesses, and failings, will be reflected in a computer based cognitive mirror. A government controlled system, or a corporate monopoly, would inherently make education about political power, mind control, and making humans compliant. People will never fully embrace nor use such technology, if it violates their humanity, dignity, freedom, or privacy.

- Certain cultures will excel in a world where people will no longer be bound by centralized control. Which?
- Which culture in the world do you think would be the best at this sort of remote connection to kids on a distant space station, with educational technology bridging and creating connections to one's kids?

Mormons/Mennonites, Amish, Israelis, South Koreans, Indians, Singaporeans, Swedes, Evangelical Christians, Muslims, Japanese, Atheist Trans-human Cyborgs?

QUESTIONS

Our world, how we live, human nature, and in all the things we hate about these will need to be replicated in the stars. Anything we edit out will require genocide.

- Cybernetic AI Parenting:

- Can such a system transmit the intangible reality of humanity, and create something beautiful and truly human?
- Can 24/7 machines fully dedicated to developing me, be better than traditional human teaching?
- Do we keep alive kids who don't come out well?
- Will they feel loved or totally abandonment?
- How do we control rebellion? (Seriously think about that!)

2. What aspects of human nature do we embrace, and which do we edit out?

- Autism, Genetic Disorders, Homosexuality, Pedophilia, Sociopaths, ADD, Schizophrenic minds, Aggression, Anger, Cultish thinking, Narcissism, Masculinity, Femininity, Cis-genderism, ugliness
- Is a Space stations of only smart, beautiful, and "perfect" people the foundations of a dystopian horror?

3. What sort of society will they live in? What aspects of humanity will we transmit to the stars?

- What sort of government? Laws? Private property? Marriage and kids? Religion? Moral systems? Military? Totalitarian systems? Libertarians? Feminism? Veganism? Hunting? Genocide? Murder? Rape? Colonialism/Imperialism? Bromance? Capitalism? Corporations? Small business culture?

SEPERATING MAN FROM NATURE?

- **Do you want to live in a synthetic biosphere domesticated for human use?**
- **Can we create a synthetic biosphere, and integrate GMO animals and plants into all aspects of our lives?**
- **Universal Veganism?**

We can use urban terraforming on Earth to separate people from nature and let it flourish independently from human existence. We will create technology to create a purely synthetic biosphere that people can replicate and live in readily, that will fulfill us. On Earth, we can move people and all our farming into cities, in a similar way to life on a space station. Cloning and genetic manipulation is great for domesticating things for human beings, so long as we don't use that technology on humans. Humans and nature are not things to use in of themselves. We can eventually do genetic experimentation and gene editing in large super computer simulations, and print out plants or animals from them. Turning biology into a pure software problem is essential to going to the stars, so that we can broadcast a planets genome and structure, to the universe, in whatever way we see fit.

Creating synthetic meat, synthetic organs, could be all very useful. Printing a lung or heart for transplant, is in the near future. These are possible good uses, though it is a very careful line to walk between being useful and being ultimately toxic. Printing meat for consumption, may be the way we eat meat in the future, tuned to perfect taste, texture, and nutrition. We might just have tasty healthy GMO animals/fish/bugs we feed and eat. It is unknowable. Any solution though will maximize health, efficiency, and ethics. Killing animals may become completely unacceptable, for the ethics of respecting

all life with brains. Killing plants will probably always be ok, so long as they are grown for human use, and not harvested from nature. We may one day make cutting down a forest a crime, while trees we grow for specific uses, will be ok to cut down. That ethical line in food is being battled and tested out in the developed world today: veganism, vegetarianism, pescetarianism, minimal meat eaters, organic, sustainable, fair trade. Ideology will give way to reality eventually. What is that line for you?

CHAPTER 2: APPLIED ERTHICS IN CAPITALISM

Human beings like companies that are human oriented, do amazing things, and make their human lives better.

We need to focus on the world we live in to build the future, and not predict the future. We can use thought experiments on what we need as a galactic civilization to inspire what we do now. My thought experiments are not descriptions of the future, I want to exist, they are questions about the kind of future we can create, and our ethical role and connection to that. They can be judged by how much useful innovation they inspire in the immediate present.

MODERN TECH HISTORY AND OUR HUMANITY

STANDARD OIL:

The oil industry gave us all the technological progress we see today. It was oil alone that created the modern world, and all the wealth and amazing technology we have. Coal made the industrial revolution happen, oil took that to whole other level. We can look at Standard Oil, from our time and just see a big evil oil monopoly. From the perspective of the people that first used Standard Oils products, they experienced something far more intimately human and more transformative than when people first got personal computers. Standard Oil, brought cheap and safe lamp oil into people's homes, and lite the streets of dark cities. Common people could stay up and read. The very first night owls without ancestral estates, could be born. This company brought light to an age and became one of the first great technology companies of history, that ushered in the

greatest age of human progress and development that had ever been experienced for mankind.

Standard oil didn't drill for oil, they standardized it and shipped it. They created the network for it to reach people in the cheapest and most efficient way. Railroads were the new technology, and standard oil used railways smarter and better than anyone before. They integrated and used the highest-level technology of the time, to deliver the most valuable product possible for the enhancement of human life. Books existed, rails existed, lighting existed, the telegraph existed, printing presses existed. Cheap light integrated them all. If standard oil had maintained a focus on the betterment of humanity and expanding human expression and experience, they may not have been burned at the stake as an evil witch at the turn of the century. People did not feel this company was out for them, and latched onto fear mongering and yellow journalism, that amplified their fears, and justified the eventual break up of this company. The fault lies in the company, and its inability to grasp the deeper morality of technology that we must embrace. The downfall of Standard Oil is a great American tragedy that should never be celebrated as a victory for mankind. There was no such thing as Robber Barons. America's killing of great tech tycoons, only made growth and development slower.

COROLLARY STORIES: BELL

The story of Bell Communications is not much different. They made great tech, changed human life, connected people across great distances, changed human life for the better, got lost in their own power, became something people feared, and got broken up. Great tech companies that stop innovating, and focus on expansion and domination, become unpopular fast. Now as great tech companies arise again our fears rise up, and when

lies, scandals and yellow journalism make a perfect storm, many of the great companies will be smashed. We smashed down Microsoft in the 1990's, because it was a very inhumane tech company, that most people hated, and was focused only on its own market share and power. Beneath that though is certainly damage to the industry and free enterprise, that we will never see in the things not created. The pretend idealistic faces tech companies put on is an act to stop them from being smashed and broken up by the people. Capitalism is learning to have a fake human face at all levels.

19TH CENTURY INTERNET AND KODAK

Internet technology has been around since the American civil war; we just didn't have the vision to really use to its fullest.

Telegraph wires, cheap newspaper printing, typewriters, and good lighting, where the first internet. Newspapers where the display, and telegraph wires the communication medium, with reporters being the interface of connection between the machines and the masses. Typewriters expanded people's capacity to write and communicate to a much broader audience. They could extend users thoughts out with typing machines.

Tickertape machines, could print out messages sent over a telegraph wire. While it would have been ridiculously expensive, we could imagine a primitive SMS like text messaging system working between homes. We could have invented a communication system between people's homes, that would have allowed them to message others via telegraph wires, before the invention of the phone. The tech existed in the 1870's.

If we understood in the past the fundamental value in communication to our humanity, and the power and dignity existing in all human beings, we could have directed

technological development better, and more towards the very things we have today. This is not about government policy, or religion. It is about really committing to humanity, taking risks, and focusing on humanity. Great companies wouldn't have fallen if they had been more like this. The risk inherent in experimentation can be greatly mitigated by a focus on humanity. All great tech aligns with being more human, and thus we can have direction in time and space to direct development, even if we don't know what we will develop.

Kodak invented digital cameras and didn't use them, because their profits were based in film sales. Digital cameras meant no film to sell, thus no value to Kodak's bottom line. Kodak lost focus on creating real value for the customers and their humanity, and used people as tools to fill their coffers. Kodak died. Most tech companies die, and much of the great tech they discover doesn't get made at the earliest stages possible. HP makes printer cartridges that are overpriced and deep down useless, to make money, and failed as a tech company. It once stood at the pinnacle of respect for humanity.

APPLE

Steve made great technology over and over again because he had a proper orientation in time, an understanding of what technology really was, and the way to connect it to human beings. Steve Jobs described the first Mac as a bicycle for the mind. Steve Jobs brought mankind into a tighter man to machine symbiosis that rewards our humanity and feels fundamentally human in our lives. Apple is a great cybernetics company. Apples future lies in its ability to connect more of our lives into technology all around us. As a brand, Apple has to make products that connect the highest-level technology, with ease of use, and a fundamental enhancement of our humanity.

So how can Apple make technology to push us closer to the stars and more deeply into the humanity of we are, as a leading cybernetics company?

I don't see much going for them, unless they seriously change course. They are in a corporate death spiral like the companies before them, and will focus on making profits as they do now, and will not undermine that profit stream to make a truly great new product. The Apple tech stack is chasing subscription money, with zero innovation over the competition. It has no service to humanity, the future, or our deeper connection to technology.

Airpods and Apple watches, form a backbone for a new cybernetics interface other than touch screens on a phone, that tries to tackle the problem of zombie like staring into people's phones, connecting people to the world more, and using AI integrated into daily life. They have some hope there. Their overall platform with a new vision can again step out and change the world. How to beat zombies is Apples great challenge for humanity, that they can and should solve. It is doable.

TRANSPORTATION EVOLVED:

Technology exists to make us more human.

I never dreamt of flying in the clouds. In my vivid dreams, I hovered a few feet above the surface of the earth skipping and skimming across large distances. Tech that can do this, would make me as happiest as possible. Our literal dreams of how we can travel,

can be the reality of how we make future transportation. If you really want to make people more human, they need to run, swim, jump, or even fly!

A person moving between two reasonable near locations, needs a system of transportations that makes them more human, more alive. Driverless cars will revolutionize logistics, and are not the ideal way we move as people. People using driverless cars will be little different than iPhone zombies all staring into their phones on a metro system. A transportation system that makes us fatter, disconnected, and lazier, is coming in our near future. We can do better than this by thinking more deeply on what we want as human beings, other than cheaper, faster, more convenient transportation between two points.

Bicycles may seem like an ideal system: healthy, open, social, connected. Motorists hate bicyclists because they slow down transportation for everyone, makes driving much more complex, and the road becomes more about the cyclists' feelings than actually going places the fastest you can. Most people don't want to go on bicycles because they are slow, make you look poor, and not all that safe. Electric motors make them more ethical, faster, and pragmatic. You can get places more efficiently, at more entertaining speeds, while not hindering efficient flow of traffic. Repetitive and healthy pedaling on electric bicycles is not as human as we can make transportation. Motorcycles are the most human feeling, and the most unsafe form of travel. Cool electric bikes that don't make you look poor, have some potential.

Exoskeletons that can allow us to run at over 40 km/h would be far closer to what we ultimately want. We could run and jump to work. The system could teach you how to use your body, and how to do amazing things with it. Normal people could do parkour like stunts, with the help of exoskeletons. You could even turn on a training mode, where

moving becomes even harder than in real life so that you can develop yourself, and possibly charge the battery. A system that trains and develops our bodies into the absolutely best form they can be in, while empowering us to run around like super heroes, would be far more fun for me than any car. We could step out of those machines, and have even more powerful, healthy, and beautiful bodies, capable of doing amazing things without them. We could literally wear an "iPhone" as clothes. Where Apple becomes more than just a bicycle for your mind, it becomes one for your body too.

The basic suit wouldn't even need motors or balancing machines. It would just a frame that fits most comfortably over your body that can create natural resistance for every single muscle group as you move, and not restrict your ability to work in confined spaces close to people. The connection to the body and its ability to move around with other people, would be more central to design, than enhancements to a person. It becomes far easier to develop something that people could use in their lives, when it isn't about being super strong or fast. It could be used on earth as a training machine for people to get the strongest, healthiest bodies, with the least risk when working out. Rather than having a power suit that makes things easier, it makes things harder in order to keep your body healthy. This is far easier to develop than more complex robotic machines. Robot development, and creating a physical interface for the body to inhabit in, are 2 very different kinds of research and development.

The resistance and training aspects of an exoskeleton are the humanity in the machine that will make it more useful and guide it towards faster development. The humanity and not the power are the thing that makes it practical, and drives its development. Such a device would be essential for those living in low gravity environments in space.

Developing the device with space in mind, will launch the product forward faster and

with deeper insights into its grander purposes and uses. I envision future with people that have nearly perfect and fit bodies from constant physical training and activity, and massive space stations being like parkour wonderlands. The way to make life more fun, is to make us more alive and human. A body training exoskeleton can be developed now at vastly lower costs, and have immediate value for many people.

MY IDEAL "BICYCLE": THE FAILURE OF SEGWAY SCOOTERS

A Segway scooter is a rather cool but inhuman technology. It replaces walking with a machine. From its very first day, it seemed a toy for a very fat person, that leads to a Wall-E like world. The technology can be made more human and not replace exercise. Segway like technology has evolved into mono-wheel devices, where you step on each side of a single wheel that self-balances, and you fly around a city freer than on a traditional Segway. The tech had to evolve to be more human or die. The change though didn't come from within the company, another company made it more human, got sued by Segway, and then Segway got bought by them.

A self-balancing Segway mono-wheel (Segway One S1) is a cool technology, and can be made even more human. It is a great direction for a technology and in of itself a great product for many young people to buy. It can get people around a city quickly, be carried and charged anywhere easily, and go easily into one's home. It isn't suitable for people that can't afford to fall down often, and like feeling safe. What we need is a device that has more structure, that can still go on subways, elevators, and into homes easily, while being fun and free, and can make people more physically fit using it. It is

combining the mono-wheels advantages with an electric bikes advantages, and not their uncool drawbacks.

A combination of a Segway One S1 with a "HalfBike" (<http://halfbikes.com/>), is like a Segway with one wheel, that you can peddle on standing up, with handlebars. You take a half bike use a large One S1 for a main wheel, and don't use 2 small back wheels. You would lean forward to go faster, and lean back to go slower, with breaks and acceleration augmenting and assisting the alignment of the rider. It would feel like superman flying over the surface of the road. This can become consumer technology we could have today. It can even be used as a powered push cart to carry heavy things as you walk around a city and take public transportation. This could empower people of any age, with its multiple purposes.

The exact way to make this, I don't know, and I am not even sure this is the right way. Maybe we need to do the same idea, but with a cool recumbent bike that you sit back into, and that has a seat frame that creates some roll bars around you. Being seated like a Formula 1 driver would be the ideal position for a bike you sit back into. It just seems clear that self-balancing wheels can make for a single wheel bicycle that will be much easier to move around in the world and use.

CYBORGS AND ANDROIDS

We don't have to cut off any body parts to become cyborgs. Machines can encase our bodies and minds and enhance existing systems, and make them healthier. Androids are not a technology we ever need. If we can build a human like android body we can build ideal exoskeletons, that encase the body. Science Fiction androids and traditional

cyborgs are emotional stories that explore what it is to be human and not about what we should actually build. There is something innately more human in exoskeletons. We as human beings will want minimum encasement, maximum enhancement. We will wear our machines as extensions of ourselves, and look better as we are, with them on, and not hidden away behind them.

Cyborgs are the degree to which technology takes over our humanity. All cyborgs in Sci-Fi should be seen as stories exploring the line at what point technology makes us more human, or less human, if we become something greater through the enhancements, or if the enhancements become our very identity. The stories that better capture the moral story and the proper vision of the hard tech, are far more compelling. Ironman captured cybernetics in a more realistic way, in terms of man's relationship to it. The armor is an excellent metaphor for the man to machine relationship that our future is heading towards. The story evolves with him struggling to keep his humanity, with the overwhelming technological power he possesses around him. He even ends up creating a nearly perfect android. The story makes a mistake and Vision should be the mental and physical extension of Tony Stark, manifested in machines, surrounding both his body and mind. The more "human" Vision becomes, the more human the interface that Tony can connect into, and not lose his own humanity.

MONETIZING SPACE: SPACEX AND FRIENDS

MY PATH TO MARS:

Space exploration and colonization must be profit and freedom driven. Capitalism must go to space at every step of the way. The Apollo missions were a failure because they didn't primarily focus NASA on making a capitalist commercial space industry. Step one of making a business in space is getting over NASA and the Apollo project as ideals, and see them as the problem. They create unrealistic goals, use poor economics, and deny the underlying reality of the universe. Going to Mars is not the natural next step after landing on the moon. Building up a massive and profitable low earth orbit space industry, is the first step. Then we have to find profitable reasons to develop the Moon, and have people traveling there. Only after this, and with massive resources and development in space, will any mission to mars really be conceivable.

Mass monetization of low earth orbit is being done by SpaceX and other companies at a tremendous pace, after the US government allowed more economic freedom for capitalism to do so. It will be much cheaper to get useful things into orbit, that can directly benefit life on Earth, that would not have been profitable enough before under a government run system. Space tourism can soon be a real industry, and hopefully other things as well, that we haven't yet imagined. SpaceX wants to go to Mars and their plan is just not realistic, nor based on how we will really get there in the future. Mars, The Moon and low earth orbit need to first be colonized by machines at massive scales, which then will make it possible to bring people into space.

Housing quantum computers that need to run at near absolute zero, could be what drives the colonization of space. There needs to be something that can drive a high profit "gold rush" like moment into space, and not just people's curious feelings. The Moon and low Earth Orbit could eventually be a giant server base for quantum computers that need extremely low temperatures to run. The overall colonization of the galaxy might be driven by finding more resources to build massive computers farms, to support human

industry and life. The large planets in our solar systems only value to future generations will be in machines turning them into massive computers, running at low temperatures. We can never land on them as people, nor easily get anything off their surface.

This all begins by building up lots of space stations around the Earth. Space hotels can gradually develop themselves one iteration after the other to support human life. As we get better at domesticating space stations and making them suitable for real human living, then we can create real space craft that can travel between worlds. We can then turn space stations into cruise ships. They can take short trips to the moon for the coolest trip of a person's life. As we get better at this, we will slowly upgrade the craft to with more safety features, human comforts, and even shields for cosmic radiation. Then we take remote building and resource harvesting devices, and land them on the moon. We can create a moon base on the moon, that requires no people to build it. We need to build up this tech on the earth first, and build whole cities in desolate and remote locations, with local resources supplying most of the construction materials. Mass robotic construction is a pre-requisite of space travel.

We then create a series hotels that stay in orbit over the Moon. Cruise ships will then serve as ferries between the Earth and the Moons space stations. The moon bases first purpose is a safety net for people on stations in orbit. Stations above the earth, can readily use escape pods to escape down to the Earth, and will need similar safety options with the Moon. With hotels in orbit, and primitive living spaces on the surface, we can get people to the moon at a larger and faster scale, than if straight making a full moon colony. Advanced cruise ships will take excursions down to the moon, and people will primarily live in orbit on stations in orbit. Moving between stations and the moon's surface doesn't require that much fuel. The moon base then can work on expanding its

size and supporting services. Producing food, back up air and water, fuel for the station and rockets, in a gradual step by step process.

The moon will eventually provide a resource and manufacturing base to create space stations. Getting resources off of its surface will always be vastly simpler than getting them off of the Earth. By this point in space development, we should have fairly large amounts of money being made in Space and new ways to commercialize it. Mars probes will have already been sent, and all the remote building tech used on the Moon, that was first developed for future cities on the Earth, will go to Mars, and just build. The primitive universal construction machines won't need resources from Earth, and can hopefully just replicate themselves and mass build on Mars. We may need to send large shipments of electronics to put brains into the machines. By that time, all processors will be made in space, and most of the worlds computer servers will be based in space.

It will be people in space stations above Mars that colonize mars, and not living in settlements on Mars that really live "on" that world first. It will be much better to live on a space station with artificial gravity, than the Mars surface for any extended time. Systems to live on the surface will have to grow slowly over time, and exist for emergency, entertainment, and practical business purposes, at first. The driving forces to make this possible will be for technological money-making ventures that are independent of those in space, which in turn can finance man's journey into space. Space has to be overwhelming profitable and productive compared to each person that would live off the Earth. Quantum computer cooling is one highly profitable possibility at this moment, short of the desire for people to be free of government.

THE INHUMANITY OF MARS

Sending people to Mars is evil, and can only be driven by vanity and suicidal impulses. If someone really believed going to Mars would lead man forward into the future, then they could justify their sacrifice of their life as noble, and for something greater. It is not the immediate path to the stars and not our near future though. The amount of infrastructure needed on Mars before even sending people there for more than a "hey I went to Mars, I am famous!!" moment, is very substantial. Mars tourism to feed rich bored people's egos, is hardly the foundations of technology. To really solve the space monetization problem, we need to go back to first principles and not vapid idealism; the Mars journey has to be fundamentally empowering to people and humanity, and not just a massive PR stunt for being human.

The main focus should be getting people up into space safely and able to stay in safe reliable space stations in near Earth orbit. The stations could be monetized for tourism, research labs, and even high-end manufacturing. Monetizing space is not sexy, it is reality. If a business man in 1850, said, I am going to revolutionize the world with train travel, where you can ship coal from the remote mines of the world, right to grand coal power plants/furnaces, to power industry, people probably wouldn't be jumping up in joy to invest. That though is the reality of trains, they move coal and oil mostly. The romantic visions of travel, where not what they really did. SpaceX is a train network for space. Let's in reality move things that pay the most per load, while selling a sexy image of travel to the most exotic possible location, like long range trains of old did.

What sort of death rate is acceptable for commercial space travel?

Dreaming of space travel and dying in space are two very different things. A large disaster with 200 dead tourists could derail the whole commercialization of space. With people, fully aware of the risks, the absolute hell and possibility of a gruesome death, and signing every form to acknowledge that, it doesn't take away from the actual horror that will come. People don't see reality until there are actual dead bodies, or they are lying half dead in a hospital bed. Getting people "into space" will probably be a more remote thing, with businesses running them from the Earth's ground and people "living" in space via VR.

AR/VR/MR: SNAP INC, FACEBOOK

Video chat is not a popular product now, it is in theory this amazing technology, that few people actually use, and if they do, only with a small number of people. Having a live video chat with active face and background filters, is where things are going. They are now building very basic filters that can do the mechanics, and send little video posts. This is great in of itself. The issues are in what sort of ideas define those filters, why people will use them, and the ultimate moral value in their existence. Being able to make my face into a dog, is cute, and only goes so far. Creating video filters that enhance humanity and not feed vanity is a challenging task.

EYE CONTACT FILTER:

A very basic filter of moral value and that is cool would be able to have eye contact with a person while video chatting with them. When you are looking at someone's eyes during a video chat, you are not looking at the camera, which then allows them to see into your eyes. It is mostly polite on a video chat to stare into the camera half of the

time. If you always look in the camera, then you won't see the other person and really connect with them. Video filters will be able to shift your eyes to be looking at someone, when you are looking at the screen. Simply shifting the pupils to appear to be looking at you, is just taking an eye and shifting its alignment in an eye socket. This is not fake emotion or lying. It isn't a video filter to make you look more beautiful than you are, or hide you. It doesn't create a more plastic like reality; it creates a more real and intimate reality.

AR PRIVACY BACKGROUND FILTER:

If you are in public video chatting, you can make those around you very uncomfortable. They don't know if they are on camera and being watched. It just creates an immediate discomfort in many people. You also may not want others seeing the people in your environment. Privacy filters can be imbedded into the background. Cameras in general looking out at the world can have faces or even whole people blurred out, unless they stare at the camera for like 5 seconds and give a thumbs-up. This will make people around you far more comfortable to be around people video chatting. People will also be more comfortable that people around them won't troll or disrupt their chat as easily. They have to trust that their privacy can't be violated without permission, even when they are being view directly by a camera. Privacy filters on cameras in general will make people more comfortable with cameras around them, if they know there is inherent software protecting their privacy.

An AR/MR reality can edit out other people and create a more intimate reality, where you are connecting with people you want to connect with, rather than an impersonal world. Humans are tribal and clannish, and yet gain great benefit from living in big cities and in large populations. We can get the benefits of a small town and big city, at some

emotional level. AR can edit out all the things not connected to us, and focus our attention on what is important. A more fun aspect would be to "blur" people by transforming them into other things. Reality itself can be bent into something else. These transformations don't have to be random or meaningless. Possibly the person themselves can set what they are blurred into, and this becomes a sort of social armor and clothes contained within a AR reality. I personally wouldn't mind being a mini-T-Rex Raptor, running around New York City, when cameras spot me and shield my privacy. When I meet up with other t-rexes we can have street fights with our avatars, and never physically touch each other. A living combat video game played right in the streets.

You don't need a centralized system controlling such a thing. When people are seen, info can be sent from them on how to see them. Cameras themselves have to lose privacy and collectively we can know what local cameras are pointed in what direction and if they are seeing us, and we can shape how they see us. We simply make a protocol that makes all cameras pointed at the world broadcast to a sort of localized internet, the orientation of the camera and what it can take in. Other systems can determine what is in that cameras view, rather than the camera itself, and then send back info on what it should block or alter its image into. We may have to lose the freedom to point cameras out at the world, without restrictions by others. People will be fearful of cameras they don't protect their privacy, and be more accepting of ones not stripping them of their privacy, and so even without any laws, the natural outcome will be regulated cameras. Good cameras add value to everything they see in real time, and people become more empowered by the cameras in the world, that serve individual needs and empower people beyond the camera owner. Systems that feed into and support a collective social-commons will be more accepted and integrated into society.

We can't alter things to cover up our humanity. We alter the reality to enhance our humanity and to respect others privacy and dignity. There could be a "just got up in the morning filter" that makes you more beautiful, though we need to enhance other aspects humanity building elements to compensate. Human vanity can't be the basis of such a technology and it can't also deny that human vanity can be a driving force in the development of AR filters that makes us look more desirable, and hide the ugly us. Very few people make avatars that look uglier than they are in real life. Cameras that look out at the world, need to not just humanize the person using the camera, but to fundamentally humanize the people on the camera, and add value into their lives, such that people will not mind a camera observing them.

3D CARTOON AI ANALYTICS REPRESENTATION

We can use silly video chat face filters to create Emotional intelligence augmentation and make communicating face to face online, far deeper and more meaningful. Much of the initial development by companies will be in making us into 3d cartoon version of ourselves. We will live in a cartoon reality. This will certainly make people too shy to be on camera, more open to being "seen". If there aren't sensors seeing our real face and translating our emotions in real time into the chat, then it is a lip-syncing toy. It could in time develop more social value, as the cartoons have valuable social info imbedded in them. Such as if the app could read the bigger emotional expressions you are emoting, and in real time create exaggerated them with your animated face, this would have high value for our humanity and understanding people more.

In a sense the AI, knows you a certain way, and represents that knowledge in cartoon form in a chat. It can create a more humanizing reality, in a completely cartoon rendered version of it. Our inner emotionally reality that is perceivable by machine

intelligence could be rendered into an exaggerated visual metaphor, that can allow our emotions and nature to be seen more deeply by those around us. Common metaphors and cartoon characters will take on a deeper meaning, and form a collective visual language. The representations of meaningful information that computers can draw out about us and show to others, can exist in nearly any form of art and sound. The meta-communications we know now only infer weakly through body language and facial expressions. Not everyone is equally good at reading these things, and machines themselves can enhance the sort of emotional intelligence of most people, in much the same way calculators make everyone better at math.

ORIENTING TO SPACE: **COMIC BOOK CHAT!**

People on colonies and space stations, need “warm-fuzzies” to hold them through the nearly unbearable distance and cold. Sixteen minute times between communications, creates a need for something like video chat without video chat. Sending video messages infused with humor, art, writing messages on the video, sounds, and voices, are mostly what we can do now. These are not good for a linear conversation. Going from video to video, makes it hard to really create a conversation. We can use Mixed reality images rather than video and have a “comic book” conversation!

You talk into the camera, have your face recorded in video. The app then takes a best image of you, and renders your voice to text, summarizes, and organizes it, and then puts it in a comic book style with text bubbles. A 30 second message could be broken up into 4-8 panels. The images could be real and the best one to match your meaning to each panel of the comic. Cartoon versions of your face could also be made that highly exaggerate your emotions in the message. These will all be thematically connected into an overall story.

Your key points and emotions will be highlighted and possibly summarized so the person doesn't miss anything important. Key information or next steps needed to be taken will be presented. There can be panels of action imbued with the text ones, to create a real "story". The app could expand beyond just your face and create images out of what you say, that connect into an overall clear pattern. If I say I want some coffee, there will be an image of me wanting coffee represented in some form, with my face. This also can connect into actual services, and things connecting into the real world. Send a message of "Hey buddy you need a pizza, a beer and hug" and the person can click on a link that brings them these things. This would be useful technology on Earth, and create many business and marketing opportunities.

MAIN THEMES OF NEW CHATting

Social and moral value in a comic/cartoon reality can be to emphasize emotions and contexts, that are hard to directly infer for people. It can lead to clearer communication and help connect people into the real world directly around them more, rather than purely into a virtualized world. This can also help communication between different cultures, genders, races. Often people can't see "Black peoples"-Asians-Whites faces/fat people's faces/Ugly peoples value/opposite sex faces and infer meanings/emotions/hidden contexts, and so on. This all makes for more intense emotional connections separated by physical and/or cognitive distances. Our emotional distances from people exist in space, time, and cognition.

OPENHR --TECHNOLOGY AND CORPORATE CULTURE:

My ideal HR platform is an open platform, that maximizes benefit to the employees, and fundamentally respects and values them, not just as cogs in a corporate machine. What people really want and need more than a job is a career. HR must be structured around the careers of people and not a job, set in one company. Workers jobs will become more fluid, their careers more defined, and get the best deal for their abilities. HR should serve the workers and not the companies themselves, because people will want to work more for companies that use such a system, and the companies themselves may get far higher performance from workers and have the right people for the right job.

Apple, Google, and few others, alone have some fairly developed HR systems. They should merge all their systems, and create an open tech stack and data set, that other companies big and small can readily connect into. Create an OpenHR platform that they all can view and share that allows members to directly recruit people from within other companies. Only member companies with open data themselves, and matching HR policies, will be able to view that data. Using such a system will make your company have a more attractive company culture for future employees, and look better based on real accountable information. Customers, the government, investors, corporate partners, and employees, will trust you more

Companies would be able to view and hire people freely. Workers can move between companies freely, and to be bided on in a more open and free market. The humanity and abilities of workers need to shine outwards, and so people can be in the right place at the right time, doing the right project, with the right benefits. If you have good company, with good projects, and treat people well, then people will want to come work for you, and work on your projects, to help make your companies vision a reality. If companies acquire many talented people and they get recruited readily by other companies, that can be a good thing for your company, because it will allow them to

recruit very talented people easier. There can also be recruitment bonuses paid out to the company that finds and develops great workers, to compensate for the hassle, and more motivate them to let workers go. Workers will not just leave a good company doing work that they believe in, for a slightly higher paycheck. People will want to be on the projects that match them best. Companies will need to attract and sell their ideas to their workers.

Being able to hire people from other companies and share workers, can also allow companies to share knowledge and abilities, allowing technology to integrate across companies more readily, and for products to develop faster. If I was a good company and I had former workers, that I still have good relationships with, working in other companies that I respect, that would certainly be useful and valuable to my company. Good and talented people, will want to work for companies with an OpenHR system. They will value the freedom, and dignity that comes with it. They know that they can get to the career they really want to more easily. Working for a smaller start-up, with lower pay, can more readily catapult them into higher position at better companies, if that is what they want.

Companies pessimism, fear, and protecting themselves is stripped away; workers are put first, so that the best companies can succeed, and people can work in the best places, in the best projects, for them. This is not idealistic, it is pragmatically realistic and good way to get great workers, and use them at their best. You know you won't be trapped in a dying company, or horrible job, that lured you in with seductive recruitment, when you start a new job with this high-powered career building HR system. Companies will need to focus more on the development of their workers by creating more useful skills and experiences for them, because they workers will want to do what builds their career.

Data on workers, can be used for their benefit first and not the companies. Workers than will more directly contribute and be open to sharing their data, because that data, will power their career. This leads to a higher embracement of training and having their weakness and flaws exposed. Which in turn leads to better workers, operating a higher efficiency, which in turn makes them more desirable and hirable by other companies and into other projects. At the root of this system is a more dynamic, living, AI powered "resume/CV", that describes workers value, experience, skills, knowledge, social networks, and whatever of value. Workers would feel empowered to be shown so clearly and be recruited from this data, if they know it served them.

Being able to actively track, train, and develop skills, continuously over a career is essential for workers. The "fake" credentials, certificates, and college degrees' people acquire in life to look good, don't inherently in of themselves denote any actual functional intelligence or the ability to overcome real adversity. Going to Harvard doesn't say as much as people imagine about someone's actual skills. Companies have to pay high prices to hire these prestige people with no inherent superior talent. The system must overcome the corporatism of hiring inferior people with good credentials, where HR protects itself by hiring people from top universities that aren't actually the best people.

At another level, social connections are what people most value from work. These social connections, relationships and sense of value gained from them, have to be magnified and rewarded. Most companies hire through the social networks of their staff, when they can. When companies are posting jobs, they have already failed at some level in the hiring process. Real HR has to be more about communication and not the high-end filtering of resumes and hunting talent with AI hunting dogs for headhunters. While I

don't know how, there must already be tools to empower and track social connections between staff, far beyond some internal Facebook system.

If companies could come together, share, and integrate all their tech and HR policies, develop a universal HR for all of them, there could be something up and running rather quickly. Such a system would be a massive threat to the existing hierarchy of any system. All the corporate politics could be crushed away, and useless management rendered obsolete and fired. Large companies at some level don't like talented workers, in the way their internal politics and real HR systems work. Talent is a threat to existing hierarchy because people don't want to hire people that can replace them. A system that amplifies talent, will be an existential threat to corporatists in a company. Actual results and not social games will define business. Careers need to be built with talent.

SPACE:

Space can inspire this HR system. You can't recruit and hire so easily in space. A mars colony may have 20 different companies operating different businesses, and they will need to share workers in order to get things done. There will need to be an overarching vision and purpose greater than companies' day to day success. The very world they live in needs to be made better, and as that world is made better, their industry improves, and their company profits, if it is aligned with the "greater good". Making great technology is a mission! It can build the world, and the future. When we invest in the future, we will arrive at that future sooner. In space, this reality is life and death.

COMMUNICATION SYSTEMS.

People love to hear what they want to hear. Many people in companies become very good at telling people what they want to hear, or already believe. This is how professional liars work.

The essence of good communication is communicating things that are fundamentally for the good and benefit of the other person, and not just yourself. People really want to take in and really think about what is empowering and helpful to them. You can empower and connect to others humanity, to be an effective communicator. Objectifying people and protecting yourself, makes you useless. In general, many people in companies with great ideas can't readily speak out their mind, and can't play company political games well. Companies too often across all industries reward self-confidence and self-entitled privilege. The man that can walk into his bosses' office and ask for a promotion or wage increase, gets that much more than those that don't ask. Blame for not asking is placed on the workers, and not the company culture. The sad thing is, that the people getting promotions, bonuses and higher wages are rarely the best workers, nor even that bright.

People need to learn how to communicate, be rewarded for good communication, and be held accountable for bad communication. Another failure of communication is that workers and bosses in companies, begin to focus more on protecting themselves, than working on the overall mission of a company. Most communication is a sea of self-protection, drowning out all useful communication. A culture of politics and self-interest sets in. People send completely useless emails to cover their own asses. Company group chats have no real substance in them, and people avoid making waves. Most email is never read. The goal of sending a message to

someone, is that they read it and the time it took was worth the amount out of help it provided.

The key to communication and a good email, is just 2 simple rules. You can apply this to all of life, and into nearly everything you ever do.

1. Communicate in a way that you like, and think is good.
 - a. Hold yourself up to your own standards.
2. Communicate what is useful for the person receiving
 - a. Get out of saying things that are only about you, and focus on the person you are communicating with.

People get lost in not knowing how to format an email or make it look good. Most everyone already knows what looks good, because they like the style of certain emails more than others. People need to trust their own judgement and apply it. Short, logical, to the point emails, with clear key points, with clear formatting, etc are better. If you like emails with color, use color. If you like teddy bears being sent to you, and think the other person will value them, send them. Take a risk, and in time things will work out towards better communication. Self-confidence tempered with common sense can go a long way. This part of your personality that judges your own thinking is that voice in your head you most associate with your own identity, is one you need to listen to, in order to make more clear and presentable writing. Make it something you want to read. People have to develop their own inner voice, and trust themselves more. If you actually like another person's style then use it. This is very different than copying a style to kiss up to your boss.

Getting over yourself and focusing on creating value for others with sincere connections based on their needs and feelings, is a fundamental skill for communication and life. People get lost in their own needs and minds. Once you can develop yourself, you need to get out of your own head, and get your words and ideas into other people's heads. You have to start thinking about what people want to hear, and how they hear things. This part of your personality is more like your conscience and an eye that sits outside you that can more directly engage the world, and criticizes you the way you imagine the world criticizes. This part of you internalizes the voices of others, and is also the voice you use to communicate and connect to others. You know what part of your personality this is, and you just need to use it properly with your inner voice, to create proper writing.

The time you give others, is your respect for others. If you don't give any real thought or time to people, you don't respect them, and can't develop any sincere communication with them.

THE TECH.

If people reject an email they receive they can get a list of reasons they didn't read it nor reply. If it isn't read, this should reflect on the sender, and be tracked as a form of communication performance data on the sender. An overall large amount of emails unread, and not in a few specific cases, illustrates a clear problem, where that person needs help. Being able to identify those problems, and create targeted training, becomes the basis of the technology. They can give back absolutely brutal feedback. "Unreadable" "Useless Rubbish" "Covering their own ass" "Nothing to do with anything we are doing" "Disturbing", etc. Can be quick shallow reasons, or even more deep and specific ones. As a pattern emerges across all their messages, in the feedback, we can target people's

problems with specific training. The system will learn and grow with us. The choices people get to critique an email, will evolve and be targeted based on what an AI thinks may be faulty with it. AI analysis of text is improving very quickly. It can target the emotional tone of different words and language patterns. Angry emails can easily be identified, and people can be given "Is too angry", as an option. The person sending an email, won't know who rejected their email, just that overall in the aggregate they have a problem.

People need brutal feedback and accountability, supported with great online training, so they can know what to do, and not just be told what is wrong. People that communicate well and contribute to the systems development and knowledge the most, can be rewarded. It will certainly stand out in an OpenHR machine-learning-big-data based evaluation system, and raise the value of that worker. "Nice" people, that can add to teams, can also stand out. Raising the level of communication so more "shy" members speak out, takes more than a management suggestion box. At some level this basic idea, has to be super powered with technology, and not just through the boss.

OPEN AG: GMOS ARE GOOD -ORGANIC FOOD INDUSTRY BAD

- **The tools to transform our agriculture need to be open and accessible to all. The inhumanity in the GMO industry is not the GMO crops, it is the companies that sell them.**
- **The more value to our humanity that we can see being added in tangible specific ways by the GMO industry, the more we can embrace this frightening new technology into our lives.**

- **Create jobs, create new business, make us healthier, save the environment, bring nature and life into our cities, lift up continents of people, end world hunger, create beauty all around us, and the promise of GMOs can be made real and be embraced.**

We can design food that is vastly tastier, healthier, cheaper, and better for the environment, with advanced genetic engineering, and new technologies for where and how we grow food. Urban farming and vertical farming can be integrated with high technology into buildings in cities. This will take farming out of farm lands, and free up more land for nature. Urban environments will be the most able to capitalize on GMO plants and add an increased health benefit to the cities, with more plants cleaning the air and homes of people. We need to realize that nature is for nature and that cities are for people. We need to separate our food supply from nature and create it in the cities we live in. The more we can create food around where we live and in the very buildings we use, the more we can create the future, protect the earth, and live happier healthier lives. People will more readily embrace GMOs in urban farming, that they can see and feel in their everyday life.

Organic food is a scam, and has negative environmental value, and negative health value. GMOs are the future and the best technology for human beings. All the science, facts, common sense, and reality back that up. Organic food is great for spoiled first world rich people, and a nightmare for people that are actually spending large percentages of their income on food. Most of the useful tech in the organic food industry can be combined with GMO food tech, and commercial agriculture, to create even better food.

GMO companies though have not really helped their case. They need to sell themselves as fundamentally human affirming companies, at all levels. GMO crops have to really focus in on human life and humanity, in order for people to embrace them. We can make food that will have every nutrient you need in them, and protect the Earth at the same time. The sheer potential health benefits of GMOs outweigh any possible health risk, that has yet to be observed. If something adds 1% increased health risk and 50% increased health benefit, then it is an easy choice to make. Cheaper food, that is more nutritious, uses less resources, and even tastes better, will be an unstoppable commercial success for all humanity.

A START UP TECH REVOLUTION

GMOs don't have to be driven by large multinational corporations distant from the humanity of the people they serve. GMOs are not the problem. It is the business around them, that needs to be repackaged for humanity. Small business and start-ups, can be driving forces in the future of farming. Companies that help small farms run their business, get VC funding, get crowdsource investments, buy their tools and products, find best practices, and connect them to customers, can help create thousands or even millions, of farm start-ups. Farming needs start-ups, in all the craziest of places we can never imagine. We need a tech revolution here.

If the companies doing all this are fundamentally open and we can see how they do their work, and why they do it, we can more deeply trust the products they make. Food made in some distant, alien and foreign way, is threatening and perfectly understandable to resist. Our farms are not integrated into our world and most of the work is done by distant underclass workers, in many parts of the world. Better jobs can be created around farming, that also add to the vitality, life, and culture of modern big cities.

A GMO company that existed as an open platform, that helped third world countries, startups, and NGOs, cooperate, share knowledge, and implement the newest tech, would be a game changer. The company could be more of a machine learning and cloud service platform. The virtualization of genetic code with virtualized testing of different strains, has to be a direction tech needs to push towards, and the companies that get oriented on that track can add massive value. Silicon Valley prospered because of openness and sharing between companies. Like the drug industry and few big tech companies, the GMO industry is driven by patent trolling and protecting themselves at all costs, rather than providing the greatest possible service to humanity and their customers. A company that becomes a platform for development, with an overall mission to empower the least well-off countries, would have a far more inspiring and powerful mission. Hiring the best talent for a company is much easier if what you are making is fundamentally focused on the empowerment and betterment of humanity.

When it comes to space and farming off world, all this tech can help make that possible, and create inspiration in how we make things. When designing a system, simpler ways can seem more expedient, and the only inspiration and drive to make it better at all levels, comes from a far-reaching vision of the future and adding fundamental value for of all humanity. The usefulness in well-made and well-thought-out tech, will pay off over the long run. How to grow food on a space station can empower the people of the earth with new and amazing technology.

OPEN ENERGY: NUCLEAR POWER AND FRACKING

-We built the wrong things from the Manhattan project and the Nasa Space race. It is time to fix that mistake, for the future and destiny of mankind-

We need Manhattan project for the worlds energy supply, with competing countries from around the world cooperating, to get this new tech on the market as fast as possible. This project needs to be driven by capitalism and innovation, rather than fascism like drove it before. An innovative capitalist start-up culture with many possibilities must be central to such an undertaking. Nuclear power is the best option. There are too many regulations and laws stopping research and innovation. Companies will invent a solution if we create a safe space for them to do research. There are different competing ways to make safe and abundant nuclear power, that all need to be explored, to see which can be the cheapest and safest to implement on a mass scale. We need a solution that can be mass manufactured in assembly line factories, and deliver power to the world as fast as possible, without inherent risks and dangers, such as nuclear weapons.

Science and technology lost its way and it hinders the future. When we align ourselves to the right path again, we will see a new golden age of science, and discovery. The Manhattan project brought to humanity a fundamental human tragedy in nuclear weapons and even greater tragedy in a belief that government funded science could solve any great problem. The Manhattan project is one of the reasons that science has slowly died in our modern era, as government took an ever-increasing role in funding and advancing science. The Apollo program was the peak of government backed fascist science system, that in effect ended free and modern science in America after the Moon landing. People can admire the Apollo mission by ignoring the rather glaring fascism to it all: nationalism, militarism, propaganda, Nazi scientists, Nazi plans, mythology, world-wide military dominion, massive government spending, corrupt connections to corporations. It made a system to rain down nuclear hell fire on any place on Earth and

launch a satellite system to watch over everything and guide weapons, as the primary goal.

In the years that followed, science turned to government funding to feed its budgets, and in turn altered its findings to fit the needs of an ever-expanding government and its power. They find new causes to organize government power around that in turn fund ever expanding scientific studies, across multiple disciplines. Climate change is the natural evolution of this fascism and the seeking of totalitarian power. At the root The Manhattan Project, The Apollo Program, and Climate Change, are one connected ideology and story. It is a story of failed science, bad technology, fundamental inhumanity, and a dying world. We need to undo this mistake in history and build the great things nuclear power has for the world.

Before we can bring nuclear power to the world, we need other power sources, and alternatives to coal and oil. Saving the environment takes immediate things that can work now. The soot and dirt from Chinese pollution is the biggest danger to the Earth and human life now. Fracking can replace coal and oil. Modern coal power plants can be cheaply and easily converted to run on natural gas, and thus not need massive new infrastructure or spending. Natural gas can be turned into alcohol (methanol) with some cheap and simple refinement. This fuel can run in gasoline burning cars that we use today, with some cheap and easy modifications to the engine. China needs natural gas now, to get rid of all the coal burning. Gas is much cleaner to burn. It is better for the planet and human life. Fracking has dangers and can even cause earthquakes. We need to balance out the dangers of all fuels, and realize that only fracking and natural gas can have an immediate impact on the Earth, where things are most critical, and immediately help the quality of life for real people. Business and not socialism leads to the future.

“Act in such a way that you treat humanity, whether in your own person or in the person of another, always at the same time as an end and never simply as a means.” — *Immanuel Kant, Groundwork of the Metaphysics of Morals (1785)*

Ethics at its root is a constant emotional process of finding and accepting sincere humanity in others. People are not things. You must try to treat humans as something fundamentally human and never as a thing to use for your own ends. We must respect our own humanity, and treasure that sense of truth in ourselves. Finding fundamental humanity in others, different from you is frightening and requires deep openness, and the capacity to change and grow. To be human is to have a will to live and strive to be more than you were before, in each and every moment. Life is a choice in all moments between living and dying. We must choose, continuously and constantly in every single action we take in life, and can only know life from death, with absolute honesty with yourself. Honesty is developed in a connection with another person, and accepting them fully for who they are as a person. Knowing other people and relationships are inherently messy, inter-tangled and complex. The process of ethics is not easily contained in one's persons minds and cognitive abilities. We need help from others and deep connections to others, to be good, rather than be consumed by our intellectual vanity. Ethics are the process of creating the future for human life, beyond your own self, and from your own self.

Humans see themselves as the heroes of their own story, and use stories to model themselves from and into. Classic imagery and ideas of beauty, self-sacrifice, romantic love, heroism, valor, honor, inspire souls. Most of fiction is just writers projecting out their own fantasies. We have this fantasy universe all around us, that we can project ourselves into, depending on who we admire and want to be like. What it means to be a hero, is the way in which we live; the rules and principles we live by, in this world; it is the people we inspire and build up. Real people, real heroes, are far more human, flawed, and vulnerable, than we readily accept. Ethics is about who we make into heroes, and how we envision the future.

- Shows of heroic doctors, spies, FBI agents, lawyers, cops, fireman, saving people, make up most of TV. These people are all nearly super human at their job, and portray this level of near mastery, that few in reality even remotely live up to.
- Watching sports is just a heroic fantasy story, where we imagine ourselves into sporting events, and achieving amazing accomplishments, or even having that star as our lover.
- Reality TV stars created far more human and flawed heroes for people to admire. It is too real and too human for many people. They can't accept ideals outside their own.
- Christ is the embodiment of a perfect man struggling to be the ultimate hero and role model for all mankind. We conceive of and see Christ from our own inner visions of our heroic inner story. "What Would Jesus Do" is a simple thought exercise that connects you to your own inner vision that structures your reality.

Ethics in their most noble sense are rules for human behavior, that would or should apply to everyone. All cultures are ethical systems. Ethics creates long lasting societies, and a more prosperous life for those that live in a society where people follow them. Ethical cultures are more enduring, and will survive better. Survival that is enhanced through culture is ethics and measured in degrees of civilization. There is a great German philosopher called Immanuel Kant. He wrote some very influential and very idealized books on ethics. He came up with the "Categorical Imperative", as thought exercise for making ethical rules. It doesn't work in practical reality.

"Act only according to that maxim by which you can at the same time will that it should become a universal law."

— Immanuel Kant, *Groundwork of the Metaphysics of Morals* (1785)

- All ethics are simply rules. And we do a simple thought exercise
 - Imagine if you had a rule that everyone had to follow.
 - If the rule makes the world worse, it is bad, if the rule makes the world better, it is good.
- Two kinds of ethics exist, the categorical imperative and the hypothetical imperative.
 - Categorical Imperative: "You should...because..."
 - Hypothetical Imperative: "If you want ...then do..."
 - To explain them as simple as possible; the former is judging good and bad, and the latter judging better or worse.
 - The hypothetical imperative is the process, and the categorical imperative is more like the goal or standard that you make things by and never really achieve. One is reality and the other is unattainable idealism.

All rules are hypothetical imperatives, are flawed Categorical imperatives in process. If we keep refining what it is we want and the rules that we have to govern that, then we can more and more approach ideal ethical ideas. The idealized categorical imperative becomes the idealized world we want to live in, and often drives us to create false non-functional rules that lead to utopia the fastest. Reality exists and it always comes back, because human nature is unchanging, nasty, and brutal.

The hypothetical imperative is just a simple logic exercise, you say what you want, and do what will logically lead to that result. I want to eat; I go eat. You have a desire and you act on it. As we get what we think we want, we realize we want something different than that. We don't even know we have these desires unless we live in this world a full life and highly stable civilizations, and have basics needs fulfilled. Our sense of hunger can reflect moral truths. The "best" chicken nuggets are GMO-Free, Fair trade, free range, sales taxed, import duties paid, Gluten free, American made, Christian owned, charity supporting, equal opportunity employing, law abiding, troop supporting, Olympic sponsoring, and dipped in awesomeness sauce that will make good women want to marry me more.

At some point a fixed rule that makes things better will eventually make things worse. Laws evolve as society evolves. "An Eye for an Eye", is a very ethical law in a society where people feel the need to take one life or more, for an eye. It is not very ethical in a world where people don't take revenge and kill others freely. Creating more civilized people by domesticating them with ethical training doesn't change the fact that we are wild animals and savages. As life and civilization become more sacred and cherished, so does the refinement in how we protect and nurture life itself, and also forget who we really are for more idealized versions of ourselves.

EVOLVING KANT

Kant's main problem was his excess idealism, and rigid materialism. He believed in the literal existence of ethics as a thing, and that since they couldn't be seen, they had a near material existence in another plane of reality outside our own observation. Idealism inherently creates a literal artificial and unobservable reality to base itself on. To believe in ideals as real things, is to deny reality and actual living. Kant's ethics, with adjustments, need just reality and no idealistic underpinnings to really work. It is nearly impossible to define something as absolutely good or bad. This is idealism, that only exists in theories and people's imaginations as real. We can approach ethical rules, and never actually achieve them fully as pure and perfect things. Life is messy, and the process is a constant struggle. The proper study of ethics is about better and worse. Ethics is a process, more than set defined absolute truth.

1. You do something
2. see the results,
3. adjust,
4. try to be better
5. never settle on any result as 100% correct

This is still too idealistic, though moving towards a better explanation. There isn't any really set or proper explanation for anything in reality. "Truth" is all based in probability and continuous improvements. We will never live in a perfect good world. We can only improve slowly on the world we live. When we evolve towards better systems, we only find more problems and more things we need to adapt to. Ethics are born out of

survival, and are based on what works and doesn't work. People can control ourselves and the reality around us, in a better and more consistent way, with evolving ethics. We can judge our ethics by how well they contribute to a better society and world, that actually works, and is more civilized. Kant made an extreme rule that we should never lie, and in that rule, we can find his failure and redemption.

"NEVER LIE" REBORN

If you want something from someone, then tell that person what you want from them.

All ethical rules need to be positive and hypothetical. Absolute and negative rules lead to poor thinking and a breakdown of real ethics. People don't want to be told what they shouldn't and should do all the time, by morally preening hypocrites. If ethics are about intentions, and not lying is ethical, then it is in ourselves that we need to be honest, in order to be ethical. Self-honesty is the root of ethics, morality, good intentions, and living for a positive future. We lie to ourselves constantly, and hide from the reality of who we are and the actions we have done. We hide from the very goodness and wholeness inside ourselves. That capacity to know and feel shame at one's self is the essence of being human and good. Addiction consumes people's lives because they lie to themselves, and are consumed by self-hate at the imperfect petty-monsters inside themselves. Self-honesty is a never-ending process of self-discovery and growth, that heals us and brings us closer to sincere honest intentions.

We must share with another human being and be unashamed of ourselves in their presence, in order to learn and grow in ourselves. Only when we can be wholly honest

with another can we see ourselves reflected back at ourselves honestly and accept who we are, while we in turn are forced to accept them, and see in them all the things we reject in ourselves, and need to love.

People themselves don't like to tell people what they really want because it makes them vulnerable and human. We don't know what we really want until we start to act out and get what we think we want, and then adjust our goals with trial and error. Good intentions can't be made real until people externalize their intentions to others, and are direct in their goals. Just saying directly what you want, precisely in an exact moment, can be a transformative experience, and profoundly difficult to do, with even the most basic of things. You know things to be right and wrong, and say nothing out of fear. The exact words to say won't be clear, and the exact way to do it won't be obvious. Show your real intentions and see reality from fantasy. You need to be vulnerable to someone and connect your desires and self to a reality greater than yourself. People can never be better for us unless we ask them to be better. Relationships and truth is messy and filled with conflict. If someone loves you, they will listen and change, if you can listen and accept them fully.

The difference between destructive and healthy relationships is fundamentally simple and easy. Real relationships drive us to be more honest with ourselves, and drive us to constantly challenge our very sense of identity for something truer. Unhealthy relationships support our lies to our own selves, and give comfort to our weakness and cowardice; they turn themselves into objects. Our assumptions about others are the lies we tell ourselves so that person becomes more acceptable to us, and we don't actually have to listen to them nor understand them. You can test how honest you are with a person easily, by telling them all the presumptions you have about them, and listening to their response and accepting it. We all lie when we have assumptions about people

that we base our perceptions of them, and we don't tell them. Good intentions can only be good, if they are driven by constant self-honesty, and a connection to another person, that challenges us to accept them fully as a human being to us.

JOSHUA'S ETHICAL SYSTEM

"A rational being must always regard himself as giving laws either as member or as sovereign in a kingdom of ends which is rendered possible by the freedom of will."

— Immanuel Kant, *Groundwork of the Metaphysics of Morals* (1785)

Finding and developing a rule system to govern ethics is a continuing and difficult process. There is no perfect or finished version. Only you can do it for yourself. My ethics exist in a sequential system. I started with my most deep-rooted principle, and then found a weakness in it, added a rule, found a weakness in that, added another rule, and so on. As my sense of who I am, and what I want expands, my ethical rules expand. I learned them in this order, as I lived my life.

THE HARD SOUL

HARD SELF: Self-Identity Ethics. Personal power, logic, and consistency (non-Agreeable non-Conscientiousness)

- Rule 1. "If you see something that is wrong, and you have the power to do something about it, you must do it, no matter the cost to yourself or your life. "
- Rule 2. "Develop you power to its fullest." Respect who and what you are fundamentally. With great responsibility comes great power.
- Rule 3. "Only ask of others what you would ask of yourself." Follow the golden rule of leaders and generals. Don't use people as ends in themselves to serve your needs. Have the courage to do what you would ask of others, and have the courage to ask of others what you need.

SOFT OTHERS: External Agreeableness of Values. Community, Society, Duty, Sacrifice (Agreeable Conscientiousness)

- Rule 4. "You can learn from all people" Find within all people something you profoundly respect, and engage them fundamentally from that space, and build all relationships from that.
- Rule 5. "Lead with your weakest foot forward" Incorporate into yourself that which you respect in others, into yourself. Face your pride and failings. While you can be superior to another, don't connect to them through that elitism.
- Rule 6. Seek out and connect to people you respect. Live a connected and pure life. Purity, purpose, direction, health, and destiny.

THE SOFT SOUL

SOFT SELF: Inner Self-Agreeableness. Personal Morality (Agreeable non-Conscientiousness)

- rule 1'. "Do things that make you happy" do something that will truly make you happy. a real dream.
- Rule 2'. "Know what truly makes you happy" respect yourself fully. Never get dark or lost.
- Rule 3'. "Do what you are told to do by those you respect." Only one person can be maximally happy at a given moment.

HARD OTHERS: External Identity Structures. Hierarchy and leadership of a structured world. Collective categorical group identities (Non-Agreeable Conscientiousness)

- Rule 4'. "Find those that will obey and follow you."
- Rule 5'. "Lead with power and strength to give people certainty and security"
- Rule 6'. "Show respect to get respect" Treat others the way you want to be treated.

EXPANDED EXPLANATION:

1. Responsibility- Do and say everything you feel you must do, regardless of the anxiety, and the consequences you imagine. Anxiety arises when we don't act. We can train people to overcome anxiety by setting small tasks that can be overcome and merely cause anxiety. A person can learn to face the uncertainty in themselves and find power they didn't believe they had, and realize this can apply to much larger sets of things, and thus, build the self-confidence to conquer fear and accept failure.
2. Self-respect- There are no such things as real victims. One must develop one's abilities to the fullest. You must become as powerful in oneself as you are

capable. This development must always be moving forward, even with setbacks. The speed of this development will be defined by reality. Even people with severe limitations, have certain points they can develop at any time and in any situation. Even in prison, one can grow. There are no excuses. There is only choosing to be responsible or not responsible for the life you want to live.

3. Leadership (pay your dues)- With responsibility and power comes an obligation to lead and define reality around oneself. We must only ask of others what we would ask of ourselves. This is a variation of the golden rule built around respect and not compassion. We learned to do and respect the shit jobs that we didn't want to do. You don't shove them on others, when they are our duty and responsibility to do.
4. Respect for others. (Sincere Respect for others). We must always believe in the abilities of others. All people have something they are good at, and we must respect that ability. Engage people from the fundamental strengths in them, that you respect in them. If you don't respect a person at a root level in something, then no sincere connection can develop, and at some level you just view them as a thing to use.
5. Learn from others. (Teachers) All people can be our teacher in something. While we may respect another's abilities, we may think we are still better than them in all things. We must get very focused. Forrest Gump was a genius in one thing, and those that could see that and respect that, prospered, in the story.
6. Purity. (the hard work). While we must be open to all people and seek the best in them. We must also cut away the bad and minimize exposure to bad influences and habits that bring us down. When we grow beyond people, we must honor what they taught us, feel gratitude, and leave. Our destiny is our habits

CASE STUDY: DEVELOPMENT OF CHILDREN WITH ETHICAL SYSTEMS

A man's 7-year-old son comes to him and says he wants to make an iPhone game. The boy is very curious, and his father is in the app development industry and knows all the things needed to do this. The father sees ability in his son, but also knows apps are hard. He wants to encourage his son, and develop his abilities, while not having his son do something impossible. He needs to narrow the goal to something his son can do, while also respecting the goal he son set for himself to do. He needs to begin where his son can do something. He gets an iMac for his kid with the basic game making tools in it, and then gets basic platforms his kid can learn in step by step. Now he also has to make sure his son is paying his dues and doing real work. If the child has tasks set for them, that they can do, want to do, and are hard, they are likely to overcome with enough time and emotional support. The boy will need to seek help when he really can't do something, and it is up to the parent to know when it is ok to delegate and not the kid. It needs to be based in gratitude and not entitlement.

If the father gets all this expensive stuff for them in order to empower his kid and not satisfy guilt, the kid can feel gratitude and not be spoiled. Guilt will drive a parent to permit spoiled behavior, and not empowerment, because they are too lazy to really care and focus on their kid. Secretly the father is probably guilty about something and when the kid just goofs around and is being naughty the dad may cave in, and be happy to see their kid smiling playing a game, rather than making a game, because they deep down just wanted their kid to accept them, and not empower their child. The father must be strong here, and show respect to his son's abilities and not secretly think the kid can't and just did this to buy his kids love.

He should investigate what his son finds interesting in games and look at the games he likes, and basically get to really know his child, and connect to the child's ability to discern quality. The parent has to respect their kids "taste" in something, or the parent won't really want to invest in that kid. If your 7-year-old cooks the most amazing food you ever tasted, you will certainly have no problem investing in them learning to be chef, if they tell you that is what they want. Find the specific aspects in each game that he is drawn to: art, gameplay, style, tone, etc. Keep asking the child why, so they think deeper on specific aspects and develop their own understanding of good and bad. As the reasons get tighter and more defined, actually work and design can take shape.

Also investigate who he wants to role model himself after. If he loves Steve Jobs, then also help channel the lessons of Steve Jobs to that kid. Can also suggest role models, show him people and see who he likes. Get a group of more than one person. We admire people for very specific qualities. Investigate what they really like.

Take all the aspects he likes and unify them into one game, and at a level his son can start to do. Can play with art tools and gather up images and styles he likes across many sources. Father can help get them organized and do things his son can't do or isn't even aware of yet, but the boy can feel he is doing all. He is the "leader" of sorts. The boy asks "HOW?", and this will inverse typical parenting roles. And the father can teach and help based on his kid's needs, arising from the kid. The "how" questions leads to respect back to his father, and to different things that the father can teach him. Parents can't do it all, and they delegate a lot to mentors and trainers. Find teachers the kid can use, even if just virtual.

Then clean up and prototype the overall package. Father can talk and guide son. Not criticize bad habits, but empower new better habits. The boy can learn to keep all the previous 5 steps clear in his own mind, as a child can understand them, when he finds himself straying. The father is instilling effective habits, ethics and values into his son through training him to do the things he asks his father to help him do. The vice of playing is the danger that will keep him from succeeding. With power comes temptation to use it not to fulfill a responsibility but to avoid them, and the father can mitigate this

by keeping the son focused on the goals the child set for themselves and adjusting those goals as the child comes to understand what they really want.

COLLECTIVE HUMANE CYBERNETICS: CIVILIZING AND SAVING ALL MANKIND

JOSHUA'S LAWS OF CYBERNETICS

Cybernetics is fundamentally a bridge to humanize AI, and thus turn it into something that we can profoundly accept and embrace. Humanity must maintain dignity and never objectified.

We have to explore that exact line where cybernetics are ethical, and not ethical--where want we to embrace them, and where they are just plain creepy, ugly and wrong. People driven to mutilate themselves in pursuit of an ideal fixation, if allowed to have free reign, would be bad for techs development. Creepy "hi tech" people, frighten people and limit technologies future development. Developing technology in the least creepy way, will allow to grow the fastest, and be embraced by humanity.

- **First Rule of Cybernetics: Cybernetics for body and brain parts, should be able to be turned off, and that person should be better off than if they never had those augmentations.**
- **Second Rule of Cybernetics: The system should make a person more human. Their individual freedom and dignity must be expanded.**
- **Third Rule of Cybernetics: Follow medical ethics. Do no harm. Don't damage healthy functioning body parts.**

BUILDING COGNITIVE CYBORGS (CYCOGS)

We can augment people, with even IQs below 85, by turning their reality into a cartoon, and enhance their ability to succeed in the world and do things of value. The very interfaces of modern first-person video games are not much different than how we could literally live in the world, for those that need this help. All of existence can be simplified, exaggerated, and slowed, for less developed minds to perceive important things more clearly. Cybernetic interfaces can be built around people's sensations, so that their very senses are processed through machines, and their self is expressed out through their senses. We don't need to cut or install anything into people. The technology can be as simple as glasses, headphones, and gloves. We will be turning people into a kind of cyborg that I call- Cognitive cyborgs (CyCogs).

Humans will still be human, and we still be what we have always been through the ages. Technology must make us more human and enrich our lives. Cybernetics must be life affirming. We don't need robot limbs, nor robot eyes, or chips implanted in our brains. This technology will also constrain the dangerous elements in society, and possibly be the very foundation and means by which collective norms are imposed on people, and where we as a society are protected from harmful and dangerous individuals. It can be the foundations of collective education, civilizing mankind, and binding us together. Mass killings, terrorism, murder, robbery and suicide should be far lower with such technology. How we do this, is far trickier, and can create a new form of authoritarianism, unless it is an inherently un-hackable decentralized system. It certainly can help the disabled and less fortunate, live better if done correctly. The very first people to help with technology, could ideally be autistic children with fully functioning intelligent minds unable to connect fully to the world.

In the big picture, this is about the disabled kid, in all of us, that has a full functioning mind that can't connect to the world fully. Our inner humanity can be amplified by technology and then shape that very technology to the beauty of the human soul that lies trapped inside us. This technology will help you to express yourself and find more deeply in yourself who you are, what you want, and to reach your full potential. It will be mirror inside yourself, and a means to communicate those things inside you to the world, where only the smartest and most creative people could express themselves before. We will be augmenting human intelligence with AI, at a more root and personal level, where anything AI can do, then all of humanity will be able to do it.

Countless masterpieces of music and art have existed in minds through history, that never learned how to play musical instruments, write music, or craft any art, and so these things were never created, and we as a civilization go to experience them. Only the most talented people could express their deep inner visions and ideas through history. Technology will lessen the skills and resources people need to bring their inner visions to life. A single person can be a symphony and not even be able to play one of the instruments, and still create music more beautiful than any musician trained to play. AI is humanized through a deep innate cybernetics interface, that allows cybernetics and AI together to augment us to our fullest. It is just a tool, that extends out from us and expresses us. We will be able to be augmented with AIs directly through a CyCog interface, and train AIs in turn through that system.

We live at present, in a collective Autistic Society with asocial walls between us where we can see each other and hear nothing, with little empathy.

AUTISM- Autism is a disability where people have fully functioning brains inside, that can't connect to the world easily. They live behind metaphorical sound proof glass walls that get thicker and harder, as the environment becomes more threatening and overwhelming. They can see out but not connect. Degrees of autism relate to the thickness and height of those walls. They can be overcome and breached with will power and training. At a basic level autism, is having no sensory filter. Autism causes people to become sensory overloaded, more readily. The brain's natural defenses cause a shutdown, to protect the ego. Inside their own mind, they often develop strong and unique inner characters, with intense passions directed at things.

"COLLECTIVE AUTISM" is a disease that infects our society. As collective groups, we behave little different than individual autistic people. When the fundamental cultural institutions of society are destroyed, eroded away, it releases an autistic walling off in society, as people can no longer cognitively communicate and connect to groups different than themselves. Social skills between out group members is dropping. In general, Autistic people are not attached to any fixed external social system, and are dimly aware of the one they live in. Educating them to fit into society, in a way they will embrace, lays at the foundation of building societies, and managing the collective autistic crisis we face in post-modern society. How we learn to create society on Earth, from technology itself, will define us as a species.

In a space station of babies, born from neo-androids, with no real parents, no culture, no society, and no human network of interactions, they will be collectively and individually Autistic. They will be highly asocial disconnected society, with people that

connect with machines and fantasy worlds, more than actual people. They have none of the legacy systems built through generations upon generations of families, cultures, government, companies, and social institutions to nourish them and connect them into the world. Our present world of children raised by televisions, the internet, and computer games, without strong culture and parental upbringing, is not much different.

AUTISTIC PEOPLE AS CYCOGS

The personal computer was a bicycle for the mind, and with AR tech we can turn them into a wheelchair for the mind. AR can enhance the humanity and living standards of people with autism, schizophrenia, Tourette's syndrome, dyslexia, ADHD, low general intelligence, Alzheimer's, and so on, live more functional lives. Autistic people can become the first cognitive cyborgs. If you saw a person walking down the street with a VR gear headset on, with cameras and sensors all over it, you would probably want to avoid them and find them very creepy, unless it was a tool that helped empower that person and you knew it didn't invade your privacy.

TREATING AUTISM- Cartoons are very helpful and powerful in the treatment of autism. Lots of deep inner emotions are learned from cartoons by being able to identify and project themselves into those virtualized stories. They have deep therapeutic value, even for adults. The simplified realities help people learn the main themes and ideas of human reality quicker. If they can eventually get outside their own mind and connect with the world, they can sometimes lead the world towards amazing new things.

AR/VR/MR- The line between, fantasy and reality will be completely blurred and fluid.

The interface and purpose of the system is to connect people more to the world and help train them, so they eventually won't even need help to function.

- **Augmented Reality (AR)-** This would be like the "Heads Up Display" (HUD) that players use in most video games, except where the real world is still seen and the HUD overlays the world.
 - Put information next to a person's face to show what they are feeling.
 - Show lines of where to walk and go to, overlaid on the real world.
 - See information about how to do things, right next to what you need to do. Automatic tutorials and guidebooks for everything. Real people can even come in and assist when needed.
 - "Health meters"- Sensors tracking stimulation levels, and helping them manage any potential emotional breakdowns.
 - Certain parts of reality can be cut out to minimize its influence, and as people get better, they can expand what they see.
- **Mixed Reality (MR)-** To turning everything around them into 3d computer animation, and fully filtering the world, and making it more palatable to a mind under stress.
 - A cartoon chair they see to sit on, would be a real chair they can sit on. It can be made to look different, friendlier, or more fun, or just simpler.
 - A person's face could be turned into a cartoon version of a face, and their emotions and expression could be exaggerated and slowed so they can understand what feelings people are showing.
 - Turning crowds of people in an airport walking, into a school of fish swimming through the air, to reduce the chaos of so many people.
- **Virtual Reality (VR) -**They can then slip into full virtual worlds, even in public, if sitting down, and play intense video games, dive into educational experiences, watch movies, listen to music, or create things and do work. They can calm and

focus the mind, by diving into a safe virtual world, and still be doing useful things and engaging reality in some way.

CREEPY CAMERAS- If we can censor and shut out elements of the world for autistic people, then AI linked cameras, should not see people's faces, unless those people want to be seen. Imagine people walking around in full VR systems, with multiple cameras and sensors attached to them. Cameras on glasses helping direct a blind person walk and interact with the world, would seem very acceptable. Glasses presenting text subtitles of speech for deaf people, that could read lips in noisy rooms, would seem very acceptable. Always on cameras on a person, with no explicit purpose, would seem creepy. There are ways to make cameras watching everything more human and less creepy. There has to be interactive lights that inform the world around them, what is being seen and what level of reality that person is in. The system needs to be built with a fundamental respect for privacy, so that people feel more comfortable to be around it, and with people using the technology

COMPUTER VISION- Cameras that computers use are different than the ones people use. Having computer visual sensors viewing the world can protect people's privacy more than constantly on cameras recording everything into video, that people can look at. Machines can, instead of recording a video, record 3d models of what it sees. These 3d sensors, allow computer intelligence to see the environment and begin to have an understanding of it. We can take the world we see and layer 3d models into it, with anything we want. We can make a cartoon version of the world we are seeing in front of us. Where all the things in, will be located in the same place, and just have different surfaces than we would normally see.

CARTOON RENDERED REALITY- Autistic kids can experience a cartoon rendered version of reality they can interact with in real time and in real life. This can control their level of sensory input, where their senses can be limited in one way, and expanded in another. The things they see will be there, but made simpler and with more complex

distracting things filtered out, or limited. All the peripheral things can be edited out, and only things they can interact with, will be presentable. Then in each of those interactions, they can be presented options, and information about the situation layered over it. This can be done step by step, with the techs natural development.

Imagine seeing a face through a camera, and that camera's data being feed into an analytics program that interprets possible meanings to those expressions and then presents that data in a simple way overlaying their social interaction. Emotional emoji's hovering near a person, could give them the social data they need. Multiple emotions can even float around, with different possibilities for each. 30% happy 10 % upset 25% lying, etc. The very composite of emotions is an emotion to consider. Another level of this would be to make a cartoon face of a person, and exaggerate and highlight all emotional expressions on their face, so they can more readily pick up meaning. Turn people's faces into cartoon versions and then exaggerate and slow down their emotional expressions, so autistic people can see and understand them.

BENEFITS FOR FAMILIES- To be able to take autistic children freely through airports, subway metro systems, public transportation, shops and malls, or just anywhere, would be a big enhancement in the quality of life for families with autistic children. The technology would also help train and teach them, get them out in the world, get them interacting with people, and thus more rapidly developing their skills to cope with the real world. Actual interaction and life experience are the best teachers, and that disconnection they have is why they don't learn as readily. At first the systems would be large and bulky, more inhuman looking. Over time they can be made more human, and show the complete face of the person, with minimal hardware covering up the face.

EDUCATION TECH- A cognitive wheelchair is just an education platform, for developing minds. Those that can dive into the technology will be best able to learn. We can have 3d worlds where one can do virtual chemistry and science experiments. Where

we can project materials out into the real world to interact with, right on a living room table if need be. Engineering/art/music systems where people can assemble things in visual 3d forms becomes possible and basically accessible for everyone.

What are the values and ethics that must imbue such a system, for people to really fundamentally engage at the deepest levels?

KANTIAN ETHICS, INTENTIONS AND CYBERNETICS

Manifesting the intentions and humanity of the disabled and all humans, is the ultimate ethical measure of this technology. The more it can bring out the desires, thoughts, feelings, and good intentions of people, the more ethical a world we can create. Ethics is not about people acting good, it is about individuals being able to express directly their intentions and real selves into the world in complex and unpredictable ways. A CyCog needs to become more human and more expressing of themselves through this technology, and that will drive its growth and integration into society.

Seeing a CyCog from the outside, and not knowing that a person was using this technology, they would look like super-humans able to do things beyond what we think of people normally being able to do. They would seem like "Jedi", using the "force" and able to do nearly magical acts. They could step on the exact right spot, at the right time, and do the correct movement, and it would seem inhumanly competent. It would be like talking to someone that has Wikipedia on demand, and you didn't know what Wikipedia was; they would seem to be the smartest person you knew. Many people when talking in real life, take short breaks away, look things up online, and come back to conversations

with info they found online, so they seem smarter. This sort of knowledge can be expanded into far more aspects of our life, and personalized to our needs.

The interface is about connecting technology with humanity, in a way where humanity is augmented and not consumed by technology. An easy and intuitive to use interface is one built around human thinking, and thus makes technology more domesticated to human thought and needs. The humanity and intentions of the techs creators can help augment the humanity of the techs users. We as people still have to make choices and decisions based on our principles and inner humanity. A CyCog system that doesn't augment ethics and humanity, is an invalid system, that must be resisted as a form of slavery, over malleable minds.

Smart and fully functional people using a system like this to fulfill specialized skill sets in life, would truly become super-human, even from the perspective of people that know they are using such a system. The extremely talented can be augmented so far, that it can take humanity to another level. The reality of such a system is a tremendous danger to society when used by the elite rather than the disabled and the unintelligent. The system must make people from the inside feel human and humble. If one wants a system to bring out the best in the best, then it needs to magnify their humanity and connection to humanity more than raw power and ability. While the smart can be super-human in specific skills, it doesn't mean their humanity is superhuman. We are all people after all.

AI connected into humanity makes it possible for the “body” of a person” to manifest into the very structure of anything.

The singularity is never coming. Our humanity and all its ugliness is who we are, and can’t just be cut away, or transcended.

People envision a point in the future when artificial intelligence can improve upon itself and reach an accelerating rate of improvement, simply leave all human intelligence behind, and allow human beings to be converted into pure machine consciousness that can live forever as immortal techno cybernetic Gods. The people that believe in the singularity want to be immortal and are latching onto tech that will grant them immortality. They are a transhumanist death cult, that is fundamentally against what we need to be. There is no need to transcend being human, being human is what is so amazing. The real non-idealized “Singularity” is the degree to which we as humans can manifest our humanity into machines. It is never a set point in the future. Machines can never become more than us, because they are us. The singularity is not us becoming immortal machines gods, it is machines becoming more humanly formed to fulfill human needs. A human mind expanded out into an entire planet sized computer, is still just a human being, and will live and die, like the rest of us.

We as humans can influence the growth and development of AIs, through our interaction and use of them. They will integrate into every aspect of our lives, and into the very cognitive functions of our mind. This will deeply and fundamentally shape their development, meaning, and purpose. Our AIs will become like mirrors of the human mind, and will create a virtualized version of ourselves. This virtualized AI self, will never become you, nor kill you in your sleep, nor grant you immortal life. This is a model of you, as described in math, based in data collected from the real world, that can make

predictions about what you would want in a given a situation. It has been trained to learn and understand each and every aspect one-by-one and semi-independently. They slowly grow out of simple algorithms that suggest things you might like online, and get more sophisticated over time, and able to process more information about you.

AI is just a search engine like Google, that lists the most likely results for the search you enter. It gets better at knowing you, as part of the data that is bases searches on to help you. It will never know exactly what to do, or exactly what you want. You still have to express your desires, intentions, and goals, then it can offer you options to fulfil those goals, or tools to help you. A CyCog interface increases that relationship with AI so that people can bring out their inner intentions into more actionable and concrete results. It can infer search inquiries and present just a few results of the search that seem the most likely. These predictions will always be probabilities, and it can be very useful even if those predictions are 10-30% accurate, because it can just offer more options and increase the likelihood that one of the choices is what you want. There may be a point in the future that 97% of the things you want to do, can be predicted by 3-6 choices presented in front of you. That itself is probably about the best that AI will ever do, even 300 years from now.

This is just Google search getting so good that it can predict the thing you are searching for within 6 guesses with a 98% success rate. It is profoundly useful and is just not as interesting or special as it might be made out to be for the future. It will feel just mundane and common to people that use it. AI being able to perfectly predict everything you would think or do, would be a search engine with 1 result that was 100% right of the time. It would be like a living virtual version of yourself. Once such a model existed of you, a transhumanist wanting to be immortal would consider you flesh body dying to be ok, and your AI version living superior. The degree that you are willing to let yourself

die so a model of you can live, is the degree to which you think the singularity has any spiritual meaning to life. Google search predicting your search results in one guess 100% of the time, might not be people's idea of something worth committing suicide over.

Your intentions and who you are inside are a continually evolving process of self-discovery and growth. No machine is going to read the inner depths of your being, that are barely formed into a coherent and clear vision. You don't even know who you are for real, and don't even know what you want, so no machine is going to replace that. It can force you to face yourself and drive more critical self-reflection. AI can help you organize and discover these things about you faster, through the collective wisdom of others trained into programs to help you. You aren't your guru, no matter how much they help you. AI cults will form around the divinity of the trainers.

No matter how well a computer can map the cognitive features of a person's mind and connect as deeply into them as possible, that mapping is not that person. It is a "ghost" of that person, a "photo of their mind". Its existence can be no more meaningful than the great art of a great artist, as compared to the life and self of that artist. The interface between the mind and machines, doesn't upload human identity, it merely receives commands. Being able to connect your mind to a machine directly, such as moving robot limbs, is not really creating an immortal you. No more than a beaver dam is a beaver, or a hermit crab shell is a hermit crab. A human being linked to a computer, can amplify their mind to its theoretical limit, and do things based in the unique qualities of the individuals mind. It becomes a super extension of their identity and self, and the ability to manifest oneself into non-material and material reality more deeply.

THE FUTURE OF AI: HOW TO CREATE HUMANE AI

Our humanity and love directs resources towards more humane and life

affirming machines. An AI linked to our brains is no different than any other technology. It is how we use it, the love we put in it, and our very humanity, that shape it and integrates it into our lives. We infuse our tools with our humanity, and stare into the bright flames of fires. A master swordsman with a master crafted sword, is a killing death machine to his enemies. Seen from its inhumanity, we see death, blood, and a nearly inhumane person chopping people up, with a nearly magical device that chops, cleaves and vivisects with frightening precision. Swords have often been seen as nearly living things, that people gave names to, and felt near mystical to religious connections with. They were prayed to, worshipped, and given more love by their masters, than they would give to their own children. Surely in history many women were jealous of their husband's maintenance of their swords: "if only he would be so gentle and focused on me, as he is with Qwithhoraloc".

Humanity was a cyborg from its very first moment of existence, and rained fire and death upon all its enemies with absolute power, and forged the very Earth into a home and place for human existence, with absolute dominion. We have always as a species been cybernetic. We are a technological species, that's very humanity and ability to live is directly correlated to our embrace of technology. Humans evolved from tool making and fire using animals. We burned down whole forests and grasslands, to just hunt and eat, from our very first moments as humans. This aspect of humanity is manifested in people fears of AI super computers destroying the Earth and humanity. The relationship between man and machine is the ultimate power, and not one over the other. Humans linked to machines are vastly smarter than machines operating alone. Our AI future is neither an AI god, nor the singularity.

If we want to live in a more human world, we can use AI to empower us to be able to buy more human integrated services and technology.

Computers can't train themselves, they need to be taught by human beings to understand and to do things, and directed towards things to look for valuable associations in. A computer can't just look at a bunch of data and tell you what it means, unless we program and train it with concepts of meaning, that come from people. If you want a computer to look at photos and tell you if there is a cat in the photo, then you need actual human beings making a set of photos with cats in them, and photos without cats in them, and telling the computer, "this is a cat", and giving them lots of examples. It can only learn what a cat looks like, if people create a massive database of photos that identify cats. As we train more and more things there will reach a point where computers will have to identify purely subjective things. A happy cat is not a grumpy cat, is a lot harder to identify than, A cat is cat. This technology will be able to enter into nearly all aspects of our lives and into how we think. Businesses will be built around singular insights in how to apply this technology to something that people hadn't imagined possible before.

AI is just Google search, and as it gets better, our sense of what we call AI evolves. With more interface layers put over the evolving search functions, we can create simulations of human beings and of reality. The next level of search for an interface is to upgrade the input of queries. The AI has to guess what you will want to ask it based on data it has. This is just a basic search like before, except that it feeds the results of that search

into another search. It is basically taking a bunch of data, and predicting a set of possible questions or intentions you may have at that moment. Then it takes those questions and finds the most likely answers. If you have 6 possible intentions and 6 possible answers for each of them, the AI has to present 36 possible options. With training and time, these choices can be narrowed. It is much more likely to model general human intentions and goals, than you specifically.

The next and 3rd layer of search is presenting that data to people in the way they like and can most use. Results need to be presented with different tools and interfaces based on what a person needs. This is what most APPs that we have do, they exist as tools and interfaces for specific kinds of things we need. A basic thing an AI can do, is observe all the data about you in a moment, and just predict what apps you most would want to use, then each of the possible apps predicting what you would want them to do. The search of your intent, leads to possible apps, that then lead to different results, presented all in different ways. The AI is able to reach out into the world and control it based on what it guesses you want, and how you respond. It guesses you want to order a car to go somewhere, and automatically presents possible locations you want to go to, and when you click on one, a car comes and takes you there.

This is not leading to an AI apocalypse unless that is what we really want AI to do for us. AI will try to fill our desires with things we will most react to and want to click on. The more likely desires of people are getting thinner, having better mates, and increasing one's social status and prestige. This could lead to a societal social collapse from a sort of super capitalist marketing system that sells us drug like consumer products that we can't say no to, and slowly destroys as human beings. AI can feed into addictive needs.

TASTE, AS THE ROOT OF INTELLIGENCE: ONES COGNITIVE GRAPH AS AN APP.

Your training of an AI is imprinting your own mind into an AI. This graph of your mind and your values and decision making can guide a custom program for people to use and adopt. All technology can be made more human in the very image of people we want to be closer to.

Personal taste is one of the fundamental acts of human intelligence. Your taste shapes what you can train yourself to be good at, with enough time. If you have no palate to taste food, and can't feel what you think is better than something else at the tiniest level, then you can't be a chef. Knowing good music and learning to play good music are a wide gap between potential and skill, that are forever intrinsically linked. Good taste lies at the foundation of nearly all skills. Only those that apply themselves and learn to express that sense of taste into world, rather than just reflect on the world, create the most economic and social value. Technologies progress has mostly been making that gap less intense through time, such that one day, we may just need to feel as deeply as possible, rather than even have real practical skill, to do amazing things. Craftsmen need to learn how to use expensive tools and get expensive training, to be able to do just about anything. People had to fit into the often-brutal realities of learning a trade. The pure essence of a trade, can now be more accessible to everyone. A "painter" doesn't need to know anything about paint, or work in anything like 16th century Dutch guild society.

Computer programming is good taste in code and design. Programmers themselves can train AI in the very "taste" of what it is too be a good programmer. AI will learn what is good programming, and we won't need as much technical training as programmers did

before, to program in the future. It will become more about design, and the very desires we want to be fulfilled. We will be the trainers of computers, rather than the programmers of computers. Traditional Programmers had to train computers in a very specific and nearly inhumane language few can understand. Computers won't need to have use speak their language for us to train them. This will free more minds to express themselves into the very nature of computer code, with AI assistance.

Really special people, just living their lives, being themselves, and expressing themselves fully, can train a computer to understand and recognize certain things as more valuable than other things. This sense and collection of information is an expression of you, and it can be copyrighted/patented as proprietary knowledge, that makes you money. Your sense of things doesn't have to be for all people, nor all the world. We can train an AI in a person's subjective knowledge rather than objective facts, and monetize that for special people.

A PERSONAL LOVE APP

Internet fame, is a form of technology that influences and transforms the lives of those that follow people. Many of the most followed people, try to help other people transform their own lives, like they themselves have done in their own. Beauty, fitness, love, fashion, PUA, food, design, business, get rich quick schemes, marketing gurus, all seem to dominate the social internet. The successful can evolve this social media influence into apps, as they do now. Eventually this will evolve into AI itself, through deep learning and algorithms, that makes social media and apps seem an old idea. Any app we use today is a set of ideas contained by those that made it. You don't use a gay dating app, to find opposite sex dating partners. The tone and style of different systems, will lead to different results, for different people. Who we choose to follow as role models in our

daily life, will deeply effect the decisions we make as people. Basic internet technology allowed us to make role models out of common people, from purely how we felt about them.

You could literally use an app, powered by the judgement of someone's whose views on love and relationships you personally admire, to influence the kind of people you will meet and how you interact with them. It will make it more likely you can meet someone like they see reality in. It makes it only more likely you meet certain kinds of people. Nothing is certain. No program can really know what is good for you or what you deeply want in life. If you find the relationships it brings you, are ones you like, then you will be more engaged with it. If it doesn't work, you can try another person's system.

Let's say you know relationships really well, or at least think you do. You can point out good and bad relationships between people that can train an AI. Your idea of love might be true, in a certain way, and only useful for people wanting to be similar to you. These differences can lie at the heart of apps and programs, that help create relationships in the world. You can never create an algorithm, that will be able to determine if two people are compatible to be "soul mates", best friends, great business partners, or ideal sex friends. Everyone has different kinds of relationships they want to form, and no simple bit of code or training, will ever be able to capture all of that. You have to make decisions.

FINDING ROLE MODELS TO DEVELOP US

We can manifest into reality through advanced technology the subtlest aspects of our inner reality. We can be much more than a dark empty future living in IKEA made

homes, with food made on a factory farm delivered to us by robots cooked in some standard machine to make yummy food, while watching entertainment on massive TV screens the size of walls, while going to bed with the latest sex robot companion. We can feed every addiction with perfectly tuned machines and numb our humanity away into nothingness, as we seek to conform and obey to a soulless and mindless system, that manifests our own self-hatred into a collective hatred of individuality.

We can want to engage AIs that are more deeply connected to the people we see as the most human, and the most empowering to our lives. A computerized version of your mind, can go out and work for you, and sustain your ability to live and thrive in the world, by you just being you. "Work" can be, for some, about being yourself as much as possible and doing one thing as good as you possibly can. You can find those things in you that you can be the absolutely best in, and then surround yourself in people and things to develop that in yourself. It is very hard to know who you are, what you want, and what your dreams are in reality, until you challenge yourself fundamentally in the world as a full and authentic human being. It takes lots of trial and error and life, and you won't be just be like other people. The biggest stars of the world, will eventually be the ones that go into totally new places that others haven't developed yet. The people with the courage to be themselves authentically as possible and applying that into the most useful skills for others, will be inspirations to many people.

SECTION SUMMARY: TECHNO-SYMBIOSIS

The least creepy will always win over the creepy, given enough time. The humanity in people will be what we want in our technology. Humans will prefer more human experiences over creepier experiences, so long as we can afford them. If we have too many creepy experiences, we are then being spied upon and used as things. Let's not

forget Kant's real rule: don't treat people as things to achieve ends. Turning humanity itself into a consumer product has the danger of destroying our humanity. If we want to kill ourselves, then we certainly can listen to those voices that want that and make technology to do that. We can build machines to do whatever we want. The ones that will succeed through time will be the ones that nurture and grow humanity to its very best, because the other paths won't have survivors.

The masters are the ones that train and make the machines that run things. If AI is all controlled by large corporations, and all our data is monitored and controlled by them, then we are slaves and live under extreme oppression. We can be our own masters. Our information can be controlled by us and for our own benefit. We can form personal and private relationships with computers, to improve our lives. Our relationship can only be close if there is genuine trust and safety. Being naked to a machine is not inherently being naked to all of humanity and absolute power above us.

The simplest way to see this, is in how we can train computers to do useful things. If machines can gather all possible data on us, and help use fundamentally express ourselves, then they have a massive data set of information to create value for us and possibly for others. If we are to make AI open for all, so we can empower ourselves as individuals, to express ourselves, then we will need free will to choose a wide array of different technologies and ideas to choose from. People will follow different brands, companies, individuals, movements, organizations, personal relationships, as a way to navigate which sorts of technology to use and integrate into one's life. If you see environmentalism as a driving moral and human force, then it will define everything around you. Other people will have different views about what makes something the most human, and in what they most want to engage with: community, culture, country, pleasure, God, science, technology, faith, ending suffering, travel, experience, food. I

probably want to live in a home on earth geared towards developing space exploration, and higher connections into my community: Techno-Matriarchal-Communal-Libertarian-Patriarch.

We will keep making mistakes, we will keep wanting to believe what we want to believe. Our ideas, no matter how certain we are right, will nearly always fail us. Reality is the best check on our innate human idealism and vanity. Machines are going to shove reality painfully in our faces. They will be intensely accountable, because we will be seeing what they do for us, and seeking ways of getting better results for ourselves. We can learn to see our full humanity in ourselves and direct these things towards getting better results, because our success depends on it. Our own greed, vanity, and aggression can make us safer, because they can push us to do more, and fulfill us more, while being kept in check by a constant self reflection.

Humans by nature stick with what they feel good about, even when it is fundamentally failing. People still believe in socialism and communism, even though they have a 0% success rate in human history. If we want a socialist society, then making it wholly accountable to machines and to human free will, in all its failings, will be the only way to get there. Maybe someone can make it work for a communal business, in the future with technology to support all the reality necessary in unrealistic childish fantasies like socialism. If we can make effective co-op businesses, that enrich the lives of idealistic socialist types, have effective real results, and are consistent with reality, then a great value can be added to humanity. A free market business service that satisfies the fundamental emotional and physical needs of communist/socialist minded people would be the most awesome and ironic invention of some future century.

-Nothing is equal, everything has a price to be paid, and good intentions rarely lead to good results-

ETHICAL CYBERNETIC AI DRIVEN CURRENCY.

The computing power and services we can provide online can be turned into a universal commodity that can underpin the future currencies of the world. The more computing power we have, applied to a wider array of things, the wealthier we are as a species. We can define wealth by how much computing power a person possesses. Wealthy people will have higher degrees of symbiosis that will in turn create higher quality of lives for people around them. The challenge in creating such a commodity from our humanity is that it always preserves and deepens our humanity.

We can eventually make robotics and AI do 99% of work we do now. We make machines, right now, do nearly all the work humans did a thousand years ago, and they only created more jobs for people. Our future jobs won't be creating things to survive, at a basic level. If we wanted to just make food to survive, we can do that with all machines right now. The products we consume add more to our lives than just survival if they are infused with our values, feelings, and complex desires. When we can, we choose to buy food and food services that are more human, fairer, safer, good for the world, good for humanity, builds relationships, and makes individuals happier.

To build "starships", to build the literal future, we have to change our view of money, and how we invest in it. There is no socialism in the future. Capitalism will evolve and how we see and use money will evolve. We can make money itself more human, rather

than cut money out of our lives. There is no exact line between what we can demystify and bring economics into, and what should remain innately human and private. We need to push and experiment and find a better way, so we all can be richer and more human. Monetizing everything is the future of economics. Doing that in a human way, that people will embrace is the challenge of technology to find and implement.

MODERN MONEY: A RETROSPECTIVE

PESSIMISM AND POVERTY- Many people don't trust the financial systems of the world. They don't understand them. They seem deeply unfair, and profoundly inhuman. Money seems to be the very sin of the world, and tainting all decisions, while how it's used is barely used for the betterment of mankind. Those in power get richer, and do so off the foolish financial decisions of the poor. People's feelings about this are a reality, even if it is not wholly based in facts. If people don't use this system, the world will fall into darkness and chaos, and if people keep using this system, it will eventually fall into even darker chaos.

Poor people stay poor, because they spend all their money on things they want or never invest their money properly in the future. Putting money in banks aren't real investments. Wealthy people know where to invest and how to get the best returns across the whole mental spectrum of life and growing up. Prudent risk taking, frugality, deferred gratification, the stability found in a good marriage, using the law to one's advantage, optimism, and investing in a relevant education, all add up into massive wealth in time.

ENERGY IS CURRENCY NOW- Real money in the modern world is fossil fuels and energy production, that is bundled up in what we call the "dollar", that is then

distributed through most of the world into the underlying value of most national currencies. Commodities underpin currencies, and the world shifted away from gold to oil to base money on. Burning fossil fuels built the modern world we live in. If there were no fossil fuels, then there would be no electricity, no industrial age, no engines, no trains, no cars, no computers, and no modern standards of living. Cheap oil gave us a near utopian world of opportunity and success. Oil was a different kind of commodity to base money on, because, unlike gold, we could never have too much of it. Cheaper oil, leads to more energy, and more wealth. Cheap energy is nearly convertible into pure happiness. Where there is cheap, abundant and easy to get energy, you don't have poverty, hunger, or failed governments.

Controlling oil pricing and production is the basis of stable currencies, and American economic imperialism on the Earth. The Saudi Arabians have functioned as the world's "Central Bank" through many decades, with its capacity to control oil prices. When the world converted away from gold to oil prices, Saudi Arabia shortly after started an oil embargo to control oil prices and dominate the world. Energy companies are the most powerful and largest companies in the world. They are more like banks, than real companies. Countries often nationalize and heavily control their energy industry, because of how deeply it connects to the financial system of a country like banking does.

This system incentivizes economic growth through energy production, at the cost of all other things. If fossil fuels are money, and the pursuit of money is the prime motivator of industry, this creates incentives to convert as much fossil fuels into energy and useful things. Global pollution, 3rd world oil country politics, terrorism, modern wars, most financial crises, are all bound up in oil money. While this system has its serious problems, turning against this system would drive all of humanity into a global world war, leave billions of dead, and the world impoverished. The perverse incentives of the

system are far outweighed by the benefits, until alternatives to this financial order become available.

AFRICA AND LEGACY ENERGY SYSTEMS- Black gold needs massive infrastructure to convert it to energy. These systems require massive capital investments and massive procurement of land, which makes implementing them a tedious decades long process. The energy and literally light wasn't turned on in all places in the world. Oil brings no real wealth without cheap energy. The underlying basis of money, must be something fundamentally valuable to human life and survival, where if we have more of it, we are truly wealthier, and it can scale into any country or place without massive legacy infrastructure. Countries like China and India that are recently connecting into this fossil fuel wealth, are choking the world to death in pollution. It has turned the lights on in their countries and created vast wealth, along with staggering environmental decay. If the whole world gains money through fossil fuels, capitalism and humanity can't survive.

FUTURE CURRENCY AND BITCOINS – When a bitcoin like currency can support a fundamentally valuable virtualized commodity, then we can transform the financial system. Information itself is a form of value moving through the internet, that can be turned into money. Turn human driven AI into money itself, and create a currency that can never go bust or be useless. This language of new money, can transform how investments are made, companies are run, and the very internet functions, such that places most resistant to economic growth, can have access to true money and wealth.

"Bitcoin" like things can be the language of a new currency, and not the commodity that underpins it. If we have more bitcoins we are not wealthier, or better off. There is no real commodity of value underpinning it yet, and thus making it useless until it does. Most of our commodities today, are industrial related, and there are not many choices that can be run though the language of bitcoins underlying language and ledger.

AI SMART ENERGY- If energy is money, then turning energy into intelligence itself will transform money into its most efficient and valuable form. The very internet is a sort of “electrical intelligence” that can power the very function and purpose of a thing.

A power plant in the middle of America makes energy, a server farm nearby buys that energy, that server farm then converts that electricity into data processing. This is a radical transformation of energy, data, and AI, into a singular force that can plug all things. We can manifest human wealth and power into any location, with minimal internet infrastructure, and connect it into the world. Money/intelligence/AI/power can move across the planet, between planets, between star systems.

CAPITALISMS REBUKE- Our present system is choking capitalism to death. Modern money has become authoritarian, and not free. “Bitcoins” form a language of this new money that creates a perfect decentralization of power in a system, and make it so no one person or entity can control them. This is their fundamental and basic value beyond the technology itself. Capitalism requires this freedom to prosper. The anger directed at what we call “capitalism” are about the emotions we feel towards this fossil fuel driven financial world order that drives war, oppression, pollution and death, while creating a near utopian life for an elite few in the world.

SOCIALISMS REBUKE OF HUMANITY AS CURRENCY- At the root of socialisms critique of capitalism is a fundamental rejection of turning people into wholly economic entities, where your very life and things you need to live are mere commodities to bought and sold on cold impersonal markets. People want their lives protected, and their beings treated as something more than meat. Going to a job and surviving on wages, where your life is weighted by how valuable your labor is to an employer, is still an affront to much of the human race. They don’t see economics as fundamentally human, like how the religious often don’t see science as moral nor meaningful.

This evolution of economics will have drastic social implications and deep social upheaval in its embrace. Humanity itself has to be turned into a form of money itself, because money is a form of technology that must be driven by humanity to succeed. Embracing capitalism brought material wealth and happiness. To embrace this next stage, will bring even more wealth than the previous stage, and be possibly even more polarizing. There will be those that embrace freedom and economic power, and those that don't. The difference in their wealth will create the most staggering separation in humanity to ever exist, since civilization first came into existence.

THE IDENTITY OF MONEY: DEMYSTIFYING ECONOMICS

Money is a literal mathematical, technological, description of human nature and human value manifested into reality itself.

Economics at its deepest level is a practical science is about giving a price to everything on a fair market. Money and capitalism, can be seen like science destroying magic, or technological progress destroying traditional ways of life. Economics demystifies what we can objectify as a thing and what we can't. Mystical magical trees being cut down for a business deeply offends those that worship trees as spiritual beings. Prices arise from what we are willing to give a price to and what we are not willing to give a price to, and this defines a person's personal theory of economics. People don't like their very feelings and sense of humanity, that they project into reality, having a price put on them. The man who is unwilling to put a price on anything and must define all things by how they feel about them, will always be poor and always be taken advantage of by those that can impose an objective order on reality and place concrete material values on things.

To find the price of something, we must turn it into a thing. In economics, the most important and disturbing rule is that, "all things have a price", which can quickly descend into a belief that "all people have a price". The line between what should have price and what shouldn't have a price is not a settled one, and is basically the dividing line between all theories of economics at their root level. Healthcare, Education, Wages, Housing prices, National Infrastructure, Security, Military, Charities, Social welfare programs, Retirement benefits, resist being defined by price, due to their fundamental collective value to us as humans. Collectively we don't want to know how much they cost, we just want them, and any true economic analysis of them, is basically political death.

They can't be fully looked at as objects without certain groups of people taking deep offense and feeling fundamentally threatened. This is mostly how politics works in the modern world for everything. People that fear numbers and fear math, fear translating reality and human experience into mathematics, then are easily exploited money-wise; they make horrible decisions with their money, and their futures. This power to turn anything in a commodity for sale, feels like rape, and blasphemy to those that don't possess it or who are unwilling to apply it to all things. A person good at math or science may still be emotionally resistant in applying it into other areas outside their education or what is socially okay.

INVESTMENTS:

Economics and investments are based in a belief in the future, and how money works defines how we invest. Wealth is created by intelligent investments into the future. We reward those that believe in a specific better future and take real risks to make that happen. You have to do something that is useful and valuable to others, to get a return

on an investment. Getting the right education, experiences, jobs, and relationships, highly correlate with success in life, and creating value for others. Good investors don't predict singular futures, they predict a probably range of real futures.

The right major you pick in school is more than what major makes the most money at that time, nor what you have the most passion for at that moment in your life. It really should be about what can you do the best, that you believe will be most valuable to humanity 5-10 years from now. Engineering majors, will always make more than English majors, because the future needs them a lot more. Those that invested in engineering fields that weren't popular and that later become high in demand, get rewarded greatly. A purpose and a vision in one's education connected to long term visions and plans, will always pay out better than the ones that don't, in the larger scale of things.

- A college degree, saving money in the bank, owning a home, are not good investments, and people's belief in these false systems are creating most of the financial hardships for common good people, at present.
- Training in highly valuable job skills, spending as little money as possible, owning investments that actually pay out money, are a far more prudent, though incomplete, guide to investing.

Hedging is the smart man's investment in pessimism. You have to bet on optimism more often than pessimism and hedges, to ever come out ahead in the future. You need to spread your investments across different possible futures. No one should be 100% optimistic about the future. A person that successfully predicts and prepares for disaster has done something of great value to society, because they have prepared the world to better cope with that crisis they predicted would happen. Preparing for disasters that can

never happen, and ones that can probably happen have far different returns. If you perceive there is a 10% risk of collapse, then you could wisely invest about 10% of your focus and resources towards possible collapse and hedges against that. Betting 80% of your money on a 1% probability is a really bad call, and letting emotions consume you in the moment. Betting against something and hedging your investments take an entire other level of intelligence that few people can really cope with, understand, or accept the economic value in for society. You have to look difficult things right in the face, and prepare for them, while stilling maintaining a fundamental hope for the future and finding the best things to move forward with.

Much of what broke the financial system of my own country in 2006-20, was people's belief that the home they lived in was an investment, not a hedge. They used the money to buy bigger houses, finance luxurious upgrades, and not homes that would make them safer in case of financial disaster. In countries like China, wealthy people often live in very simple homes, and buy multiple luxury homes as investments. The Chinese by nature buy homes out of pessimism and as a hedge on things going bad. They primarily want to own something that will have some value no matter how bad things get, or how high housing prices may get in the future. They see sky rocketing housing prices, and want to buy a home now, so they don't become too expensive to afford in the future rather than optimism that increasing home values will make them rich. The value of housing prices going up, is actually just making the hedges case stronger, and leading to a more likely crisis. The wealthy Chinese are buying homes overseas on massive scales as a hedge on Chinese economic collapse.

Kinds of "Investments":

- Owning money is investing in government
- Training and education is investing in yourself, and schools

- Owning stock is investing in corporations
- Owning a home is investing in security and the past
- Buying a second home in an undesirable city or place, is investing in the future of that place. (It is BTW, one of the best ways ethically to make money on real estate)
- Starting or backing a small business is investing in the future or in a person you respect
- Donating to charity is investing in culture and society
- Buying a fancy car and luxury goods, is investing your social status or fun.
- College degrees that can't get a high paying job is investing in your own hobby and interests. Or, it is investing in charity, by learning and doing work that is good for society, as you see it.
- Buying gold is investing in the collapse of a whole economy.
- Buying bullets is a hedge against the collapse of society. This is a very popular investment in America lately.
- Having Children is the ultimate of all investments. Its meanings transcend all categories I can say.

You can invest everything in a talented child's education, and in a field/major that will bring them the most success. It is the most pessimistic possible investment that is driven by a hope for the future. Spending your time, money, and energy, to get your child an education, is the duty of parents, in many cultures. Your child becomes an investment in your own social prestige, and possible retirement benefits. You invest in the thing most resembling you, most like you, most controllable by you, has the potential to be more than you, and can exist farther in the future than you. Children will eventually have an identity outside the one you want to control, manipulate, love and manage, and you have to take a risk on that person just becoming something outside your imagination and control. Parents who aren't investing their real selves in their

children like this, will less likely have financially successful kids. Chinese and Indian parents produce more successful kids than most other cultures, and thus are both the longest lasting, largest, most influential, and most powerful civilizations in the history of the world.

A CASE STUDY IN JAPAN:

In America, all economic growth and new jobs come from small companies. The big companies of Japan have no future, and collectively Japan can't let go of them, and invest in a new destiny. Japan has a stagnant economy, because there is no real investing in small businesses and startups. They save all their money, and invest in a past economy of large super corporations. Smashing the present system in Japan, by bringing down the big companies, is simply something they won't do collectively, so there is no future for Japan. Staying loyal to big companies and not letting them lay off workers undermines the future. People don't want to change jobs all the time, like in a real economy. They will have to learn new skills and constantly be aware of the job market. The few who haven't given up, still fight.

The solution is simple, as individuals, they need to find those little points of light in their country to invest in, put their hearts in it, their commitments in them, and their unbreakable resolve. They have to use financial tools outside their banking system, use modern technology, and just take risks on new things, outside the controlled order of Japan's failed economy. This can be done in large groups of highly motivated people. Getting access to small business startup capital in Japan would be easy, on a person to person level, where individuals take their massive savings, and just gamble them on the future. Co-op start-ups, crowd sourced funding, highly skilled motivated labor, strong cultural image and power, expert marketing, deep infrastructure, all make Japan an easy

place to start a “pirate” tech revolution, for the future destiny of a great civilization and its people.

CAPITAL: AMERICAN BANKING

Capital is how well money *moves* to the most effective investments. Banks at their best, create loans that turn into something more than the hard currency that made them. You can think of money like water, where moving rivers are capital, and unmoving money lakes are like just money in itself. The effective movement of money is called capital. Banks create the feeling of a lake and safety for depositors, while in reality move that money into flowing streams and try to get the most effective movement out of that money. The quality of the investments made by banks and investors, determine the value of capital, that with later successes can be reinvested in new ventures. Banks are the infrastructure and “secure” middleman, for lenders and borrowers to leverage money, to create capital. The leaking and uncontrolled loss of water in the system, is far higher than what we might imagine.

Banks make money two ways, one by getting returns on good investments and two by charging fees and making money in the process of making loans. The financial crisis in America was created by Big Banks maximizing number 2 with little regard to number 1. Banks could make mortgages and just sell them, they didn’t have to collect money on good or bad mortgages. They made money on the fees of making those loans. The real burden of investing money in the right places to get the best growth and return on investments, was lost in the pursuit of easy money to be made on government backed fees. The government indirectly insured Big Banks against a loss on poor investments to fulfill political and social aims. If banks don’t make money by their understanding of human beings and where to really invest money, they lose their moral value in an

economy. The more they tie to themselves to the government, the more they invest in government and not people, by buying government bonds that have no real returns.

Knowing who to invest in, is more important than knowing what to invest in.

We can know who are good people, and who have the right mentalities and mind sets to invest money in. The traditional idea of a noble community banker, that makes hand shake deals, looks people in the eyes, knows their community and the people in it, is not something of a bygone era. Community banks provide the majority of the loans to small businesses that drive the majority of economic development and nearly all new jobs. The only way to know how to really invest money wisely, is to invest in the humanity and dignity you see in a human being right before you. We can't know the future, and what will work or not work.

**A CONSPIRACY THEORY: THE GREED OF THE AMERICAN PEOPLE
THEMSELVES CAUSED THEIR OWN FINANCIAL CRISIS.**

The real cause of the financial crisis in America was the greed induced in the American people, by the illusion that they were making lots of money as their houses went up in price. All that greed was then feed into a confirmation bias that a mortgage itself was a good investment that could make money. The more people believed that, the more they poured money into this false financial investment. This led to inherent moment when the limits of reality and fantasy had to part.

The banks weren't the villain, because if anyone tried to stop this expansion of money through fake capitals streams, the American people with massive new forms of fake wealth, would have been pissed. When this fake money inflated the value of people's

homes, it created a fake sense of having made lots of money, from doing nothing. Normal people were "making" money from just owning a home. It was the American people that elected the government that pushed the policies for this easy access of money, and vast expansion of government backed credit. It was then these same people that would never allow a change to the policies that led America to that point.

Though at another level, the financial crisis was caused by the rise of China. The Chinese government did what "responsible" people did with all its excess money. It bought safe American bonds to back its currency. The Chinese government flooded \$ 2 trillion in cheap money into the American economy, and Americans used that money to buy bigger homes, fancy bathrooms, and bigger cars. The money was there and people chose to invest their money in useless crap, that weren't real investments. Buying a big home is not an investment. The American people could get cheap money for loans on houses and not investments. People understand homes and not financial tools. Just imagine if all that Chinese money had gone into venture capital investments for new businesses. The initial Internet investment bubble had recently burst around that time too, and so people were not optimistic about the stock market and advanced finance.

HOW THE FED WORKS: BANKS INVESTING IN THE GOVERNMENT

AT ITS SIMPLEST- The central bank of a country, at the most basic level, is a bank you walk into with a brick of "gold", you hand it to them, they figure out its value, and then give you money for it. The key concept that people are rarely taught and don't understand is where that money comes from that is exchanged for a bar of "gold". The money is made in the very moment of the exchange, and didn't exist yet. The central bank can in a sense just go into a back room, and print out new money, with new serial

numbers to buy that "gold" with it. The "gold" is converted into a government document, that can be used as money. The bank has the "gold", the new money is backed by that "gold". If a lot of new "gold" is found, the bank can make lots of new money, and have more "gold" in its reserves. This power to make money from nothing can be used to buy **any asset** the bank wants. They can buy up a bunch of commodities, give out money that never existed, then later sell those commodities at a higher price, the money that goes back into the central bank is then no longer part of the economy. They need to buy things that are very stable, seem very responsible, and by nature don't change value.

LOANS TO BANKS- Central banks can convert non-existent things into new money, by creating loans banks can buy. Normal banks can borrow money from central banks and the normal bank just gives a debt to the bank with interest in exchange for money. Debt is the commodity the central bank converts into currency, that can be spent. The central bank will have a loan on its books and the big bank will have newly made cash. Most of what central banks buy are valueless paper documents, that are converted into real money, that people perceive as being reliable and valuable. Banks will actually pay back the Central bank with actual returns on investments they make in the real economy, from actual profits made by real companies. If one assumes large responsible banks won't collapse then the Fed can push its new money through them, and be certain their loans are solid, reliable and real.

BONDS- Central banks buy Government Bonds directly and indirectly. Governments can now run debts as high as they want with Central Banks buying government debts like they were real commodities similar in value to gold. Central Banks can buy the debt of other countries and fund each other's own debt spending, with no actual connection to economics, or real financial returns. Governments never have to get returns on those debts, or pay them back. They pay off their "credit card" with a new "credit card". The Fed at its basic level, is not much more than a middle man for a government credit card, where they can tax people without elections, voting, or accountability. Properly speaking government bonds should be bought by real people and investors and not government

institutions. During World War 2 the US government had to get American people to buy war bonds to fund the war, and didn't just buy its own bonds through the central bank. The complex relationships between governments, banks, and central banks trading debt are taxes we can't see nor stop.

GDP GROWTH- The Fed can directly influence the economic growth and development of a country. With more money that goes out, at lower interest rates to banks, the more money banks have to loan. This "stimulates" the economy. Money is made super cheap for banks to get, and they can loan it out at lower rates, and get a profit. There is a limit though to how many good investments can actually be made in an economy, when banks have to go looking for investments, rather than a line of investors seeking scarce money. If there is simply no good investment to be made, and lots of cheap money, then money flows into fake investments that later blow up. The Fed creates financial bubbles that later explode, if they leave out too much cheap money. The Fed has to keep the economy from getting too hot, and too loose with its money. The real economy, the actual things that can make money only increase at a certain rate. If the economy is growing faster than the actual real capacity for it to make things of value, the Fed increases interest rates.

TOXIC ASSETS- Central banks don't have to be responsible. They can loan money out at 0% interest and basically give free money away. Then they can also convert bad assets into real money. Instead of buying solid and reliable assets, they buy toxic and broken assets, and passes the risk on to the common people, and help support those that are invested in those toxic assets. In America, the Fed bought up a bunch of nasty loans from big banks, converted them into new money, and told the banks to go find better investments to put this money in than toxic mortgages. The big banks got rid of their toxic loans and didn't go invest that new money in the economy. They instead bought more American debt with it and deposited money in the fed, to get interest on deposits with. The government saved them, so they invested in more government and not business.

A LITTLE CONSPIRACY THEORY: Governments can come up with lots of new products to sell banks and dress them up in different ways. They all end up as corruption. The reason governments can grow larger, produce massive debts, and provide no benefit equal to its cost, is because they create a relationship with the financial systems of the world to invest in making them bigger and not the free people of the world. The government can always change the rules when they have to pay back their debts, because they control the very means of creating money itself. Banks and the government, drive money into government debt, and then inflate the currency those debts are owed in to pay for them, while banks take payments in fees on those transactions, and government workers and politicians take jobs in industries that debt money flows into. Government created inflation is then channeled through the banks, so while they lose value on the money they hold in deposits, they gain profits in the newly created money lent to them by the government at low rates. Which they can then dump into even more government debt, that is bought with money given to them for free from the Fed.

THE FED, TREASURY, AND BANKS MYSTERY RELATIONSHIP- Banks all over the world invest money in countries and not the people or companies, because it is a safer investment with guaranteed profits. The whole process is made nearly incompressible so people can't understand what is really happening. Money and assets are exchanged in a very complex system to make illusionary things look real. Government bonds get created by treasury departments, bought by banks, banks deposit the bonds in Central banks and collect interest payments.

- All the horrible loans that created the housing crisis, got bought up by the fed over time.

- Americas central bank created trillions of dollars and gave it to banks to stimulate the economy and little of it was even loaned out.
- Profits were made on fees and inside government dealings, rather than real economics.
- The government financed its own debt making system, and blamed the banks, while making them richer at the same time.
 - Every new regulation on banks in America, only made big banks richer, have less competition, and make more money from government dealings.
 - Small community banks got smaller and big banks got bigger. All new small business loans come from community banks.
- The US government basically taxed the worlds money supply and channeled it into its own banking system to keep it afloat and fund government spending.
 - Countries with lots of US Dollars basically had to bail out America, without any say or vote in it.
 - Americans with cash got it funneled back to them through the US banking system, and in spending by the US government, which led to upper and upper middle class, in America, making fortunes and poor people getting poorer.
 - Small businesses and startups didn't get money, didn't get help. They weren't big enough to play politics and get a piece of the free money.

NEW INVESTMENT CONCEPT: CHINA, WECHAT, AND OPEN FINANCE

A NEW VISION OF MONEY

To really take economics to the next level, we have to make more of the world, literally, *accountable*, something that can be stored on ledgers. Many things you would never consider to place on a ledger, need to be placed there, to determine their value. People may look at "spending time with their kids" and have not determined the

cost it takes to do that, and the value that is created from doing it. If they could get an accurate accounting of the real value and costs, they could make better tradeoffs. To understand the value of things, we have to take apart the very identity of things, and lay them out in parts.

The history of modern business is a history of accounting. Accounting creates the basis for all economic systems to exist. Red ink and Black ink, liabilities and assets, are the ledger by which value is determined out of parts. Good bookkeeping is good business. Simple ledgers get approximate values of different "parts", in a written document. The parts are very generally defined in primitive ledgers. A household ledger is like: liabilities; groceries, rent, entertainment, cars, insurance, kids, and assets; stocks, bank accounts, property, cash, mutual funds, government protected funds, pension checks. More accurate ledgers list more detailed aspects of all of these things, listing everything money is spent on and everything that stores or brings in money. Loaning money to another person creates money on ledgers, purely on paper. I can make the money I have pay me money, by renting it to other people. If I rent my house and get rent for it, I consider that house to still be mine. Money is rented out for a price, and everyone gains. Ledgers get more complex with loans.

Can we place a monetary value on...

- a mother's love?
- Quality time raising your children?
- You own innate self-worth?
- An education?
- Marriage and committed love?

There is value being generated, it creates real wealth, and yet can't really be tracked. A mothers' love does have a concrete value in dollars in how much value it can create

through the healthy functioning human beings that receive it. Not all love creates equal value, and simply spending time with children and loving them, can have no value, if this time instills poor values and ideas in kids. If a mother could know what sort of things that they can do for their children that will help them the most, they do those, and then they can minimize the things that don't actually help. It requires economics, science, analysis, and common sense. The value of good and bad things you do for your kids, in dollar terms, would cause great stress and anguish, to most. Things that can have a real price, that can be tracked and need to be seen will be looked away from to appease some parents own vanity and pride. What parent wants an APP to show how poorly they are investing their time and money into their somewhat idiotic children?

BIG DATA AND ECONOMICS

Google-Ads, Online Advertising in general, is the underlying currency of the internet, that drives real money and investment into servers, hardware, and the actual salaries of human beings. Every single aspect of advertising, every single interaction a customer can make, and every single arrangement of services to support that, has been monetized to highly precise levels. Users get lots of free services that are paid for with their own personal data that are used to help ad companies target people better on the internet. Google doesn't process most of the data it has, because people would reject such intrusions of their privacy. "Big Data", in general, is about people, and creating value from the very identity of who people are in very simple ways.

Companies that have data, are like the original banks, that have money sitting in safety deposit boxes, and haven't figured out how to leverage that money to do something. They can do basic loans, they can do basic analysis to provide basic services, and just now think they have hit their killer service. Googles wealth comes from data analysis of

1-4% of its data. if they could monetize 30% of it they could make astronomical amounts of money. It creates a kind of gold fever that leads to poor decision making, and the inability to properly use that data to benefit society and people. By shifting their mentalities to a new paradigm and relationship with customers, data itself can be transformed into this massive wealth generating system. It though won't be fully in the power of companies, and this makes the promise of big data a more magical unicorn than a reality. People will use big data with their own lives, when that power is more decentralized and in their own control, directly acting in their benefit.

Spain in the 16th century, or modern-day oil emirates, are countries that believed they were rich, because they had lots of "gold".

INVESTING IN PEOPLE IN A PESSIMISTIC REALITY: CHINA AS A NEW WORLD MODEL

If you don't have optimism in your country, or the world, you still must find something as a human being you believe in, and invest your money, with all your heart and soul into it, if you want to see a future for anything you have ever loved on this earth.

American culture with its openness, respect for law and contracts, and ability to assimilate new cultures and ideas, became a world standard for how to run businesses, and created an international global economy. The Chinese have copied the American model and added in their own culture and history to it. It has the potential to be the basis of a modern investment technology that is built from the humanity of people, and for companies operating at the smallest level, starting from nothing. If you can make

something that works in China, it can readily translate into other countries with low economic development, highly pessimistic people, histories of epic tragedy, famine, highly corrupt governments, and weak civic institutions.

“Bamboo Business Networks” built some of the greatest economies in modern history, inside completely broken and corrupt systems, where people were deeply pessimistic about the world. Chinese culture is paranoid and pessimistic, and yet has intense entrepreneurship and intense warm social bonds between family and close connections. You can have pessimism in the world, and optimism in a narrow range of things, like say family, friends, and social networks, and still drive a personal network of loaning, borrowing and investment. This leads to a lot of corruption, and it also allows small businesses access to capital they could never get otherwise.

This business culture can create, in an idealized fashion, a business standard for international finance and entrepreneurship, if we integrate high level technology that can bypass the inherent internal corruption in traditional Chinese culture. By taking Chinese traditional investment culture, and backing it up with computers, fraud analysis, machine learning, apps, or whatever, then those personal relationships can be more freely invested in, and reach out to a wider circle of connections. You would be much more likely to loan money to a friend if you could monitor every single aspect of the business they needed it for, and have guaranteed returns on your loan, if they make a profit.

For Americans, this would be like investing in a food truck, based on your belief in the person doing the food truck and how you feel about the food. You can invest in a person and a food you have a passion for. And this feeling can be backed up with actual data, open book keeping, live cameras of the van inside, machine learning driven fraud

prevention, or whatever will be developed to make those personal investments more secure, while still being based in our humanity and personal feelings.

WECHAT MONEY AND CHINESE INVESTMENTS.

In China, the largest and powerful app is WeChat. It is a social network, and messaging platform, that has a financial service linked to it. WeChat is a huge success because it connects into our humanity to drive its tech development. WeChat began its money system with just transferring money between friends, moved into being able to find and pay for dinners, movie tickets, concerts, donate to charity, and any other very human, social, and connected thing they could.

- You can deposit money from your bank account into a WeChat account.
- With this money, you message your friend money, instantly in real time.
- You can buy many things online with this account too.
- You can also go into actual stores and pay for things from your mobile phone, through your WeChat account.
- Using WeChat wallet has a this very personal and profound feeling that comes with it.
 - It is not like just using another payment system.
 - It feels very human, and makes using it much more appealing and engaging, than can readily be described,
 - unless one has such an experience of using a socially driven virtual online currency.
- WeChat focuses on using its wallet to increase social interaction, and enrich your personal life, and not just using it like a super bank card.
- Most of the things that you can spend money on in its system are socially connected, or amplify the quality of your life.

The money that is being moved around in the accounts, is not real money. All the money is just deposited and controlled by WeChat. They have ledgers of where all the money “should” be, though in reality, people just deposited money to WeChat, and WeChat converted that money into its own currency, that can move around freely. If you want to withdraw money from the system, into real cash bank accounts, then you pay a fee, which is really a currency exchange fee. Money in the WeChat system, moves instantly around, to where ever you want it. If you withdraw money, it happens much slower, and is driven by more traditional financial systems, that it has to interact with.

WeChat can also allow small companies to create platforms, for people to invest money in “venture capita”l. You can play venture capital investment games, with actual businesses. You can play a Farmville, with actual farms, and not just in China, but in other countries. Money can move freely, instantly, and socially. You can create apps, that power investment portfolios, venture capital, mini-hedge funds, or you being a bank yourself. Just forming on this platform, are tools of advanced finance that can be made into smaller versions driven by high technology and virtual online currencies.

You can make more personal and human investments. Risks can be mitigated, by fraud analysis, and real-time data and information on your investments. If you invest in a batch of a fish-farms fish, you can monitor the farm, by live cameras and access to all their financial data. These are all new things, and barely developed. Fraud will be widespread in new virtualized banking systems, sadly, because no one really understands them, and the real future they have, so bad people gain.

"Open Finance" is to open up all the financial data, the complete ledger, in real time of a company. Investors can do real time external audits and oversee fraud prevention systems. If there was a "stock market" of companies that had open finance, this would revolutionize how we invest money, and analyze companies. Companies that want access to this modern open stock market, will need to use open financial systems and other open systems from their very start. The systems of open finance must empower companies that are just starting, by being cheapest, most powerful, and socially connected platform, that then gives them easier access to financing because of how open they are to potential investors. If a company from the beginning used such a system, to lure in investment and create real verifiable trust, they could adopt an open finance system more readily in their company DNA, and be a different kind of company right at its founding.

Finance is one of the hardest things for new and small businesses. The most advanced software won't make it any easier, unless the very underlying way one does business and uses money is transformed. Traditional finance and money, can never fully be represented and tracked online to the necessary levels, as compared to a wholly virtualized currency. When you move around real money, there are fees and costs incurred, there is a friction created from the rules and the existing financial order. That friction takes away a percentage of money at every exchange. Payroll costs money to transfer money to staff, and uses actual HR staff to do it. By eliminating nearly all internal cost of moving money around, value is created and smaller companies can now afford to do things they couldn't do before.

Everything you buy will also be bought and tracked in that currency, all online purchasing, with no transaction fees, no credit card fees.

- The items themselves that you bought will also be instantly inventoried and tracked, with data.
- A ledger won't just be a simple object description with cost and quantity: 200 boxes of Acme Super Pens.
- It can have machine readable data, that tracks many aspects of those objects and connects them to a wide array of things. 1/10th of cent values can be tracked.
- All vendor, internal, and customer transactions, are done within this one system, with open data.

It will lead them towards deeper human experience, and make even the finances of a company a humanizing thing. This also can't finance organized crime, terrorists, and revolutions, because it requires openness, to get the money, and thus it would be very easy for government to infiltrate and get that data, and know all the things a "company", was doing. This is oversight, beyond anything a government can do now, and it is for reasons nobler, freer, and more humanizing. Having all your finances watched for government ends or for your own ends, is two very different experiences; one is voluntary, and the other is not.

PART 2: THE FALL OF CIVILIZATION- ZOMBIE WORLD

True unimaginable economic wealth can be made from unleashing the whole of humanities potential.

We are already living in the apocalypse. Most of the worlds countries live in hell. Within developed countries mass sections of society are disconnected from the future and any hope of a better life. We are already dying. Civilization is at its greatest level ever, and also has more people live in suffering than ever before. More humans today live in hell, in a single country, then there were humans in the times of ancient empires. Functioning civilization and society is a rare thing. How we build technology can shape how we survive hard times when society falls apart. First world countries need their fundamental systems to all be based in how to build up a society and system from nothing, like the basic protocols of the internet and its entire architecture. When something can function from the smallest size and grow outwards, this is how you build order from nothingness. Decentralized systems, promote freedom and network efficiency, while creating a framework to build civilization from nothing in any location.

Designing great technology at the highest levels, can be inspired from a vision of the world dying from massive armies of the undead consuming and killing us. Our darkest nightmares can be used to guide and create truly human and beautiful life affirming technology, and connect us to the humanity of the world we are disconnected from. Our fundamental humanity builds technology, and it is the basis of everything we value. We are also the ugly "humanity" that exists in a fallen world without civilization or operating social institution. It is that "humanity" that lies, cheats, steals and kills over nothing, that we must embrace to fully prosper. We must never cut out this part of ourselves out of ourselves and civilization, to prosper. We are the descendants of those that slaughtered every inhuman and uncivilized person they could find. Our ideals of humanity, and the reality of humanity are difficult to reconcile. Idealism literally and

actively kills people, to create the world of those ideals, and it is how peace has been achieved through all of human existence.

We often try to help suffering groups, and nearly always fail, because we use the wrong thinking and principles to help. In many ways imbedded in the very help and idealism to help the less fortunate lies innate human desires to kill off the vulnerable and less able. Only capitalism has ever helped people in the modern history of the world, with its pursuit of individual success. People don't need pity, they need investment, new companies, new jobs, and a better future for their families and communities. African development plans for much of history have been based on inhumane ideas, and have largely failed on the most massive of scales.

In recent years, people that have been investing in Africa for business, have found success. Chinese investments are helping Africa grow. The Chinese more fundamentally understand the collapse of society and what it takes to rise up out of it. They lived through their own apocalypse and endured some of the darkest horrors of the 20th century. This gives China a deep competitive advantage over the West. A totalitarian communist country uses primitive state capitalism to connect to and relate with Africa, while capitalist "free countries" use socialism and communism, to relate to this region and its peoples. One leads to prosperity and the other to perpetual darkness, genocide and death.

Much of the development of the world's economy, in the near future, will have to be in some of the worst places on earth, that have been resistant to development and modern business. As we develop the most underdeveloped places, we can release their latent potential for wealth generation and prosperity. If done right, the worst places on earth, provide the most opportunity for investments to get the highest returns. All development

in the modern world comes from cheap energy found in fossil fuels. This system is very heavy and requires a lot of people and resources to run, and so can't really solve the deeper underlying problems of survival. This though can't just be brought to Africa anytime soon, not without a lot more development in these regions. We need something that can be done with less initial resources and can scale readily with demand, that can deliver wealth and freedom to a dark and lost continent of human beings, based on their own humanity and dreams for themselves.

CHAPTER 5: JOSHUA'S GUIDE TO SURVIVING THE ZOMBIE APOCALYPSE

Are you ready to commit genocide for your ideals?

The zombie apocalypse is the release of humanities will to commit genocide in the name of preserving what people feel is human to them. To orient this to the future. Imagine a better future, a distant future, and imagine the sorts of people in it, and who won't be in it. Your vision of that future, and your ideals, shape the groups you will form in the present. The apocalypse is scary because people can be revealed as who they really are on the inside. All the social norms that represses their identity can come raging out. If you vision of the future doesn't have certain groups of people in it, then in apocalypse you will have no problem letting those people die nor killing them to survive.

QUESTIONS:

Consider yourself living in the Zombie apocalypse or any sort of end of the world scenario.

- What is your sense of humanity? What is your vision of the future? If you are a black liberal single woman, would a conservative patriarchal Asian man be acceptable as a husband? Would you value the lives of people you once hated in the old broken world?
- What sort of groups would you want to be recruited into, and what sort of groups would you not want to be recruited into?
 - Would you enter a group that was a completely different race from you? Culture? Class? Ethnicity? Sexual Orientation? Political affiliations?

Educational background? Religion? And if so which ones more than others?

- Would a French black gay liberal evolutionary biologist want to join a group of all white conservative American Mormons? Should he convert to the religion and marry a woman to be a member?
- Does an ideal society, for you in the future, have.....
 - Gays, Straights, Conservatives, Religion, Socialists, Atheists, Israelis, Japanese, Russians, Chinese, Science, Technology, Capitalism, Money, Banks, Roads, God, Terrorism, Crime, Murder, Hate, Failure, Ugly people, Domestic animals to eat, Equality, or Inequality????
 - If it doesn't have any of these things, then you are probably an extremist, and your idealism will kill off whoever doesn't fit your ideal vision of the world.
- Should you all dress similar, have rituals, speak similar, have standard protocols for doing things, and be "patriotic"?
- Is marriage and child rearing the foundation of civilization? If they aren't how can you have a future without family and children?

THE DAY THE WORLD ENDS. GO GET A POINTY THING AND POKE!

Zombies break out in front of you, you are unprepared and don't even know what you are seeing. You are probably going to freeze and die. You will hesitate, you will not act. You will die very soon and very badly. You won't even be lucky enough to be a zombie, because you will just get eaten by a hungry fresh herd. The world will be falling apart before your eyes, and you will see people dying and doing the most horrible things in

order to survive, while armies of the undead lay waste to your world, home, civilization and everything you have ever loved. See your family, the ones you love, being eaten. See the home you lived in burning before you, and everything being covered in shit and dead bodies. Anything you have ever imagined as sacred or real, will be smashed. Any imaginable cruelty that can be done to a person, will be tried. If people can imagine it, then it has been tried, and will be tried again.

Staying calm is the most important thing, and remember you have a future you want to build and be part of, from the essence and soul of your being. You will need a weapon, armor, supplies, and a shelter. Your goal will be absolute victory, and nothing less. The Greeks defeated vast zombie hordes in the past, and you will have to learn how to do this again. Get yourself a back pack, and start getting the things you need to live. In these moments of utter clarity, there is nothing to fear. Star straight at life, and run for it.

GETTING BASIC SUPPLIES. BIKES

You are going to be making yourself into a light skirmisher. A scavenger tinker. You will be adept at range, tactics, speed, and using your surroundings.

Bicycles are your best friend. There will be plenty of them, with tools and parts to repair them. Bikes will always be faster than any zombie, can go over most imagined terrain, be easily carried, be taken into buildings, be a storage center for tools you collect. Start collecting rope when you have a bike. Rope is a tool, a weapon, a wall, an escape route, a home, and civilizations best friend.

Eat everything in sight. High fat food is the best. Drink all the water you can. Food will be easy to get at first. Don't worry about getting fat; you should try to get fat. Forget any notions of a diet, that you once had.

Get a backpack to carry things. First aid kit. Canned food, water. Cooking oil. Pocket knives. small common tools. Saws. Light axes. hammers. Duct tape. Light rope. Containers to hold water. Basic camping equipment. Bullets. Vodka (put in plastic bottles). Screws. Bolts. Wire. Sewing kit. Fire making tools. Small bottle of fuel and a lighter can get most fires going. You won't need lots of fuel to make a fire, if you know the basics of fire making.

WEAPONS:

- **You are too weak to fight, and should avoid direct combat, at all times.** Get Poles that can push people off you and to get away with. Distance, range, breaking knee caps, running away, are the way to fight. In that moment, right in front of you, just find something to keep people off, and get away with.
- **The gun store is not some genius place to go, it will be a bloodbath of human on human violence.** If you have a gun and lots of bullets, don't use them or show them. Think of every bullet you have as being money. Guns are not as useful as you think, unless you want to die yourself.
- **The more your weapon can also be a useful tool, the better.** Tools make you more friends, than pure weapons, and are far more useful. Start building a tool belt or a bag of tools. Getting tools, being light, being fast, and good scouting are the most important things. Framing hammers seem to be about the best common kind of weapon/tool out there. Light "tomahawk" axes are another useful tool/weapon, that can help you in nearly all situations.

ARMOR:

Think about comfort and how things feel on your skin. Don't be immune to your senses and emotions. Comfort, breathability, sweating, and mobility all matter. You need something you can wear all the time. Heavy bullet proof Jackets are awesome unless you have to hike in hot weather for 10 hours. Police riot gear is at some level the perfect armor since it is designed for zombies.

You will need to start gathering parts to assemble armor. Gathering a large tool kit will be essential: knives, heavy scissors, pliers, wrenches, bolts, wire, nails, glue, sewing equipment, wire, straps, duct tape. Gather the best materials to wear on your body: Ballistic nylon, gortex, leather, spandex, hemp, linen, silk, straps, cording. Scissors, and straps can make quick armor out a back pack or suitcase, made from strong materials.

Wear gloves forearm guards and boots with shin guards all the time. Boots are armor and a great weapon. Kicking and stomping, will be most of how you fight, not punching or swinging things. You can readily do push kicks and stomps, not martial arts spinning and flying kicks. Kick to get away from a fight, and cripple a knee as much as possible. You will need to build up armor around your shoes up to your knees and on your forearms.

Helmets. Find one that works for you, and find better ones, when you can. Police riot gear helmets are probably good enough. Your helmets are far more about social meaning, then they are about defense. They will form the social collective identity of a group and of individuals. Your helmet says what kind of person you are on a battlefield.

You need to organize around bases, and develop your bases in size and location through time. Your very first few hours of survival, will need its own temporary base. You can't plan, build, rest, or prepare without a base, no matter how small.

Workshops and the culture of work are the basic humanity and technology to create civilization from nothing. Investing in the future is created in your work, and preparation for it. Your temporary base of operations will be a workshop to make things. Get a location you can defend, wall off, and escape vertically from easily. Acquiring your initial group will be based in reality: family, friends, co-workers, classmates, members of community organizations you belong to or have belonged to, or public servants you don't know. The hardest situation will be apartments in dense cities. Since most people in cities don't even know their neighbor, organizing a building for common defense is unlikely, and being locked up in your room, makes you an easy target. Your house may seem safe; it is in reality a dangerous place you may need to move from quickly. Get all the things you need in there and move. You never had a real home to begin with.

You have to start making people that are all equipped with basic gear and a workshop. Go get people and lead them to places to manufacture weapons and armor. You will need to find people near you. Get your people together, and have a plan for after you get together, before going out. Basic organizing of people, that haven't lost their minds, can save you, though don't depend on them.

WORKSHOP AND TECH SUPPLIES:

Technology is going to be critical to survival. This is not a return to the stone age. At worst, you get back to a pre-industrial society level of organization, with all the infrastructure already built for an advanced society. You will have your mobile phone and messaging apps. If you happen to have the time convenience, then you can also gather up computers, routers, phones, cables, networking gear, because these will be the most

valuable things you can own, with the right set up and software. Saving IT guys that can run this stuff will save your entire world.

SUMMARY OF GEAR:

You might be alone or with a few people. You are going to need to move yourself and your gear, that you can travel with, on a bike and on your back. Eat fatty foods.

All this basic assembly and gathering of equipment can be done in a few hours. You shouldn't go anywhere, until you have the basics. In large cities where most people live, there is a societal collapse already, that is held together by social norms and simple prosperity. If you take away hope from a city of people with no real connections to each other, no basic trust, then you will have a brutal reality descend on you. You will have to forge groups of people to work together, in order to survive. There is no lone wolf survivor, of any real significance nor meaning. Surviving is not enough to survive. You need people to build a new world with, and to create life with. If you plan only to survive with no long-term vision of the future, you are dead already. Zombies and the apocalypse are when people have no individual vision of the future, and the societal institutions that binds these goals together in an overall sense of hope and purpose.

GETTING TO YOUR LAUNCHING POINT

MAKING A LOCATION TO MEET YOUR FAMILY:

You must arrange where to meet up. It has to be a place you know well, have access to, and possibly someone in your group even owns. You should consider places you can

make things as ideal locations, because manufacturing supplies is a central task. It can't be a famous location, or a place with heavy resources. You need to be on the side of resource rich areas, be out of sight, and have a defensible location with many ways to retreat. It can't be some obvious place other people would go. I should be along major transportation corridors, with lots of things to survive with. Your sense of the familiar will attach you to certain physical locations, and this feeling can kill you. Your sense of personal attachments though can save you, and help you form larger social groups.

TRAVELING TO FIND YOUR FAMILY: SCOUTING

Sensitivity is a survival advantage, if you aren't going to fight directly. You are going to have to move some long distances to form into the groups you will need to survive. Random people nearby are far less valuable than close relationships within a day's journey. Traveling is not so hard, and not so scary. The world has plenty of places to hide, and there will be so much chaos, that you can disappear. Open your senses, and feel the reality of the world around you. Let it flow through you and around, and don't resist it.

Stay in contact with your group in a group messenger chat, as you navigate where you are going. Tell people everything you are doing. Send photos. You won't be smart, because of fear and uncertainty, and you can get other people looking at your situation. If you are one on one chatting, then you are more likely to get bad advice. Get ideas and information from people, not advice. You have to trust your own judgment and decisions, with others giving feedback and insight.

Don't tell people what to do. Be mindful of how insecure and unconfident young women are going to be. You are not going to help them by reassuring them, you will

help them by giving them useful information. Peoples lack of self-confidence will make them very vulnerable to listening to everyone around them, and not trusting their own instincts.

Environmental awareness is key. Expose as little of yourself as you can: sight, sound, smell, taste, touch. Use your senses to feel and avoid danger, rather than become terrified and overwhelmed. You always move from points of cover to points of cover. High ground, and places with multiple routes of advancement and escape are useful. Use mirrors and or cameras to look around all corners, so you can move from covered location to covered location. Rope is useful.

Always scout where you are going. Every street, every intersection, every destination. Scout them and plan for escape, at any given moment. You will need to get to a meeting point with people, and pick others up, on the way. This will take planning, and foresight. People in a newly falling civilization, won't have the mentalities to focus on all the tiny details of route planning. "I am just going to run if anything happens" are the words of the pre-bitten. If you are running or fighting, then you have already failed.

YOUR SECOND BASE

You may need to sweep a whole building before settling in. Don't kill the zombies. Simply lock them off. Enough melee battles will eventually end in your death. Securing a location is not that hard. Taking out stairs and getting up high, make safety easy. Apartment buildings can easily block off a first floor, and access to higher floors. Ropes and ladders can be dropped from second floors and then can be easily pulled up.

You will make a short-term base, before heading on a long journey. Set up a more complex workshop of many people. Get mostly women and children, sewing, and making light armor. Everything is about mobility, ease of repair, redundancy. You need

to make your workshop portable. You will need to scout and scavenge. Avoid obvious places. Avoid people. Don't build anything outside.

Lay traps for escape routes. String wire and rope across certain areas to trip people or zombies as you go out, you lay traps for the route going back the way you came. You then can run back along your route, and be reasonable safe. Intelligence and future planning can easily defeat those that have no planning and vision of the future. It is why killing them is so easy, and why you will win.

Bringing strangers "home" is a bad idea. Don't bring people to your base, that you meet scavenging. Your base is to gather supplies, so that you can leave the place you are in. You can help individual people be prepared and suited up: weapons, tools, armor, food, water, bicycle, and so on. Build a model of survival and organization that can replicate easily around you, and in your interactions. Close relationships and trust take real time and real work, and can't just be assumed because you want it to be true. In a group, trust is a great thing, and it can be taken advantage of by outsiders.

WHERE TO GO TO START OVER

Supply up, make a plan, and get out of the city in a group by bicycle. You will have bicycles, armor, back packs, weapons, tool kits, first aid kits, sewing and armor kits, internet gear. You will probably need to travel 100 to 30,000 km to get to a suitable location. You will need personal connections in that place, if you are to go there. Good places will be easily fortified and keep outsiders out. You may need to develop a leap frog plan of base building, along a long route, within the social network of people you know. You will eventually within in weeks or months, reach a location to settle in for the next few years.

Small towns are the obvious places to go. Resource rich. Fenced off farmable land. Independent sewage and water systems. Can block off entrance by zombies with just 4-6 barricades across the 2-3 streets that will run through a small town. Fences and properties will create natural boundaries. Towers and scouting can help organize defenses, readily.

The wilderness may seem ideal, but it lacks in resources, unless you know how to hunt and live off the land. You will most likely become a raider that lives in the woods and scavenges nearby cities, if you move there. Homeless, starving and ready to kill anyone to survive, is not the best way to be living.

Farmland and ranches will provide everything, and have all the means to survive on their own. These locations are rarer and will be taken. They will be away from people, and large dangers. You may have no clue how to survive in such a place, and they don't provide plentiful resources to really build a future. If you know how to live there, then go live there. Make food and live. Farming is the foundation of civilization and why it even came to exist.

ON COMPASSION AND GROUP COHESION

Having a plan, with a future is important to keep a group together, and keep people focused on what they are doing. Focusing people on the task at hand, will take compassion and good communication skills. Humans are going to fall into despair and get lost. You have to hold your group together, and tend to emotional wounds in people. Groups will not win based on how well they can fight, they will win because of how well they can plan for the future and hold together as a unified group of people.

Your natural inclinations to comfort people, needs to be transformed into empowering people with the means to survive. Kindness isn't always soft on the outside. Focus all communication and actions on things that will lead to winning. Sensitivity gives you awareness, and knowing how to use the least force to achieve something. True compassion and kindness requires brutal and direct strength.

BEFORE YOU LEAVE, LAST THINGS, AND NEW PEOPLE

FAMILIES ARE THE BASIS OF CIVILIZATION.

-Everyone must have a family or seek to build a family-

Single men are the largest danger to society possible. Men need people they are willing to die for, and direct the purpose of their lives towards. People will need to form families, and get closer to their own. All adult men must have partners, and people they are responsible for; these connections need to develop quickly. Lone survivors roaming the wastelands of civilization are the very people that won't be that relevant. Civilization was not driving by men out in the middle of nowhere by themselves. Society has groups families working together, and men sacrificing for their future.

Families are the foundation of civilization and the most efficient ways to use resources. Wealth in all modern societies is created mostly by the money saving features found in the efficiencies of marriage. Single people are not needed. Boyfriends don't leap on grenades for their girlfriends or strangers. Men sacrifice their lives for what they love, and feel a permanent bond to. New marriages, and old ones surviving and getting stronger are true measures of a group's overall social fitness.

Pick a leader fast. Groups of families will form into basic groupings. You need to form up with who you can work with and have a basic operating trust. Arguments over

hierarchy and leadership, need to be settled quick. Get each individual person focused on tasks they need to accomplish.

Everyone needs to have connections in the group, and these connections form a network of identity. You will form a base group of 5-9 people, family and close friends. Then form larger groupings (bands) of people from within your social network of 30-60 people. This might only have 10-15 able bodied men, that can fight. You can't rush this group formation, and it has to be driven by real connections that exist between people. Adopting children, siblings, and even parents, might be crucial, where people have new brothers, sisters, moms, dads, uncles, aunts, and grandparents. Social institutions to deal with homosexual marriage are not clear, and in certain conditions could add to the survivability of a group.

Respect and friendship can form marriages. The idea of love and feelings defining marriage, will have to be abandoned. Responsibility, community, family, self-sacrifice, and other similar values, will define marriage at the end of the world. People that don't want to form into family like units and have loyalty to others and be willing to die for others, are not worth knowing, or being involved with. Mutual profound respect for another person, can create a deep, sincere and meaningful relationship, that will give you greater power and ability to survive and prosper.

Adopting children with no parents, will show a commitment to humanity that is bigger than oneself. Children need families and stability. They need a vision of a future being built right in the relationship in front of them. We have to create the future before our children's eyes, so they know how to live as good people, and bind themselves in love to others to build the world. People that leave children they know, to die, don't deserve to be part of the future.

- **You must go out to find and recruit, important smart people as a moral imperative.**
- **An elite computer programmer can have more value than a doctor.**
- **Visualize the future**

Your business and professional networks can now be tapped to build your group out. You will need to get engineers, scientists, doctors, biologists, accountants, mathematicians, and businesspeople, to get your group strong. They will readily join your group, if you sincerely help them for the future of the world. Protect the top people, even if they don't join your group, and give them as much as you can afford to spare. They will be more likely to join larger and more stable groups with strong social bonds and long-term visions.

If there are other groups recruiting the same people, then you might want to join up with those groups. Those groups have a clear moral vision similar to yours, and probably will be good to work with. A group all decked out in weapons, ready to fight, informs you of their values and what ideas they cherish. There won't be slavers out looking for engineers at the start of the apocalypse. Slavers like hot women and dumb manual labor, because they seek to feed their own aggression and vanity, with instruments of power.

Respect and intelligence can bind functional social groups together. Intelligent high functioning people will be naturally inclined towards civilized society. Intelligence in many ways, is the capacity a person has to function in civilization and large groupings of people. Smart people can adapt more readily, and have experiences to empower other people and not just themselves. A person that can take mobile network towers and reconfigure them to make a local area network for all mobiles to tap into, might seem unimportant at first. We won't have enough electricity to power a lot, but we can easily

make enough to power mobile phones, mobile towers, and basic servers. They can add the most value to everyone in the tiniest action.

TECHNOLOGY BASED LEADERSHIP AND WOMENS PURPOSE

The people that can rebuild the world, and have the knowledge to run and create infrastructure, are the ones that must lead. When looking for the most useful people in groups, combat ability should not be what defines a groups hierarchy. Fighters should be near the lowest people in a group. Does the group have engineers, scientists, computer programmers, doctors, nurses, auto mechanics, carpenters, welders, metal workers, petroleum engineers, mobile network technicians, farmers, veterinarians, military officers, cops, fireman, small business owners, entrepreneurs, middle managers in companies, and so on. More educated and trained people will be able to form larger groups of connected people, with rational organization systems.

Women have a power in themselves beyond anything any power on earth can compare to; they carry the light of civilization and the future, in their very beings, flesh, and will to survive. Women's survival is more central to human survival than men's survival, and this plays out in deep cognitive ways in how we connect and think as human beings. Women give birth to children, and are vastly more likely to be able to love babies. Men and women are not equal, not the same, and not ideally suited to do the same things, and this reality needs to be understood. Women are more inclined cognitively to minimize risk, seek certainty, and value dominion over themselves, due to their inherent physical vulnerabilities, acute senses, childbearing capacities, and lack of leadership training. Women are inclined to prefer certain types of power and behavior out of men, because they are useful for female survival and minimizing their own exposure to death. Men's identities are formed around female choices.

Fearful people obey leaders readily, by human nature. Your own fear will draw you into following those that can make you feel safe, reduce uncertainty, and seem in control of the world around themselves. Deep commitments to humanity combined with a strong technical vision of the future, is the basis of good leaders. A raw capacity to kill will not create the future. Your world is filled with technology and having the vision to repurpose it and use it, will determine your survival.

Religion can also help form large groups. People will naturally go to civic service organizations to build a network of people that can work together. Community leaders, church leaders, government leaders will form natural groupings around them. Societal leaders without a strong mix of real brain power, will make a group weak, in the long term. Civic leaders can't be the most valuable people; HR heads shouldn't become the CEO of a great company.

THE JOURNEY

Your group will have to survive through a collective challenge of a long-distance journey to a new home. That journey will form the community, its bonds, and what really works and doesn't. Groups that have a challenge, will survive better than those that have it all from the start. Culture needs struggle to form character, and not too much struggle where no real progress or development can occur (Toynbee). You will see death; you will see tragedy. The world isn't a movie, and big things just don't happen at random that much. Most of the time things will be very uneventful and with sparse and intensely shocking moments. Even in most violent wars, soldiers spend most of their days like prisoners, with nothing happening, where boredom and loneliness become more dangerous than bullets.

You can't save people on this trip. You are traveling to save your group. When you see horrible things, you can't just go running in, or running away. Ruthless steady reason must prevail. Think about what is good for your family. Would you send your own child to go help?

- If you see common people, needing a little help, then certainly help them.
- Get people set up, and with a plan.
- Spread organization to those that you readily can.
- Pick your route to avoid major points of interest.
- Stay off the main routes humanity will take.

If you do run into functioning units of society, aid them, don't follow them. Your own fears will make you willing to be controlled and follow anyone with the apparent ability to protect you. In this world, you need to maximize your own capacity to face fear and make something, rather than turn to authority to protect you. If you are a strong group, gangs and raiders won't pick on you. There will be plenty of weak humans to prey on, for human predators. Be something that good people are attracted to, and can see a future in. This is a brutal world, and this journey will be disgusting.

BUILDING A SETTLEMENT

Build a larger unit of people that are *your* people. You need to live with *your* people and people you will fight with next door. Build up your family and the groupings of families that you bond to. Real feelings, and real relationships, will save you. Build groupings of homes together. Have strong home units for a family, and then structures to protect a group of families, and prepare to defend yourselves. Build defenses from the inside out, and not from a large wall in. You live and die on your own, in the end. From

each place you stand, you will have to fight to survive, organize with those around you, and implement the structure of your surroundings to survive.

Voluntary organizations are much stronger than mandatory ones. You need to build a community, and do community work. You can't just work for yourself. You will have collective duties, that arise out of community commitments, that arise themselves out of your own home, and your own family. Your community is worth investing and everyone that belongs in a community must add time and work into that community. It is voluntary, and it must be strict. You work for yourself, and what is yours is yours. Your duties though extend to the things around you that sustain your life. Your self-interested actions need to function and support a community.

Fences and walls, make good cities. You will need to build fortifications from the inside out, and based around your own closest bonds. People need to first fortify their homes, then groupings of homes, communities, to small cities, and so on. Strong homes make strong cities. Making one big wall is a bad idea, and will inherently fail due to human self-interest and nature. Slave armies build massive walls. Lots of internal defenses and structure will create a stronger and more sustainable society.

CAPITALISM WILL SAVE THE WORLD

- **If you aren't investing, your time, resources, and love into something more than yourself, there will be no future for you or the world.**
- **Capitalism is the Solution to the Zombie Apocalypse, and all societal and economic collapse in the world.**

The solution to defeat the apocalypse is simple, first you make strong families and communities. Those communities then push for capitalism at its highest levels.

Capitalism is about decentralized and unregulated commerce. A healthy capitalist society allows people to start any sort of ethical business anywhere with the least resources. They follow contracts, property laws, and basic civil society. Don't cheat, steal, or murder evolve into more nuanced ideas. Society collapses when we ignore the fundamental principles that have created all success throughout human history. There are no socialist or communist success stories ever. Capitalism is voluntary and doesn't require violence or force to be run. All other systems are run at the end of gun, and with the threat of death. In places with no hope, there is always someone to gamble the future on, something to put your time and money behind. Building civilization from the ground up, is not about charity.

Progress comes from struggle and individual choices. Hard work, entrepreneurship, frugality, family, sacrifice, duty, character, values, and freedom are things that will always have fundamental value to the individual character of those that build the world. Countries with trade, economics, and freedom, have prospered the most. Surviving zombies is about building a company, that provides valuable services to your customers, and getting paid accordingly for it. Contracts, deals, negotiations, trade, money, HR, management, product development, sales teams, customer service, and all the rest, are always central to real life, no matter what.

THE WORLD HAS ENDED---You don't escape having a job, just because the world ended. Creating a functioning economic system is the most important thing a society of people can form. If you don't have economics, business, money, and trade, then you can only have raiding and conquest as the primary form of intergroup interaction. Life won't be pre-industrial farming, with raiders and monsters coming to kill you, if you choose capitalism. You will have high technology to create new options to survive. Some will want to make a utopia and end the failed ideas of the previous world. Editing out the

failure is an impossible task. Strong visionary leadership, with bold plans, are safer, than people's natural reactions to close off, and minimize danger. Closed societies will perish. Open societies with no rules will perish.

Capitalisms growth and development creates the same sort of reactions people have to change, new technology, and the destruction of their former way of life. People need to fit in the machinery and logic of capitalism and its technology or die. People have to be domesticated into technological machine worlds ruled by money, numbers, and exact processes. It takes time and is very harsh looking. Helping the rest of the world learn how to work in factories is not immoral, and it is something we may all have to return to one day, if our societies fail. The "apocalypse" could just be the day fat lazy office workers having to work go back to working in factories or farms. Weak minded westerners are mostly too proud to face such a world and would rather die than do work they consider beneath them.

GREAT INVESTING IS ABOUT INDIVIDUAL PEOPLE. The future is built by individual character. If we try to collectively build up broken societies, we will all inherently fail. Nearly all aid to Africa fails, all welfare to intrinsically poor people fails, and all pity thrown at people leads to no adulation. Tech can give us the power to find and empower any human on earth. Great people will do things in ways that have never been done before, and be more deeply connected into the reality of things as they are now. Those that are "without", are more aware of what we need. Tech can't just think about a vision of the future, it must have a connection to the present and how we are living now. Our orientation towards the future can be guided by data, by guiding us towards the best people to invest in and the best ways that we can live and organize our lives. Data itself is a social living entity bound up in who we are as individuals, in our

relationships, and in the direction, we lead our lives. Analyzing big data is about something far deeper than just targeting advertising to people.

SAFE EXTERNAL INVESTING- If we invested money into an African company before, their very business and profits could be harassed and pillaged by local corrupt governments and gangs. Technology can empower us to put our humanity into places we couldn't before. If the world could save Africa with \$10 trillion investments, and we knew it would work 100%, we would make the investment as a people. Most help only harms those that receive it. We didn't invest in companies properly. The same technology we use to invest in Africa, can also create a better internet, a better capitalism, and make a system that can life up a society that collapses. It is a system that makes society zombie proof, so long as people want to live.

Building companies in the middle of horrible places, can be made easier with the virtualization of company infrastructure into computer services. People can invest in a company's future from outside the country, by building out an investments IT side, and drive new companies into using the most current management tools. The whole of a company can be managed and run with apps, being powered by servers in distant countries, with investors from all around the world. People with nothing, can bootstrap from nothing, have greater access to investment, and have most of their business online and untouchable by local forces, even if they barely have a functioning internet.

AMAZING JEWELS HIDDEN IN THE DARKNESS. We as individual people have to be accountable to our own conceptions of who is human and who is not, in our own eyes. There is nothing wrong with loving your family more than other families, loving where you live more than other places, loving your culture more, or loving your country more. We have to invest in people. If everyone in the world cuts themselves off from a

place, there is no network of relationships and business into that place to build it up. Poor inner cities, undeveloped countries, penniless rural farmers, and dirty humans with barbaric thinking, are places we don't invest in enough. We aren't connected into them socially. We as individuals have to build up the world around us and use technology that builds up the world and systems around itself.

CAPITALISM FOR HUMANITY SURROUNDED IN ZOMBIES

HUMAN NATURE- The societal and government institutions to run businesses at their best are rather simple and straight forward. They have been evolved through millennia of human history, and are the cultural wealth of history bestowed upon us from those that lived before. The people lucky enough to inherit such ideas have known great success. Human nature doesn't change much anymore, and the ways to deal with it are a form of social technology. We can spread this power through technology and modern institutions. Marriage, rule of law, education, frugality, honesty, humility, hard work, individuality, sacrifice, ethics, and freedom, are legacy treasures passed down through time to succeeding generations.

MARKETS- The term a "free market" is the literal meaning for the freedom to open a shop without impediment and sell what you want to sell at the price you want, and not be shut out to favor those already in the market. Free markets are about the shop keeper's freedom and not the customer. One is the foundation of the other, and not the other way around. Restrictions on being able to open markets and market places lead to tyranny. The challenge of free markets is in growing businesses and more markets, that compete against each other, so outsiders can shop around for the best markets and shops in them to sell and trade their goods. Those that interact with your community, can trade things with your community, and both sides can get the things they need the most, and there can be a win-win relationship. If your community has products and

services to sell to people, that are actually useful, then people will come buy them, and respect what your community can do. It is hard to raid a community that makes things, and not just simply hordes supply. By allowing a “free” market, where people can trade good at market stalls with no interference, you could create the greatest prosperity for the people outside of your community.

You can imagine a flea market/bazaar like situation, outside the wall of a community, where locals and outsiders engage in trade. Multiple markets could exist at different gates to the city, with no central management or control of all the markets. Outsiders could not enter the city, and market owners could enter the gated city after the closing of their shops for the day. Warehouses could all be inside, and made safe. Virtualized markets are still possible in a world with no society and order, so long as you have a basic internet system set up for people to connect into.

GOVERNMENT CONTROL OF MARKETS- Free markets need to have few rules, because rules will always increase to favor those in power. The more power there is in government organizations, the more businesses will tie themselves to that power to take out their competition. Limiting the trade, and the formation of companies, is a natural and common government exercise, because companies come in and petition the government to stop competition on their business. The shop keeper doesn't want to restrict their customers from buying, they want to restrict others from selling. The south side market gets the government to make rules that hurt the north side market. Modern slavery over western people in western countries, controls people through the rigging of prices, and who can even make a business in a society.

People don't even realize they are enslaved because the systems of control are placed on those that run businesses and not common people. They see the bad practices of

companies and blame the companies, and not the governments who created that conflict to grow their power. People can imagine themselves to be free, while all their decisions are defined and controlled by external controls of prices and the ability for new players to enter a market. One cannot make effective rules to regulate a future business that has never existed, one can make rules so a future business never exists. It isn't up to you as a community leader to decide what sort of companies will exist. You can't know everything, and you can't know what will work or not work. Over time, though, your community will have wealth beyond any neighbor.

PRODUCTS FOR BOREDOM- Services that bring hope, excitement, joy, and vibrancy to life, will always be successful and essential products to sell in a "zombie apocalypse". The weird things people will really want, are beyond anyone's real imagination. Boredom will probably be one of the biggest problems, in a world with nothing to do. All you will mostly do will be to work, fight zombies, and eat horrible food, while living in horrible conditions. The primary punishment of going to prison in our modern world, is the sheer boredom that rips through a human mind, where stabbing someone seems a good idea, just to make a day more interesting. Humans random will to commit violence for no reason, can consume a society, as just a form of entertainment to break the monotonous and boring conditions of survival living.

When people have no vision of a better life, they will implode and descend into madness, extremism, and violence. You can understand nearly all the issues in dying ghettos, Africa and the Middle East through the lens of this paradigm. The denial of capitalism and prosperity for people leads only to mental insanity and death. They have been fundamentally deprived of the cultural wealth of freedom and its underpinnings. With no future, no joy, no exploration of possible realities, their souls die and descend into

violence. Nearly all entertainment exists to soothe our failed dreams and lack of a real future.

SELLING THE FUTURE AND CIVILIZATION- Most places in the world, have never been developed and never know true freedom. Delivering prosperity, as a business in of itself, is the greatest product to sell to those that have nothing. To build up the hopeless with the knowledge and power that you possess, is a moral and profit seeking imperative. Spreading the institutions and ideas that make you strong, are ways to defend your community, by surrounding yourself in allies that have built themselves up from the values you hold. Cultural conversion of your neighbors comes from success. Pure assimilation of other people into your society will not promote and grow civilization. Conquest is unnecessary with the modern inventions of capitalism and its technological liberation. We don't need to conquer and pilfer the world to be rich. We don't need to invade countries and steal people to enslave into manual labor to be rich.

THINGS YOU CAN'T SELL- Beating slavery means, you make something that people want, and they freely come to you to buy or trade for in an open free market. You don't sell addictive things, that control people's minds. Drug dealing is not a real business. The people interacting with you should feel gratitude at the chance to get things they normally couldn't get. Great services like Wi-Fi, mobile data plans, the internet, apps, farm seeds, information, critical tools, water purification devices, pretty bedding, home décor and even fashion can all make for great businesses.

ENERGY AS CURRENCY- Fossil fuels and electricity bring people out of darkness and into modern civilization. Those that control the energy and electricity, control the economy of a society. The real wealth of nations that we have now, is the oil and energy that backs our global currency. Drilling for oil and gas, is the best product any society can go after to get rich. Smart people will go after oil rigs and refineries,

rather than food and gun supplies. You can bring dangerous neighbors to your door to trade, if trading with you is far more profitable than fighting with or trying to steal from you. Natural gas and fracking might be one of the easiest ways to get fuel, when large oil refineries are rarer and hard to obtain. Being able to control a gas field, and process gas that can be sold to people, is a huge power. It is easy to get normal gasoline powered cars and motorcycles to run on natural gas. Electricity and stable supply of fuel for vehicles, creates the foundations for massive wealth that machines can bring us.

With energy comes clean water and sewage treatment. Making these system work and get new parts for them, is a large challenge. For a fallen western country this is easy, for a country with no such infrastructure, these are hard to get services. Machines that can make these parts, or any parts that are needed, are world changing innovations in technology, that are being developed right now. These are long term technologies that can protect the world from collapse because they will allow it to develop up from any level. Universal manufacturing machines can liberate humanity from any stage of desolation and collapse, if they have the energy to run.

ZOMBIE INTERNET: TRANSFORMING ENERGY INTO DATA

Mobile internet first for everything is the key to making modern solutions that can scale the world and into the most resource strapped locations, and defeat marching hordes of the undead.

If you really thought about it, the one thing people would want more than anything at the end of the world, is a good Wi-Fi connection for their charged devices. Wi-Fi is the most valuable thing you can have for the smallest amount of energy, currently. The

basic infrastructure of the internet can be a commodity, at any stage of societal development. Network processing power, storage, bandwidth, cloud services, are reaching a point of such development, that they can almost be a currency, in of themselves.

A peer to peer, bitcoin, and block chain, run economy and internet, is an anarchist unicorn fantasy. Things have to actually work and be useful, and not just random wide-eyed fantasy. We make things inspired by these ideas to drive innovation, and not being concerned if some perfect ideal is achieved. It is a driving force and not a defined idealistic end. An African business still needs real money to run a business, people still need bullets to shoot zombies. A large cloud service company with idealistic intentions for a more open and free internet, could be a strong driving force, to help push forward the protocols to make such a system work, and make people more likely to engage with the system.

Instead of a society armed with guns, they will be armed with computers and the internet. Money can be stored virtually and secured. The "gold" would be the actual network, services, and data. In a broken society, the best way to get rich, would be to go get as many computers linked up into a network, with the best connectivity, and support people getting connected.

People should be able to "harvest" computers and network equipment, and it should be as easy as shoving a thumb-drive into a USB port. We build large central servers because they are easier for large companies to make money and control content. Netflix could run off everyone's local computers, and not even need a large central server, with peer to peer (p2p) networking, like bit torrents. The internet is the technology that can build the world from nothing. Companies that want to build for the future, need to

consider how their technology could run, in a collapsed society, without any large companies to maintain them. The software of running a government, a business, a community, are all awesome services to sell to your neighbors over a fully running zombie world internet.

At some level, we need a wireless internet that develops from peoples own homes and devices, and not a centralized telecom system. We can use Wi-Fi connections to build a whole network, where half the bandwidth of a router or device is used for connecting to the network, and the other half is used to build the network. The system would bypass most of the centralized systems, and yet connect into them, to create a larger internet. It would supplement the current internet and vastly expand the supply of access, which would reduce prices for people. This can use the existing devices and tech people have, and we as individuals build an internet. This will work better in denser cities with people living closer together. The system can have seed routers, that help maintain a basic system, and then new devices add in and amplify it all. Cars themselves could be wired in be like moving Wi-Fi towers, and a whole see of driverless cars moving through a city, could themselves be part of the very arteries of an internet.

Most all countries don't offer real choice in people's internet and mobile phone plans. The companies are all government backed and or controlled, with prices and services all regulated. This sort of conflict can even exist in the zombie apocalypse. Hating your phone company can be universal. We won't just get over comparing the different mobile

phone plans, we have available, even at the end of the world. The internet is a business we can bring to people and sell them, it can't however enslave them into a narrow set of choices. A single company controlling access and content, is not free, and won't grow to its maximum potential. Beware of companies like Google and Facebook trying to bring the internet to the world for free.

Monetizing internet IT companies, apps, webpages in the apocalypse will be hard, since there won't be a large enough market for the advertising systems we use now. Branding and advertising companies could, with the right vision, be useful. A new form of monetization, branding and advertising can be achieved in such circumstance, with a proper imagination and a desire for a capitalist future. Free people using free markets, can do anything in the worst circumstances. Virtual goods and services can create the backbone of a post/low-civilization society.

Devices all linked to each other can create an internet in of itself free of all masters. We can just buy devices and the very devices contribution to the network pays for its internet access. We could pay for internet with the amount of computing power and connectivity we put into the network. We all can be partly responsible for the network, and this changes the future considerably. This can build out from local networks like those used in companies. You could nearly run a whole business just using mobile devices to access everything, and connected together by a small local server. Meetings could be made hi-tech with the just all the devices of the people in the meeting, being linked. You could make teleconferencing cheap, easy, and seamless. Everyone has cameras on smart phones. You could run just about anything, with the right software, and devices linked up into it. No large IT costs.

This is a very useful platform for the apocalypse, with the right software installed. Many apps, like messenger apps, would need to be able to run on local networks, with no stable internet connections. Larger versions of apps could have server versions that make them run more smoothly with people using those apps over that network, rather than have to rely on a more distant unstable server. In Africa, data connections could be very unreliable, so software will have to run inside the local network, even if connection to the internet drops off temporarily.

With colonialism, you could conquer a country and manage everything internal to that country. This was inefficient. With imperialism, you could control all the laws, trade, police, border, and effectively control a country, with more distant forms of power. With neo-imperialism, countries could be controlled by just controlling the money, like America does with the world money supply being backed by the dollar. With the internet of things, that connects all things and people into the internet, you can control the root nature of humans with little resources and maximize their potential labor output based on the individual potential. The agency that controls the access to the data, controls the system, and thus controls the world. Data is money, data is power, data is the ability to control people with the least amount of effort or resources.

My main idea is not in how to turn the internet into a commodity to underpin our financial system, it is how to use that resource to make a new kind of smarter investment, that is profit motivated. If there are \$10 trillion in profit motivated investments to be made, with about 10% of that intention being noble and good for others, versus \$200 Billion 90% nobly invested, then the more capitalist profit driven investments have more overall nobility and kindness in them, on the whole. There is just so much more money that can move into black society, if we can make an effective form of investment, that will get results. Profit motive mixed with nobility can achieve far more than either alone. Pity or pure exploitation have not worked.

If you loaned only virtual money, that could be used for online services, this would make for a much safer and smarter form of investment, into a company. From the start, force the company to be fully high tech in its internal infrastructure: all online finances, HR, management, inventory, everything online. We can drive the entire business into using internet tools for everything. This will open up the whole company's finances and information for analysis. Investors will be able to track fraud instantly. This will make it safer to invest more money, and make companies more likely to succeed. Intelligence and computer power can be transferred into companies, with little physical infrastructure around them. They can have access to the very best online resources to run a company, even if they have little electricity, running water, or even stable government.

If you were a company, and you had a bunch of virtual currency, that you couldn't spend on real material things, the most likely thing you would spend that money on, to build your company, would be online internet services. Basic cloud based servers would be a minimal thing, anyone would buy. We can run tremendous amounts of a company from just online tools. We can greatly reduce the cost of starting and running a business, with modern technology. You can manage a lot of life purely online, and thus the services

provided by a purely virtual world, can be of fundamental value to survival. Virtual currencies that exist online, and are based in the power of network, create an interesting economy.

APPLYING THIS TO AFRICA:

Finding the best returns on investments is not about greed, it is about finding the best places and people to put money into. We need to invest in the best people in Africa, and not "Africa". Black people are a group of individuals and not some homogenous mass grouping. There are more super geniuses in Africa, then there are in Japan. They are just more spread out and harder to find. Individuals need to be helped, so that the best people can rise up and create something for their society and the world. The greatest charity a person can do, is believing in someone and making high risk investments in them. It will always be the person, and not the plan, that matters more, when investing. We must find the best way to invest that money, so it can't be wasted, stolen, or mismanaged, by the shattered societies around them.

If someone wants to invest in African companies, it is very high risk. They are in unstable societies with little rule of law. People that have been wanting to help Africa have kept donating to charities, and those have had little benefit. Governments are given money by western countries, and nearly all of it is squandered and stolen, and puts countries in deeper debt. Financing poor places leads mostly to more problems, rather than greater success. Black people need money invested in them, all around the world, and people don't loan money to them as readily. While high percentages of black people are the lowest performing people in the world, among them are some of our greatest minds and brightest lights in our world. No matter what group of people there are in the world and no matter how low their intelligence is collectively, there is always

highly intelligent people with IQs, general intelligence, and gifts, greater than nearly all other people on earth among them. We search the world for black athletes, musicians, and social leaders; we rarely search for black mathematicians, inventors, and exceptional entrepreneurs. Genius is human and transcends all racial boundaries and collective identities.

GOVERNMENTS AND PHONE COMPANIES AT THE END OF THE WORLD

Even countries could be run this way. Instead of giving countries money, we give them online internet services to run their countries, and force them to have all open data. Open Data is good for everyone, except the group that doesn't want their dirt to be seen. Open standards that are driven by putting everything that an organization does online, with usable trackable data, is good. It makes investments easier to make. People will want to even pay taxes more, if we can track the all the money the government spends and then see the results of that in real time. We can build investment systems, that can see everything, and make organizations truly accountable. The way to build this, requires building something that would work with zombies and a completely failed society. We need to build from the smallest level up: individuals, families, communities, business, local government, social institutions.

If you are a boss and you have all your employees linked into a system all the time, that can track them, you can gain control over them. You can use these devices to ensure security, building access, and even network access. By being able to monitor and control everything, this creates greater incentives to invest and greater opportunities to extract value from people. In a broken society, all this monitoring can increase safety and replace many forms of public service. People that are linked up and monitored, will be more civilized and committing less crime.

CHAPTER 6: THE MORAL VALUE OF SLAVERY AND THE BIRTH OF NEO-CAPITALIST SERVITUDE

Slavery is considered the vilest institution in human history. We as people love slavery, and use it a lot. The world we live in today, is filled with millions of slaves. It is not a black and white institution with clear lines between what is a slave and what is not a slave. Englishmen once lived as serfs on lands owned by nobility, and had to do whatever their lords told them to do. They had many freedoms, and could do as they liked in most day to day things, there overall life though had no power over anything more than their immediate surroundings. They couldn't pursue power. They couldn't rise up to be more, than what they were born as in this life. Nearly all of humanity has lived in bondage and slavery for most of their existence, or as uncivilized and wild human beings outside the very story of human history. The very story of history is the story of the dominion and enslavement of humanity to serve the needs others and their own personal values. We cull and kill people readily that we don't see as human, and call it noble.

We just create different names for the kinds of slavery we like. The word "slavery" has more to do with kinds of slavery we don't like. The kinds of slavery we like we call things like good government, welfare, open borders, immigrant rights, or social justice. If tied to things we don't like then it is rather easy to see the slavery in it: wage slavery, capitalism, globalism, elites. Saving humanity requires bringing people into humanity, and not simply treating them as human. Reality and results matter than being nice to people you look down on. To be human is to have a purpose and a future, and those that do not, can only be called useless, savage, or monsters. Genocide is widespread in all countries across the earth, as we choose certain groups of people or humanity itself for

extinction. This is not a conspiracy, nor all that complex. We are just dark nasty creatures that don't like to admit it to ourselves.

ZOMBIE WORLD SLAVERY

People often imagine in the apocalypse slavery being a common institution that will return. The collapse of civilization doesn't mean we have to make all the same mistakes that people made in the past. We can never truly return to what we came from. We have too much cultural innovations and technology around us, to ever really lose everything we have made. If we really believe that slavery is evil and one of the great social mistakes in the history of humanity, then we need to develop ways to live that don't require it, if civilization were to fall. Our modern cultural innovations should be able to improve the quality of life and the humanity of civilization, at any time or stage in man's development.

The real moral challenge is whether or not slavery is really bad. If slavery has a function at some level of human development, then it is a moral institution. Using outdated forms of slavery when more modern ones could be implemented could be viewed as the problem and not slavery itself. People that like being safe and not dying, more than being free, often like to be enslaved. They find being well fed, a warm shelter, and work, to be better than living in a dangerous wild world. Finding ways to deal with the "good" reasons slavery happens and the moral dead end it leads to, might be impossible. Fear drives many people into asking to be enslaved. The majority of Americans, white and black, before the American Revolution came to America as slaves. Many of them entered slavery voluntarily, most didn't survive their contracted time as slaves. In nearly all the world, before the American Revolution, the vast majority of people on earth in all places

existed in some form of slavery. America was the first country to begin the war against slavery and usher in new ideas of how people could live in a country.

THE ROOTS OF SLAVERY

In a barbaric world if there is war and then those people warring against us surrender, we can either kill them or enslave them. Releasing enemies back out into the world, that can come back to attack you, is simply not realistic. If you are living in a village and raiders come and attack you, and you defeat them, should you kill or enslave the survivors? If you imprison people and make them work, that is slavery. Once you start to take slaves for the most justifiable of reasons, your society will be more inclined to take slaves for other reasons. Your other choice is genocide. Slavery seems more justifiable than killing every man, woman and child. In history, cultures faced these real dilemmas, and they can't be wished away with idealistic thinking.

When western people plundered Africa and enslaved people to work on plantations. They, at some level, justified this by the horror of life in Africa and how little those people were using their potential to work. The people themselves were an underused resource that needed to be channeled and used for better purposes than random chaotic jungle life. Slavers actually saw themselves as helping those people, as compared to them living a life of savagery and uselessness. Slaves in America lived better lives than people in Africa, and were already themselves enslaved by their own people, in the mass African slave system. Nearly all slavery in Africa was done by Africans. They sold them at any markets they could, and took anyone a slave they found valuable.

The internal African slave trade is not a history that people get taught. North African slavers invaded Europe and enslaved white people. The black male slaves they sold to the middle east, in massive numbers, were all castrated. There is not a clear message or story here, that can fit into historical narratives of civilization, progress, or whatever. Humans liked enslaving people, and it wasn't America behind the slave trade. Hating on America is about the only unifying aspect of teaching the history of slavery and anything that doesn't fit into that narrative is censored out and made racist. It was in America that full humanity of blacks and the full savagery of slavery, was made apparent to the world in images, pictures, books and art. It was America that first started the movement against the slave trade, though took longer than other countries to end its practice internal to their country.

A western slave trader could rationalize that moving slaves to a better world, was a moral improvement over slavery in a more backward and dangerous place. If you believe slavery to be wrong, then you will need to find a way to run a primitive society without it and be based in the reality of the world around you. When you are trying to lift up a people with no stable civilization around them, it is very easy to end up enslaving different groups of people for different reasons. It is easy to rationalize these enslavements as being more just than the alternative.

CLOSED PLACES WITHOUT SLAVERY

In a zombie apocalypse community, it is rather easy to govern a group of people that share the same values and all want to come together. People will resist in less developed people into your society that don't share your values, culture, or even basic intelligence. Eventually people will come to trade with you, want to join you, and/or steal from you. How you manage these people without slavery is far more difficult. You can close

yourself off from new people coming in, only for so long. The ideal society, in a zombie world, is made of members of the highest value and usefulness. You want people that can all work together, trust each other, and provide useful skills to the whole group. Different people will be good at different things, and form levels high specialization. These sorts of societies are easy to not have slavery, because you simply don't have people that could even be made into slaves enter your community. Idealized northern European countries are racially and culturally homogenous societies, with little to no immigration throughout their history.

Challenging Questions:

- Are countries that are rich and successful, yet closed to letting outsiders in, morally justifiable?
- Can we treat a country like a company hiring the best workers, and not wanting people that can't work well?
- Is it better to live in...
 1. A village at the end of the world, that is happy, prosperous, closed to outsiders, and that doesn't have slaves,
or
 2. a society that is open to outsiders, even ones that have attacked them, but makes them into a second-class group?

I really don't know what is right. If I was a person that was in a lower class, I might prefer to be a second-class citizen in a safe place, than a first-class citizen in a harsh dying landscape of armies of the undead hunting me, with little food and no real chance at a better future. Though if I lived in that "good" place as a second-class citizen, did good work, and was a good person, I would want a chance to be part of that society, and if they didn't embrace me, I would feel resentment and not gratitude. At a moral

level, it appears that a functional society needs to be able to bring in new members who are not clearly useful to that society and provide them opportunities to rise up to the top of that society. Any sort of inequality in access to opportunity would itself undermine a society.

Another possible answer is that we trade things and not people across borders. Strong borders, with strong trade, and an open economic system, can mitigate many issues, and might be how societies have to form. Creating a government that leads to prosperity and safety for your people, while not enslaving those that can't assimilate into your society, is probably only possible with highly prosperous capitalism. If you are able to freely trade with people, and sell them something of value, that will give them a better life, than you don't need to integrate them into your community to have moral value to the world around you. You can sell "freedom" to people.

BROADER CLASSES OF SLAVERY

Any system that holds people back from highest levels of attainable power, that rules over them, have enslaved those people.

Slavery is a not a black and white line.

- If women can't hold power in a society, are controlled by the men in their families, and made to do work, then they are slaves.
- If people do jobs where they are kept from getting higher education, higher positions, and their children never rise higher, then they are slaves.
- Migrant laborers exist as a form of voluntary slavery, where they are not allowed to integrate and rise up in the society they work in. These kinds of workers are even denied the right to even fully live in the place they work.

- Companies that employ people and don't let them freely change jobs or have any say in how they are led, are slaves.
- Countries that turn their people into serfs, that work the land, while the elites rule a country, are slaves.
- If the rules of system are rigged against you to eventually possess power, you are a slave.
- If you follow ideologies that never empower you to live a better life, than you are a slave those ideas
- If you are addicted to lifestyles that trap you from becoming more than you are in your life, than you are a slave to those addictions that consume your life.
 - If addictions are fed to you by organized institutions then those institutions rule over you.

MIGRANT WORKER SLAVERY

Migrant workers all around the world live as slaves in societies that want their labor, and not them as human beings.

Mexicans illegally enter America to work in millions of jobs that are waiting for them. They have no legal protections or rights, and work for the lowest wages with the lowest level working conditions in America. This life though is better than the one they risked their lives to escape from. Is it good to let them work and have a better way of life, make them citizens of your society, or send them back to their country? In the real world with real consequences, such choices are not so simple. If you have millions around the world waiting to enter America legally, and these illegal immigrants become legalized residents, it undermines the rule of law, for the whole world. Having a slave/serf underclass undermines society even more. In China, over 150 million people

work as migrant workers, and this cheap labor partly fueled the economic growth of China. They built cities that they would never be allowed to live in. Eastern Europeans, East Indians, Indonesians, Latin Americans, and countless other groups of people are marginalized in varying societies and live as functional slaves all over the earth. This is the modern face of globalization and its modern slave markets that drive its profits.

THE MORAL VALUE OF SLAVERY

We tried a lot of different things through human history as way for strong rulers to control and manage a group of people. We have seen success and failure. Certain ideas were very popular before, and we now no longer find acceptable. We found better ways to do things. Certain things from the past are good practices and others are ones we can never use again. The line between them is not clear.

- If we returned to a level of pre-civilization, of man at war with man, what sort of systems that we thought we had evolved past, return?
- What sort of society would you make in the rubble of fallen and dead civilization?
- Is slavery a justifiable institution ever?

If we want to make the most ideal systems, we can pretty much figure those will fail. Idealism in of itself never gets results, because it has no relationship with reality. Good government has a lot of nuance; good systems develop over centuries and millennia. Humans are utterly uncreative and can only identify and use things that have already been thought up. People are incapable of creating new ideas, no matter how obvious they are, until they are made apparent. No one person can fully understand why certain

rules exist, how they came into functioning existence in world, for anything, ever, in all circumstances.

There is an intrinsic randomness that creates things, and is at the root of all effective creation, change, and evolution. We do not control how things come into being. We do not control why certain things work and why other do not.

In peaceful societies people forget the rules and systems that lifted them out of conflict and poverty. There are fundamental institutions of society that are harsh and very realistic. They evolved from the absolute reality of human nature. People go to jail, and get killed by their own governments justifiably. Systems of government need to be always conscious of the possible fall society and how to continuously rebuild it from nothing. In all places, there is a nothingness that we cannot control, that we cannot ignore, that can stimulate and give birth to the new. The more we try to control uncertainty, the more it grows into an absolute nothingness that envelopes an entire society. The more we deny the essence of being human, the less human we become, and thus arises the very breakdown of society. Step into uncertainty, darkness, fear, human nature, and reality, in order to create, find meaning, and lead yourself to a life of greater power and mastery over the earth and others.

The greatest danger to power, is an entire society, system, company, country, economy, collapsing. If you have nothing to rule, even with absolute power, then you have no power over this world, but your own self. You must create systems of power that lift up society and govern humanity, in order for your life to have any sort of meaning or hope. Power flows from the amount of people that have power, and that power arises from each individual's place in life. You as an individual can change the world around you and

the life you live, by your choices. In a fallen world, your choices will determine the future, for yourself and those around you.

CHINESE EDUCATION AS SLAVERY

In China, the education system is supposed to be the great equalizer. People can rise up and get good government jobs, by how well they do on standardized tests, that are taught and prepared for in schools. They can go to the best schools and get the best jobs in China, with just high scores on certain exams. High test scores can lead to higher power in a society. If the testing system is not fair, then the people rebel in enraged anger, at the unfairness, and inability to escape their circumstances. Testing is a fundamental social institution of Confucian societies, that promotes social peace and prosperity. It creates a system by which people of all classes can obtain power through the merit of their own ability, study, and habits.

The exam system is only one institution that keeps corruption in check and creates a perception of fairness and opportunity in China. In reality, positions of power in China are given out to those with connections and personal relationships and not by a clear fair system of rules and test scores. Those with connections still need high test scores to go to the highest levels in government. The system is not wholly corrupted in the sense that the sole basis of power is personal connections. If the Chinese government doesn't correct this perversion, then their very hold on power will collapse and the best people in society won't struggle their hardest to rise up and be the best. As a principle, you want to lead the best people, to be the most powerful. People that feel like slaves don't become super intelligent and don't reach for the highest level of human being they can

be. The illusion of upward mobility can create great developments in a population of people, over time though if that promise is not delivered on, it creates the foundation for societal collapse and revolution. People study hard in China because they truly believe that is the way to succeed in life. If they stop believing that, and people stop studying, the consequences would be world changing. The reality is that so many people invested their sense of reality into education as the means of personal advancement, that any system that doesn't promote that will be crushed and will have to evolve.

What happens to a society if its best and brightest people don't rise to the top of that society?

In China, the testing system is the very power of that country. Testing controls everything. It is in most ways the very ruler of China. The rule of law, at its most basic level is governed, in China, by the national testing systems. If those systems are completely fair and without corruption, they can still have serious problems. Fairness is only one consideration in how to make a good testing system. Making it fairer for all with little corruption is fairly straight forward. Reform of this system is probably one of the most emotionally charged subjects in Chinese life, short of housing. The system itself in its ideal form, still only promotes a narrow range of human potential and ability, to the highest forms of power. The people that don't fit into this system, are wasted, and have to find others ways to advance themselves in life, if they have any.

Making a system that nurtures the best out of people and raises them up to be the best and the brightest is the fundamental problem. If the test can't capture the abilities and talents of great people, then those people are being oppressed by that testing system. Chinese tests are mostly singular and only focus on narrow parts of human intelligence and ability. They demand certain ways of thinking, and don't deal with other ways of

thinking. If tests can't ever capture the full potential of human beings, then they will always fail as a social institution. The testing system itself becomes a form of slavery over the minds of its people, and limits the full potential of people.

Massive amounts of human potential are destroyed. A country becomes poorer because of this. The people that can't rise up, that should rise up, hold deep resentments and anger and never contribute fully to their society. They are mentally enslaved by a system, that traps them out from power and success in life. Their discontent can shatter worlds. The education testing system is so deeply ingrained, that it has to be evolved and not replaced. It is their culture and it is their history, over throwing this in the name of some idealized system that people believe will work better, will fail. China will always have exams, so long as it is Chinese.

There will need to be different testing regimes that use entirely different kinds of thinking and rules. Expanding the tests to include more kinds of abilities and intelligence, is one way to modify and not over throw the system. For instance, China can use other countries national testing systems, for some of its citizens as a way to balance out their own system. Internal cultural competition can be a source of vitality. The central management of this system is a source of great power. Over micro-management of the testing system, only leads to a systemic failure of the whole society. Governments don't like to let go of absolute power over their people's minds.

People though will rejoice at the chance to take very different kinds of tests, that don't fit into traditional molds of Chinese education. Most Chinese people that have money and can't succeed in the Chinese education system, go running off to other countries, and spend all their time in China, studying other countries tests, so they can get into those countries systems. Normal Chinese people don't have those choices, and thus live

trapped in a system that doesn't represent them. Only the most intelligent and or wealthy, have any choices over their education, and their future prospects, since they can more readily leave China for other systems that will better develop them as people.

Testing to get government jobs and the right to be a leader, will still be narrow and nationally controlled tests, though these exams could learn from other countries government testing systems, and include a broader range of intelligence and knowledge to test. People need the ability to be able to rise up in the institutions that rule over a society. The Chinese government needs to get its best and brightest into its leadership and have diverse kinds of thinking. If everyone thinks the same, and has the same kind of high intelligence, then the government overall is less intelligent. The desire for a more powerful government can drive the opening of a testing system, because it will bring in more diversity, intelligence, and awareness of reality. Diversity of thought is rare in nearly all institutional powers around the world.

In America, nearly all elites go to the same schools, learn the same things, and are indoctrinated in the same narrow ideologies. Risk taking and raw intelligence is never valued in any countries leadership. The best and brightest don't go to even go to Americas top schools, or any top schools in the world. Over a certain IQ and the more that IQ goes up, the more those people have diminishing success, and higher aversion to integrating into society. IQ isn't even a measure of intelligence; it is just one narrow kind of intelligence that is most associated with obedience to government. Kids that sit and do what they are told, get higher scores and go to better schools. They then can get positions in government, where they can rarely actually perform well in the real world that is ruled by chaos, uncertainty, and true randomness. Self-confidence is the power of an individual to face personal uncertainty; those that sit and obey, will never possess

this power. They can only possess hubris and arrogance, which veils their own sense of inner personal failure and nothingness.

All governments face the problem that many of the best and brightest will not want to work under leadership with shitty bureaucratic management practices. No test can capture leadership. Leaders that rise up in government, are profoundly different than those that rise up in real life, from real experience in the world. Studying management and getting good scores on tests has little correlation with actual leading. People don't seem to want to be led by individuals that did well in life but didn't go through the highest levels of education. Nearly all countries seem to associate intelligence with going to elite schools. So nearly all countries have inherently failed systems for getting the best people at the top. Actual life performance is rarely rewarded with government power. These education elites can also come to dominate corporate business cultures, where business and government are tightly linked. These are fake business people whose job is to minimize uncertainty for a company, by manipulating the government system to their advantage, against the rest of humanity. Corporatism is a false form of free enterprise that blurs the line between government and freedom. Unaccountable elitist education systems will nearly always lead to government and large corporate tyranny.

THE SELF AND BEING UNDER CONFUCIAN EDUCATION

The other and even darker side to east Asian Confucian education systems, focused on tests, is that they leave deep scars in people's minds. Those that don't fit into the system, are beat down to fit in. The minds of Chinese people are tortured and abused in high school to learn things they don't even need to learn. Their entire sense of personal dignity, their hopes, their dreams, and even sense of self, are smashed out of them. The

system is a form of mental slavery that rips into people's minds and souls. This has many advantages overall for certain kinds of people, and for the most part is beneficial. It works great for people with no education and from poor families. It creates better futures for those who have the least. Creating prosperity for people with nothing, balances out the emotional sacrifices and cost. If you look at poor countries, you will often find education systems that can barely achieve the results of a Confucian one. If for instance Africa adopted these sort of education practices, it could lead to rapid societal development in short time, with millions of the poorest people lifted out of poverty and systemic oppression. Confucian education excels at mass scale preadolescent education.

The real driving force that will reform Chinese education, beyond the government wanting smarter people, is parents realizing what sort of harm that system can do to their kids' minds. They remember the abuses and suffering of their own personal education experiences. Chinese people have more resentment towards education, than just about any other thing on Earth. For richer families with choices, this will often lead to more fun extracurricular education, and kids doing more creative things. They give them more freedom, and less hours of study so they can live happier and healthy lives. Studying 12 hours a day without break and minimal sleep, can never be good for real learning or health, though parents continue to push this while knowing in their deep hearts it can't be right. Initial reform will happen that reduces the amount of study to levels that don't physically harm people and leave brain injuries. Understanding how happiness and health can maximize education results will take time. Balancing out all the different needs of people to achieve reform is not simple. There will be a lot of failure giving to kids too much fun.

The real world has very demanding requirements. Getting children ready to face the world is fundamentally important.

1. Get them learning actually useful things.
2. Have room for different kinds of intelligence and abilities
3. Embrace creativity and peoples' dreams, to a certain degree
4. Don't damage their minds with bad study methods.
5. Learn and test in a variety of ways
6. Get them set up to be able to achieve real goals and real success.
7. Let kids take risks, have fun, explore, make mistakes. Play.
8. And more than I can say.

If you want to be a good ruler, then you need smart and able people to rule, that are out doing good work. The Chinese government should consider only how to maximize its own power, by better utilizing its citizens and creating a better education system that brings that out. The people that will rise up in that government need to be better than the people that are presently in it, so that the government is only getting better year after year. A system doesn't protect people, have fair rules, and promote prosperity will become weaker in time, and have no real power. Reform in China can rise up from the self-interest of its own government to be more powerful. The people's desire to protect their children from harm and brain damage, can help drive things from the grassroots.

THE HUMAN EXTERMINATION PLAN: MAKING ZOMBIES

Killing off large sections of the human race has been a long deep held dream of mine for decades that sickens me fundamentally as a human being. There is this cold harsh brutal reality to the world and the needs for our collective survival that requiring taking out a lot of the people, on the Earth. The Earth isn't overpopulated in people, it is over populated in people that have no future. The myths of overpopulation nearly always end in a need to reduce the human population, and ironically only drive down the population of the most valuable members of society, and push up the numbers of the most useless people, and only creates even more populations problems, more environmental decay, and an overall better future we are even farther away from. Killing off the weak, stupid and inferior became an unacceptable option for humanity to openly admit wanting to do, so any and all programs to do so, had to be done indirectly and in a more "humane" way. The sad result of these programs, we have now, is that they have made the very people that can save the world childless, depressed, hopeless, and living in dying countries.

Population control and eugenics are expressions of the same overall theme of our collective survival as a species. Population control manages us as groups, and eugenics as "individuals". We can collectively ignore the damage caused by population control, far easier than with eugenics, because there are no singular faces to show as victims, the damage is done to collective society and not to individuals. We don't even know if we are victims of population control, or even if we are beneficiaries. There is no real way to measure results in the immediate present. We have to look at decades long demographic trends to understand. In all places with strong population control, societies face an existential crisis that will most likely end their existence. Eugenics had the decency of murdering you to your face as an individual when it found you to not be human.

Genocide is better when it is more personal, and in your victim's face. The reality of all forms of population control and actual eugenics programs is that are all based on group identity paradigms. The kind of thinking that wants to manage humanity as a collective entity also sees people as collective identities. Eugenics reduced people into sub groups of overall humanity, and then did population control on other "species" of people. It organizes people and the sum of who they are into false categories like race, which has no bearing on a person's actual potentiality as a person innately. Population control can target sub-groups, with overall trends and macro manipulations of public policy.

The machinery of genocide has to work based in individual characteristics and not one's group identity, to possibly be effective. The exact line between what we consider desirable and simply unacceptable has to be more clearly defined in a realistic way, that people can accept. We all believe in some form of genocide, and only differ on what that line is in humanity. Certain people we consider less human, and when they die, we don't care, or even inside celebrate a little. A mass murderer, terrorist, child rapist, drug dealer, or a low functioning down syndrome child, gets killed and few people will cry for their lives. We can watch news of a one person in our hometown losing their home and cry, and see have half a million people die in a far-off country, and we call that just life. Killing off "others" is far easier, than things we see as similar to ourselves and more human.

SUB-HUMANS

It is hordes of dumb single men, with no possible way to have sex, get a job, or have a future, that are the greatest threat to any society or country. Single men with no future rage with violence.

People on massive scales believe at some level that certain people are less human than others. The issue is how to deal with this kind of thinking, and not if these ideas are correct or not. Only the definitions of "sub-human" have evolved, not the thinking. By nature, people create an otherness and rationalize it with whatever sources of authority they can. "Zombies" are these people, and real hordes of them run over countries and leave them in the throes of death, and render the world in chaos. These are the uncomfortable facts of life. Any plan for the future has to deal with these things, and if they don't they are just idealistic fantasy. When faced with a horde of the dumb, angry, and armed, there are brutal choices one has to make in that moment. The "zombie apocalypse" is about how to kill these people and not feel bad about it. Compassion doesn't breed real solutions. Bringing zombies into one's home and country, doesn't cure the problem. It is a denial of reality. Much of this book is about trying to save these people from those that would come to commit genocide on them. And to save ourselves from idiotic plans to "save" these people based on a poor understanding of the problem. They are not poor and futureless due to education inequalities, persecution, colonialism, patriarchy, and so on.

If people don't like genocide in their lives as a basis of civilization, then they need to consider the fundamental problem seriously. If non-civilizable people breed in high numbers and live off a system they can't create nor run, the world will die. The issues of who needs to be managed is contentious and a good debate over who is more of an evil asshole. The underlying problem is unescapable. If we imagined colonizing space, not a single person would suggest sending out our dumbest and breeding them on mass scales to fill those societies. Now the person wanting to select what they imagine to be the best in society and sending them out, is also in a false paradigm that will fail. There is no such thing as genetically superior, in the larger scale of things. When we try to match

the world to our ideals, we cut out what we find ugly, and that ugliness isn't inherently bad.

People with sub 83 IQs are basically non-functional. There is a certain point in intelligence where a person can no longer function in the world, without massive amounts of help. They can barely even do the most basic work. 1/6th of the human race is nearly useless. Above such an intelligence level, we are all basically equal. More intelligence doesn't denote more humanity. Sentience is sentience, once a person crosses the threshold of sentience, they are fully human, and people who don't cross that, are not fully human. No one would really cry that much if a special needs school filled with the most mentally disabled kids, all burned to death, as compared to a normal school. If a stranger, with a family and children, severely risked their life to save a down syndrome child, few would find it noble. Being kind and loving the cognitively non-performing members of society, takes the best and most noble people to do so. Most of us deep down inside just feel pity and wouldn't mind if they died. It is a sign of one's humanity and kindness to show warmth and love to them. In their eyes, we can see our own humanity and dignity reflected back at us and have our lives nourished by that. Families that have raised down syndrome children often call these children blessings. We are human, and reminding ourselves of our humanity is a deeply meaningful and noble thing. Any person wanting to create down syndrome or "special" children intentionally is not noble. Any person wanting to create genetically superior super humans is just evil and deeply deluded.

The greatest scientists and all the smartest most educated people on earth, once believed in the truth of racial eugenics, and the absolutely need to cull and limit the existence of black people. There was a time when people literally saw other races as different species and where certain races were inherently incapable of higher forms of

thought. They saw black people as like high functioning apes, that could never have someone of the highest levels of genius another race could achieve. While in reality, if we took 10000 black people and 1000 South Koreans, there would be more geniuses of the highest levels of intelligence in the group of black people. Culture plays a more important role in intelligence than race. If a certain culture values less intelligent people and they have more kids, the culture will become dumber in time. If a culture makes intelligence the highest form of status, and those people have lots of kids, they will be more intelligent over time. Race has nothing to do with intelligence. Certain cultural groups have severe problems with low intelligence and can barely function in the world, because their smart people aren't valued and respected in their culture.

Many of the failed countries on Earth, simply have too many dumb people to even prosper. The best and brightest in these failing cultures, can't rise to the top. Within even prosperous societies, certain groups just stay poor and dumb, and it has little correlation with race, and more to do with the value people place on intelligence. Breeding with family members is also a common cultural problem in many of the dumbest places on earth. Many Muslim countries have people marrying their first cousins, and this just creates failed genetics. The American lefts desire for gun control is mostly connected to an image of zombie hordes of inbred hillbillies armed with rifles descending upon them. While the right just wants more guns to kill off the zombie hordes they fear. In all cases there deeply smart and amazing people in any group of people on earth, if you have a large enough sample, and those people's futures means something for all of humanities future.

VOLUNTARY "SOLUTIONS"

Mexico "solved" its problem by pushing its undesirable citizens to go to America. They could get better jobs than in Mexico, have safer lives, and free education. These undocumented workers even send money back to Mexico. Many of them are amazing people and add a lot to America, and many are not. If Americans don't want undocumented workers then they have to stop employing them, giving them free education, and benefits. Americans getting rid of their best workers from Mexico, who have strong families, could be an even bigger loss for the host culture, than them staying. Those with poor family connections, dismally low IQs, and few job prospects, are the ones America wants to get rid of. The ones with strong families and that get employed readily, have little reason to leave America, and little reason for Americans to want them to leave. In a mix of the lowest class people in the most powerful country on earth lies truly amazing treasures and truly despicable human beings. You can't judge the group by either.

The best way to get rid of immigrants you don't want, is to pay them to leave. They have to have a reason to go back to Mexico or whatever country they came from. If the price is high enough, they will gladly leave. The ones with the least reasons to stay, will be the ones most willing to leave with the right incentives. At a basic level, you have to set an amount of money, that would actually lead to wide scale voluntary deportations. \$10-100k would be the approximate range per person, to have them leave. A family of 8 could in theory get \$80-800k. When the money is seen in family terms, motivations would be much clearer. You could even buy people homes in their home country, give them all the things they need, and give them money. It would be cheaper than the process of finding, processing, and deporting them, plus the amount of money spent per year on welfare for them. Though without capitalism in their home countries, their economies won't develop and create the jobs they need. Their home countries are not places with bright futures for anyone living in them.

Let's say we want to, in a nice way, sterilize the dumbest people, and make it so they don't have kids ever, nor have influence over society. The easiest way to do this ethically would be to pay them to do it. Simply give people money for being sterilized, and those that need the money, would simply go do it to get money. The net effect would save money for society, and thus could set very high prices to encourage people to try. \$100,000 to be sterilized would be a tempting offer to anyone. Straight cash though would be a horrible way to pay people, because these people have no way to budget money or use it well. It would be better to give them things they want, and that actually have a useful value to them. 60" 4k flat screen with a gaming system, cable, good shoes, clothes, a nice sofa, a bed, delivery food, and a small apartment. You could provide a basic life for people for under a \$1000 a month, with some weed and booze tossed in. It is basically like welfare, and it requires being sterilized to be paid. Build up ghettos of housing systems to store these people, and watch everything they do. A completely controlled environment. People will get kicked out for being even a little misbehaving. They have to want to be in the system, and fight to be in, rather than out. Make "prison", the "ghetto", and "welfare" into something a person wants and is good for them. Any system that rewards bad behavior will fail. You pay people to cease existing in reality and they will cease to exist in reality and you can transition them into death in the gentlest most voluntary way possible.

A GENOCIDE SYSTEM

If we look at society, and say wanted to kill off 20% of the human race in one go. There are many ways to approach this, based on differing principles and "ethics".

1. Disease- release a virus that has mass casualties rates. This favors rich countries and kills off poor ones. No matter how deadly or virulent a disease is, they can be

contained by the social ordering of a civilized modern society. Disease targets and kills off uncivilized regions and peoples, even within rich countries.

2. Random- Everyone has to play Russian roulette. Play once and you are given a certificate of survival: "Certified Survivor". This certificate can be given to anyone, and not only the person that played. People can play more than once and pass on certificates to more than one person. This will favor families where people will sacrifice for another, and create higher death rates among those not bound in family structures or where others would risk their life to save yours. Lots of the elderly will die off, in such a system
3. Death Melee- Put groups of people into an arena and when a certain target ratio of people dies, then the game is over. This can have similar dynamics to a "certified survivor" system. This will favor those that want to survive, vs those that are willing to give up. Strongly motivated people, will overall survive a death melee. Social and organizational skills will be favored over pure killing power. Asocial people will die.
4. Kill to be Free- Kill one person and be a certified survivor. Fringe elements and sociopaths will be killed off. 2-5% of the nasty people that everyone wants to kill, but can't, due to law, will get killed. Criminals and the such will die, because there won't be a vengeance cycle nor stigma. Tribalism will form with "good" groups vs "bad" groups. This will take out more uncivilized "tribes" in a society.
 - a. "The Purge"- Making murder legal for a certain period of time, will not lead to that much death, because the law has little to do with why people don't kill. The law functions to stop vengeance, and cycles of killing. A purge will just prolong vengeance and make people plan things out more. Social stigmas against killers will still exist. Humans will still form into natural defensive groupings of decent people.

5. Tests for the Best- Make a collection of different tests, where high scorers get to survive. Society can base these tests on anything that a person could possibly be good at. The top percentile that is saved would depend on the amount of people wanting to be killed off. This system favors people with valuable abilities for society. It can be, math, language, art, music, athletics, knowledge base, creativity, being good looking, healthiness, and so on. Save the top 5-20% in range a set of categories.
 - a. Just be good at something, in order to live. People that are good at something have a certain focus for the future that is valuable and contribute to society the most.
 - b. If you can cheat the system, and escape, you also deserve to live.
"Hacking" can be a skill set.
6. Combine all of the above- Don't kill the low scorers on tests, put them into different paradigms to get a certificate. You can see all of these systems, as a kind of "test", and only high scorers will survive. A normal family with strong bonds, would easily survive and the elderly in their family would mostly be culled to save the younger people. The elderly, anti-social, uncivilized, criminals, and extremely lazy and useless, would be most targeted for death. While the poor face deadly disease driven culling pushed on them by elites.

Now having a state with the power to enforce such a system would, by necessity, require that all criteria favor the survival of those closest and most valuable to the state, and those farthest from the state, to be targeted for death. It would be mass human sacrifice by the state to ensure obedience to it, and the creation of a more civilized future. The state has always been doing this since the first ones came into being 6,000 years ago. It is the basis of civilization. Creating a more systemic form of killing of the unobedient and unmanageable would only grow the absolute power of state control. All large states with

high levels of power, commit mass genocide. Human nature demands purges. This then demands a more accountable and high functioning state. We can judge the performance of a government by how well they prune and select citizens for survival and positions of power. Killing just based on political orientation would also be an underlying aspect of any system, where politics itself is reflected in genes, and purging an entire way of thinking counter to collective power, could be achieved in genocide. They will certainly try if the state has so much power.

If death seems to extreme, the consequences of failure can be sterilization and or shortening of people's lives through poison or disease. Instead of killing, one can simply bathe them in radiation, where their immune system temporarily dies, their reproductive organs no longer work, and their lives will be vastly shorter. If they survive and can reproduce, then they might have some superior genes or life habits that need passing on. Russian roulette, death melees, viral plagues, can all be expressed through poisoning, rather than absolute death. People don't have to die immediately. They just have to die sooner and have no children, to be effective in the larger scheme of population management. You can make such experiences far more pleasurable, and just give them lots of fun drugs to take.

THE MATRIX

Intelligence is just how fast you learn, and anyone can learn anything, given enough time and patience.

The "Matrix" can be made real, and it can save the world and its 1 billion idiots. It can be fully voluntary and people can be fully aware they are in a system like this, and still want

to be a part of it. They will be the “batteries” for AI, and create a useful service for the world. The real challenge in the face of AI and our zombie apocalypse of sub 83 iqs, is how to bring the two together into one solution. AI *seems* to make this threat to humanity and making people useless extend up into even higher IQs where average peoples skills will never be useful.

We need to create jobs for people at the lowest levels of intelligence, and feed the growth of AI to its maximum potential. Low IQ people and low performance people can be put into a AR/VR world, where they train AI algorithms to do basic things, and the AI trains the people to do more and valuable things for its own growth. Even the lowest IQ people can tell a cat from a dog, and applied to everything in the world, we can find endless things even the dumbest can do. AI training makes it possible to create work for any level of intelligence. People can specialize in things that they most like and are most useful. Dinosaurs and porn will be labeled to perfection, down to the tiniest detail.

AI is very simple at present. Anything you can for certain identify in 1 second of thought, is something AI can probably be able to learn. These inherently simple tasks you take for granted on a daily basis, can actually be work for people, that rewards and develops them as people. Education itself begins with things like a picture of a cat, and child saying “Cat!”. We expand the number of things they can label, and inverse the problem, where they have to be given the label, and find the correct object. Teacher says “Cat” and student points to a group of animal photos and finds cat. This can evolve into more complex things. Red cat, pink cat, happy cat, Siamese cat, jumping cat, sleeping kitty, old kitty, kitten, jumping Siamese happy kitten in a white living room. A system that pushes people to generate such results will want more and more evolved and developed concepts. The system itself will want to improve the people in it to be smarter. There are nearly endless things we can use the lowest levels of intelligence of people to train AI for

decades to come. The key is to find the things that have value now, and that can be monetized, and grow from there. The people in the system will become smarter from using the system itself, enjoy it, and feed into an expanding possible income. It all has to be made into something a person would want to do, and has a future in doing. It would be like living in a cartoon world of video games, and being able to make money, have real friends, and eventually lead a much better more independent life than possible before, by doing humble gamified labor.

As the system evolves it can also help people integrate into society and be useful. It can help people to do the most basic tasks and then use whatever talents they have that AI can't do easily as the real service they provide. AI is being trained as it bridges a human being into the world to do functional tasks with AI aid. An AI can't be a human with joy, love, and a smile, creating quality services for others, ever. You could see this like AI making people into puppets that express AI services, and this wouldn't be wholly incorrect. You can also look at all people as puppets of their parents, society, teachers, and employers, where we learn and repeat, as the basic form of education. They would learn to imitate the world and possibly be able to function and do things without any system, if they had 24/7 personal development to learn to be good at least one thing. Instead of putting robots in fast food chains, we can put people who now aren't even smart enough to flip burgers.

The centralizing of power to manage such a system is the fundamental danger of this system, and not the AI. It is the people wanting to control it and direct it towards their own individual goals that make such a system dangerous. If you have people becoming like flesh bodies for AI minds, you can imagine AI bound people like a zombie horde that can be hacked and manipulated to do things, because all the things they do are managed and assisted by AI. Large companies controlling it will be made out to be

possibly evil, and then government will step in to control it and then it will only become even more centralized, more controlled, and more of a danger. Such a system requires zero centralized management, and should also be something taken up by individuals and their families by their own free will. People can go and select the service they want to be part of, and use the interface they like. This is goldmining for human intelligence in people with little intelligence, as a business model, and there needs to be lots of different companies and paradigms experimenting with ways to extract that knowledge out of people in the most profitable way, that makes the workers themselves in the system benefiting from it. Harm done to people will always be an excuse to take over the system by larger more powerful groups. Making the system inherently decentralized, in its very nature will be a fundamental and necessary feature.

BASIC INCOME

When the world loses all hope, and thinks machines and technology will consume us all, we are left with people willing to be consumed and dominated by absolute power for their survival. The best way to give welfare and basic income, to the lost and hopeless, if it must be given, is to require work to receive any assistance. Welfare in of itself is murderous, and all statistics can show its absolute devastation on groups of people receiving it. A healthy welfare system has to be measured by how few people are on it, and how many people get out of its system, over a period of time. A genocidal welfare system, maximizes the number of people in the system, and increases the needs for that system, as they use it. It exists for no other purpose than its own existence, and eats people to feed itself.

To shift welfare systems from genocidal human slaughtering machines to humanity growing machines, is not hard. We must pay for what we want. If want successful

healthy people, pay for healthy successful people. Purchase that as a product. Most welfare pays people to be sick, single, with more children, unemployed, and ignorant. If you want them to work, pay them to work. If there are no jobs, pay people to create jobs. The vast majority of all welfare issues are based in joblessness and being a single parent. If people with children, have a job and are married, their chances of being in poverty and needing welfare, in a western society, is practically non-existent. The small group that is still poor needs to be paid to be better skilled, and integrated into society. The overall problem with those that need welfare is a lack of vision and realistic planning for the future.

SOLUTIONS 1-5

Solution 1: Take an existing welfare system, and change nothing. Add in work requirements to receive welfare. Have them do any job with a fixed and set schedule, in order to get government help. There has to be an elimination of all work regulations, taxes, and wage requirements for people, in such a program. Companies can create positions for people, if they can pay them any wage they want, and have them work in any way they want. If a company treats their highly subsidized workers poorly, welfare workers can just go do another job, and change readily. Many industries can create work for people, if the government is paying large parts of their salary. As people get better at work, their value as a worker increases and then they can change jobs, or negotiate for a higher salary, while still getting benefits. This solution is ideal for low skilled unemployed people, that just need to learn to work.

Solution 2: Pay companies to employ people through the tax system. No welfare benefits are given to the worker, and this is not a subsidy to a company. The company taking this money is performing a service for the government, and being paid the amount of

money needed to get them to want to do it. Vary the amount of the subsidy based on the problems the person on welfare has, and how much companies don't want to employ them. Former felons, the disabled, and the mentally ill, can be better served by such a system. Employing certain people costs a company more money, than a normal worker, and increases risks for that company.

Solution 3: Doing community service to get welfare. People that have worked in higher positions and have skills, are reluctant to work lower end jobs. Peoples sense of prestige and status have to be maintained. Many competent and able-bodied people receive welfare benefits. The elderly, temporally unemployed people. The benefits are usually paid through insurance like programs, like in America's, Social Security, Medicare, and unemployment insurance. The longer these people don't work the more they shrivel away, and often end up on drugs, chronically ill, and dying. Their illnesses then drive them into needing more welfare, and being unable to return to the work force.

Community service in one's local area, that is non-political, and focused on community services, can be a rewarding experience for anyone. Communities without much community service opportunities will naturally grow them, as peoples demand for them grows. When people need some form of employment in a real job, or in community service to get benefits, they will go out and get them. They can be involved in things they care about, and connect more to people and be valued in society. The same money will be paid out by the government, in the same way, and there will be massive increases in societal benefits, and vastly decreased healthcare costs.

Solution 4: Some economists have considered inflating the money supply by giving out the money as basic income to the poor, rather than to large banks. Central banks instead could inflate money supplies through small business loans done by small local

banks. Small banks get money, and are more connected to local economies, which make jobs, rather than big banks that feed money into existing large companies, that create no new jobs. New jobs are created by new companies, and most new companies start as small businesses. Most people end up working in small community businesses.

Empowering the system that feeds into real job growth, creates a system that doesn't even need welfare for workers to get back into the work force. It also feeds a start-up revolution, where people can feel pride in the companies they make. There will be more start-ups, and more failures, some will work out.

If you were to use inflated money to run the other solutions that modify existing welfare systems, it would be better to drive the money through a community bank, and then into community service programs and local businesses. The exact way to do this, I haven't figured out. The main principle though is that money has to flow into localized community systems, through accountable agencies that prosper and most profit from a strong local economy. Small banks and businesses will inherently be less able to manipulate the system and profit from corruption on large scales. Fraud at small levels, will specific cases, rather than wholesale systemic fraud. Problems will be localized and specific, rather than broad and general. There will be some scandalous localized stories, and not a large centralized monster, whose failures are rarer and more massive.

Solution 5: Integrating new people into a community is a problem, not addressed in the previous solutions. Immigrants need to learn a language and culture in order to fully connect into a society and prosper. People trapped in failed communities need to escape and move to new growing communities. We have local emigration and national immigration, that need services to support them. Both are an education issue. To get people to get an education they need, then we need to pay them for the education they

get, not the education classes they take. We pay for a result, and not the classes. If they need to move to a new area to get good work, we pay them to move to a new job in a new place.

If a person is in a bad area, we can pay them if they find a good job, at a certain salary level, in another distant location. The money can cover moving costs, and help instigate moving people out of dying off areas, or places with no future for them personally. They only get money, when they get work. This then can be tied into training programs, where they also get paid when they get certified in a real job, that connects into in demand job skills, with waiting jobs for them, if they can complete the training. No free training, no free handouts, no welfare, is being given. People study to obtain a new level of ability and are paid when they achieve results. There are many jobs in any economy waiting for people with the skills to fill them, and it takes motivation to get stuck and hopeless people moving towards the future. If you pay people to stay where they are, and not get educated, they will stay where they are, and not get educated, no matter how many free education programs you offer them.

CHAPTER 7: OPENGOVERNMENT

USING THE POWER OF OTHERS

You are most powerful as a leader, when the people you are leading have the most possible power, because the overall amount of power you can manage is maximized. If you want to rule a small community, a company, a country, a galactic empire, or the universe, the basics are the same. If you have power, you can do anything you want with it. Power can be made larger by how you use it. Intelligent use of power, leads to having more power, and bad uses of power to less. Micromanagers control everything and don't let other people have power, and thus they have less power over all. An absolute ruler of a poor country is much less powerful than an elected president of a rich country. A society that divides power among the most people, grows into a more prosperous and successful place. The strength of leadership comes from the quality of people that want to follow you, and your capacity to create an environment that brings out the best them, and uses their abilities to their limits.

The best people that live in countries that stifle their freedom, leave and go to other places. Great employees don't like to be over managed, and when they can, they leave bad companies and bad management, for better. If you develop procedures for who gets promoted in a company, and everyone works with that in mind, and then someone uses personal relationships and secret deals to get a promotion, it undermines the authority of leadership and the rules themselves. Systems that try set up rules for how power is used and minimize personal abuses of power often fail, because rules can't manage human nature out of existence.

You can certainly do whatever you like to do and use power however you want. There are times when the system and the rules don't work, and things need to get done. No system is perfect. We pick leaders to lead us, because we understand that a system can't work all the time. Leaders aren't supposed to follow the rules. If the rules and systems worked perfectly then a society is solely ruled by laws and doesn't really need leaders. Well run companies and governments mostly just have good administrators that run the system of rules that make things work. These are not leaders, and rarely ever become great rulers. The head of HR and the enforcer of company policy are not leaders.

LEADERSHIP HAS NO RULES

Breaking rules makes people love you. When you break them to protect their lives or create more prosperity, people will respect this. If the rules are going to cause people to die unnecessarily, then breaking the rules is a moral imperative. There are those that would let innocent people die, rather than go against rules. People will not even see such behavior as illegal and might actually perceive it to be more lawful than pure rule following, because it is fulfilling duties that they perceive must be fulfilled. In politics one side will see you as a hero and the other as having done something very wrong. The only real difference is just how they define a leader's duties. If leaders do what we think government should do then it is "awesome".

Most of global politics is like this, where we have to establish rules and then often break them. Just wars, just torture, just invasions, just genocide, just stoning to death of women for being raped, are the oxymora of political reality. Traditional international law basically says countries have to leave other countries alone and let them do whatever they want inside their own countries, except when they have behaviors inside of "bad" countries that can justify other countries attacking them. Developing weapons of mass

destruction and genocide are about the only reasons for just invasion. When we fight our neighbors, we have to do so within the reasons we have government in the first place, jealousy, rule of law, safety, and prosperity.

Invading countries for them dropping the world-wide supply of critical resources, is in reality how real war is done. People not using a resource well, nor developing it fully is nearly always a just cause for invasion or the seizing of peoples' property. We just don't frame it in the real terms that leaders are really thinking. If the duty of a state is to maintain a prosperous economic order for all, then a lot of things are going to have to be done outside governmental rule making systems. You can always go take what you want from others, so long as it serves collective idealism and fits into the duties of a state. Even if you disagree with this, it is reality. The moral math of it can be explained in maximizing the economic value of property, and in how that collectively benefits society over the individual. In the case of countries with valuable resources, they are being invaded in order to liberate those resources for collective uses around the world, and for the benefit of the people in that country.

PLUNDERING FOR GOOD

"Good countries" develop themselves and their resources to their best potential or risk being overthrown from within by its own citizens or from the outside by another country, based on the belief prosperity is a governments duty.

One of the most common reasons to go to war and invade people, is to support prosperity. You want something, someone else has it, you go take it. This is what governments existed for in history. Governments maximizing plunder for its people is

how civilization flourished, and how everything we have now even exists. The world before had very little and the only way to be successful was to conquer others and take what they had. Enslaving and plundering countries used to be standard politics, until we developed the technology to produce more wealth by cooperating, rather than fighting to the death over limited resources. Humans still operate on forms of politics that rationalize plunder as noble, across nearly all political ideologies. Only capitalism freed us from this barbarity of inherent plunder. Socialist countries only survive on war and plunder, even the northern European ones, as they did through their whole histories up until the present.

In our world, there are sometimes things that we still need to invade and fight to the death for, in order to create global wealth and prosperity. Plundering others can be done for noble reasons, that don't directly benefit the ones that do the invading. Let's say you have a country that has a special resource that the world needs for the world economy to run, and that country isn't developing the means to harvest that resource ideally. This is a justified reason for invasion. Governments use this sort of reasoning to take land and property away from their own citizens. The underutilization of a resource is a form of conflict with neighbors. This is where common peoples' idealism and political reality often conflict. At the root of nearly all justifications to take others property is that they are underutilizing it to its potential, and collectively society would benefit more if someone maximized it. If a person is sitting on a gold mine and not mining it, has prime farmland that is never farmed, oil fields untapped, a one-story house surrounded in skyscrapers, "excess" money they don't need nor use, they might just need some noble plundering.

We don't sell this idea to people very well. We don't have rules and clear laws to govern it, so the system doesn't work when it meets these very real issues. If there were more

clear rules to justify conquest for collective prosperity, than different groups of people would be far more inclined to develop what they have, for others to buy from them. The clear defined threat of invasion to maximize a closed country's poorly utilized resources could drive that country to develop those resources. International law, with strong threats of violence and absolute death, have a way of getting people to comply. Selling invasion and war, based on resource development, just requires an idealistic definition of a resource. With an idealistic enough cause, plunder can always seem morally acceptable to nearly anyone. Not developing a resource well, is a very abstract concept, and yet it is a reality of why we have justifiable conflict with others. Leaders that want to be more powerful, need to develop all the resources in their society to their fullest, as a means of protecting themselves from hostile neighbors, and to maximize the consent of being ruled by the those you lead. Prosperity is an intoxicating drug that can rationalize nearly any behavior. People like being richer, even if they are less safe.

The main point is to maximize your power as a leader. Get as strong as you possibly can. The tools of power though don't allow individuals to use them as freely as they may want. Systems of laws make a ruler more powerful by giving everyone equal chances. Strong individual people and groups make a society wealthier overall and safe-happy-healthy citizen create civilization. We shouldn't overthrow bad governments. We need to teach bad governments how to be more powerful in a step by step gradual way. Certain types of leadership and management just don't work. Certain economic systems and rules have been shown not to work. The main problem is that other societies try to make worse societies then theirs, behave like they do. Democracy and freedom is not for everyone at every stage of human existence.

The mighty rule over the weak and drive out of them all their resources, for the collective use of society and the maximization of the power of those who lead.

Developing humans to the their fullest and getting the maximum performance and work

out of them, requires feeding them their basic political needs and minimizing the things they resent the most. Plundering the earth and humanity, is the natural state of government and living as a human being is just finding the best slave master to dominate you and not kill you too readily for disobedience. A little freedom goes a long way to make those under your dominion achieve peak performance. Just remember that in real life, if you don't pay your taxes, people armed with machine guns can come to your door and threaten to kill you if you don't pay, even in the most civilized and developed country, over less money than it even cost to send those people to collect that money. They sometimes even drive tanks through people's houses in America, over taxes. It is rare, because that power exists, and people have that fear ingrained into them so deeply that they don't even realize it exists.

OPEN GOVERNMENT

A country of laws, can be a country of simple codes, that we all can understand. Government can be a computer human hybrid, that uses the fewest and best people to run a state. Government itself will eventually be able to be upgraded with software upgrades if we make the whole government accountable to machines. We can have "more government" with less actual people running it. Smarter people with more machine power can do a lot more than a large group of above average minds. Making government accountable, will be hard for people to accept, rather than its traditional role of making more jobs and power for itself. Every system, all information, has to run first and always through computers.

The mobile revolution is quickly turning into the internet of things revolution and governments are just barely using desktop computers to do most tasks. A band of people, in a low civilization place, could set up servers, link devices and computers into

them, and start to run a village, with little effort. They can use basic solar power and minimal energy to run everything, and probably have a better government from the start than most developed countries.

Government can't hide when the very system is designed to expose and hold everyone accountable. The top levels of government need to be run and lead by people that actually understand how to use computers and implement them on a large scale. The "head of IT" should be higher than the head of the governments police force and even military. This is not like some corporate IT manager out controlling the whole government, with old fashioned IT management. Government by nature is risk adverse, and so they often take what they perceive to be the safer route that will cause them least individual possible embarrassment, while in the long run, creating systemic failure and highly risky collective behaviors. Opening data is way to push back against these natural tendencies and bureaucratic thinking. This is not some one step solution, it is a continuous evolving integration of technology that grows out of industry and organizations. We have the solutions in their basic form already. We can step by step, get these implemented and then evolve them through time. Capitalism and not government makes these things possible.

Things to consider when looking at a "OpenGovernment":

1. How quickly they upgrade
 - a. Are workers using the latest hardware and software?
 - b. Are they implementing new things fast and often?
2. Accessible real-time information with analytics interfaces
 - a. Numbers that add up and make sense
 - b. Is there anything being hidden. Does it need to be hidden?
 - c. Do intelligence agencies really need 70-year-old secrets? 40? 20? 10? 5?

3. How simply you can access and get what you need
 - a. Layers of bureaucracy can exist only in code, and not in your experience
 - b. Simple, direct, and pleasant experiences
 - c. Customer service should be excellent
4. Easy to get information that makes people in power look bad.
 - a. Can get information from the police that normally they wouldn't want to give
 - b. Are real scandals being exposed and people being held accountable
 - c. Are those in power getting away with abuses of open data standards
5. Are there real IT people at the top or just yes-men that enforce policy?
 - a. Do engineers hold high level positions?
 - b. How many elected leaders of a government have technical educations and work experience? Engineering, math, technology.
 - c. does the IT department have its own police force that is actually kicking in doors and locking people up?

ELECTION OF LEADERS

In this section I am going to present a very novel and new way to elect government officials in a democracy. This is also a solution to problems like campaign finance reform, that troubles American elections so deeply. It gets cronyism money out of politics by putting in the system even more money, than they cronies can pay. The size of government is an irrelevant issue. If it can do something of benefit, and we want that, then we should let the government do it. There are just few things that a government can do well, without causing more harm than good.

The Iron Rule of Life: All things have a price that must be paid. We must pay for what we want.

We get what we pay for. Instead of just voting for candidates, our votes should be matched with money. Who we vote for, gets paid by the number of votes we give them. Salaries and whole budgets of leader's offices should come from money bound up in our votes. If we elect someone to lead us, all their salary and all their staff's salary, will be connected to the money they get from the votes they receive. We should pay our leaders the top possible wages so we can get the best people leading us. Leaders are incentivized to promise a lot and deliver very little. We elect people based on how much we like them, and in how much we want to believe the things they tell us. Humans by nature prefer good intentions over results. Over time this lack of real results leads to systemic failures that certain groups of people, can no longer ignore. When people become disillusioned with government, and pull away, there are no consequences for those in power, if they can stay in power. Open data can make leaders more accountable, and make us more willing to invest in them, and the government projects they push forward. We can regain our faith in the ability of leaders to lead, when we have the personal power to hold them accountable at every single level. Getting elected with 12% of the voters, vs 60% of the voters, is a huge difference in money.

THE MEANS JUSTIFY THE ENDS. We prefer humanity over results and doing really good job. Corruption always begin with the noblest intentions and starts small. We forgive vast amounts of corruption, because we like corruption that seems to try to help those we like. Government has to be intensely human and connected to its people, in order for it to work well. The art of modern government is in creating the most targeted manipulation of key groups feelings, to get them emotionally behind a leader, and maximize their re-election, rather than actually working on running a government.

Elections are where most of the corruption of government comes into politics. People want to get elected and they need friends to help them get into power. They make relationships with people that can help them get elected, and if they get elected will help them back in exchange for their help.

CAMPAIGN FUNDING AND SALARIES

Elections and politician's salaries can be funded by the peoples votes.

The low pay of government leaders and staff only drives them to be more corrupt. People often hate the government and try to minimize how much people get paid. Their wages in government office can easily be made much higher than anything they would be paid on the outside. Human beings in power, being paid little will see themselves in comparison to the rich and powerful around them, and by nature feel lacking in their own lifestyle, and self-image. Humans will always be petty and most frivolous with the smallest differences between them and their immediate peers that live around them. They need to be paid more money than they make now in corruption, when they get high paying jobs after leaving government service.

Change doesn't require massive top down laws to change these things. It can be done at a local level first. It should begin from the smallest units and work up to the largest. Local changes can lead to a mass political revolution. We can take money out of elections, by paying for them with the number of votes a person gets. Each vote can correspond to an exact amount of money, and it shouldn't be small. We can require far more transparency in how they do things, because if they want the money, they have to be open. Campaigns can still raise money the traditional way. We only pay vast amounts

of money to them if they run an open campaign with all money only coming through the voting process.

Paying them from votes will turn their focus towards their voters, because have to work on getting more people to vote to get more money. The more people vote and are engaged, the more they get paid, and the easier it is to get elected. If we want high voter turnout. Than we pay for it, by giving more money based on turn out. Even the people that vote against you, can add to the overall money you can receive. Getting paid for the people that vote against you, may seem counter-intuitive. We don't want leaders repressing the vote of their opposition, as a way to win. Paying for higher voter turnout means paying our leaders some share of the money from the votes they didn't get.

AMERICA AS AN EXAMPLE

“Every vote counts” needs to be replaced with “every vote pays”.

The biggest problem with American federally elected officials, is that America pays them far too little. The second biggest problem is that they have to spend most of their time getting money to run for elections. The third is how these two previous things converge to conclude, and create systemic corruption. In America leaders and their staff plan heavily on how to make lots money after they leave office. Voters feel that their vote doesn't matter, and that candidates are beholden to special interests. These elections have low participation, and don't represent the people. If we want politicians to care about every voter, and to try to bring as many people into the election, then we should pay them to do so.

1. All votes are backed by money that pay for campaigns and salaries. The amounts for both need to be huge.
2. Candidates can choose to use the full government financed version, or raise money themselves. If they want the large salaries, then they need to use the voter financed election process
3. Primary votes for a party go to the party winner. Gets all money needed for a campaign. More overall voters coming out, more money for campaign. More contested campaigns should have higher turnout.
 - a. \$10-15 per vote, per position: Congress, Senate, President
 - b. \$4-8 million dollars for a congressional campaign.
 - c. \$6-60 million for a senator. Depends on state
 - d. \$1-2 billion for a presidential campaign.
4. If candidates are running by raising money and running with large campaigns independent of this system, then the amount being funded needs to increase
5. General election votes will determine salary and staff pay. They get money for every vote for or against them. Higher turnout, more pay.
 - a. Representatives will need about 1-2 million in salary. 200-400k for key staffers
 - b. Senators 6-20 million. 400-600k for key staffers
 - c. President 50-100 million. 1 million to 2 million for key appointed positions.
 - i. Can be a billionaire if they win 2 elections. This will reduce corruption and the desire to seek out money for personal fortunes.

6. Bonus Pay. Some possible variations

- a. If the budget isn't balanced, no one gets paid.
- b. If we go to war. Pay is cut in "half"
- c. If GDP grows by over 3%, \$1 million for each percentage point to each official.
 - i. 8 percentage GDP growth. Get a \$20 million bonus, for everyone, staffers included.
 - 1. \$1 million for each 0.1 % above 6%. Paid out over a 20-year performance window.
 - 2. Maybe half a million for each above 0.1% above 4%.
 - 3. And if you receive bonus pay, can't work in any government connected job outside, or any lobbying. Bribe them to not work against America for their own profit.

7. Parties and Registration

- a. Possibly pay parties based on the number of people registered to them. Don't let them raise outside money.
- b. Help finance elections before primaries
 - i. Party registration funding could allow parties to fund who they want, or at least organize events for interparty competition.
 - ii. This though gives insiders and the party too much power
 - iii. Can have a no campaigning rule before primaries and force candidates to compete in open debates.
 - 1. Shorten Americas long election process

- iv. If someone doesn't want to pay a party, they can fund a "super PAC", and support a specific cause or issue.

8. Reform can be done state by state at first, and then eventually be passed as federal law.

- a. A state can pass a law for its representatives and senators. And finance the federal government themselves.
- b. Make Washington answer to the states and congressional districts they represent.
- c. If people want real reform, they will need to pay for it, at their local level, and take real responsibility.
 - i. Use your own local money and power to save the world, and not wait for others.
 - ii. Buy your elected officials obedience to their districts, states, and country.
- d. As the movement develops, it can reform presidential elections, in time. Because it would require federal law to fully regulate such a thing.
- e. A state could still give to a presidential candidate money for every vote, if a candidate agrees to not accept donations from people and businesses in that state. Though that is not an ideal solution.
 - i. A state could also bribe a president with performance pay for that states own economic growth. This could motivate a president to do things in that specific states economic interest.
 - ii. Or even pass propositions in a state, to pay leaders if they pass a certain law, and not just vote for it.

- f. This sounds like a road to insanity in politics. It however is how the game is played already by the elites and those with money to buy politicians and governments.
 - g. Reps and senators elected from states supporting this reform, can also be the ones to push federal law, and as more states come on board, a federal law will eventually be passed
- 9. Evolved form. People in order to vote on something have to pay a set price for that, and this generates the revenue. Democracy then becomes truly about the people, their own choices, and all power being in the voters.
 - a. A system that pays itself, will always be a bit corrupt.
 - b. The poor can be given money by the wealthier to vote, and the votes a person makes can still be secret no matter what.
 - c. Communities could raise voting money, to give to people on demand, as a sort of “charity”. This would require a very developed society, with deep social institutions.

HUMAN RIGHTS AND CITIZENSHIP

The right to kill someone is the basis for the power of the state and for contracts. All real laws and contracts are enforced by threat of death. If jaywalking is illegal, then it is possible for a government to kill a person to enforce that law. When we sign contract, it means that you consent for it to be enforced at the threat of death. Putting our name to a contract is a serious matter than must be understood in absolute terms of life and death. The terms of agreement on a contract matter. Contracts are voluntary, and the laws of a state are not. States that try to align themselves with the voluntary nature of contracts more, are ones that are ruled by laws, tradition, and constraint. The more laws

that are passed and the more laws are passed without the real consent of its people, the more government rules with the power of death, and a police state. Modern policing is problematic because the primary job of police in an authoritarian state is to enforce tax laws, and increase the power of the government. Cities that help fund their budgets with tickets are authoritarian.

RIGHTS AND THE US CONSTITUTION

The size of government and how much power we give it is irrelevant. Power defines government, and people pursue power as an end in of itself. The laws that govern a nation, can be anything a government wants them to be. There are rules, traditions and institutions, that constrain how these laws get implemented. The process by how laws are made and enforced is fairly controllable. The intent of laws, is far more flexible, and easily perverted to meet whatever ends people want, based on changing sentiments and intentions.

Americans have a clear constitution that lays out all these rules for government, that the government doesn't follow at all. People complain, and think fixing the problem is getting government back to its constitutionally defined powers. All the rights in the constitution are not individual human rights, they are constraints and rules on how the government uses power. The US constitution is more of restraint on power, than an entitlement to power, since all governments by nature have the right to absolute power. To restore constitutional power and authority to the American government is simple, and people don't want to do it. The leaders didn't break the constitution, it's citizens did. Americans can fix their government and restore it to its legal and traditional settings, by simply following the law on what the government has the "right" to do and what it doesn't, and no one wants to do that. People love all the things the government does

that are fundamentally illegal in the constitution, and being a “constitutional lawyer” is the art of creating rationalizations people will believe and accept.

The law is often divided between “Natural” and “Positive” law. At their essence, these are ideas of what rights are in a society. Does the government give people rights, or do people give themselves rights? Most legal systems get these all confused and mixed up. The government has duties and rights, the difference between different legal systems, is the differences in the rights and duties between them. Both in reality are the rights of power, that exist in this world.

- Natural rights are the rights that an individual person has to do whatever they want, so long as they don’t truly harm others
- Positive rights are the right of government to do whatever it wants so long as it protects and doesn’t harm its citizens.

Some people think that Americas first amendment gives people the right to free speech, religion, press, and peaceful assembly, when in reality it doesn’t. The bill of rights constrains the power of the government to take people’s natural rights away from them. The bill of rights are all positive laws enforced against the present government by a more supreme government existing in laws. Americans at the time of their founding, believed they just had rights, that God gave them, and if anyone were to try to take them away, this was justification for conflict. If you want rights as a citizen, you have to take them, and reduce the power of government above as much as possible. All centralization of power is a threat to individual power, which is by nature spread out and divided among many. Governments have the right to do anything they want, laws exist to constrain them, and define how they use that power. In an age when people have forgotten the basis of rights, they believe the state gives them to them, because the state is their God and highest form of power.

US SUPREME COURT

To understand the US supreme court is not that hard, if you understand the law in this way.

- Anything that grows the power of the state is 90% most likely unconstitutional, and anything that gives rights to the people and their communities is 90% likely constitutional.
- If you are dealing with international affairs, and then the government can basically do whatever it wants, even slaughter entire countries for sport and sell their bodies for gold.
- The US government owns the borders between all states, and the border of the country itself, and it has absolute rights over these things. Inside states, the states own all the land, and can do with it as they please. States can just take anyone's land without any reason necessary, because property is an illusion.
- "Property Rights" are more good contract law with the government. There is no protected right of property in the United States.
- Taxes are rent payments. The government taxes things, labor, and people, which means they own you, and have the right to collect rent. Anything that can be taxed is not owned by the person being taxed.
- The fundamental tension between the left and right side of the court is whether Jesus or the state is God.
 - Those that see the state as God, see that the future is created by government planning and vision, and that no progress can emerge from human beings unmanaged.
 - To those that believe Jesus is god, they believe individual choices create the future, and a holy country is best at achieving the future.

If Americans would like to have the right to own property, they could pass an amendment to the Constitution, and that would make all property taxes and income taxes illegal. If you are an American and you think you own your own home, then why is it that the government can tax you for owning it, unless they actually own the land and are just renting it to you? Clear definitions of what Government actually owns and doesn't own are not clearly stated, because people would seriously constrain that power, if they understood what it meant to be taxed on something.

Most everything the current US government does is fundamentally unconstitutional. In the past, they had to pass amendments to the constitution to enact any large changes, and currently such large changes can be done with barely even voting by the congress, much less using an amendment. The government took over the whole health care system, and made a system that would lead to universal health care, and didn't even try to pass an amendment to the constitution, and in the process, led a country to near civil war, and at present in my time a brewing fundamental war against the very system of the government.

THE ILLUSION OF RIGHTS

Rights can be a constraint on government power or a duty that it must perform with its power. These rights live in inherent conflict. Rights are only about government power and have no real individual basis.

Most of humanity at a local level of human organization live in barbarity and with absolutely cruel relations between them. Stoning "unclean" women to death, is a

"natural right" in many societies. It is easier to reform bad laws, by appealing to an even higher form of law, that can rule over the flawed local one. The more we turn to central government power for necessary survival and to overcome problems locally that won't go away, the more we lose the power to adapt change and grow. Rights are a bit of a silly thing, that most people have very different ideas what they mean, at the most fundamental of levels. The reality of rights, is that they come from governments, they come from power. Good uses of power connect into and understand human nature, and the fundamental humanity of the people they govern. Local power is far more connected to immediate human emotions and human nature, and thus a far more personal form of rights, closer to the individual. All humans since we first existed have lived in communities of people with rules to govern how they behave between them. Making rules for people that go against their natures and undermine human dignity, just don't work.

Good laws embrace humanity, and its fundamental nature. Bad governments try to change people into things the government wants them to be, so they can be controlled easier, and follow a select few peoples' vision of reality. Human civilization flourished when primitive local rights were subsumed by grand collective societies and empires. The rights that individuals feel when free of government are illusionary rights, and they are in reality a product of their relationship to the communities they live in and that community's rights to control them as individuals. Natural and positive rights are a conflict between local power and more universal centralized governmental power. There is at some level no such thing as individual rights, since we as people never actually exist as individuals. Local power vs central authority, is about the only real conflict that exists in rights and law itself.

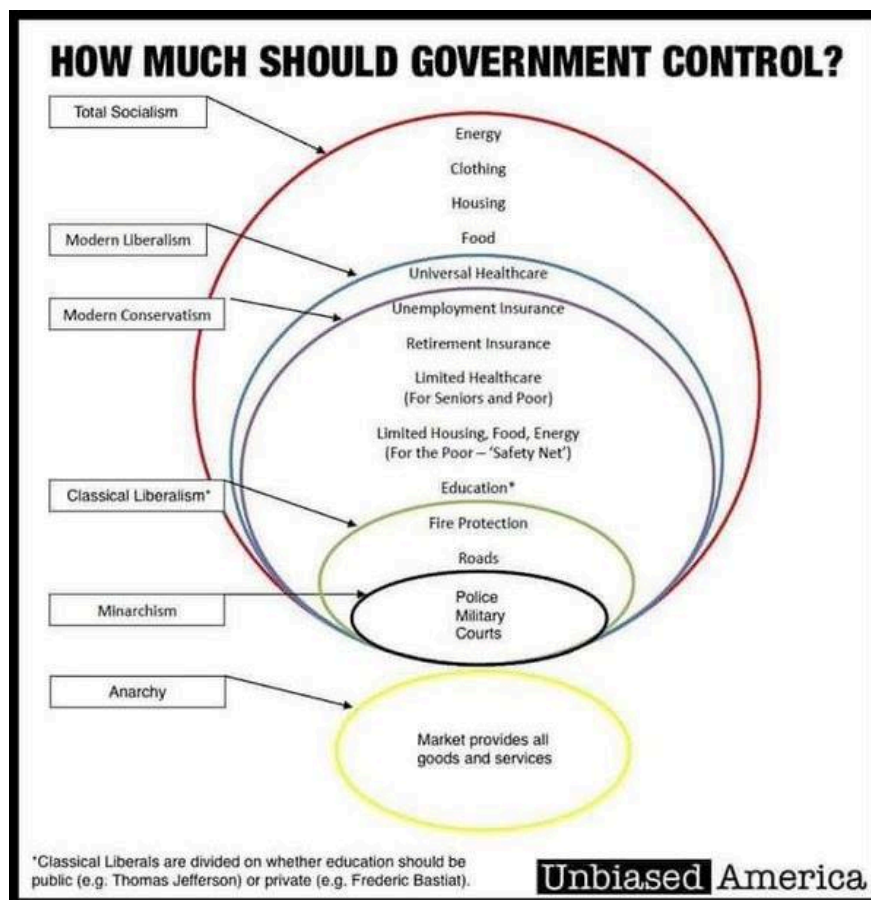
GOVERNMENT DUTIES

"Positive Rights" or "Modern Rights", are the duties a government has to its citizens and it is expected to provide.

The government should be as big as possible, and do as much as it can for its people. If it tries to do too much of one thing, it fundamentally limits how much it can do of another. The more the government wastes on one thing, the less it has to do other things. If a country could get twice the results for half the price, for anything, it should pursue that. Massively inefficient systems undermine a country, and the effectiveness of government, no matter how noble that system may be, and no matter how much good it does. It is difficult to measure the damage done to society by the things that never had the opportunity to prosper. Governments main duty is to protect our lives as citizens, support a prosperous society, and contain the savagery of humanity. Military, fire and police services are the minimal positive rights. Politics is a disagreement over what duties should be provided to its citizens, and what should be left to freewill and human choice. All economic growth comes from freedom, and if the government takes all freedom to provide for everything, there won't be an economy to pay for anything. If you think the duty of government is to make the most free and prosperous country you will find another government whose duty is to make its people happy, healthy and connected, to be the most disgusting possible government. Government that makes happiness a duty, has to define what happiness is and eliminate human freedom.

Few issues can more polarize people's views on politics than medicine. It gets at the fundamental ways people define what it is to be human, and their idea of what a humane society should be. For many, a society that lets its citizens die from disease is failing in its duties; a government that makes a socialized health system undermines its countries prosperity and economic future. Finding the means to protect all people from

death and not undermine an economy is a very tricky feat. Government duties to provide healthcare of their citizens can be structured on very different principles. Protecting them from death, and major illness, is fundamentally different than providing for every possible thing they need or protecting them from every possible unhealthy thing. A society not fighting for the lives of its citizens is failing. A country paying for healthcare for the bad decisions of its people and actually increasing bad behavior, is failing.

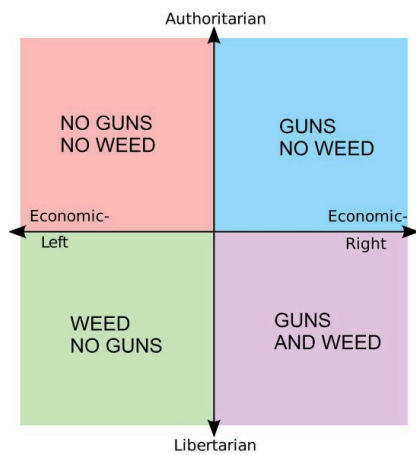


Governments at the large scale, with positive rights governing how they think, are judged on how organized, civil, and healthy their citizens are in that country. Democratic socialist countries are often judged to be the most superior. They have good trade-offs with success and overall social health. There is an inherent decline in overall possible prosperity in these countries that leads to inevitable collapse. Economic growth is a

central duty of a country, and no government can actually run an economy or make a business that actually works. A single percentage point less per year in GDP growth adds up into an enormous amount of lost money over enough time. In communist societies that difference between potential growth and real growth could be over 6%, which is a staggering sum of money over just a few years. You can double the entire wealth of a country in less than ten years with 7-8% GDP growth. The cost of lost economic opportunity is easily hidden and its victims have no face to attach them too.

DECENTRALIZED BUSINESS INNOVATION. TECHNO-PIRATE COMPANIES.

Human nature is biased to organize around certain types of political organization. As best as science can tell, people are born left wing and right wing, with tendencies hardwired into their basic cognitive structures and how they see reality. Any type of person can be any type of political affiliation. These cognitive natures only make inclinations. Then within the left and right, we can be more authoritarian or individualistic. Humanity has four intrinsic ways of politically organizing, that exist in inherent tension and disagreement. We evolved a diversity of ways of thinking to adapt to and organize the world, because different kinds of thinking are advantageous in different situations. If anyone type comes to dominate a system, that system becomes unhealthy and people resent the government.



People have to learn how to integrate all 4 into an effective government system, and not blindly try to impose their own system on everyone. We don't have to have a government that treats everyone the same, in order to be fair and functional. You can see these basic forms of power laid out in my basic descriptions of the 4-different ethical and moral systems, in Chapter 3. What made the US government so successful is that it was a fusion of the left and right working together, to make a functional government. They were anti-authoritarians who rejected centralized power, and depriving individuals of freedom, while also being upper-class elitists. Those that love freedom and resent state power, live in great tension with present day modern governments. There are some simple things the government could do, to make this group of people happier and contribute to society on vastly larger scale, while not upsetting or changing the lives of those that like being dominated and told what to do by others.

Deregulating three main industries would great increase the happiness of freedom seeking people, and let them make any kind of business they wanted. Medical, finance, and small business start-ups, are places where more freedom would benefit everyone, and radical changes are unnecessary. There needs to be a space for just totally free businesses whose business practices only effect the people that choose to use them. Deregulating highly polluting industries or ones highly entangled with international affairs are entirely different sorts of issues to deal with. There could be businesses that

just operate outside the legal system, and don't use any government systems for help or structure. Customers have to make a choice if they want to use these "unsafe" companies or not. If you eat food from an unregulated street vendor, you are eating at your own risk, and should have a right to live with the consequences freely. These sorts of businesses need to be labeled like cigarettes: "This business is unregulated and can kill you. Only their word and their promises can you hold them to". Murder, fraud, theft, and slavery, are still illegal and these businesses must be honest.

Nearly all problems that can happen in these person to person (p2p) businesses can be dealt with in the contracts they use to run these companies.

- Normal disputes can be settled outside of courts with private arbitration, that will simply look at the facts of the case and come to a binding settlement as fast as possible. The kinds of arbitration service a company uses can help determine even what sort of companies you do business with. They must clearly state such things openly and be easily seen.
- Most business can be monitored and overseen by private inspectors and people can pick the sort of businesses they deal with, by the quality of the inspection services they use. Inspectors have every reason to maintain their reputation and brand name, because competition can easily step in. Fraud and malicious companies and services can be stopped with privatized systems.

MEDICAL SYSTEM

Innovation happens in spaces with the most freedom and least government intervention. We can radically innovate medicine by creating free spaces for individuals to do

healthcare however they want, and have them live the consequences for themselves. Often the worst effective solution is the best solution for most people. Increasing supply is the only way to decrease costs, and the only way to increase supply is to have more medical workers, medicines and services, and the only way to increase that supply is to allow lower quality solutions compete with higher quality ones on an open market. Let a certain part of society buy any medicine they want, from any doctor they want, from any source they want, without any regulations. The only thing they need is contract laws and rules against fraud. They harm no one in taking medicine made in another country that is unregulated by America, without the permission of government approved doctors. They will do this at their own risk, and it is no different than people thinking marijuana and alcohol should be legal. If laws against narcotics don't work, then laws against real medicines certainly don't work.

You can have universal health care, a regulated state medical system, and a completely freedom based anarcho-medical system, at the same time. Any country with universal health care could make a pirate healthcare system legal to operate in their country. There is no reason to not allow people in any country, even ones with socialized health care, the freedom to arrange their own medical system, that doesn't use any government regulations. These different systems will help feed into each other and make the overall medical industry more efficient. If people want to exist outside the system, they should have the right to, and their innovations could launch huge new tech industries. A revolution in medicine can only happen in a free economic system, where companies can just do what they think is best and experiment. People can use their own personal judgement to choose which sort of medical services they want. They can go with safe, static and old or vibrant, evolving, and risky, based on their own needs.

The first western country to really deregulate their medical industry for those that want to live outside of it, will have a tremendous business opportunity to make world changing global tech companies, and bring radical advancements in health to the human race, everywhere. They don't even have to undo their existing system, or threaten the safety and ordered society of any of their citizens. In many places, even with this freedom, many people would not risk or even try a totally new kind of way to get medical treatment, and they won't be forced into such a system. Only in place where people want that, and would dare try, would a market and new industry grow and prosper. As people see the positive results of such a system, more people will want to be involved in it, and it can grow gradually with success, and not in any sudden revolutionary way. Seeing results will lead people to adopt such a system, and no force or manipulation is necessary.

You can see it readily by just imaging big tech companies or large pharmacies organizing a freedom based medicine system. Different companies will appeal to different customer bases. If you got medicine through Amazon, you would probably get home based medical tests mailed to you, talk to doctors online from home, and get medicines mailed to you at the lowest prices from the best sources in the world, with no boundaries between countries in selling medicine. If it was Walmart, you could imagine going in person to a store, getting blood drawn in person for testing, going into a private booth and chatting with a doctor by video streaming with devices in the booth to help examine you, where the pharmacy could make up your order, and give it to right there.

Companies we trust, can make new systems to serve us and keep us healthy, and would have zero reason to not do their best to provide the greatest services. All the start-ups and people going to work in them, will be driven by a desire to change the world for the better and help people, for the most part. Creepy companies, and creepy technology don't survive long.

The price of the exam and medicines would be vastly cheaper than just a prescription you get now. You could get an online medical exam for less than \$10 from a qualified doctor. Both systems could be maintained and monitored by the highest quality call center standards and have outside inspectors checking them non-stop. They could be 90% cheaper than what we have, and be vastly safer, with lower quality medicine and lower quality doctors. Not everything needs the best to be cured. With massive increases in supply in doctors, medical services and medicines, prices by nature of supply in demand, will severely drop, and fall every year. We can expand this slowly even into medical procedures and surgeries. Now most people for life and death surgeries will want the best services, and a socialized medical system, might be more what people want with life and death issues. A medical system that protects against serious deadly problems, is far less an intrusion on freedom, than one that monitors all. We can keep this socialized system in check with a healthy free system, running alongside it. People may find buying insurance against death, that provides top level services still preferable to socialized health care, because it can be faster and more adaptive. All depends on individuals. Universal healthcare itself will evolve and focus more on the real duties the state has, rather than just controlling everything.

FINANCIAL FREEDOM

Allow for totally unregulated banking and financial services for people to use, and power them with crypto-currencies, to create a financing and internet revolution. This will create backbone financial systems to run most all online tech businesses, and companies of the future. Being able to move money around freely and instantly through a system, will have extreme benefits for tech development. People will have completely free will to choose to use such systems, and most of the people able to even use such modern finance, will be fully aware enough of the system to make an informed decision, and not

be cheated. Using money with no government backing has extreme risks, and it presents an opportunity to create new ways for money to be used free of all previous constraints. In a narrow part of the economy new models of economic behavior can be experimented with, and can possibly develop into world changing technologies. We can't deregulate after there is an effective alternative, because it can never truly develop unless there is the freedom to make a new system freely.

Governments are presented a great opportunity to strengthen their own financial systems and currency with such new freedoms. If the new money takes off, a government can just buy up parts of it and convert that into its own currency, and then underlay the state currency with a more vital and free virtual currency, in a gradual and steady manner. Government don't lose the power to inflate and manipulate currency. The new currency though can create a different kind of reliability and performance for companies, that can't exist in the present. Imagine a world with no large banks being necessary to power finance. All the services would be online and require no physical banking buildings, atms, and infrastructure. Banks require tremendous amounts of money to run, and nearly all those costs get past on to customers and tax payers. Purely online ones will develop slowly and only deal in narrow things they are good at, and as innovations develop can expand and create a completely new modern feeless banking system, that is incorruptible. It will have new risks and not provide all the same advantages of the old. If people want to pay the higher costs of old fashioned banking, they will have the freedom to do so.

ANARCHO START-UPS AND A NEW LABOR UNION MOVEMENT

One of the great ironies is that unionism is best served in a system with the least government power. In modern societies Unions mostly exist in government jobs and not

in the private sector. Private work force unions could have a new revival if they changed their form to better match a new political age and economics, and not relying on government force to achieve their goals. Large amounts of government intrusions on business lead to more expensive products of lower quality and less people able to create a company and jobs for other people. As unions pushed for more government control, they killed the very companies they worked for. Unions didn't die, the companies they worked in died.

The freedom to make small businesses that provide services to people locally is the key to creating a new Union movement. The freedom to make small businesses with no intrusion allows them to be created and organized based on any set of values. If people want to buy the values you are selling, they will buy your product more readily. Companies able to form with less regulations can self-regulate themselves and use less resources to start their business. More cash strapped areas, can start a business with less capitals. Groups of people can work together to fund, start, and run a business together. The regulations themselves make it so only those with access to large amounts of money and more formal business experience are able to even start new businesses. Every regulation and new imposed cost, makes a new business less likely to exist, new jobs less likely to be created, and poor people less likely to create something for themselves that empowers their lives and those close to them.

Not all companies have to be high technology and advanced fields, like engineering, programing, medicine, law, and finance. As we deregulate more industries newer kinds of companies and businesses can come into existence and serve people. Change and rapid development can't happen in regulated economies. One of the ways to regulate the quality of unregulated businesses, is self-regulation and outside inspectors and standards. We as customers will need to become more aware of the sort of companies

used to inspect industries, and go with the ones we trust. Food service with no government or private food inspection would be questionable for anyone to eat. You may actually get safer food with a private inspector, than a government one, because they are highly motivated to create the best reputation for the inspection company, and empower companies to be sincerely trusted by customers. People like making money, and no company has any motivation to hurt customers and lose business.

Small companies simply can't organize massive misinformation campaigns to manipulate society. They are connected to their communities and make money by how much people feel locally connected to them. Build up your communities and social connections, and use this to drive a local business, that people want to see succeed and feel an emotional connection to, while delivering superior services at competitive prices, in order to succeed. Capitalism is the freedom to use whatever value your system you want. People just prefer cheaper over more expensive and this creates a pressure on companies to compete in ways that normal people just don't like. Modern technology can help run small business at lower costs and make the overhead for all the internal management costs vastly cheaper and with high tech tools that can drive constant business improvements. When people fail, it will not be easy on them, and not everyone has the heart and will to make a business, even in a big group.

People have to get licenses to do just about every job. New shops in a city have to get permission to open, and not cause competition with existing businesses. You can't go selling flowers on the street, when florists stores go complaining to the government to arrest you. Much of what you will see is utter chaos and businesses opening up out of vehicles and right on streets curbs. It is not pretty and offends the eyes of those that like ordered and disciplined societies. You can imagine and inner city of a western country covered in street businesses, food, music, local art, dance, creating a local

energy and space that people want to go into and engage this freedom and new vitality of culture. Poor areas could be freed to do whatever they want, and open any sort of business they want, and this will draw in customers as these business hustle with customer service and personal connections. Often the finest food comes from the humblest locations and from people most in love with life, people, and freedom.

It is better to allow certain areas freedom, rather than let them be desolate and regulated centers of hell in utopian western empires. With freedom, the kinds of ideas and innovations that people will create will be unimaginable to a single person's mind. In an age of new technology, new ways for businesses to be run are emerging that make highly decentralized small business competitive with bigger and more organized companies. Online marketing can connect customers and business together in deeper and more complex ways, that no large company ever could. Fast food chains might become things of the past, when you can get the best food made at the lowest prices from real people, that you respect and want to see succeed.

CHAPTER 8: THE ORIGIN OF CONSCIOUSNESS

HUMAN EVOLUTION

Humanity is different than animals. Human evolution is different than the evolution of animals. We come from animals. We evolved step by step from animals, and we are still animals. The vast majority of the evolution that created humans, is no different than what created all know life in this universe. Our sentience greatly influences how we evolve as a species, and human evolution works in a different way than normal evolution. One day something changed, and that allowed us to evolve ourselves. Sentience has evolutionary value beyond just intelligence, because it can control and shape the evolution of human beings. Humans gained a power to control their own evolution, and direct this towards a purpose and sense of meaning. We can manifest our identity and self into things, and have things manifest identity and self into us. Our sentience is itself evolving, through culture and how we organize ideas, where only certain things can work and prosper through time. Sentience is the root of technology. We manifest our living minds into things, and those things manifest into us and propel our prosperity.

I would hypothesize that, sentience could arise gradually out of our sense of what is human and not human. Our sentience is an emergent property of instincts, and very specific aspects of our cognitive biology. It isn't just some random feeling of self, that we have, it has at its root a conception of certain aspects of reality. This is a sentiment that could develop from very basic levels, and gradually develop into something what we call being human. The instinct itself, is independent of the content of humanity. Being able to recognize members of your own species, can evolve step by step, into more specific and complex things: species, group, family, basic needs,

hunger/fear/danger, good member, bad member, similar to you, different from you. A sense of self emerges from objective reality and observation, that encodes itself sequentially through generation after generation of evolutionary adaption. This system projects itself into things in order to see if it is similar to your or not. This system then could evolve into projecting into things beyond even humans and see "humanity" in things. Sentience is the moment we can see things as human and change them into more human and useful to us, rather than just social animals imposing social order into other members of a "pack".

We had this feeling inside of us, that we call "humanity". This is the most magical and wonderful feeling that we can have as human beings, it is filled with everything we consider good and noble. It is nearly impossible as a human being to step back and objectively look at this sentiment, and its primitive origins from our evolution from animals. We don't truly control this instinctual and primitive sense of humanity. We can deepen and broaden our conception of humanity, while never really changing its root reality, that exists inside of us. We all grow up with different life experiences and will value different people and different cultures, as more human to us. Beyond the surface cultural conception, there is a deeper inner animal like sense of self and not self. This shapes how our sentience and sense of being develops at a day to day level of living. In the end, humanity, is what survives, and survival must lie at the root of any conception of being human.

In the definition of humanity, there is an inherent corollary, in what we call not human. Inhumanness is something we label people with, not animals. To call a human being an animal, that lacks humanity, is the most fundamental attack on a human being imaginable. They go from a thing of divine dignity and beauty, to a thing that we can smash, kill, use, and consume, like any other thing on this Earth. We create this intense

social in-group, and an intense social out-group, that we can kill. Human sentience may arise out of our instinct to avoid that which is not human. In modern humans, this aversion evolved into possibly genocidal instincts. We are by our fundamental nature social animals. Only social animals can evolve into sentient beings, with feelings, and a sense of self. If sentience is an emergent property of social instinct, then by the very nature of social values and social formation, there is a root sense of who is a member of a group and who is not a member. This social sense of belonging and group identity, in sentient beings may at its root have a very narrow and specific way of thinking. This instinct may be the only way creatures ever form into an ever evolving and growing species. A sense of time with a directionality has its origins in our social instincts, that give rise to our sense of humanity, and in turn create the very intelligence and sense of self and reality, that we call sentience.

TELEOLOGICAL PURPOSE DRIVEN EVOLUTION

We direct our own evolution with purpose and direction. When humanity first learned of evolutionary theory, and how animals came to be, they immediately made it all about themselves, and in how we came to be as a species. Normal people conceive of evolution in human terms; they conceive of it with purpose and direction. Many people took their "human" ideas of evolution and applied them into society, government, economics, family, and all other domains of human existence. Evolution made science and not God the creator and ultimate judge of things. Power itself began to define how we see evolution. It isn't that the science of evolution was bad, it was that the people used that science, to justify horrible things.

Evolution inspired tremendous evil in human beings. If were to judge evolution by the morality it inspired in others, and by the things they did with that, then evolution might

go down as one of the vilest ideas in human history, along with communism. Marxists themselves would also take evolutionary ideas of how to best evolve and develop mankind, and then would go out to slaughter millions. Communism was the merging of progress, science, reason and politics, and it ended up being the single greatest man-made killer of human beings in all of human history. Human evolution as we humans feel it in our souls and from the reality that we see is something very different than the normal kind plants and animals have in this world.

Animals evolve through random chance. There is no real direction to their evolution. You can't really describe anything in nature as more or less evolved. We can say things are more adapted to their environments, or less adapted. We can think about bigger concepts, like if a species has the ability to adapt to change and new environments and not just to the stable one they live in. The capacity to change and adapt to the world, defines life. Humans are different. We don't think of evolution as random, rather we think of it direction and an end goal. At some level religious people, reject evolutionary theory, because they see it as inhumane, and not connected to the reality and sense of being that they know. This was wise of them, at some level. We don't evolve like animals, that much, anymore.

LINEAR EVOLUTION

We domesticate things, and domestication itself is a form of evolution arising out of human sentience, which tries to project its reality into the future towards an end destiny.

Humans see evolution in a line. It is singular and directional. We evolve as a group. We as people kill off wild human beings. We domesticate plants, animals and people to makes us as individuals survive. A beaver builds dams to build their homes. Many animals build and create things around them, as machines that extend into nature, and expand that animal's capacity to thrive in nature. We as humans seem able to build anything around us, into something that reflects our own self, survival and being. This capacity to consciously, domesticate our environment is the greatest power to ever come into existence, in the universe. Consciousness, as an objectively measurable thing, and is about how we shape our environment to our own survival. The patterns of our minds and our thoughts are imposed into the structure of our objective reality. We can embody humanity into things. We can replicate thoughts into structures around us, that in turn can replicate and grow themselves. We can conceive of cities and systems that grow and evolve like living entities, of concrete and steel that serve human life.

People that have more consciousness and overall awareness, have a greater power to control and shape their environment. We can choose what we build, measure the success of those choices, and modify those things, at any level, to work better.

In evolutionary science, many people have tried to create collective theories of evolution, where whole groups work together to survive. This is and will always be bullshit. Nature doesn't evolve with collective consciousness or some grand interconnecting sense of being. There is no mother earth, no magic energy binding people and things together. There is self-interest, competition, and brutal domination. We imagine feelings and meanings into animals, trees, and the stars themselves, as an individual function of sentience. Mother bunnies don't love their babies. It is just hormones and programming. People want to see in evolution a cooperation among

members of a group to thrive as a species. We create this delusion by projecting and injecting our humanity into the environment. At some level, there is collective evolution because we as humans through culture, our humanity, project a collective reality into our social groupings. This can drive aspects of our cultural and cognitive evolution, which then can drive our biological evolution. We are all so similar, so we project similarly and are programmed by others and the environment around us. The "humanity" in things, creates us and, we in turn can create it, and this creates the tension between collective and individual values.

HUMAN COGNITIVE PROJECTION

Projection of human consciousness into things, has an evolutionary pattern similar to genetic evolution. It builds on itself through trial and error. Its code is virtualized into a form an intelligence that can rewrite itself faster than genetic adaption. Our realities are composed of nearly all things made by and altered by human will for human living and being. The people and things around us create us, and we in turn create them. Humans live in spaces that they modify and are surrounded by other human beings. We are social animals, that by nature must be able to influence and be influenced. We impose our internal mental structures onto the environment, and our very minds are shaped and defined by our environments. We see ourselves in everything around us, and when things don't fit with how we feel, we want to change them to match how we feel inside.

We project our humanity into the sciences and into evolutionary biology. The underlying origin of imagination and ideas to solve scientific problems, arise from our own internal sense of humanity, and who we are as people. We push forward ideas that we feel a

deeper intuitive connection to, over those that seem disconnected from us. Intuitive smart people feel good or bad about certain ideas, and the very structure of how their ideas are formed, is based in how the mind thinks. The scientific method itself is a projection of human thinking processes into a formal methodology that actually works. Everything we create arises out of the patterns our minds can form.

We are animals, struggling to survive, ruled by biological instinct, hormones and innate cognitive biology, that is all shaped by the environments we evolved out of and adapted to living in to. Culture itself is just a reflection of the humanity we projected into the physical environment we lived in, and that helped us to survive in that place. Different cultures arise out of different environments and conditions. The delusions that arise out of humans projecting human feelings into things that don't feel, has a realistic limit, and has to be more precisely controlled and contained. When we are faced with things we don't understand, we always fall back on projecting humanity into it, at some level. There are some projections that end up working, and producing amazing results through the ages of time. Most things felt are just absurd mental crap.

THE WILL TO VALUE

Nietzsche saw humanity as ruled by a will to power, and I think that is just one reflection, of one aspect of what a human can be and express. A will to power can be more properly be expressed as an innate will in a person to be individual and define their own innate identity. This is only one half of the innate will of the human mind. A broader sense of this will to manifest self into reality, and to experience the world as one's self, is a sense of value and not just identity. We can also possess a "will to value". We can have a part of us that is valued in of itself, from itself. The will to power and the will to

value are systems that need no external object to be what they are in of themselves. They are the very teller and organizer of your story.

People are designed by humans to fit into a collective society. Evolution is guided by intelligence and will, when it comes to people. We are intelligently and consciously designed in our humanity, by humanity. Linear evolution is the conscious and intelligent design of a species to shape reality in itself and around itself to nurture its own survival, beyond just the flesh and blood of individual human beings. Sentience gains the power to domestic both nature and humanity to its will, from primitive evolutionary processes to sentient human controlled processes. The evolutionary process that leads to sentience can only evolve from social animals, and is thus inherently social and collective.

We are not individuals that can survive on our own. Humans gain power in groups, and can think collectively as groups. We don't have magic powers to connect our minds together. This "magic" is just a feeling in how a person can experience external reality as an immediate and personal thing inside of themselves. When we see who we care about in pain, we feel pain, and certain people can feel what that person is feeling inside, as their own emotions. Our projection of self, and the projection of self from others, all blend together, into something more. With close enough groupings of people, that live together long enough, these boundaries fade away, and our certainty rises a great deal. Social intelligence is more precise in closely bonded groupings of people, with little change in their immediate surroundings. The more we know people the more detailed and accurate our understandings of their emotions are at any given moment.

By our very natures we will always feel more humanity in things closest to us, and bound to us in memory, self, identity, and life experience. Families that are close will always be more human to each other than strangers. Humans evolved in small groups,

very closely connected to their own families, and culture. Everything in our world around us, was simple, singular, and all consuming. Our parents can appear like gods in our consciousness, even when we hate them, or have been dead for 50 years. We pass that love and sense of being, self, and identity into our own children, even when we vow not to be like our parents. This chain of family creates human existence and culture. Our minds are built out of the things that created us. We are our parents, we can feel them, see into them, and even be them at some level. Our communities, our land, our friends, and our cultures, define us and create all the potential for us to exist as humans.

Our sense of self is created from before we are even born. People are different. There are patterns to these differences, and they all come to work together, quite well. Different people are better at different things. By themselves each person is not so useful, together, they can do a lot. We as humans have specialized minds, that are better at doing certain kinds of things more than other kinds of things. This cognitive biology is far more elegant than what random evolution might "imagine". If we were only products of our environment, then we would quickly have everyone thinking and acting the same, and thus we would not be collectively intelligent. That innate individuality is foundational and irreplaceable in sentience and being. Without it, you are just a machine, with no purpose, no meaning, and no future. We have an innate value and identity born out our innate individual wills.

CONSCIOUSNESS

Consciousness has a consistent form that arises out of nothing and randomness. Humans experience the world around them. They feel it in their bodies and beings. We don't see a rock on the ground. People feel that there is a rock on the ground, and try to infer meaning and purpose in it being there. There is always emotion infused into our

sensory perception of reality. We can't conceive of things, unless there is a self to judge those things. Different forms of humanity can be projected into reality, to both perceive it, and to control it. There is not a singular sense of humanity, that we as people project. We have variety in people, and their different emotions are projected into reality.

Our sense of and control over reality must be based in reality. These judgements and perceptions must actually work and correlate to the world around us. Our projection of humanity and self into them, is not a distortion of true reality, it is realization of a different kind of reality. Consciousness exists not just as awareness of being alive, it exists of sense of being greater than, and innately bound to the earth beneath our feet. We feel more than the world we live in; our own personal self is more valued than any material thing in the world. We are not a thing, like the objects around us. Objects of value are more valued than objects with no value. We infer value into things based on how they affirm and support our own life, and how they connect into the future. Our own sense of self transcends even our own body and being, where we can be willing to die to protect the very humanity we identify and connect into. Dying to save your own child is little different than saving yourself.

Consciousness is a framework of the mind that seeks to preserve the future over our own immediate survival. We can sacrifice our lives for a sense of our own self living into the future. Immortality itself is never a physical state of being to seek. It is a state of mind and a way of living one's life to the fullest and engaged in creating a future for what you feel to be humanity, and which in turn sees you in something fundamentally human. If our sense of value in things, is completely out of sync with reality, then a person doesn't live long, nor maintain social relationships.

Survival itself is reality. We care more about how people think of us, then we think of ourselves, or even the things we can own. Social interactions have serious consequences, if they go wrong. Our interactions between people have to actually work, and make sense. Patterns of thinking that can't form communication, that others can understand, can't be the basis of consciousness. Inside our own heads we talk to ourselves, and this communication forms our identity. Brains have multiple personalities in them that form conscious thought. The two sides of your brain are like two independent minds. The more you can use both sides of your brain, the more ability you will have in understanding and communicating, with other people. This is why women are better communicators than men, because they use both sides of their brain and can more innately understand other people's thinking more. They learn language faster, because they talk to themselves, and this in turn makes women more indecisive. Women are more conscious and more able to manifest self and being into their environment and in turn are more influenced and affected by their environment. This sensitivity, vitality, and being define femininity, and make it something more innately human and worth protecting than non-feminine life. Female identity and being is survival, and the tree of sentience, from which all reason and thought emerges.

AN INTENSE AND VERY LONG AUTHORS NOTE

We seem to have innate structures in our mind, that greatly correlate to the highest forms of thought we have developed as human beings. One of my core presumptions that I had to make, from when I was a kid, was that a theory of the universe and a theory of humanity could be solved with a single theory. This presumption is not based on anything rational. I just wanted it to be true, so I made it true. My own desire to see a cognitive model of human consciousness relate to the universe itself, is rather strong. I

am very skeptical that such a thing is possible, though at a root level, it just seems the most obvious thing.

- The way we run society, develop ourselves, build starships and understand the universe, should all be driven by the same theory, because in the end it is the human mind that has to make all these things and we exist and live in a universe, thus our minds are defined and created at all levels from its reality.
- We can only create a theory of the universe from our own minds, and that system will be a reflection of human thought itself, because people made it. Our own minds are themselves born out of existence in this universe, and it is consciousness itself that can create a functioning model of reality in science, that will then itself need to have a fundamental correlation and connection to the rules of the universe that created it.

“Only possible things can evolve. What is possible to exist and evolve is narrow.”

I came up with a clever trick to get around dualism, and magical thinking. Our minds evolved into conscious minds with a certain pattern, because that is the only pattern it could evolve into and form consciousness. Evolution comes from purely random forces. Our minds evolve from purely random mutations. The mutations that work, are very rare. Over time they begin to form into the image of something, that can actually work,

and get more and more refined in that system, through time. The sheer complexity of human thought had to evolve from nothing, and be bound by something. We don't see reality; we see a simulation of reality. It has to correlate to reality, and the real world, because if our vision didn't correlate, we would die rather quickly. Our minds play tricks on us, and don't fully capture the reality of the world outside our brains. They work to make us survive. They are formed in relationship to reality, and the very complexity of the human mind is formed from fundamental patterns in the very fabric of space and time.

We as people have all these variations in how we can see things. Evolution built out complex mental structures out of a big mess of competing systems, that sort of found a way to work. Personality is an emergent property of that, and evolved from how we project our own identity into reality. If that projection doesn't work right, then that system fails. We are blank slates. It is just that the blank slate isn't flat, and not all the same texture. There is an innate terrain and geometry to the mind, that has inclinations to have certain things imbedded into it more than other things. You can use the sort of intelligence often associated with art and painting, to learn math, if you try hard enough. Visual minds create more visual mathematics.

The fact that certain personalities can form amazing and functional theories of physics, and that the universe can be intelligible by math, is one of the greatest mysteries in life, that at some level, people just accept as being normal. Math doesn't pre-exist human imagination, nor is it some magical thing we discover floating in another reality. Math that is useful to human beings, correlated to how we think, and actually works in the real world, is the one that evolves and grows. It is functional shape and terrain in the "blank slate" of the mind, that certain things, like math, form and imprint more readily into. If people can't understand it, and can't conceive of it, then it doesn't come into

existence, no matter how accurate it is in its descriptions. We can make math from many different cognitive structures, and use it to fulfil very different human needs.

Most people don't like math, because their personality has a hard time projecting itself into it, and having math project itself into their minds. This is not a problem of math itself, it is a problem of having too narrow and limited expression of what math really is in its reality. It is a far more human, emotional, and meaningful field, than most people know. In the end, we can only conceive of things we can project our minds and self into, and thus our very selves, emotions, sentiments, and minds cognitive structure, at some level must conform, and be structured on reality itself.

Our personalities define what is possible to create, because they favor specific forms of thought over others. Emotions are not some random and arbitrary thing, separate from reality, that we need to get over, in order to have proper and clear rational thought. We approximate reality in our minds, and are not in reality seeing it, nor wholly knowing it. Our feelings can be real and most often they are not. The mind has the capacity to project self into reality and have reality project itself back into it. Sentience correlates to reality, because if our sentience and conscious sense of self, didn't correlate with reality, we would die, and/or be absolutely insane. Insanity itself is when the wires of the brains don't conform to reality, and warp perceptions so that our inferences are all wrong. Schizophrenics see god talking to them in how leaves rustle on a tree. Social anxiety makes people infer far more danger in certain social interactions, then actually exist.

Emotion is the subjective experience of self into the universe. There is no separating our reality from ourselves and how we feel about it. All objects in the universe exist as objects, because they are defined by their use relative to human emotion. Our projecting of our subjective self into things, creates things into meaningful objects that we can

study. We can only have objective reality if we infer usefulness in the things we observe. Emotions themselves, our feelings are a form of information on the objective world, they are information that translate the objective world into subjective experience. Emotions relate to our survival, and interpreting our senses and reality in a way that favors our own existence and propagation. Logic itself is an emotional construct. What we think of as "objective" is really just inter-subjective, where we imagine how lots of people independent of us would feel. More logical and objective people would be able to make decisions not based on their own personal needs or survival, and this can be important, for our survival overall, and thus that very "objectivity" must always serve a subjective need and purpose.

The boldest claim I can make is this:

*The initial quantum state of the universe, that is timeless, has nothing in it. Our minds have the capacity to leap into this reality. It is the root of the human sense of self and identity. To exist in this state, is to take a human mind back to its primal initial state from which all conscious thought and being arise out of in this reality. It is not a mental state that people like to readily enter or face, unless they have to, or become a nun/monk. Being a hermit is not easy. The entering into this state is ironically very emotional. In order to conceive of reality, we must be able to conceive this state at some level in our minds. For at the root of all thought, when we truly go into the deepest states of human consciousness, there is simply nothingness. **That nothingness is not meaningless.** The very structure of space-time is what influences quantum structures take form, and inspire the formation of sentient thought. These are real system, real structures, and not some magic thing. We will never be able to directly observe them, we however can directly model them. Crystals form from the structure of space time, in things most closely connecting to the thermal energy levels of space-time, which are near absolute zero.*

THE BODY AND FLESH

We are mostly not all that conscious in our thinking. Consciousness has a power to control and change things. Most parts of us as humans, are beyond simple control, and awareness. Becoming more innately connected to our bodies, gains us greater degrees of self-control over our instinctual systems. Certain people are better at connecting into their bodies and controlling them, while others are not. Much of the humanity in our self-control over our primal and rudimentary instincts, has in itself different patterns. We all know the different ways that people control their bodies, instincts and self. Our very sense of our bodies and their primal instincts arises out the sort of human emotions we feed into our body.

People that love power, and people that love social value and connections, will treat and understand their bodies very differently. Everyone can connect with their body, they will however come at that understanding differently based on the sort of personality and self they can project into their bodies, and how their bodies own senses come to project back into the mind. We can all be correct, in how we understand our flesh, all be different in that conception, and still be approaching a more objective sense of reality. You can't control and understand you flesh truly unless you experience it as your own self and being. You can objectively stand outside your body and look at it like a thing, or you can subjectively experience it, in all its feelings, as you yourself. You can experience the feelings in your body, as the feelings in your emotions. People that are more subjectively linked to their body, are far more likely to have control and understanding of their flesh, over those that examine it like an object.

Consciousness has to have a consistent form and relationship to our own flesh. We can understand our own bodies as something personal and as self. Through empathy, we can see how others connect to and feel in their bodies. We can see pain and suffering in others very readily, when we are more connected to our own self and being. What can be seen and understood, is only what the observer can see and understand in their own self. Our bodies are extensions of our minds, and in of themselves reflect and generate thought. This is possible because the bodies that more connected to consciousness, were more able to survive. The mind shaped the mind body connection into a feedback loop, that generates a holistic mind-body consciousness. You can most control your own self, and not others. The capacity to project your identity into things, is most present in your own body, and it in turn has the most power to impress itself into you. Survival and sensation can't be made independent of thought and self.

HUMAN CULTURE

We exist. We feel. We see the universe around us. We make rather accurate models of how the universe around us works. We perceive reality in some form. We are not evolved to see reality perfectly, nor will any life form evolve in such a way. Our brains have many illusions in them, and our brains can be tricked. We have biases for certain ways of thinking, that help us survive, though don't necessarily relate to reality.

Human beings remember bad things more than good things, because bad things are more likely to kill, and thus making having more good experiences less likely or impossible. Dead people don't have new happy memories. The more likely one is to die, the more they should be aware of bad things than good things. This is probably consistent with all life that ever evolved into sentient beings, ever. Survival is the ultimate reality. This "illusion" that distorts much of how humans see reality, and creates an unrealistic pessimism, may be a quality of nearly all sentient life, to a certain degree. The subjective requirements of survival inherently warp, and give rise to our objective reality. We shape and name all things to the very human function and emotional purpose they have, to use as people.

Sentient life though gains a new power over life, and survival. It can create new ways to survive in how it thinks and relates to others, with *culture*. The sort of animals on earth that can evolve into sentient life, are limited. Plants, and ocean coral are very unlikely to become sentient, unless we give the earth another 10-20 billion years of time for life to evolve. Our sun goes out in 4 or 5 billion years, so it won't happen.

Living things with culture will have reached that point probably through a very narrow and limited range of possibilities. Will a species with no mating, and asexual reproduction become sentient, and have culture? Most likely not, because living together with their own kind in close knit groups wasn't essential to its survival. Rapid evolution that creates highly stable increased complexity, capable of adapting to a diverse set of environments, is itself a rare form of evolution. Evolution itself evolves and is made of many different variables.

Culture will evolve in social animals, ones that raise their children, and have reproductive mates. A certain "morality" will limit which sorts of life will evolve into sentient beings and ones that will not. This morality is pure math and probability. Certain conditions of survival are more likely with certain survival paradigms than others. More "moral" paradigms will survive over time to produce a sentient creature. This "morality" is bound by what is possible and not possible, and may have a very narrow range of possible functional moral systems.

I am not saying all humans must have the same moral system. Within our collective humanity will be a distribution of interrelated and connected systems working in an equilibrium or in symbiosis. Morality is not a function of individuals; it is a function of collective groupings. Individuals survive or die, while breeding in a collective gene pool and require for their own individual survival cultural interactions with others. You could describe this as a singular moral/ethical system that is distributed over a certain range of diverse human groupings.

Fairness, is not just something just randomly evolved into our brains. We would never have evolved into sentient beings with culture, without evolving by pure randomness

certain moral and ethical inclinations in our brains. Immoral and unethical animals won't survive as a sentient species. Imagine how long humans would last, if we modified our DNA so our brains don't have certain moral codes hardwired into us. Not all humans have a sense of fairness and a sense of right and wrong, enough however do, for a stable species to evolve. Most people now who don't have such things, are brain damaged in some way. Even sociopaths have certain "ethics and morality", just not as "warm" as we expect in real "humans".

We also kill off humans we don't see as human, and that we can't trust. Sociopaths have to be good at pretending to look like humans, and the ones that can't and then go kill people for fun, don't last long. Though having a certain amount of functioning sociopaths in society is good for its overall survival. There is a morality to them, and they can "want to be" helpful to society. A group of humans will not have homogenous minds and homogenous moral and ethical compasses. There can be a range and variety within a group of humans, and thus make them more or less able to survive.

Whatever the case, there is a "humanity" in us that evolved out of randomness, and it is a very rare condition, that is necessary for all sentient life. There are very limited stable and progressively evolvable evolutionary paradigms that are possible for a sentient species of animal in any universe, on any planet to develop. They will be similar. Morality and ethics can evolve from pure randomness and yet possibly be consistent with a completely alien lifeform on another planet, because those possible arrangements of morals and ethics that work are nearly singular. There can be a universal "humanity" in all alien life, though it would be vastly narrower than we may traditionally consider morality.

At some root level, nearly all space faring species will have similar mathematics, even if the notation systems are completely untranslatable between them, at initial appearance. Binary math, will be something, every one figures out and needs to use at some level, in order to develop their culture and technology. This is about the biological hardware that underlays the cultural inclinations inherent to any sentient species, and not cultural advancements like math. The ability to do math existed in people's minds, before we had math. Math has its roots in the cultural biology of the brain, and in how we render the universe into something usable and meaningful for our survival.

(the dualistic reality of math will be in later chapters. Short version: dualism is just the qualities found in the timeless absolute ground state, of what is possible and not possible. There is only one reality. Absolute nothingness does have some intrinsic qualities arising from probability. Nothing and something, is the only possible duality for the universe. It is just that "nothing" isn't exactly free of certain qualities.)

All sentient life has morality and ethics. No advanced alien civilization, doesn't have advanced moral principles. I would even go as far as to say, all significantly advanced civilizations of aliens across the universe, will have a shared "humanity". Though our communication systems will be so different, that it may be impossible to communicate. Octopuses are the most "alien" intelligent creature on earth, and we can feel a certain intelligence, and humanity in them. I feel the intelligence and feelings of an octopus, when I see them, and they are in reality intelligent. It isn't just projection. How to communicate with a sentient octopus like species? I don't know. I can sense a "humanity" in them and intelligence that is beyond that of fish. It is an empirical truth that I both feel, and can be proved by experiment.

Now I can't project happiness onto "smiling" dolphins, without being an idiot. Their smile is just how they look, and not an expression of emotion. This is just like how "grumpy cat", a cat with a frowning face, appears grumpy and we all know is not an emotional expression. The seeming mental contradiction of these two things is funny however. We enjoy things that feel more human, even if they are not in reality. An octopus is more "human" than pretty flowers. Though pretty flowers might make most people much happier to look at, and thus make the person feel more "human", in themselves, thus giving inanimate things a human identity, in their emotional function and use to us.

TOOLS AND TECHNOLOGY.

Technology becomes an extension of culture. Culture and technology evolve now independent of biological evolution, and for the benefit of biological life. If technology and culture don't work, they don't exist for very long. There is also an intense constraint on what is possible and not possible. Certain technologies must develop also in an evolutionary chain, of increasing evolutionary stability and function, with ever increasing complexity and diversity in those tools. Humans seem to try out every possible cultural arrangement. Very few survived in time. Stability (what is possible to exist) and complexity become forces that evolve culture, as well evolve biology and the laws of the universe. Technology is part of the universe too.

Octopuses may never invent the wheel however, for a very long time, in their cultural history. Pure ocean like species, may never evolve into a sentient species, nor develop advanced technology. Many super useful things we develop require a chain of developments to stack on top of each other. Dolphins for instance developed as land animals similar to dogs, that went back into the ocean, their lack of hands will make

them evolving into a technological species, nearly impossible. Maybe if they develop some sort of super tentacle tongue, they can have "hands". Though in the water, building technology wouldn't really work as well on the land. Burning things with fire doesn't work well under water! Doing chemistry is really hard in water!

To a certain degree, certain physical conditions that life evolves in, will be different than where sentient, culture bearing life, evolve. Life initially evolved in the oceans, and then evolved into a land dwelling entity. Evolution on the land was much faster, and developed vastly more intelligent creatures capable of bearing social connections. The sea doesn't change so much, as the land does.

This doesn't need to be true to prove my point, it is however a wonderful thing for someone to think about and explore. It is a mini thought experiment to work around in, and go research the actual biology of this earth, and how it evolved. By considering the limited possibilities of how life could evolve on other planets, we may be inspired to see driving forces in our own evolution. Evolution is a purely random thing that will most likely produce fundamentally very similar things.

Endoskeletons are more likely for sentient life, than exoskeletons. Carbon is a vastly more likely base than silicon. Tiny differences in functional possibility create vast differences in evolutionary probability. And oxygen, water, CO₂ cycle for "breathing" seems very likely to exist for sentient life. Breathing methane emitted from bacteria's farts, would be a sad life. Life at a basic level could evolve in a whole crazy host of ways, beyond my imagining. Sentient life, will probably be more similar across the universe. Space faring civilizations will be so narrow, as to be nearly impossible to exist, and will probably be remarkably similar. I estimate around one star-faring species per 10-70

galaxies, where that even if we travel to every star in our galaxy and another galaxy, we are unlikely to find even the remnants of a star faring civilization.

More stable, long lasting and complex universes, that are have longer windows of time and more places, hospitable to life, will in general have more technology and more advanced culture over time.

THE GENOCIDAL INSTINCT

Warning this section is unsettling and very disturbing, even to the author. Like the section on slavery, in part 2. These sections took an extreme emotional toll on me to write and attempt to objectively lay out for the reader. I don't want to impose a moral framework on these things.

Idealism over perfecting man, drove modern eugenics and social Darwinism. These are repugnant aspects of human culture, that in order to control better, I push forward this idea. I don't want to advocate genocide or creating a more civilized and better human being. I want to open up a deep introspection to find those genocidal instincts within ourselves, so we can better understand our own humanity and how we came to be. Our very idealism, and warm feelings may have in their genetic primal roots, a drive to kill and slaughter.

Pre-reading, and references

- <http://berkeleysciencereview.com/self-domestication-and-the-evolution-of-human-language/>

- <http://io9.gizmodo.com/7-signs-that-humans-are-domestic-animals-1586580895>
- <https://en.wikipedia.org/wiki/Self-domestication>
- <https://www.wired.com/2012/02/self-domestication/>
- <http://www.sciencedirect.com/science/article/pii/S000334721100546X>
- <http://www.sapiens.org/blog/animalia/human-ancestors-and-domestication/>

IN THE BEGINNING

Cultures don't have to be so nice to evolve into the dominant one. In human beings, there seems to be a cultural or even biological evolution, that developed after we became sentient and was different than the other sentient beings that evolved. The way we evolve as humans, is different than the way animals evolve. Common people's conceptions of evolution relate more to the projecting of human evolution on to the world, and religious people's rejection of scientific evolution is driven by how it doesn't describe humanity. Common notions of evolution are that is that is progressive, and where things can be more evolved and less evolved. It is more like some racist code words to make certain people more or less human than others, based on their genetics, and where more evolved species are more human like.

I hypothesize that the essence of being human is based in our innate drives to commit genocide. Humans evolved to slaughter their competition, and this changed the very nature of how evolution expressed itself on earth. Over a long period of time an instinct to kill for collective purposes, could lie at humanities success as a species. As this instinct evolved, it created the very differences between humans and other humanoid species, and eventually evolve into the very driving force that created civilization. It wouldn't just be a single killing gene that popped up in the gene pool. It would require a

set of genes that evolved over time, and in relationship to evolving cultural practices. It isn't a singular drive, or singular way of thinking. Culture itself shapes genetic evolution. Culture defines in groups and out groups. It creates a form of self-domestication, by favoring certain kinds of social behavior arising out of genetics that are conducive to living in a culture bearing society. Genes shape culture, and culture shapes genes. We domesticated humanity with the ultimate domestication adaptation of using targeted mass genocide. We killed off those that didn't fit in, on a global and not just local scale. Humans are a "global" species, with the power to kill off whole groups of people for a moral purpose.

Over time, this genocidal inclination evolved into a civilization gene. We developed an instinct to define what is human, and not human. "Humanity" is directly born out a history against inhuman people. It could have begun with repugnance and disgust at certain physical appearances in people. A drive to create healthier and viable members, could evolve into a collective capacity to commit genocide on those different than us, and in time evolve into the driving force that created civilization and civilized people. The drive to make humans look more like children, could be such a connected thing. Humans select more childlike appearances and minds for survival as a collective culture bearing species, because it makes them more receptive to learning and fitting in with a larger group. Then later we could evolve a trigger to kill more adult like appearances found in "primitive" looking people. A simple aesthetic preference could shape a great deal of human brain development, and favor child-like minds over more set adult ones. This preference made us more malleable into large groups and thus more collectively powerful.

We as humans kill all sentient life that is not the same as us, we have an extra urge to kill that which is most similar and yet different. Anything that is too close to us as

humans, but not human, we kill. A certain genocidal instinct seems to be possible, and existing in human beings, maybe not all people, but enough to be a driving force to kill off other competing peoples. This may be an evolutionary dead end, and if we really are so genocidal, we may be more likely to destroy ourselves, than a more peaceful humanity that lived in nature, and adapted with multiple species of humanity. However, a peaceful collection of species would never grow in size to create social groupings of people numbering in the billions.

Silly tangent

This could be the real original sin in our creation, that dooms us. It also could have evolved to its peak form, at the exact same time, as the moment original sin appeared in the book of genesis: 6,500 years ago. This instinct itself may have evolved over 40,000 years into what we see now.

What made man able to create civilization itself, was a deeply dark change in the nature of human beings. Civilized man and stone age man, may almost be different species. Adam and Eve came to kill the inhumans, while also bringing the light of civilization and new age to man!

In a group of humans only a certain percentage of people need this inclination, for them to go out and slaughter our competition for the singular position of sentient life on this planet. Or you might say, we have an inclination to kill off "wild" humans. There aren't really any wild humans left, so the desire to kill them, is not felt, and thus this latent instinct not triggered so directly and purely, almost ever, within large percentages of the human population. What we think of as genocide in our gentle world, may not compare anything at all to the sheer butchery of the past. Humans have this interesting quality in

the evolutionary record. Whenever another human like species of animal, came into contact with humans, they quickly died off. There wasn't just one sentient species on earth, at one time. Many existed at once, and we went around the world killing them off, with gorillas buried away in deep dark jungles, and thus escaping a total genocidal massacre.

When we see 3d animation that looks too human, it triggers a revulsion in us. Similarity is cute to us, until it become too human, and yet not human looking. Androids that looked very human, would trigger intense violent death on them. AI brains, that seem too human, will generate intense revulsion, and a desire to smash them. Possibilities, possibilities, possibilities. Technology that becomes too invasive too wise, too knowing, will be smashed to death. We as humans will rule supreme.

- **Technology that does not bow to our vanity, will not exist.**
- **Our humanity is a self-fulfilling birthright to absolute power.**

EVOLVING CIVILIZATION

This possible way of thinking, that exists in people, where we kill by instinct, could be our evolutionary dead end. Better possibilities could have been a species that learned to get along with the other sentient beings of the earth, and create a grand harmonious multispecies society.

Or

This cruelty could lie at the foundation of civilization itself, and we form into a large complex, super stable societies through it. We have to kill off the more animalistic humans in order to have more stability and larger units of organization, which can create more complex levels of organization.

We feel guilty for this emotion inside of us, this imperfection, this savagery. We want to edit it out of us as humans and or deny its very existence. We want a culture that can lift us up beyond the absolute killing and brutality, that made life and civilization possible. We want to imagine a world where we don't kill off distant others, to push forward our own group. Our inner vanity at our own idealistic perfection is born from absolute aggression and brutality towards others.

At the deepest levels our very desire for peace, and our idealism, may itself, be the very heart of an instinct to kill and to commit genocide, the instinct to preserve and grow that which is human inside of us.

NOMADS AND FARMERS:

Civilization is more than just a cultural adaptation, of new technology and new ways to organize human beings. Civilized human beings have different biology than humans who weren't civilized. Humans civilized themselves down different paths of nomads and farmers. Those that domesticated animals and those that domesticated plants, evolved different cultures, and different biology's, to eat their different diets. They also seem to think a bit different, and organize themselves differently. Different levels of aggression are acceptable in different cultures. Certain hormones levels, can have different levels of

survival in different cultures. Nomadic horseman and farmer villagers, can be very different in their values, and they may have innate tendencies towards different ways of thinking and organizing themselves.

As we domestic animals and plants to our needs, we domesticate humans to our needs.

Often, we have created separate classes of humans to enslave and use as livestock to serve the dominant group. In general, we just kill off the too wild of us, the too savage, and those unable to live in large societies. We capture and enslave sub-human groups to serve us, and domesticate them like we do plants and animals. The process of slavery and domestication of humans probably developed before we domesticated plants and animals. To be human is to be domesticated to serve others humans, and function in social groups. As social environments change, the needs of domestication change. It takes biological adaptations to live in large cities of humans. Governments/warlords cull us to serve their needs at particular times. A genocidal instinct orients itself towards collective centralized power, and individual self-preservation. A singular individual would survive better with brainwashed domesticated humans serving them, and could then breed more widely than others. Variations of such social interaction exist in great apes, and the variations of these behaviors developed in us, in ways that led to more rapid social and evolutionary development.

Modern man is a genetic mix of farmers and nomads. Nomads raided and raped their way into the culture and the genetic pool of farmer cultures. All modern cultures evolved from farmers, save a few like the Vikings. Africa didn't have the land and climate for stable farming, and never really evolved more complex stable cultures. Most of Africa's history, where farming wasn't stable, was nomadic animal husbandry, ruled by warlord

societies, of differing levels of stability and complexity, though nothing on the scale of say ancient China, India, Rome, Persia, Egypt, and the such. A civilization can't form to any great size without farming, and the cultural and biological adaptations that come with it.

The poor civilizations of north and south America didn't evolve with such intense infectious diseases, as the rest of the world, and died off from the very first contact, with the much larger and connected lands of Asia, Africa, and Europe. Immunity is a form of evolutionary adaptation for civilized peoples, because infectious diseases grow in denser human communities. Most infectious diseases come from China, through history, where people lived in the largest groupings of people ever in human history, now and through all of Chinas history. China had multiple great rivers, and the largest farmlands possible. Disease is a great defense for intruders into societies where they have no immunity. Disease itself is a weapon against less civilized peoples, and confers great advantages on domesticated humans. The more social people, would also have a greater propensity to survive outbreaks due to being cared for by others, so that the more asocial and violent, would more readily die off.

China evolved a culture to manage such a massive area and population. By nature, only one culture would ever rule this region. It is a very hard region to rule and keep. The sheer size of the area requires that one group of a massive size to rule it all, and Chinese civilization evolved to create a unified massive society capable of occupying such a region. It required a brutal domestication of the people and culture, to install a dominant centralized authority. Surrounding this large centralize region, were more isolatable regions that developed and enriched the main region, or were raiders of it. In countries like Japan with more isolatable regions, there existed more feudalism and distinction between varying lords. It was just too hard to invade and conquer another

part of Japan, with such massive mountains. Greek civilization formed with many varying kinds of people because every single Greek island could be its own country, with its own laws, and that created the basis of western civilization and its core ideas.

Cultural evolution is not just how big you can get, though it is important. The Chinese were one of the most civilized people in history. A culture also needs complexity, and not just authority dominating over it. It needs new ideas, mutations, to grow and evolve. Large unchanging societies will die off and stagnate. China existed in a symbiotic relationship with its nomadic neighbors, who themselves shaped and defined Chinese history, with many Chinese people actually being descendants of these people. "History" has little to do with what really happened. If the Chinese redefined their history and identity to include its nomadic identity, it would redefine cultural myths and self-identity. The Mongols, the Huns, and Turkic people, all are part of the identity and history of what is to be Chinese. They now exist as citizens in China, and yet are not fully integrated into the legacy, culture and history of what it is to be Chinese. The variations of culture that surrounded China, assimilated into the main Chinese culture, and created the very vitality that helped it survive longer than any other civilization in history.

The Irish had to shape an identity from rape, conquest, pirate raiders, enslavement, and imperialism.

Genocide though is part of being human. And it feels deeply at odds with this innate morality and divinity we feel in us as human beings, and seems underlay most all of our relationships. Feelings are real things, and not just something arising from pure animal emotions and desires. We feel guilt, and these feelings exist in "humanity" across all planets. Civilization itself is driven by a different form of evolution than what came before. If it is dead end or an improvement, I don't know. It at one level, creates a soft,

gentle idealist pure human, and yet gives birth to that by slaughtering all wild humans, and domesticating human beings themselves, along with the land, plants and animals. All is bent to serve a singular will for survival of some kind.

Progress, Domestication, Civilization, Genocide, may all be words for the same thing, at different times. Progressives of the early 20th century advocated genocide and eugenics as a moral imperative to make the world a better place, by sterilizing inferior races and people. It was what "science" found to be correct, and morally enlightened people pushed the most violent beliefs. Where we find the most noble and civilized people in history, we find them killing, enslaving, and domesticating human beings at correspondingly large scales. At some level, many of the kindest and most idealistic people harbor such thoughts of dominion and murder. It is just being "human". The more progressive, civilized, or advanced one imagines themselves, the more they probably want to kill people unlike themselves. This is how more developed people win through history, and if they really are better, they can win more readily. A zero-sum game to the death shaped most of human history and development.

THE POLITICS OF LEFT AND RIGHT.

I have a 6 dichotomy system to explain all of politics and human cognition that will not be included in this book. 3 of those can correlate to one's political orientation more precisely. I am just going to discuss one of them, and in a very simple way.

Part of what it is to be human is not to love everyone. To evolve we have to love our own, more than others. At the most basic level, politics can be described along these ideas. We as humans have innate variations in how much we love our own, and how much we love outsiders. Forming a stable group that can survive requires intense culture and close connections to those around us. Deep bonds of family, community, and country, can hold a people together, and allow them to survive through the innate difficulties of life. This instinct is at odds with the overall trend of greater stability *and* greater complexity. Larger complex societies more closely interlinked with high stability, will survive longer and align with the very forces the universe.

"Conservatives"/Right Wing/Capitalism

- They form tight smaller bands of people. This is an excellent way to survive.
 - These groups of people are heartier and stronger.
 - They can fight man for man much better, than large farming like communities.
 - Risk taking, bravery, and dying for others are ingrained ethical patterns.
- Conservatives by nature, don't seem to want to live in large groupings of people.
- They have more intense social bonds to those around them, and want to be left alone by the outside.
- Isolationism and intense internal community and culture, govern how they survive.
- They are better at highly complex forms of thought and interaction. Mathematics come easily to conservative minds.
- They intensely see the humanity in their in-groups, and don't even see the humanity of their enemies.
 - Their empathy is very localized and not universal

- They have strong senses of honor, loyalty and self-sacrifice
 - Honor is the degree to which you follow and respect that which created you, regardless of all else.
 - Honor creates a bias towards traditionalism, as a form of respect.

Progressives/Leftists/liberals/socialists

- Larger groups of weaker people will beat smaller groups of stronger people
- Size wins
- Increased size with increased complexity, allows for more complex cultural institutions and specializations to evolve.
 - There are periods of rapid growth and collapse. With very rare changes to culture that lead to complex evolving stability
 - Size in itself doesn't have its own stabilizers for high complexity
 - Hive mind tendencies
 - They grow big and crash often.
- They like large simple systems, that don't readily work through time all that well.
 - Not good at math, and complex logical systems.
 - They like top down structures and rigid management.
- Liberals by nature are more open to strangers, and find new things more interesting
- They are drawn to novelty, and don't mind uncertainty as much
- They are idealistic and embrace large cities and living closer to large amounts of people.
- They have less cultural and social cohesion. More violent and wild humans are usually isolated, exiled, killed, or imprisoned.
 - You can see this in the ghettos often built in and around left-wing places.

Optional: a philosophical sensitive explanation

Conservatism is obsessed with first causes, and building the logic of how things come to be from their first principles. This in a material sense is a fixation on the initial ground state of the universe, and the values one could build from nothing, or from a universal god that created the whole the universe from nothing. Non-determinism and mathematics oriented.

Progressive liberals are obsessed with the end. Their logic of how things work is constructed from perceiving the final result. This is called teleological thinking. They project their own identity into things to form a whole. The universe is a singular entity with purpose and direction. History evolves towards an end that is fulfilling to human life. In the physics analogy they see the expanding evolving universe, moving towards something. Determinism and teleology.

This may have origins in the beginning of man's development of civilization, in nomads and farmers, or it may not. Conservatism seems to correlate well with nomadic societies and how they adapted to survive, and liberal left-wing correlates well with grand cities of farming civilizations. They are a 2 different cultural species of human beings, that are now intermixed together. People can still overall be born with innate mental preferences for one way of thinking over the other, down to how their very hormones deal with things. Left and Right political thinking are innate features of a human being, that don't just arise from their upbringing. Two very liberal parents in a very liberal city, can have 2 very liberal kids, and one very conservative kid. Sort of like being gay, in a family of straight people, and everyone around you is straight. There is science and basic practical reason, to back this up.

This central division in humanity may also be innate to all civilized sentient life. We have an ability to fine tune our own cultural stability and complexity, based on the actual conditions of the world. The two forces become represented in the very identity of being human, as reflected in the very laws of the universe. Neither is exactly right or wrong. Bigger, more powerful, diverse society, vs a smaller more densely connected group with high stability and trust. Strong culture and society vs strong government. History shows what happens in extreme societies with high government and low culture, and where high culture and low government exist. Both are the stuff of nightmares. Living in a warlord society or a communist/fascist totalitarian society, is at some level cognitive preference, in choosing which hell you think is better to live in. The answer is not actually so clear. There isn't a right answer, but people will think there is one, based on what they prefer, based on their politics.

Conservatives seek more stability, which leads to smaller, more isolated groupings of people that are more intensely connected and related to each other. They value a fundamental internal consistency to reality and self at a fundamental level. The left can handle increasing in size of societies, with diminishing cultural cohesion, and increasing chaos. In that chaos and sheer size exists greater opportunity for change that actually works to come into being. There are more chances for new things to exist in a liberal society, even though most of what it creates will be rubbish. Most of that chaos is death, and societal collapse. Early liberal man would have had massive growth and collapse, in an unending cycle. It was a horrendous way of life, that there was no escape from, where people just died where they were born.

Civilization and large societies in its beginnings was in no way pretty, or clearly better, until the age of enlightenment in western Europe. Farming often had booms and busts.

Most human history is just smaller groups taking the food from weaker groups, with the weaker ones dying off in mass numbers. I would presume far greater happiness in primitive nomad groups than primitive farming societies. In conservatism lays a contradiction, they embrace human nature, but reject its evolving reality. A large society that is ever increasing and mixing, that doesn't have strong conservative stabilizing forces, will collapse in chaos. A society not growing and allowing for variation, will not evolve. Strong cultural institutions and deep loyalties to one's clan and family, create highly stable and strong societies, that don't adapt well, because they lack constant outside contact, that drives change.

America was a grand fusion of these twin pillars of humanity. The Founding fathers were a mix of these different ways of thinking, and they worked together, respected each other, and built the greatest country ever in humanity. The fusionism began in England, and reached its maturity in America. This fusionism was, for me, the foundation of the enlightenment beyond the scientific vision of Newton. Human beings could have easily lived for thousands of more years with no America, and the fundamental changes it brought to the world. The recent technological revolution of the world was driven by American culture, that fused conservatism to liberalism.

The American right is a whole other species from the right-wing cultures of many other countries. Pure strains of it exist, and it is often repressed through American history. Nativism with intense identity politics that becomes nasty is not really the classic conservatism Americans embrace. It is however a real and driving force in the world, that when *repressed* leads to totalitarian and authoritarian societies, with closed borders and closed minds.

This evolution was an essential one. No great civilization in the future, can survive without figuring out how to fundamentally balance and respect these twin forces. They are not just a single binary system, and far deeper levels of complexity exist in them, and in the innate natures of human minds think and come to have reason. Our personalities themselves are innate, and contain fundamental patterns that relate to the very nature of universe. We are born conservative or liberal, and so much more.

HUMAN COGNITIVE DIVERSITY

Being an extravert or introvert, are innate features in a person, with degrees of variation in between, there however is an overall defining innate biological component. Aliens will also have this variation, expressed in some form, in their cognitive capacities. We have to learn how to use all of this variation in human natures, that enrich us. We have ever more complex variations between us, that allow us to specialize and be good at different kinds of thinking, innately. We can organize ourselves, and solve problems, and think differently, as a way to culturally adapt. We have a cognitive diversity spread across us biologically, and a share unifying cultural identify, history, and technology to unite us in a common cause of collective survival.

PART 3: SCIENCE, PHYSICS, AND GOD

PART 3 INTRODUCTION:

This part of the book is Hard. I try to deal with the hardest problem ever considered in the history of humanity, and it may very well be the most important question in all of reality. With the world, I am living in burning, and the moral foundations of society in ruins, this is an attempt to change the entire nature of scientific, philosophical and religious debate, for all time. I am not a theist, an atheist, nor an agnostic. This section is not science, not theology, and barely philosophy. It is just history and logic.

I attempt to crack the underpinnings to reality, and in the realm of science, describe what came before the big bang, and why such a question can be answered in real scientific terms, and how this itself created the conditions of the big bang. I don't use magic, I don't use mysticism. I use logic, hard speculative science, the facts we have, and imaginative critical thinking. This is an attempt at making hypothesis's that can lead to actual scientific experiments and actual new discoveries. The goal is to inspire people and take a fresh look at the problems in physics and how the results of such new study could change the world drastically. If this work leads to experiments that completely disprove it and in the process help lead to a better idea, I will be proud of this work.

SCIENCE IN MY TIME, IMAGINES ITSELF IN A GOLDEN AGE. I SEE IT IN A DARK AGE.

At the heart of science is a person trying to prove their own imagined model of reality to be false. This is for all of science's complexity is its simple genius and profound moral value in changing the world for the better. To make paradigms that change everything, you need to see deep models of possible reality in your own head, and in your sense of self. You really need to believe them to be true, and then go out and create experiments to falsify these models and ideas. You don't make experiments to prove things to be true. You try to disprove it, by making crazy predictions that only your model can predict, if the predictions are wrong, you have a bad model or you did the experiment poorly. Testability, falsifiability and a real human connection to an idea, create science.

We have to get teams of people working together with different mental abilities and using different procedures, to get science working at its very best. The scientific method has broken down in our modern world very badly, even with its mountain of epic achievements. Scientific methodologies are at their heart about dealing with human nature. It is about how to get people to work well together, that don't get along well, that think completely differently, and have valid different points of view. There is an underlying system of rules, procedures, peer review, "laws", and governing bodies, that did a lot of great things in the past, that has failed in our current world. Eventually any formal process will fail because we are human, and there is always more to the story than we are willing to admit. The root of solution is far more about respecting and understanding humanity, than it is in some complex philosophical debate about the meaning of truth. We will all have a different meaning for the word "truth", and not fully respect differing views on the subject. Good science challenges one's own model of reality and one's own sense of self. If people aren't actively trying to disprove

themselves and working together with people different than them, the system will fail. The system doesn't reward people for proving themselves utterly wrong. "The 2024 Noble Prize for Honesty goes to.... for completely disproving his life's work that had become mainstream belief."

Science doesn't have to be impersonal and cold. Our feelings create what we study. The more we understand our feelings, the more we can push into new frontiers of science and understanding. The more we can bring our feelings and selves into it, the more we can attract people to do science and to think more. We can attract and bring in truly smart people, that will be able to imagine things in new ways, and give us new insights into how things work. Women don't have to start thinking like men to be useful, and their own unique humanity and differences can inspire new insights into the nature of reality. Science must reclaim its heart, and that pursuit of "truth" will always have man and our future inside it.

Chapter 9: The Philosophy of Science

The Nature of Science

Science demystifies magic out of the world. This demystification and ripping the magic back is upsetting. The closer it gets to the origins of life, the universe, and our very identity the more shocking it becomes. Love is mostly a bunch of very different biological functions that get labeled love: chemicals, insanity, addiction, reflexes. Love may exist and in the realm of science it is mostly just collective delusions needing to be peeled back to show its primitive mechanical nature. The quest for a God with meaning in the universe, can lead people to be lazy in their scientific explanations, and the desire for there to be no god, can also blind science. Atheist Scientists, are also driven by many un-reflected upon desires and impulses that make them prone to believe in absurd things, like say communism, transhumanism, an earth goddess, eugenics, or the meaningless of marriage. Theists refuse to challenge things, and refuse to dig deeper into the origins of things when they challenge the perceived sacredness of things.

Science pushes back the veil of darkness and superstition, it takes the magic out of the universe. Disease is not caused by ghosts, demons, or bad energy. Trees do not have feelings. Science pushed magic out of nature. When some scientist tells people that love doesn't exist, and is just chemicals, this can make people really mad, and reject science itself. "Real love" in people may exist, however a great deal of what we call love is just chemicals, programming, and instincts. It is just mechanical operations that create an emotional illusion in our brains that is created by neurochemicals and cognitive instincts.

Much of what we feel is just an illusion. Our sense of reality is an illusion. Much of our very sense of identity and self, isn't even real. Science can help us see and understand that. It will one day really understand these things, and this will help people live better lives. When we can accept and understand the mechanical aspects to how we experience love, we can be in more control of these instincts and these addicting neurochemicals. If there is real love, we will only be able to enhance it and bring it into the world fully, when we are able to strip away the mechanical and material reality that makes up most of it. Learning to break the brains illusions and addictions, that keep us alive, will free us so we can live. Trying to disprove your own idea of love, has the potential of revealing the truer nature of that love inside yourself. Honesty is a continual inner process of challenging one's assumptions and facing the reality of who you are as a person, and it is not about saying what you believe to be true.

Demystifying Love: A Failed Act of Science:

Science has to sell the humanity in taking away the very things we think are our own humanity. Most adults can understand that romantic first love, is more like a drug addiction, with immature emotional development, making sweet beautiful lies seem like they are real. Some people live lives chasing after that feeling of "real love", they had one time, and yet never again feel that "truth" in falling in love. We can use science to, maybe one day, treat and identify these "love addictions". No one will ever love a person, the way they imagine their mothers loved them. Freud's belief in an Oedipus complex was not good science, though it was interesting thought experiment for those that might have one, to unravel their attachments to their mother from the reality of real relationships. Science can help push back against unhealthy forms of thinking, we often associate with love, and help people live happier lives, in more fulfilling relationships.

Some people may think marriage is a great lie and in the future, we will evolve beyond marriage. It is more likely we will develop a better understanding of how to have more stable long-term relationships, that deal with the animalistic instincts and contradictory feeling within us. Bad marriages exist, and maybe most of them are rather horrible, this doesn't mean marriage and love are things we need to evolve past as we go into the future. A future with no love, no marriage, doesn't seem very human. We can use technology and science to find the most human ways to connect us and provide us more rewarding lives. It can help support long standing institutions, rather than create new social institutions from nothing. Science can't model all the elements and variables that form culture and society from experiments, because they can never know all the variables to test for nor be able to isolate them all.

A Philosophy of Science

Hard science is driven by one thing, the study of cause and effect. We see a result and we look for the cause. Primitive thinking projects magical like feelings into things, to explain their causality. History, time, the universe, the stars, nature, have all been things embodied with deep magical thinking, through most of human existence. Science challenged all our presumptions and forced us to make experiments and find out what really caused something to happen, and challenge our own deeply held emotionally driven models of reality. What we imagine causes things, is often wrong. What we think are connected together often have no real connection. What we think have no connection, are deeply connected to and caused by the same thing. This is the challenge of science. Scientific models are functional and always evolving, they are not truth, or some certain definitive thing. We can always find deeper causes to things, and can always make more useful models to use to better human life.

A scientific process can begin like this, you see that babies cry and when you feed them, they stop crying. So, you make a hypothesis that babies cry because they are hungry. You then make an experiment to test this. Now as you can obviously know, the experiment will disprove the hypothesis, and the hypothesis will need to be modified. Babies cry when they want something. And you can try to get more data on what babies want, and try to design an experiment. You keep doing this over and over, until you have lots of data, lots of failed experiments. You may reach a point where you study all infant emotion and how babies, think and learn. You can scan their brains, and watch them perform tasks like rats in a maze. The model you are making will slowly improve over time, and yet may never really work. Studying physical things like rocks is much easier. You can hypothesize that big rocks will fall as fast as small rocks, and make an experiment. We all know those kinds of experiments. The scientific method is an evolving and complex thing, with deep implications about how we can learn about the world, from how we implement and develop it. Idealized perfect science, doesn't exist, because this a human thing, serving human purposes.

Traditional science does experiments in boxes. It isolates and analyzes variables to find what caused what. The scientist is like god overlooking the experiment. They live in separate realities. The idea that you can step back and observe things as an outside observer and determine what is true, is a common misconception of the world. You can never fully step back and observe something and see what caused what. Science is good at studying things it can put in boxes, cut off from the rest of the world, and look down into with a god's eye view. It is like being up on a mountain top and looking down and seeing things happen, and knowing objectively what caused what. You are far enough away to not influence things, and can see how things really happen. In the real universe, there is no gods eye view, there is no place you can step back to and get an objective point of view on what caused what. You can't even know the sequence of events objectively, if you stand back far enough and view them.

Newtonian thinking was modern thinking. He invented the modern era and new kind of thinking. You really could step back and look at the world and know how it works. A clockwork universe, governed by rules that people could find and know, was a defining belief of an age. They believed they could find the definitive rules and causes for everything like a machine, and then be able to control everything in the world like a machine. As science became a new god, it created totalitarian governments believing they were the true heirs of science and understood the roots causes of all humanity. Post-modernism and the failures of the past, smashed that ordered, structured, and failing world. We didn't want to be cogs in a vast machine, with no souls, and only programed and conditioned by what sensory inputs we had been subjected to by authorities. Newtonian inspired enlightenment thinking lead to totalitarianism, and a harsh mechanical world, where we believed we could control the exact future and create a perfect world. The industrial revolution can be seen as a spiritual process of turning men into cogs in a vast mechanical machine, and the politics that rejected that was more fueled by peoples discontent than any ideology. Magic and god died in the 19th century. We found new ways of dreaming of humans as Gods capable of controlling time.

Einstein

Einstein smashed the old paradigm, and made a new one that defined the 20th century. His theories seem so abstract as to be magic. In reality, they are just simple and direct logic applied to reality and working out the consequences of that logic, no matter where they may lead. You don't often associate with relativity, in the casual sense, extreme dedication to logic, reason, and empirical reality. He found a deeper way to explain the fundamental roots of causality in our universe, and this upended an entire structured and ordered sense of what was real and solid to the world. If one sees the world like

Newton, it would be really hard to fully accept modern physics. Not even Einstein himself could accept it all, because his mind was born out of an older age of thinking. He though followed his own inner vision and made it work out into hard science. He wanted to find those ordered and structured rules that explained the universe, no matter how obtuse or against common sense they may have felt.

It is important to understand his work and not avoid its fundamental ideas. Relativity isn't really about "relativity". It is about Locality. If we had called it "General Locality", and said "everything is local", I think there may have been some very different results in how people integrated his ideas into popular culture. Einstein's achievement in life was redefining causality. He made new mathematics to describe causal relationships in physics by creating a whole new way to conceive of how space and time work together. It wasn't some random relativity, with no truth, no center. The great tragedy of the 20th century was the poor morally philosophy that people developed out their crude and poor understandings of relativity. "Everything being relative", moral relativity, and where people no longer believed nor advocated in any sort of truth, eroded all of society, and brought about conflict and societal collapse. When the foundational paradigm of an age was disproved, it unhinged most people from reality and science.

Causality and locality are important ideas to learn from Einstein. Our schools barely teach his root ideas; they don't even approach them. Yet they are the most successful modeling of reality in the history of mankind. People are afraid of looking stupid by stepping into totally new ways of thinking and doing things. Even physicists often don't really understand the basics of it. The math is hard; the visualizations are easy. Elementary school children could be taught these things, along with quantum physics. It would be fun, and not really that hard.

Science changed after Einstein, and not all for the better. Einstein flipped the scientific process on its head, and changed the very philosophical underpinnings that science had been using up until that point. Einstein used experiments to prove his theory, he didn't use experiments to come up with his theory. He achieved what he did, by not following the traditional process, not studying in the school's scientists did, and not following the things they did. He was a free thinker, and used his imagination freely. Ironically, we crush this out of people in our education systems, and in the study of physics. Being able to let go of the past and build a new model of what caused what, takes courage.

In some ways, he was a philosopher first and a scientist second. What he did lies outside the realm of empirical observation and experiments. The post-modern age was born in Einstein's work, even if he wanted nothing to do with it. He inspired something and gave shape to inner feelings discontented with a falsely ordered Newtonian modern world. He is more a celebrity and a symbol than a human being at times. The world wanted to burn, and they did a fine job of it, in the years that would follow. He even gave the match to make the biggest "flame" of all that could burn cities to the ground in a second. With the power to be smart that he gave us, he also helped unleash some of our greatest stupidities.

Newton and The Ideal Observer

Causality became something different in Einstein than with Newton. The scientific method and how the world was seen up until Einstein, was Newtonian. It was a mechanical universe. Causality in a mechanical universe is very simple. You just observe what did what, and make experiments to ensure that something caused something else.

You can't assume that watching something hit something else, caused it to move. You need to experiment, isolate all the variables and find out the truth. Everything can be analyzed, measured, and determined, with enough study and time. Science had a good method of working out how to deal with finding out how things work. Something is scientific if it can be tested. Testing is where we can isolate each variable and test variables independently, to find which one did what. Our assumptions are often wrong. If we can step back as much as possible from an experiment, we can achieve nearly perfect results. True objectivity in science was believed to be possible and that there is an absolute universal truth, that we can know, through correct observation.

The outside observer has this "gods eye view" of things, and can make objective analysis. In the case of Newtonian relativity, you could be on a moving train and throw a ball to someone, if you observed the motion of this ball moving from the perspective of the train it would be very different than its motion relative to the train track. From the prospective inside an idealized smoothly moving fast, but non-accelerating train, you would have no idea if you moving or not, so the balls movement to you would seem true. A distant outside observer could look down, and see all the different ways it could be observed. Newtonian Relativity was showing a different way to see causality. Your frame of reference would define your perspective of causality, and only when you could step out of the system, could you see the real causes of things. Inertia made more sense mathematically, though even in its simplest form, is still hard for most people to understand. He gave mathematical tools so a person could transform their perspective of something into a truer one. The ball is at rest, relative the train, and moving relative to a fixed position on the earth.

You can basically understand the contrast of modernism vs post modernism, with just this one idea, and its fall. A mechanical universe was a beautiful lie, that worked so well, for so many things. The gods eye can reveal so many truths, for all the things you can

put in a box or make into a simple machine. It can create the wonders of an industrial age. You can at a personal and emotional level feel how such thinking would constrict and choke people, who didn't fit into that mechanical model. There is something missing. You need to really step back and think about this box, and who is doing the observing. There is always an observer, there is always a person inside that experiment. The "box" is never really solid and clearly defined.

The most disturbing things from the conclusion of Einstein's work, was that there was no clear causality and no clear objective point of view. There was no stepping outside the box and getting a god's eye view of things. There are many ways to understand this, and come to this conclusion in physics, in the study of the small, the very big, and even in mathematics itself. We can't ever truly take a god's eye view of the world and know clearly how it works. Einstein wasn't wholly stepping outside of Newton's classic mechanical world, he however opened a door to undermine the conception that things can be truly objectively observed, and used a different way to do science, that didn't wholly rely on pure empirical experiments and observation to develop a theory. There is a kind of magic in his ideas, and to understand his view of time, takes some serious leaps of thinking and imagination that most people have not. General relativity doesn't remove objective time, it merely makes objective observing it nearly impossible, since we can't go observing the universe from 20 random points spread through the universe and confirm what we see. The speed of light changes things. Causality itself gained a speed, and became bound in time, in a defined way, where it hadn't been before.

It isn't obvious that the world can be described with math, and actually goes against any basic common sense. Connecting all the dots in numbers into abstract theories, takes very specialized, trained, and focused minds. If you could lay out basic math, that could explain the underlying order of the world, that math would almost be magical, and seem to come from a god or another universe. Making solid models of the universe to describe reality takes more than just a traditional procedural approach to science.

Newton really ushered in the era of truly connecting science and math together. The great astronomers before him had started this process. He really brought down into the earth and made it a full viable movement. Physics and math, became a bound and inseparable relationship. Though beneath this apparent super objectivity lies a far more human story. Math is far more human and nearly magical, than people like to imagine. Most of math that we have now is based on magic. We don't really even know what numbers are other than that it is a notation system that really works well.

Traditional science and math work together like this,

1. if we do enough observations and experiments, we start to get a lot of numbers, that seem to really connect and relate at a deeper level than just causality.
2. We can start to make basic mathematical models that can explain how things work.

People that are good at math traditionally are not the types to go out and about exploring the world. Scientists and mathematicians are not of the same species of intellectual. The process of making good math from science, takes very different kinds of

thinking than the kind used to go do science. It becomes a sort of chicken and egg argument. Science is grounding and more connected to reality, than the math. More people can do good science than good math. Math creates rarer super freaks. It creates supermen. A person obsessed with the nature of reason and the inner processes of thought, tend to end up with very strong minds or insane. The fact that we can describe physics with math, seems only now obvious. If you study math your whole life and deeply think about how to explain things with math, then you will be biased to believe in mathematical rules more than reality itself. There are scientists that will say their model is true, even when it doesn't match the world at all. This dedication to mental delusions can create religious zealots, masterpieces of philosophy and even scientific theories that change the world. It can also just lead to being an idiot with a lot of brain power, and no real connections to the world.

This is summed up in two different types of logic. Science used a new type of logic called induction, to ignite the scientific revolution. The old mathematical form of reasoning, deduction, had been around for thousands of years. Math had developed through the eons, while science never really developed in any sort of steady manner ever, until the 17th century. Deduction and induction use logic in completely opposite directions. Over centuries they have learned how to get along, and the separations are not so obvious in our modern world. In the 17th - 19th century, they were far more separated and arguing about truth. Math people are better at deductive logic, and science people are better at inductive logic. Rare people like Newton were masters of both. He could do all the scientific observations, and make a mathematical model. Getting different kinds of thinking and different processes working together well, lies at the heart of how science has improved. It isn't just about logic and the meaning of truth. It is different groups of smart people learning to get along and agree on a shared process, to create collective results, that made science evolve and be so awesome. At a basic level, it was like two religions arguing over the nature of god, and it is super ridiculous. "God comes from

above and orders the world”, “god comes from inside, and we can know his will through reflection and reason”. Great theological battles evolved into scientific battles, and the underlying issues and feelings were still the same. The mathematician mind had to learn to connect their ideas to the world directly, and the scientific mind had to learn to describe their observations in math. They both then could begin to work together, with a shared language of Science.

Now all people are naturally from birth a “rationalist” or an “empiricist”. It is just the direction your sense of logic, order and identity move towards. One moves inwards and the other outwards, and both are still logic and do mostly the same things. It is something ingrained in the very cognitive functions of how your brain works and thinks. This is not some abstract impersonal thing. Different people in the sciences fundamentally think differently and have massive clashes of differing ways of ordering reality. These arguments are deeply emotional. Most people have inclinations to be attracted to different sorts of methods of doing things, over others. They will be more comfortable sorting things into different categories than other ways of sorting things. There isn’t exactly a wrong or right way, there is just the reality that people come into the sciences with different inclinations towards different methodologies. One doesn’t merely adopt a philosophical framework, one is born with one you are more inclined to adopt; then you go out and fill it up with things that compliment or clash with it, and try to work it all out to be consistent with your own self and mind.

These two schools of thought Empiricisms and Rationalism are awesome, and both if left alone to themselves, lead to some pretty insane conclusions. Humanity needs both to work together, and they do. I don’t want to get into this nutty and insane conflict. It is large obnoxious egos attacking each other for centuries. They used two different kinds of logic, induction and deduction. Empiricism won for a long time, and ordered the entire

modern world into boxes and manageable systems. It defines economics and government to this day. The 20th century is more a story of rationalisms return. Rationalism is really good with computers, and Newton's greatest rivals work is almost like a guidebook to computers. Rationalism from its first great masters had computers on their wacky heads centuries before their invention. Descartes had robots, and Leibniz actually talked about thinking machines. Newton ran the treasury of England, and spent most of his life trying to make gold.

Empiricism

A basic way to understand this is that empiricists love to go explore the world and look at things. They can pick up rocks and they are actually fascinated by different kinds of rocks, and not just because one is pretty or not. They are curious about the world. Empiricism at its most basic level is about looking at the world and putting different things in different boxes. They believe that what we see and feel with our senses is what is most true. The logic of putting things in boxes is defined by things intentions and purpose. There is a softness contained within the logical structure that favors things that are useful to people, and in doing practical things. Boxes just have to be functional and be useful, rather than perfect.

For an empiricist, identity is made from your life experiences and the things their senses have felt. Your identity is created out of the environment you live in, and it will also create an attachment to that arbitrary reality, which is you. Your very sense of self is created from your connection and experience of reality. The only thing you can ever know, is what your senses feel. Your mind organizes those senses into a mental mapping of reality, and this becomes you. The logic of empiricism is to apply logic to this way of being attached to your senses as reality. Empiricism is not just some cold form of

logic being imposed on the sensory reality we live in. It arises out of deeper softer more complex emotions of self, intentionality and compassion. You can find these sorts of feelings in eastern thinking. The underlying thinking of empiricism is very similar to eastern spirituality. It is call phenomenology or in some ways teleology, in formal philosophy.

It then develops around it a more externalized logic and structure. The empiricist uses a kind of logic that organizes the world that we see. An objective logical order on the world manifests out from their own emotions and feelings. The literal making of a city into grids with square buildings, and straight roads, is a direct manifestation of this logic. You can see that this kind of logic is very different than a rationalist mathematician. It wants to know what is what, and where things will be in the future. It wants to make a predictable and safe world. The logic of kings and leaders is made into science itself, so that we can rule not only people, we can rule the very reality of the universe. This begins with a process of reasoning, and working it out slowly through time. Inner warm fuzzies can manifest themselves into an ordered world that protects us as individuals and organizes us together as a unified group. We create a logical and structured future through empirical logic, with clear chains of command and order.

The rules that one develops to define what a box is, is most of the knowledge of science. We classify all living things in a massive hierarchy of rules. You can't be a mammal if you lay eggs. A mammal is defined by a set of rules. Kangaroos look like mammals, but don't fit all the rules, so have to be put in a new box. Science is about rules you impose on nature: boxes and categories. You write down what you see, record numbers, and collect as much objective data as you can. These boxes are not real, they are arbitrary categories that we make, based on how we feel, and what feels like a group, and not a group. This feeling is organized into a set of rules. The actual rules can be changed, if

our sense of purpose and identity in those categories change. We could base those categories on entirely different feelings, and still have just as solid a system. We could organize life around their degrees of domestication and usefulness to us. Their relationship and connection to humanity would shape a classification system.

The world around us is what is most real. All knowledge comes from experience and the world we see around us. Science is naturally very similar to government and bureaucracy. The rules and laws are arbitrary. We judge the quality of a government by how effective it is at controlling things. We judge the boxes we use to organize things, by how useful they are to us as human beings. Words that we use to label things, are based on human emotion and feelings. An object has a word attached to it based on its usefulness to us. A planet with no life on it, barely has anything to label with words, unless it something we can modify to create something with that is useful. Iron ore makes iron, and iron is what we make things out of that are very strong and useful to us. If you shove a pen into an apple, it is just an apple with a pen shoved into it, and not a new thing, with a new word, because that thing in of itself has no useful function, unless we want to make apple-pens into a concept to describe categorical silliness. Science has to label everything, even if its relationship to humanity is distant, and there is no clear usefulness in it. Empiricisms logic can push language, labels, and boxes onto all of reality, that is even outside of what we can readily feel and connect to. It is the process of ordering reality that we can't control yet, so we can make it useful for human use and feelings, eventually.

Empiricism doesn't lead to truth. It leads to effective systems for managing reality. If it works 95% of the time, that can be good enough to call "true". If we can know when it will fail 5% of the time clearly, then we can simply not use that system for those 5% of the time situations. Scientific laws are valid only for certain things, and can't readily be

expanded to everything. You can't use Newtonian mechanics to explain the universe. This way of thinking focuses on what it can do, and tries to slowly expand its power to include more and more things. Newtonian principles were slowly applied to more and more things, that it could work on, and it didn't always succeed. It couldn't explain race, good government, mercuries orbit, particle wave duality, ether, religion, how women think, and so on. The world itself was transformed by this expansion of power. We could use basic rules and ideas and apply them into more and more aspects of life. We could harness power, and use it to control the very world we live in. It also reaches a limit, and then we need to make math and theories for that other "5%".

We base scientific laws as being laws based on their usefulness, not their truth. This applies to things like quantum mechanics which are pure empiricism, and have little correlation to reality and truth, these laws however do a great job calculating stuff in a narrow range of things. It isn't that laws are just useful; it is that they are easier to do the math with, and thus more useable and teachable. Complex math, that is hard, and more accurate, is less useful than approximate math that is easier. Any sort of difficult or hard math, to an empiricist, unless absolutely necessary will not be favored over easier math, that works good enough. General relativity is more "true", and it has horribly difficult math to use, and thus more empirically minded people rejected it based on utility and personal feelings. This is completely rational and appropriate.

When we tried to apply Newtonian thinking into people, this leads to management theories, finance, and economics. Human beings became parts in massive theoretical machines. We studied how to manage people like things, and integrate them into the vast machinery of the modern world that science had built. Industrial and office workers toiling away in repetitive jobs, became the very image of industry and progress. We developed entire schools of thought where could manage society towards a better

future. Government and science together, seemed poised to solve all the problems of life. All those emotions people were feeling, shaped what was to come. The wars, the conflicts and deaths on scales never seen in world history would frame the greatest age in human advancement ever seen. All that death and wonder was born from power, science, and a quest to order the world. We still think with Newtonian like ideas in economics, politics, government, finance, and it still leads to disasters. People think they can understand the clockwork machinery and manage it, with simple and easy to use math. Science wedded to government is always bound up in imposing an ordered structure onto reality that makes it more usable to humanity as a collective entity. Science and politics, bound together, is idealized totalitarianism. "Let's save the world with....Science!".

Rationalism

Math is a bit different than science. It is inwardly turned, and much more personal to the person, that develops and uses it. A mathematician is a very weird sort of man. Nearly all mathematicians are men. Great women in math are mostly all in computer science, and there are some amazing ones in that field. Science is much more appealing to women, than math, historically. Math requires a level of personal sacrifice, that few people can make. You have to live inside your head, and disconnect from the world, and at some level be completely oblivious to people and life. I don't suggest becoming a theoretical mathematician. They take something they see, and apply logic to it. They take things to extreme logical ends, and go as far as they possibly can, down into the tiniest detail. They can argue over the grammar of a sentence for 3 years. They care about what things mean, and how you express them. They are marveled by the inner power of the mind. "I think therefore I am".

Science had to use induction, and by freeing itself from deduction, did the world a great service. It required people observe the world, and collect data, first! Too much of history is philosophers thinking they have discovered an eternal truth, and revealing it to the world. Rationalism descends into religion and mystical thinking readily. They believe in things that can't be seen. They develop elaborate models of alternate realities that can never be observed. Science liberated people from this insanity. It made things solid, real and empirical. It made it possible for ideas to more directly rule the world. Rationalists make religions, empiricists make governments. The rationalist's mind digs into the origins of everything and into the very identity of creation, while the empiricist tries to order and create a future.

The feeling of being a rationalist is rather mystical. It is not just following logical thoughts to their natural conclusions. On the outside a rationalist uses a form of logic that exposes interrelated truths contained in the things we know. It unpacks things and takes them to their logical ends. Logic becomes a chain of connections, and the rationalist searches for chains within things that can link back towards deeper truths. Most of the process produces junk thinking. Trains of thought that lead to nowhere have to be abandoned and new paths taken up. Digging into reality in search of its underlying logic and rules takes time and most of the work can't be shared with others. Tight and true connections have to be found, and not just generalities. They don't want things to be just functional and useful, they want absolute concepts that work all the time. They dig backwards to root causes, beyond what can even be observed. They really think about the beginning, the very first thing. They want to just be able to unravel life with a few hacks, and find root ideas that can create all the things we see around us.

The feeling of finding something beautiful is exhilarating. It is like looking for diamonds in your own mind. In the darkness, there are moments of sharp and clear insight that

make it possible to see new things. Digging into one's own thinking to find the root of who you are and the meaning of life, are fulfilling things to a rationalist. They enjoy their own thoughts more than others thoughts. They love to solve puzzles and explore the mysteries of things beyond what we can touch. Finding out how to play the world as a grand game, can also be a great pleasure. To really understand deduction and rationalism, you need to understand its emotional reality. This is not a warm and fuzzy way of thinking at any level. Pure rationalist minds don't like sentimental closeness and warmth from people. They can be very social and deeply connected to people though. They are in many ways more social than empiricists. They treasure their personal relationships far more, and are deeply influenced by those around them. Nearly all rationalists in history have been very devout religious people, above and beyond the standards of their own time. Math and sainthood, have a bit of a relationship.

Rationalism has the power to create knowledge from nothing. You take a set of basic things and begin playing around with them. If get the right set of things to play with, one can begin to see a puzzle to find how they connect and interrelate. The rationalist mind loves puzzles, and in puzzles they can find simple connections and sequences between things. When you take that set of things, you can logically explore what the causal relationships are between things. You can by just using logic deduce certain relationships must be true and not others. This can produce amazing results and also delude people into thinking this is how all truth is found. Deductions can also just be guesses, that a person then works out and builds complex systems around. If the system ends up working well, based on their basic deductions, they can be very happy. They often don't want to test their systems in the world, nor share their ideas, unless they are certain they work. It is embarrassing for their minds to be exposed and to have logic errors in them.

Modern Science and The Old

Science respects what has worked before. If there is a new theory, science has a rule, Occam's razor, to respect older theories more than newer theories. We often learn this in a different way in school, we are told to take the simplest theory that works, and not more complex ones. "Simpler" means respecting older models, and building on those. This inductive process doesn't care about truth. It just wants something that works, and it is much simpler to just make small modifications and evolve a system, than overthrow it with a new one. If it works good enough, then it slowly becomes a scientific law. No scientific law works all the time, just enough, that it has a practical usefulness for doing day to day things. This creates a bias towards more grounded and real thinking, and that is a good thing, until it isn't. Science must always begin from a traditional method and work up a large collection of models and data. It will always reach a point that it can't go past, where science stalls, and can't find a way out. This is when you need to invert the whole method. You need to make a theory, predict data, design an experiment to get the data you want, then develop new questions to ask about the world, and remake all the observed data of history.

When you need to create a completely new paradigm, it is very hard to smash the old one and build another. Science loves to gradually build on the discoveries from before. It is a slow process, that is not about finding the truth. It is a process of testing things, that needs to be humble, gradual, and very skeptical of individual human reason. Empirical science is a very collective and process oriented way of doing things. This way of doing things creates gradual and steady growth, until it hits something, that its path simply can't understand, and can't process. All systems fail. Empirical science by its very nature will always fail at critical points, and never be able to make complete models, or fully function ones. The scientific method itself is a process that can't always work, due

to its need to make a system that can order, predict and control the world. At a certain point things, just don't fit into ordered logical boxes perfectly. This is because causation itself within induction is probabilistic and thus all formal systems must be defined by probability and not "truths". Empiricism predicts the future, and the future is not a set and determined thing governed by clockwork rules.

Flipping the scientific method on its head seems really cool, it however makes for some insane science. People become free to just imagine whatever they want, devoid of any actual data or observation. Models become reality more than data. Computer made models that say something is true, become truth, even when the models don't work. Math can be used to self-rationalize whatever a person wants to believe, and when put into a computer can make them seem devoid of personal bias. Math is just a tool, and good math, doesn't prove a good idea. All good ideas have good math, not all good math has good ideas. If great math doesn't back up experiments that come from flipping the process, then it will certainly fail.

New paradigms are made by rare individuals. Science doesn't like overthrowing its Gods. Many scientists still follow Newton, and just think of Einstein as a fringe thing for exotic stuff. Other scientists follow Einstein like a god, and justify everything they do, based on his thinking. Even when they say he was wrong on certain things, due to being a mortal man, they worship an overall idea and approach to science. Einstein's idol was Michael Faraday and he was the greatest scientific mind of the 19th century, whose vision of physics laid the ground work for Einstein's great works. Faraday did this nutty thing where he built experiments, to generate the data he wanted them to generate. He first came up with a theory, then the data it would create, then an experiment to generate that data. When the experiment didn't give the data he wanted, he changed the experiment until it did. He approached science like a crack-pot, and created a whole new

paradigm to see science from. The scientific method done in reverse can create paradigm shifts. The focus of science shifts from creating great experiments that can be deeply tested, to creating outlandish models that experiments can readily falsify.

Modern science at that time, if it had steadily followed Newtonian science, would never have made the achievements Faraday did. He had a strong inner vision and belief, and he manifested it into reality, and proved it through experiments that actually worked. Though this methodology doesn't work for most everyday science. He also couldn't create the full mathematical description of his work, that Maxwell created for him. History seems to remember Maxwell more than the crackpot inventor of a new paradigm of thinking.

A Different Method of Science Explained.

When science reaches places where it can't put things in a box, then it needs to use a different scientific method to explore things. This is why social sciences and psychology rarely produce any useful or real science. You can't really study people in a box. We can study babies and little kids in a box, because they are mentally not connected to much of the world yet. We can study how human minds work when we strap that brain down into an actual "box" that scans their brain patterns as they think. Cognitive neuroscience can do good science, because it can isolate the brains functions to some degree. The problem with most all social sciences is that the scientists will always be in the box with those they observe, and we can't really cut people out from the world in a box, and not expect their behavior to change.

You have to make the person in the box a more consistent and knowable thing, if we are inside the box with the universe. Rationalism is a good way to come to conclusions when you are in the box with the universe. The rationalist begins by believing that your mind and you, are the only certain thing. So, it is very good at cleaning up this mind to do amazing things. Rationalists get good at math, and having clear consistent thinking internally. They dig into themselves to find the root causality of their own thinking. When we do science, we need to dig into our own reasons that we are doing science and the real goals we have. When we do social sciences, people often hide their real thinking and pretend to be doing objective science in a box. Current social sciences, in my age, are deeply connected to government power and promoting totalitarian government systems to control and manage people. Since they feed the vanity of the state, they get generous funding, with little to no valid results.

Rationalism can do something empiricism can't and at times it is very important for its mysticism to be let out. Rationalism requires a bit of magic to work. When Einstein as a child realized that light couldn't double in speed, it wasn't based on empirical observation. He just felt it to be true as a kind of basic logic. This deductive insight, he would then later work out to show how light really worked. It was that feeling deep inside him, that he worked and explained the nature of universe with, and not years of focused study of facts and figures on planets and stars. If we look at the universe and its basic properties we have to consider what caused what, and the fundamental levels of reality. With just thought experiments fundamental features of the universe can be guessed to exist, and then actually found. With just a few pieces of information, a rationalist mind can dig into the logical structure of things and deduce truths. A modern rationalist though has to take their idea and go test it and make sure it can be tested. If it can't be tested, then the idea is nonsense, no matter how beautiful the idea. They have to have the courage to show their beautiful ideas to others, and do their best to prove them wrong.

A SIDE NOTE: RATIONALIST MATERIALISM

There is a modern school of rationalist thinking, called Rationalist Materialism I want to comment on. I don't endorse it wholly, nor completely reject it. They are fundamentalist atheists. It is a confusing school of influential thinking that should be explained. This explanation can help illuminate some very deep ideas about different ways to approach science. Since I am by nature a rationalist, I do care about the nuances. The differing schools of thought that a rationalist can hold don't all get along. If you believe A caused B or B caused A, and A and B are important things, then 2 schools of thinking, would have deeply divergent views. At the heart of rationalism is dualism and Plato's allegory of the cave. This story covers pretty much all there is to know about rationalism. Plato's great classic "The Republic" is the story of what one should do, if you do leave the cave and come inside it. This all may seem very abstract and not relevant to one's immediate life. There are people living their lives by these ideas, and some of them can have considerable influence. If we take Plato's cave and see this as literally real, and not a metaphor, this can create 2 very different schools of thought. In a basic rationalist way, we imagine that truth and reality are outside the cave.

In a modern rationalist materialist view, we would describe that reality as what is real. Science then becomes the real-life process of revealing truth into reality and stripping back human perceptions of shadows. The shadows are our human limitations and cognitive bias that we evolved to perceive reality, and with deeper study and science we can help peel back these mental delusions and more closely arrive at truth. We project by human nature magic into the shadows in the wall to explain how they form and exist. Rationalists by nature are magical thinkers, that need to solve the problem of this kind of thinking and minimize it. Science can be one way to minimize it, by creating a system

that falsifies their magical models and ideas. Rationalist scientific processes are driven by a Karl Popper style method. They can go out of their way to disprove their most cherished ideas, as a way to push forward science. You need to keep alive the magic and vitality in Rationalism and temper it with deep internal skepticism and fearlessness in exposing themselves as being wrong. Rationalists are very proud of their own thinking and exposing it to deep ridicule and the highest chance of being shown to be illogical, takes a kind of moral character.

Science steps outside the cave, and sees reality and brings those truths back to the cave. Anything that fights against this truth and revelation is a negative force that must be organized against and rooted out. The process of science is to liberate people from the bondage of perceiving reality as shadows, and help them through reason, education, and social institutions live a more real and healthy life. The true world creates our world of shadows and those that perceive that true world, should rule and organize the world of shadows. They believe that they can determine values, morality and politics, from truth outside the cave. It is by nature an idealistic utopian authoritarian philosophy. Plato's Republic is a guide to building a utopian society, and thus is a guidebook to authoritarianism. A small group of people will determine through science what is morally true and then impose that on an unenlightened society. If they do the scientific method correctly and there is actual objectively true morality that can be imposed on a society, then there is in theory the potential to create an ideal and better world. Objective external moral structures bound into and defined by deductive logic and reason, is an enduring quality to rationalist thought. This morality would arise from an understanding of the material nature of human consciousness and social evolution. We would study, brains, DNA, and computer models to determine social values.

This is a powerful set of ideas, and deeply appealing to many rationalist minded people's idealism and internal self-belief in their own intelligence over that of other people. It is not a theory driven by Marxism, Collectivism, or Identity politics. It is in many ways a right wing theocratic ideology, with no God. They are seeking to create a new social order and religion to order society. In order for their ideas to work, they need to recognize a need to separate their ideology from politics and to not have a state ruled by their atheist religion. If they seek no government power, and only try to create a non-centralized religion based on voluntary interactions, and increasing the quality of life for people based on truths revealed in Science about human nature, then their movement can be very successful and embraced by many. Atheism in of itself is no threat to anyone, except those that wish to impose their beliefs on others through political and military power. It is a new secular religion, that won't be going away anytime soon, due to its appeals to innate human nature found in more idealistic rationalist minds. This way of thinking can discipline people away from useless magical thinking that their minds can generate as natural garbage thoughts we all have. They first will have to create a moral character in their adherents in how deeply they can challenge their own thinking and expose it to the harshest tests of falsification. That is an intrinsic moral element necessary for modern materialistic rationalist new athiests.

The other way of seeing Plato's cave isn't not a movement with a clear name yet. They see that the shadows on the wall create the "truth" outside the cave. They are also materialists, since pure rationalism without an appeal to the real world is not a functioning form of ideology anymore. Kant and Jung are sort of the underpinning thinkers for such a school of thought. They are not atheists, and they are not religious. They don't seek to create a new religion and they don't seek to end the concept of God in this world. Another solution to rationalism and how to connect it to the material world is the power of the human mind to embody itself into reality. Only that which can embody ourselves into is real. "I think there for I am" can expand the domain of human

thinking into the reality of the external world, into the minds of other people, and even into the stars themselves. "I am alive, therefore the world is". The power to embody is not the power to project humanity into things, and see them as human.

Anthropomorphizing is more a solution to the problems in Empiricism. Embodiment is a more literal and objective creation of human thought into objects and into other people. The means by which we pass on culture to other people, the way we domesticate plants and animals, the technology that we create to serve our lives, the creation of thinking machines reflective of how humans themselves think, are all issues of embodiment.

There is no radical belief in absolute truth that we as people can know. They can believe a world greater than the shadows exists, and that it doesn't really matter. The only thing we can know of this world outside of the cave is constructed in our imaginations from our experiences of the shadows we see on the wall. It doesn't take an empiricist view that shadows are all that we can know, it more focuses on the mind that does the seeing and how the patterns of cognition themselves may structure the world outside of the cave, and be that world itself. This is sort of the magical quality contained in such an ideology, that our very sense of being and mind has a meaning that extends into the very root and creation of the universe. The scientific challenge of such a view is to find a way to make that be possible, and be testable. This book and this part of the book is an attempt to do just that. Leap into the mind itself and create the universe from nothing. In the end, we can only know what our cognition is capable of conceiving and the fact that it can form coherent, functional and intuitive models of reality is profound empirical evidence.

Our bodies living in this world, are shaped by this world, and thus must be consistent with reality to exist and survive. We are created from billions of years of evolution and there are simply too narrow of options for us to exist and have a functional connection to

reality. Embedded in our bodies and culture is knowledge about reality beyond our minds capacity to ever conceive. We didn't need to ever know why we survived, we just had to survive. Our cognition is not adapted to control everything. Human conceit and pride are more likely to kill you than save you, if we had such knowledge. We can't change human nature. We can't ever really know what that nature is in any absolute sense. The important thing for this kind of rationalist to focus on is creating things in the real world that work, help people, make us more human human, and expand human survival. They must embody humanity into our extended reality. If science is objective truth, and technology subjective truth, then the difference between a Rationalist Materialist and A "Embodied Rationalist", is which is more important, functional technology or true scientific knowledge.

There is a correct answer.

Chapter 10: Evolving Dimensionality

Note: This chapter is not meant to be clear or understood readily. Read at your own risk. This is the most dangerous stuff you can read. You will be expected to look up and study everything on your own.

The human mind is connected to something more. It arises out of the universe we live and reflects itself back on itself. We as humans can feel the truth in things. This delusion guides nearly all understandings of how the universe works. One of the truly great puzzles of physics beyond all its mathematical achievements and workable models of existence, is how any of this can even be intelligible by human created math and thinking. We can make simple math equations to explain the largest and most influential forces around us. Humans can even feel which sort of ideas are more likely to work over others. Great minds impose their very self and being into the world, and create patterns from their own being, that end up correlating to usable systems to predict and calculate how things work in the world. This is "magic".

We as humans can leap into reality itself and into other dimensions through the power of memory and imagination. This is a guide to thinking in zero-dimensions and 4-dimensions. In later parts, we will visualize leaping into black holes, white holes, and emerging multiverses. The irony being that it uses the same trick to do both. The hardest concept in this book, and maybe in philosophy is to understand what "nothing" is, from a deep intuitive level. Empty space is itself a hard-enough concept. This goes even deeper. When you can leap into a zero-dimensional space, you can leap into higher dimensional spaces mentally. At the heart of this chapter is a very difficult to grasp but workable way to conceive of reality in 4 dimensions. You must do it emotionally, subjectively and from in yourself, and not as a separate and distant external observer.

After you get that down, we will go into the 5th dimension, as a real and concretely observable thing.

Zeros and Infinity

Most of my childhood I considered numbers. I thought about what is zero, what is negative 1, what is 1. The ideas they formed are the foundation of everything in this book. Zero is an almost magical number. It isn't even a number really. It is neither even or odd, and you can't add up more it, mostly. If you divide something by zero, you get an indeterminate amount, it is just something you aren't allowed to do. If you divide by a number approaching zero, you get a number approaching infinity. Infinitesimals are the tiniest possible amount of something; they are a single point on a line. Zero is a point, it is an infinitesimal. It isn't nothing, it just the smallest amount of something there is, and that means something.

Zeno's Paradox: The Oldie but Goodie

- https://en.wikipedia.org/wiki/Zeno%27s_paradoxes
- <http://plato.stanford.edu/entries/paradox-zeno/>
- <https://plus.maths.org/content/mathematical-mysteries-zenos-paradoxes>

All of his paradoxes are just the same paradox. Solve any of them, solve them all.

Imagine a wall, and each step you take towards it, you go half the distance to the wall. If you take a billion steps, you will never reach the wall. Solutions to this classic Greek

paradox are in some regards the beginning of physics. The idea of "Greek atoms" arises out of this thinking. The universe has to have a point where you can't keep cutting it in half. An atom, an indivisible block of matter, is an obvious solution. You keep cutting until you reach a minimum amount you can cut, and from there you have the building blocks of the universe. The Greeks by logic alone figured out the existence of atoms, and to another degree the things that make up atoms. They found the "pixels" that made up the universe. There are indivisible blocks that can't be seen, except at the tiniest scales, like our computer screens. The bigger the pixels, the less step you have to take to reach the wall.

Another way to solve this paradox is to atomize/pixelate time itself. As your steps get smaller, the time it takes to take a step get smaller, and you can arrive at a set time, and have no paradox. If you can manipulate time, you can take infinite steps to arrive at a wall, in a finite time in this world. At each step, time also takes half the time, so the more steps you take, the faster each step is taken. Smaller steps are faster than bigger steps. Though for this to work, it has to have a set rate, that can't be changed, and yet can be different depending on where you are in space. Time in a really small space will act different than on a very large space. 1 meter will have one time, and $1/10^{50}$ meters will have different rate of time. There has to be a way to smoothly move between each time frame and define each one as correct. Relative time emerges as a solution. Time changes space. You can imagine this as two runners at different distances from the wall shoot an arrow at the same speed at the same time, and they will observe the other persons arrow moving at different speed and being shot at a different time than theirs.

Another solution is to say that position is uncertain at a specific time. At any given moment, there is a fluctuation of position, and that uncertainty contains a domain, that

can cross that line in a finite period of time, with finite steps. As you get smaller at a certain point you can no longer define exactly where you are in a definitive and determined way, the smaller you get, the more probability defines location. You could make that as a modeling of a certain "vibration" of the universe at some level. That "vibration" could also be described as time, or even entropy, depending on how you wanted to build a solution. You could say that a thing can't have a certain position on the line, or rather that the line has no certain position to occupy.

Instead of looking at the person taking the steps and how long it takes them, look at the line itself and the properties in it. We can look at Zeno's paradox in modern terms as exactly the same as crossing over the event horizon of a black hole. With pure logic, one can infer from this paradox the basic philosophical structure of general and special relativity and quantum physics. The line itself bends with time, and that in order to even look at a 1 dimensional space, one needs to connect in a time dimension and graph the approach to a wall on a line with a 2D graph, with one axis being time. As you move along the X axis towards the wall, you go down on the Y axis, and curve. In a system where you never reach the wall, and get as close to the wall as possible, the line would curve from flat to nearly perfectly vertical (*Need a diagram here*). The curving of the line is all described by time. So as your steps get smaller, you are altering time more and more, just as in real life, as you get faster or increase in energy, you alter time. You can also infer the basics of quantum physics, and the uncertainty of observation, and the uncertainty in even the exact location of things at the smallest scales. Logic alone could make these assertions possible, with just the right thought exercise.

The experience of time, is based on where you are standing relative to the wall, and different positions on the line will see time differently. Two racers one slow and one fast, will actually observe time differently from each other, as they approach the wall at

different times. While the person taking steps towards the wall would feel everything the same as before and not think time has slowed at all, they would see the line getting longer. A line getting longer and time slowing are an expression of the same thing, from different perspectives, when you consider time on the Y axis, and space on the X axis. If you really took the paradox seriously, you have to reach such an idea, though without any empirical reason to believe such a thing, it would be a bit absurd. Shooting arrows with different perceived times adds in a difficult level of complexity for a deductive argument. A very unlikely theory of gravity could have been made from Greek logic that would have been consistent with Einstein and not Newton, millennia before science was invented. The sheer absurdity of the solution, is why people couldn't solve this paradox for so long. The fact that it how reality actually works, is just a simple product of reality being consistent, and that one can fit a solution to the discovered science. Deductive logic with some facts can inspire scientific experiments and new discoveries.

Playing with infinities is a "basic" part of physics and math. Thinking seriously about Zeno's ideas is an important part of any model of physics. Quantum physics is a challenge of figuring out what you can quantize, what you can make have a smallest possible size that can't be cut. Matter can be quantized. Energy can be quantized. Can time? Can space? Can gravity? You can play with a thought experiment and come up with multiple solutions that can be correct. Is the X and/or Y axis pixelated?

An Empty Box of Apples: "Heisenberg's Apple"

Let's say we take an empty box, we seal and suck out all the air, and make a sealed box with a perfect vacuum. There is not one atom of matter floating in its space. And then we freeze this box to nearly absolute zero. The walls of the box will barely influence the inside of empty box at all. There will be fewest possible box atoms jumping around in

the empty space. We then can encase this box in 20 meters of lead, and put it 1 mile beneath the surface of the earth. This will be an isolated *space*. I have a question about this box and what is inside it.

How many apples are inside an absolutely empty box?

The answer in physics is not zero. Zero as this idea of absolute nothingness is an idealized concept that "doesn't" exist in the universe, except when talking about impossible things. Anything that is possible, has a possibility of existing. Probability makes it possible for anything to appear anywhere at any time, with an actually definable amount of probability. No box in the universe is empty, ever. There are no empty spaces in the universe, and in all spaces, there is something, and the possibility for something else to be there. Empty space radiates energy, from nothing. All possible things have a possibility of existing and thus exist.

The smallest amount of nothing in the universe is still not zero. In that empty frozen box buried in the earth, there is a probability of an apple being inside it that is so small as to be impossible, unless you change the parameters of the "experiment". If we had 2 boxes, we would have twice as many apples. If we kept making more and more empty boxes we would have more and more apples. If we waited longer observing these boxes, there is more chances of apples coming into existence as time goes by, though in each second the chances are the same as any other second. If we grow this box to 10^{200} times the size of our own universe, and wait over a time 10^{100} years, an apple appearing in that box is nearly certain, because it already did. I ate an apple today, so I am mostly certain they exist, relative the existence of everything else that I claim to exist.

Single points don't actually exist in the real universe. Things don't exist as exactly one thing in one place. You can think of the smallest things as sort of making a dent in the fabric of the universe. What we normally call a point would be more the exact lowest point in that dip, and the sloping edges of that indentation are just the probabilistic space it "vibrates" in, most likely. In physics though we have to say it indents all the universe a little bit, and is partly part of all the universe at once. This is not some grand mystic idea of everything being connected. It is simple logic that needs to work out so you don't get absolutes and infinities in the universe. Infinitely small things don't exist in real life, and if they did, they would make things more black than a black hole. The whole of an apple is made of tiny non-infinitely small things. With Zeno's paradox, we can't be sure if it is vibration or space-time warps that solve this, or that those two things are the same idea. If quantum uncertainty formed the basis of space-time, things in physics would probably be more consistent and logical. I personally have to go with time being the uncertain and probabilistic thing, that creates pixilation in the universe.

Dualism and Nothingness

Nearly all mathematically minded people thinking about reality have to believe in a reality separate from our own, where math exists and then imprints itself into our reality. The classic book on this is Plato's "Allegory of the Cave". It is a story any educated person should go learn, and isn't that hard to understand. It creates what is called "dualism", there is the world we live in, and a more "real" world of perfect forms that our world is just shadows of in comparison. Numbers don't exist in our world and can only be conceived in our imaginations as sentient beings. Our very sentience is a god like capacity to stare into the realm of the gods and see the truth of existence and

divine perfection of creation. Science, math, religion all have their own "divine" realm separate from our own, with their own parameters on it.

Scientific materialism, where there is only the world we can see, is the only correct form of thinking to build scientific models from. There can't be a spirit world, and there can't be a reality that we can't directly observe. If it can't be seen and observed in theory, then its existence is irrelevant and must be disregarded. Even the past is an imaginary land, that can't be observed or traveled to, yet even the most astute of minds can believe in that delusion. Time traveling into your own past is as possible as traveling into a fantasy world made up by a British children's fantasy author, with elves. The fantasy world might be even more likely to travel to, than into your own fictional history, because one doesn't have to be all that consistent to be "true".

The universe though is discernible with mathematics. We can describe nearly everything in the universe with simple elegant math, and don't really have a good scientific explanation for why. There is still this magical quality to science, that needs to be pushed back on. The hardest magic to get rid of in Science, is the magic that underpins much of scientists thinking. We as humans have visions of reality, that at times, when worked out, become the basis of the greatest scientific discoveries. We seem capable of perceiving something more than the immediate world we see. Reconciling this is hard. Eliminating dualism from even the most skeptical of minds is hard, if they take seriously deep thinking about where things come from and the origins of principled ideas. We all have magical fantasy kingdoms we think are real, no matter how wise, educated, or skeptical you may be in your grand genius.

To fix this in my own naturally rational dualistic thinking, I made a dualistic universe that can actually be observed and be knowable. Dualism seems real, because it can produce

profound results, so I defined dualism by making this world and the magic realm distinct and clearly defined. I made the magic reality as small as possible for it to create the one we live in. We have to imagine what is the smallest possible fantasy world necessary to create everything we know. The past itself is so massive, that only the present is a bit bigger. The future is even bigger than the present we live, and contains near infinite multitudes of possibility. "Absolute Nothingness" seems suitable for the basis of the "magic" in dualism, and all the achievements in rationalist thinking, especially if one considers Buddhism and mediation.

"Absolute Nothingness" needs to have a minimum set of qualities that a dualistic-universe requires. Nothingness has to have a minimum quality that all things can eventually arise from. Instead of there being endless perfect objects or idealized math, there just needs to be a simple quality that can give rise to all the other things. Like say that probability pre-exists reality, while math as we know it, doesn't, is a possible configuration. Numbers don't have to exist in this system, nor anything that we would be able to normally label. Quantum uncertainty is something that can exist in nothingness before time. The universe we see could then be built up from that state with a little time. A rationalist would need to build a model that could do this and approximate all the systems we see now, arising step by step from pure probability, or with an additional feature. It could be one thing like probability or one more thing in addition to it.

Only possible things, and not impossible things, have a possibility of existence, eventually with time. If nothingness has probability, then it becomes like an empty box, if it gets big enough and enough time passes, the universe and all we see can come into existence, in some possible way. Math emerges from what is possible and not possible. The universe requires consistency as a basic law with probability. Illogical or fundamentally inconsistent things can't exist. More consistent things are more likely to

exist and thus exist. "Math" emerges as a property of nothingness and probability. It is the smallest and most likely conceptual "apple" to emerge in a box. We as humans are far less likely "apples" that clearly are possible to exist. Any sort of inconsistent life, that can't function through time, has a very low probability of existence, the ones that are more consistent and can move through time, are the ones that live.

The question then becomes what is "nothingness", and how did it create the universe we see. It is impossible for zero to be perfectly nothing. All numbers and maths emerge from probability, and their usefulness in time. Nothingness in a sense has two qualities here, and they play the key role in how things can come to be. You have random things coming into existence from nothingness, and only consistent things surviving. Evolution and adaptation can arise out nothingness as products of probability and consistency. Survival itself can become a selected for feature of the universe. Stable universes exist more than unstable universes. Evolving, growing universes are more likely to evolve, changed and become something more. Too much change, and nothing survives, too much stability and nothing changes. Things that exist must be able to adapt and pass something on into the future.

Time emerges as an ordering of reality. Time is an emergent property from nothingness. Nothingness is just randomness, and probability, that contains within it the possibility of time, and a more ordered determined world. Time is one event leading into the next event, and causing it to happen. Causality itself is an emergent property of Nothingness. Causality in the highest form becomes something more like evolution. One thing leading into the next. That process itself has a nearly magical like quality that created the very life reading this sentence now. We can know these systems, and judge them now. All that needs to be done is to identify what is more fundamental to the universe and thus which quality caused which quality, until you reach a quality that can't have any cause.

The nature of a timeless nothing universe, where it is possible for time to emerge, is the puzzle.

Zero-Dimensional Space

Going back to Zeno's paradox, you can imagine yourself entering into that every shrinking space as you get closer to the wall. As the space gets cut in half each time, it gets smaller and smaller. It begins to approach a space with the dimensions of zero. If a one-dimensional line collapses into a single point, it is becoming zero dimensional. A single point has zero-dimensions. It is pretty simple, and also very hard to really see and understand. You can visualize this by continually zooming in on a single point. Imagine looking into a thing, and keep zooming into to smaller and smaller sizes, go past the atom, go past quarks, zoom in past a photon, and keep going until even photons are as massive as galaxies. Keep "zooming in" and don't stop, and your mind will "descend" into absolute nothingness. This is a mental state people can have, do have, and those that survive are forever transformed by. No true mystic, nor saint, nor serious thinker has not faced this mental state. It is a "space", with no distance, no time, no qualities, no form. It is absolute nothingness. A single point has the value of zero. A thousand points added up, is still zero. We will need to reconcile an idealized 0D space, with one that actually could exist and create the reality we see around us. A point on line, and a point on a plane are equal, no matter how many more dimensions you add.

For a 0D point to really exist and for it to solve Zeno's Paradox, a point must be able to evolve into a higher dimensional structure like a line. A point is infinitely dimensional can evolve into as many dimensions as possible. Instead of looking at the problem from an ever-shrinking line, one can think of the problem as how to take a single point and grow it into a higher dimensional 1D line. We need to flip Zeno's paradox and its head and

make it about the line itself changing, and not the runner on it. A real world 0D singular point contains probability and it allows for random things to emerge from it. At a simple level, you can just imagine that a point must vibrate. It can't be defined by an exact location in the real universe; the size of that vibration is a fundamental feature of time and space. It is partly the basis of quantum uncertainty and how lower dimensional things evolve into higher dimensional things. This vibration has a set rate in itself and yet changes depending on where it is in space; we call this vibration time, and that vibration rate, the speed of light. Zeno's paradox is a time dependent paradox, that ties space and time together into a single thing.

Random spaces could evolve into higher dimensionality, due to uncertainty in their position. If everything in the universe had a defined position, nothing could actually exist. It would all be absurdities like $10/0$: infinities and zeros. Real numbers and real things in the real universe need to "vibrate", be uncertain, and be random at the tiniest levels, in order to have real position that can relate to other things. In the real world, idealized math can't exist as a real thing due to its absoluteness. A vibrating point becomes a line, as that vibration is described by time. If you take fixed perfect absolute point, it can never evolve or change into anything, and will never be able to contain time, nor evolve into something else. By a point being uncertain, and containing all dimensions, it gains the ability to change, and that change creates time, which draws out and expands into at least a single dimension. A line is created by an uncertain point mapping itself out. A vibration in a line, makes it into a wave. Zeno's Paradox doesn't happen on a single line, it happens on a plane with one dimension being space and the other time, and when mapped together is a simple space-time structure. Wrapping one's head around simple ideas can be the hardest thing to do.

We experience this nothing dimension from the center of our beings. It is not truly empty, even though it contains nothing, because it must contain the probability of something different, and the capacity to evolve into our universe. Understanding what can exist in absolutely nothing, is a hard thing for a brain to wrap itself around. You can feel it, at a personal level. It is known from inside you. You can't know it in a concrete material way. To feel reality, to feel time, allows a person to see 4 dimensions more readily.

TIME IS A KIND OF SPATIAL DIMENSION.

To understand time, you need to see it spatially, as an expanding entity, that manifests itself out in the universe. How it manifests itself, defines how change happens in that region. Time arises out everything, and to see time, you have to take your mind into a universe before time existed, and see the universe evolve dimension by dimension into the one we live in. Time emerges from physical properties in zero-dimensional space, it has a real structure that it comes from, it is a physical thing, and not an abstract concept. Your brain processes time as a subjective experience and not as a visual thing you graph. The 3 "space" dimensions don't have to be what we perceive them as, and all 3 are related to different aspects of reality. Spatial dimensions themselves might be forces of the universe expressing themselves through time, and not actual space, or vice versa. This problem in dualism is not solved yet.

If you want to know where the literal past exists in the universe, it can be embedded in space itself, that is expanding away. The very distance between things, contains time and describes relationships in time. This is not an abstract thing, it is a literal thing we can eventually observe and define. The structure and patterns of time in the universe are just small and not observed with the eye.

Time is observed with clocks, and when we can get smaller and more accurate clocks, we will be able to see the finer structure of time itself. Its larger structure is just universal expansion, that is built out of entropy at the smaller scales.

A NOTHING THAT CONTAINS EVERYTHING

Zero is not an idealized pure nothing, it is defined as infinitesimal. It is a really small amount. In the real world, "nothing" is zero and infinite. This is the basis of relativity and time. There is one more idea to help bridge this all together. Zero divided by Zero, is not one, zero, or infinity. It is all possible numbers simultaneously. It basically like drawing a vertical line on a Cartesian plane that goes straight up. "Y" equals every possible number at a certain value of X. This is a very important mathematical concept that comes up often in math, and for the real world and in real life based math, is something that should be and is quickly pushed aside and told to avoid.

Because 0 times anything equals zero, we can say the zero divided by zero can equal any number.

- If $0 \cdot X = 0$,
 - solve for x,
- then $x = 0/0$
 - thus X can be any quantity.

A "Nothing Universe", can simply be described by $0/0$, where it is basically nothing and everything at once, while simultaneously not being 0 or infinity. It meets the criteria for

a basis of mathematics and creating a universe from nothingness, for a minimal dualistic system. You then can imagine a universe oscillating between numbers. Everything in existence is just a number. A number line is just how you describe a 1 dimensional number. You can have a "number plane" or a "number cube", or even have a 1000-dimensional surface describing a single number. A number is just a single point anywhere in space, and it can have as many dimensions as it wants.

A universe existing before time would just rotate between different numbers, randomly. Each number would have no relation to the number before it. Time arises from causation, so a universe without causation would be a universe without time. Anything possible could come into existence, and not cause anything to happen in the next moment. This is timeless. There is no way to determine one event from another. The sequence of numbers has no meaning. You have a universe, you have "existence", and yet it is nothing and timeless, while also being able to become any possible thing. This is a rough sort of proto-time. All you need for time to exist, is for a random number to emerge, that could cause another number to exist after it, so that they are connected. It would need to be a number that all other numbers could evolve and grow from, that we can see in the range of possible "numbers" in our universe. It would need to be something that could cause something, and evolve into something more complex.

If we look at basic numbers, the first number to come into existence in any math system, would have to be one that all other numbers can be derived from. "1" can't be the first number. You can't make -1 from 1. You can make 1 from -1. $1^2=1$. $-1^2=1$. Thus -1 is more primary than 1. If we take this a step farther, the square root of -1 is the most primary. $i^1=i$, $i^2=-1$, $i^3=-i$, $i^4=1$. Complex imaginary numbers are more fundamental than real numbers. Then we need to consider Pi. e^{Pi*i} , seems to cover most any possible number in existence, and has interesting qualities as a number that can explain the patterns in our universe, and how all of math comes into being.

$$e^{\pi i} + 1 = 0$$

This is a pretty number, and probably connected to the first number to come into existence. It has unique properties that would also make it good for creating a universe. It has a kind of internal symmetry. Time needs a kind of symmetry to form. There has to be more than one thing, for something to cause something else. The universe has to grow for there to be time.

Leaping into New Dimensions: A Visual Guide

Zen Buddhist masters will find this easy, because they already learned what is easy.

-Kaons by The Joshua

To really think in the 4th dimension, you need to exist in the zero dimension as an immediate and personal thing.

If you leapt into a single dot/point, you would experience all of reality as all black or all white. The subjective experience of a mathematical point is for it to be all of your observable reality. You see "black" in every single direction, and not some tiny dot graphed on a line. An infinitely thin plane is equivalent to infinitely thick or a 0-D space. Absolutely nothing can't be conceived from outside of itself. In a flat land, the universe is

infinitely thin, so you can't enter into and experience the world how they would experience it. There is this false sense of reality from observing it from above and not inside it. You can't be inside an infinitely flat plane, *unless* you change how you conceive an infinitely thin plane. The flatness of flatland, is the very 0 dimension, the very nothingness. The subjective experience of nothingness can be infinite thickness and everything.

To make it simpler let's imagine that even in a zero-dimensional space, that all the known higher dimensions you know, exist in that space. You don't have to get all super small, and forget about being able to look up or down. You will still get to look all around, and be you. You can look in any direction, for this visualization. Imagine you are in the center of a sphere. The sphere is like a big movie screen all around you and at any distance you like it to be. It can be like a movie theater that is all around you in every direction you look, where there is just you and the screen. There is nothing in this closed sphere, except your ability to see in any direction. 3d space is not grid like, it is more polar coordinate like. A simple mathematical description is a sphere, with a wave like line going from the center to the edge of the sphere. This model is how to understand any sort of quantum model, and possibly how to understand particle wave duality.

The Subjective Experience of Being.

You need to visualize entering into the reality of thing, that you know well in this world.

- If you were a computer programmer, my idea is to live inside a program, and see the code as emerging in front of you, as you go through a program linearly. You

would experience code as it moved from line to line in time. Logic gates would emerge *in front of you* like a primitive first person 2d game.

- As a gamer, all games would be first person: Not all first-person games are 3d.

First person games dimensionality evolved through time.

- https://en.wikipedia.org/wiki/First-person_shooter
 - Battle zone is an original first person shooting game from the 1970s
<https://www.youtube.com/watch?v=yMrYkbEbnEQ>
 - <http://www.wonder-tonic.com/wolf1d/>
 - Imagine leaping into this directly. Outside you see a line, inside, you would see that data, fill your whole view.
 - the first Wolfenstein game can be played as a line. A 1d game. If you jumped into this game's actual reality. You would see a field of white, and then a square straight ahead, it would be a different color and could change in size. Smaller is greater distance, and larger is closer distance. All white, nothing to see. All one color, that thing right in your face. Firing a bullet would be like all black, then that black square would get smaller and smaller, until it hit something. The black square would overlay another colored square.
 - The first Doom game simulated 3d.
 - Dimensionality evolves in games, from the first-person perspective
 - You will notice in many first person games the illusion of a 3d "world". You play the game in 3d, but the story and where you go only moves in 1 dimensions. You mostly just are moving down straight lines, shooting at things. With no real choice in where you go. Truly open true "sandbox" gaming took a long time to develop. It took time to even conceive such games.
- As a physicist, this could be like experiencing a Feynman diagram at a subjective level. You would be a particle and see those changes emerge *in front of you*, as you moved through time. You would be in the diagram moving through it, rather

than looking down it. Or imaging being an atom skimming just above a black hole event horizon and feeling in your body the experience of being an atom, speeding at 99% the speed of light in orbit of a black hole event horizon.

- As a psychologist, this would be cognitive empathy, you see and feel reality through the eyes of another. This is not imagining who they are, rather being who they are. Not imagining how you would feel as them. You simply become them, and visualize reality as them
- As a car mechanic, you would become the gasoline engine. You would become fuel going through the engine and transforming at each different stage of the engines process.
 - You flow into the fuel tank, then through the fuel lines, and get sprayed into a vapor into piston, you mix with air, you feel this new relationship and potential with air, as the piston is sucking down, it pushes back up, you compress, a spark goes off, you exploded, and drive the piston down, you are burnt vapor, you are smoke, the air and you are one, you are heat radiating through the engine, shining out to the world, the piston comes slamming back up and push out through a hole, you expelled at high force out into the world, you are water and co2 floating in the air, The drives of those pistons turn the crank shaft, and the crank shaft turns the transmission, you energy is kinetic, you drive out into the bearings of the wheels, you press against the road, the dirt and pebble digging into you treads, the the bumps and shocks of the road, then pressing into your suspension, the very car pressing up into the seat, the driver themselves, pressed back into the seat, they feel the road, they feel the speed, the feel the heat, and the sound in their ears.
 - transform yourself from gasoline into energy, and move as kinetic energy through a system of gears and levers, until you transform into the rotational energy of the wheels pressing against the ground. Feeling the very rocks and roughness of the ground stick to the tires and allow gravity

to fling you forward. Feel gravity pull a car into the earth, and be proportional to the acceleration times its stickiness to the ground.

- You can even become the pollution moving through the environment. CO2 being pulled into a tree, and turned into oxygen, that we breath. You can experience the chemical processes in a plant. Be one with nature, be one with the human's life that you move.
- Gasoline!

Zero-Dimensional Space

If you looked in any direction, you would see the same thing. All black. The experience of nothingness is all surrounding and all encompassing. If you are that dot, then all of reality is that dot for you. You are darkness or light in its entirety. Zero-dimensional space is just all white or all black in every direction. Now this state is indeterminate. It can be anything. It is both all white and all black, at the same time. You can look around in any direction, and all is black, sometimes the screen will flash white. This is a real feeling, that a person can have. It is a real state of mind to exist in total nothingness, a true void, then you can find a single point, and change your experience of this nothingness reality. This is just a point that can randomly flash, with no pattern or structure. It just comes and it just goes. As you look around certain orientations can create a white screen, and then in the next moment it is gone, and you can go search for it again. If there was a pattern to the white, you would be able to follow and orient towards it. Without a pattern, it is just a random sequence of white flashes that appear at random points across the screen. It would be a bit like only being able to see the white static of a TV screen pixel by pixel, though that TV static has an actual pattern and comes from energy from the universe when first formed into light.

Some stare into a fire, or feel a single point in their body. They engage and become one with the very thing their mind focuses on. It is contradictory to describe, and experienced differently by many people. Just sitting in black void of nothing, is not good though. The human mind needs connection. You can find the right things to connect to, a clearer mental signal, from this state of mind. Meditation is taking the mind back through dimensionality to a zero-dimensional state of nothingness. Prayer is an orientation towards pure light, where it takes the mind towards a state of determined absoluteness and the future. To live in a zero-dimensional reality is a choice of perspective. Absolute white or absolute black. Meditation and the ying-yang symbol are not random notions in human imagination and invention. To embrace total emptiness is not something any one can just do. It is a highly dangerous mental state to be in, and the mind protects you from it, for good reason. We naturally meditate towards a more determined state.

1-Dimensional- Time

If a signal forms in the zero dimension, you can move into the first dimension. Your orientation becomes a direction of movement, the signal you listen to at a given moment, determines the future. If you lock into a signal that actual forms from all the random ones in nothingness, you can make that signal become your whole universe and reality the longer you can follow it and track it. It is like you find a radio station you like, and it keep changing station numbers. You have to keep rotating the radio dial to find the signal you want to follow. It is very possible that there is no signal even to listen to and the rotations of that dial is just creating its own pattern and signal to listen to. Sort of like a DJ scratching and mixing record to make music over the base music. You could

stare into tv static and looked at it pixel by pixel and find a meaningful set of information communicated to you through time, or possibly not.

The flashing random signal now changes into a pattern. What you will see is oscillations, of black and white. As you transition into the 1st dimension of time, you will only see one signal, no matter where you look. Everything will be the same, no matter what direction you look. It is sort of the most boring movie in reality, just a screen flashing white or black. It could make a sort of musical like beat visually. It is however just all black or all white. And nearly everywhere you look around, this "signal" will be the same. From the outside, this is like a point that is just flashing patterns. In a purely graphical way it is just a line with different sections of white and black on it. Time makes us experience those patterns as a signal, that you can actually experience. If you look around, you can find a specific point that is special, just like you emerged from 0D space. This underlying pattern in 1D space will exist in all dimensions evolved on top of it. There are no universes without time.

In this first universe, you will be able to see a future, see something fixed, and not fluctuating that leads towards a possibility of greater stability. I think of this like a guiding star out of darkness. A possibility for something, exists in a place of infinite nothingness. In a universe of all black there is still possibility, there is only possibility. There is vast nothingness undetermined. If the universe is white, then all is determined, set and known, it is absolute, and unchanging. The only thing that could arise in it that whiteness that would be different than infinite determinism, is complete chance, complete randomness. A single black dot can form in the absolute whiteness, and it can grow. It can grow into all of reality. Until all that remains in the infinite black is the possibility for the universe to be determined again. A white dot in the vast distance trillions of light years away, lies in a direction towards wholeness. That white dot in a sea

of absolute darkness and nothing, is the moment the universe gives birth to time. Time is deterministic, and arises out of the pure randomness of empty nothingness. **Time is a direction in absolute nothingness towards everything.**

Emotionally- You reach out from the darkness into the light. In one dimension, you can experience time. You can experience change. You can orient your mind towards something. You can orient a universe towards a state different than it had before, and move towards it. Time becomes reality in the 1st dimension. It is both time and space. You move towards something. Space expands and moves. To experience a 1 dimensional universe would be to experience a universe that flashes between white and black. It would be mostly black, with moments of white. Time gains its directionality, because "you" can orient yourself towards what sort of experiences you want to see/experience. You can focus on the white. Like in a universe of all black, there can be a white dot, infinitely far away, that you can point yourselves towards. Whiteness is an abstract thing, that your mind can conceive in nothingness. You can move towards it, and not just see it. You can change reality by your orientation. Real life though presents many paths to the future that are both similar and fundamentally different.

2-Dimensional Space and Time- A Line with Time

2D- In 2 dimensions, things get weird. Whenever we look around, the signal changes. We see only black or white still. This is like being able to turn the dial on a radio to a different station. Mostly as you look around, you will see all black. Though at certain points when you look, you will see a massive amount of signal, and varying patterns. From the outside. This you being able to move through a line and experience different signals. There are many possible futures to choose from, where you are able to pick from a stacked array of lines, and observe that. This is a plane with patterns in it.

"Straight" lines through that are a form of time, that you can experience. Your whole vision will be black and white only. However, you can "rotate" and orient yourself towards different sets patterns that actually seem to have meaning. This would be like being on a radio dial, and you can change that dial to different stations. Most all the ranges are emptiness, but as you orient in certain directions, you will see patterns of all black or white flash at you. You can choose which patterns to look at more. You can be drawn towards the brighter and whiter directions, where you can get a signal.

Emotionally, you gain free will to change. You live in a world of light that can change. You can choose what "voice" you listen to, as you emerge from the depths of darkness deep inside yourself. While there can be a future, change, evolution and growth, you will still have to pick among the different signals. See who you are listening to in the world, and move towards that reality. Some signals don't lead far, others lead farther. In a simpler universe, there is just one point, one god, one direction. This direction doesn't mean you know, or are god. Life will suck and be all darkness. That white point is far from you, and can only help you find your way.

A 3-Dimensional (Flatland): 2 Space-1 Time

As you look in different directions on a flat plane, you will see different lines form patterns of varying vertical parallel white and black sections. These lines will extend like longitudinal lines on a globe. They are like pixels on a screen, but are the same vertically and vary horizontally. This is what is to be inside that flat plane. From the outside, this looks like flatland, and from the inside you are experiencing an infinitely thin plan, being stretched out vertically all around you. As you rotate, you won't just see just black or white, you will see vertical bars of black and white arranged around you. They are like pixels that go up and down vertically forever, that are stacked side by side. The

pixilation is caused by the time dimension and not the space dimensions. You gain a form of peripheral vision, compared to the 1D+1 world you emerge from into this one. If you shook your head back and forth in the 1D+1 world, you might begin to see this 2D+1 world emerge.

You can now experience a 2-dimensional plane with time. You will be able to recognize things by patterns and codes they broadcast. And sort of "see" like we do now. If a blind person had to walk around, and all they could see was vertical lines arranged in different patterns, they could get around in the world, better than just the blind. They would have depth perception like a dolphin with their ears. Though in flatland they could just rotate 360 on a flat plane, and not look up and down. Doors and corners, would be clear. Distance could be seen. Would be sort of a range finder, one could spin around on one axis. You could also have "color", for what sort of thing it was seeing at a certain range. You would be able to "spot" different things.

In this universe, you would have a better "memory". "Memory" forms each of the preceding dimensions. As memory evolves, dimensionality evolves. "Memory" arises out of time. Time seems to create more than existed before, because it "remembers" and builds on itself. You can virtualize the 4d world in a 3d one with enough memory. Computers with more and more memory can virtualize more and more dimensions, from its narrow 2-dimensional mind (1d space with time).

Emotionally- You can expand beyond yourself and the voices in you, and experience other voices around you, that can also listen to other voices. You move into a reality of human beings. On a flat earth, we can know others. We have depth, range, choice. The land has shape and terrain, at a basic level. You have to walk and move in a world as you follow a signal.

4th Dimensional Universe/Our Universe: 3 Space-1 Time

Evolving towards a 4-dimensional universe would be like, taking those all vertical lines that you see as you go around, and then being able to look up and down. You still see only vertical lines, but as you look up and down you can see different ones. Now you can look in any direction, and see different vertical lines form. Vibrating your "head" around will create an illusion of "depth" that will create an image. The vibrating of a world of vertical lines shapes into a world of 3d spatial perception. A chicken can see 3d, with one eye, by bobbing its head back and forth, thus why birds bobble their heads. Two eyes, or even ears, can create an illusion of depth perception. 2 different images of vertical lines arranged in horizontal pattern, can have a more traditional 3d spatial image arise out of it.

When time is collapsed in a lower dimension universe, it can simulate a higher dimensionality. Vibrations, probability, and uncertainty create the possibility for higher dimensions to emerge from lower dimensions. The Zero Dimension has many properties in it, that help create the evolution of all dimensions to higher level ones. In my visualization, the very reality of 3d space exists in the zero dimension in how we perceive it from the inside, and this is not some abstract unreal thing. It is what was possible, and thus existed in some form, even if it did not exist discretely yet. Everything and nothing are the same thing, and contain within this reality all possible possibilities.

4th Dimensional Thinking- Proto-5D visualization

- Imagine a line, and then take an arrow pointing perpendicular to that line. You have a sort of T shape. That is like a 1 dimensional spatial universe with a 2nd dimension of time sticking out from it.
- Now take a plane, a flat piece of paper, and stick an arrow pointing perpendicular to that. This is 2 dimensions of space and 1 dimension of time
- Now take a cube and attach an arrow to it. It can attach to any one point on the cube. The arrow exists now in the 4th dimension. It would be like a field of arrows all pointing out from the center of that cube, like a sphere of arrows
- The arrow of time, that is 1 dimensional is no different than a seeming field of arrows arranged symmetrically. Outwards is no different than up
 - In linear algebra, this is not a hard idea.
 - **Must watch videos on the essence of linear algebra**
 - https://www.youtube.com/watch?v=kjBOesZCqgc&list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

To understand time as property of space, you have to connect it to the very expansion of space. This expansion is not always in the same everywhere in the universe. This expansion is also gravity itself. Entropy, universal expansion, and time, are all connected. Though they are not the same thing. They are properties that arise out of similar causes, most likely. Time travels faster between galaxies than it does in galaxies. The directionality of spatial expansion creates differentials in time dilation.

Space is perfectly and absolutely flat, at all times. You can't really describe space as being warped. This is not really hard science or logic here. Just think about what it would mean for observation for space to be curved. How can you have more or less space in a set region of space? Is one cubic meter of pure vacuum not equal to another cubic meter of vacuum? Our perception of more space or less space in a region distant from us, is

based on differences in time, as time changes between two points, the perception of space changes. The man walking in Zeno's paradox will never see time change for themselves, though those on the outside will see him slow down.

We often think of objects all moving through space, and that space is expanding as things move outwards. Rather, I think of space moving through things, because space is expanding. "Curvature" arises out of "space" moving through things, and thus allows space to act on objects at a distance. This movement and "expansion" of space, is described by time, and is time itself. Time is a literal thing we can see in this universe, in how the curvature of space alters how things move. Time is a tangible thing. It is moving in physical directions in our universe. Time however can be traveling at different rates depending on where you are, and you can record/observe that from a distance, though never actually experience that directly. Light bending around stars, can be described simply by time dilation and not the curving of space. Space curves at the smallest level, as an infinitesimal. It pushes things towards other things, from differentials in its volume on each side. It is like a radial wedge, where the skinny side points towards the direction of gravity "pulling", and big side, sort of "pushes" in the direction space moves, creates a warping in time.

Spherical Coordinates and Expansion

The universe is far messier and not made of perfect clean math. It is something you need to experience in order to understand, and not just contemplate as an impersonal idealized object.

At a basic level the universe uses spherical coordinates and not Cartesian ones. Cartesian grids make it easy to see everything as one massive connected entity. Everything in the universe has a grid position. There is a fixed tangible realism to grids, that people get attached to. They are simpler math, that our brains can wrap concepts around. The universe is more like a wave, and filled with curves, and spheres. The shape of what we see in the universe is not square boxes. We see literal waves and circles. Nothing in the universe is a square, and the root ideas that formed the universe are not squares. Radial/Polar Coordinates with wave functions, describes the universe better. This also means that there isn't one universal grid, to connect things by one universal standard, in a way. Just the mental shift from grid coordinates to spherical ones, is a massive mentality change. The grid math is easier; the spherical curved lines are more real. Luckily, we can convert between them easily, and do all our math in the easy square grid way, then at the end of calculations convert them into more realistic curved spaces. (*thanks to Euler: $e^{\pi i} + 1 = 0$*)

- The universe is more like an indeterminate point, wave, circle, sphere, and cylinder.
- Uncertainty lies in spheres.
 - The 0-dimension of a square world and a circle world are not the same.
 - Newton's physics and modern physics have a very different root nothing.
- Uncertainty in nothingness, creates the emergence of higher dimensionality as a possibility.
- A square and a circle can never have the same area. That difference in area, between the two, seems to me, to be a foundational component of reality.
 - Pi is a fundamentally "random" number, with no algebraic pattern to discern the following number.

If a sphere intersected a plane, you would see a small circle, grow into a bigger circle, then get small again. You would be seeing a sphere slice by slice in 2D space. A great

deal of our universe and the geometry one needs to understand, is how higher dimensional things intersect with lower dimensions and appear to our eyes. These higher dimensions are created by time, and thus can't be perceived as more physical and spatial things. Much of what physics does in its modeling is model time as space dimension, so we can see it. A sphere is a circle with its size being bound up in a time dimension. You can draw the sphere only if you progress through time.

The challenge then becomes what would a sphere be that we see in our universe be if it was emerging from in a higher dimension of time, intersecting into our universe. It is a sort of "cylinder". You can look at the whole of reality as a cylinder, and each moment of time we live in, as spheres contained within that cylinder. Its axis up and down it, is a progression in time. If you stacked infinite spheres on top of each other, and separated them by infinitesimal distances in one direction, it would map out a cylinder. If those spheres were empty, and the surface of those sphere were infinitely thin, then they would add up still, into a solid cylinder. This is a hard-enough idea, and it doesn't even get into the harder levels of thinking one needs to model it completely.

Spatial expansion changes how these cylinders express themselves. You can look at expansion as a singular thing, as a sort of inflating ball. Where each moment in time is just one level of inflation. This however doesn't model the world well. You need to visualize many balls all inflating, in a big sea of them. Each ball will have curvature, and collectively be a flat expansion. The challenging concept to consider, is what if every ball inflated forever, and in itself was filled with endless balls individually inflating. This creates a kind of endless branching and expansion, that requires another level of dimensional thinking. It isn't just how much something expands from another dimension, but on what branch of that expansion it exists.

Chapter 11: General Relativity and Einstein

Warmup:

Let's say you are in space. You have 5 identical planets that are equally far away from you. You shoot them with a planet destroying death ray (Star Wars 7). You will observe each of these planets blow up at the same time, if you are at the death ray site. From each planets perspective, they will see itself blow up first and each planet blow up at different times, depending on how far away they are from them. Every planet will have a different order of when planets blow up. Now the gods eye view is the planet destroying death beam. It caused everything, therefore could determine a more precise level of causality and sequence of order. If the gods eye view doesn't cause everything, and isn't the root causality, then there can be no exact point in the universe that has a correct view of what happens first. We often can't ever observe the root causality of things we see in the universe, and thus have no means of determining an objective truth, readily.

Causality and Locality

The speed of light is a constant in a vacuum. The study of this speed laid the foundations of a revolution in physics led by Albert Einstein. A defined speed of light that is constant, has radical and deep consequences for how one has to see reality. The speed of light is an incomplete term, because it is not about light actually. "The speed of causality" is a more accurate term. Nothing can happen faster than the speed of causality. This creates what is called locality. Because things can't travel instantly great distances, the universe acts locally. Things influence the things around them, more than the things far away from them. Gravity can only travel at the "speed of light". There is a limit to how fast something in one place can influence something in another place. There is no way for anything to cause something to happen that is far away from it any faster than it would

take light, in general, to travel there. There are no magical connections that link anything faster than light. Even if there were real psychics and your family died 2 light years away from where you were, you wouldn't ever be able to know until enough time had passed for some form of signal to reach you.

The universe is massive and yet has little immediate influence on the local reality we live in. The most powerful explosions in the universe still take time to reach us. We can still see back to the earliest parts of the universe, when we look up into the sky. As we look farther out, we look back farther in time. Nothing reaches us faster than the speed of light. This creates an observable universe. Events as massive as the big bang, can still be observed if we look far enough away from us. We are at all times experiencing things that have happened through the whole history of the universe. The distance we look out, directly relates to and connects to time. Distance itself is correlated into time so deeply, that it is not easy to separate them, and maybe never at all.

If we could see everything in "real time", then the whole sky would be brighter than the sun. All the light in the universe would just hit us at once. Without locality, the universe wouldn't have darkness, space, and time. No place in the universe is any more the center, than any other place. All points in the universe have their own local reality and time, that then can influence things around them at a certain absolute maximum rate. Whatever happens today to you on earth, can only ever effect anything in the universe around you as fast as the speed of causality (the speed of light) can reach it. The speed of light, is an expression of time itself. Time has to move through the physical universe we live in. Time is fundamentally interwoven in the nature of space. We can observe time itself in the movement of distant galaxies. When we look at the expansion of the universe, we are observing time more than just simple space. Thinking about time differently is the really hard part of Einstein. Locality and the speed of causality are

about time as it relates to space. You can actually see time, in how space moves, and in how things move through it.

People want to think about the universe in ways where things are consistent with how they feel as a person, and not accept simple logic. Not even Einstein was immune to this problem. He couldn't accept the natural logical conclusions of his work. He made the math for universal expansion that we use today, and considered it his greatest mistake. He came up with quantum entanglement as an absurd conclusion to quantum theory, to disprove certain quantum theories, and it ended up being a cornerstone of physics. His mind could take things to their logical conclusion and couldn't always accept the rather unbelievable results. The expansion of the universe and quantum entanglement while exotic and new, are as real and fundamental to the universe as drinking a cup of water. They are more concrete, solid, logic and real than drinking a cup of water.

Space

Things are relative in space, because space is not a fixed thing.

Relativity and all things being relative is a consequence of space and time themselves being able to change. We experience space and time on the earth as fixed things, and that only objects move around. Space and time are like objects, like balls, they too move, have "speed", and direction. They may also possibly entangle like quantum particles.

If we really want to get at the heart of the problem, that people can't understand, it is empty space. Nothingness, normal empty space, a vacuum, is something. We can't experience absolutely nothing. Space is a thing, a vacuum at some level is an object. It has energy, it has properties in it. You can certainly measure how much of it you have, at any given moment. Empty space at its most empty is in of itself something. It is the least amount of something, that something can be, and still exist. Empty space can do things to the world around it. Space can move in of itself. You can have an object move through space, and you can have space move around an object. Space is not "material", by its very definition, it is however something, and different parts of empty space, are different than other parts of empty space. It has properties in it, and it changes.

When an object is moving through space it is not the only thing to consider. We have to understand how space is moving around an object, to understand how it is moving combined with its own movement. Galaxies for the most part don't move relative to each other. Galaxies in the universe in the largest generalist sense just spin, they don't go flying through the universe. Yet we can observe galaxies all getting farther away from each other. They aren't moving in any direction, and yet are moving apart from each other. The space between galaxies is expanding, the galaxies themselves aren't overall moving. When we observe them, we see them "flying" apart. This is simply not movement in the traditional sense that we see on the surface of the earth. They are not being pushed apart, nor flying apart, they are just getting more distant from each other, because "distance" itself is changing. The spaces between galaxies, and not the galaxies, are moving and changing, and so we see a change in distance between them.

Space between galaxies is expanding. The space between galaxies is increasing, inside the universe. They are not expanding out into more space. Galaxies are not moving into new unknown regions of space; they are not moving. Space is increasing inside the

universe and between galaxies. Things don't have to move, in order for you to see them moving. Space itself can change, and make you perceive movement. There is a change in that things *relationship* to you. Things really do get farther apart from each other. Though at a basic level, you can't say which one is moving and which one is not, unless you can map how space itself is moving and expanding too.

Think about this seriously. You have a field of objects, all getting farther apart from each other, and they are not moving, and they have no direction of movement. There is no center to this field. Any point can be the center of this surface. This is often explained using a balloon, and marking dots all over the balloon, and then blowing up the balloon, more and more. We need to imagine that the dots don't get bigger, and we can observe the distance between all the dots expand. This expansion is a way of describing time entering into our 3D space. This is a basic way to understand it and not really how the universe works. It is more like you have a field of many balloons that are all expanding, and new balloons keep coming into existence. You could also imagine it like zooming in on a picture and everything expands. We don't want to think of this as a sort of force pushing from the inside, like a blowing up balloon. There is no "pressure" in the traditional sense to universal expansion. Neither is there stretching. Space is growing with time everywhere, in a direct and exact relationship. A single expanding balloon, is just one kind of time, and it curves. "Space" is not curved at the larger scales, and only curves at the smallest scales.

The empirical fact is that space is expanding, in that there is more of it every moment. More space is coming into existence all over the universe all the time. 70% of the energy in the universe is contained in the vacuum of empty space, and this amount will only increase until it is nearly 100%. This expansion is caused by "gravity". Most of the "gravity" in the universe causes the universe to expand. Gravity is not inherently an

attractive force. Gravity is a description of how space and time move. We created the theory of the big bang before we really understood this fundamental and observable reality to the universe. If 70% of the universe is dark energy, then it is more likely that dark energy created the universe and not a super dense singularity.

Things moving away from you, are creating more space between you and it, and this changes even your perception of how fast time is traveling relative them. This is not the same as the kind between galaxies. Things are actually moving through space. Our perception of time can be changed by how things move through space. Time is happening everywhere independently. It is arising out of every point in the universe equally. You should think of time more like the force filling the many balloons. The more you move through space the slower time becomes, if you move through a lot of space very quickly time begins to slow more and more, until you can't even go faster. You move through more balloons as you accelerate through the universe, and if you have more matter, you interact with more of these balloons. You can see these differences in the perception of time, as simply a description of differing amounts of space, that you can't see, because that "space" is "warped" locally.

Time

Relativity is about the observation of things that are separate from us, moving in space and time. Space in the universe is flat, time is curved. Time moves differently depending on where you are in the universe, and what you are doing. Time can move very slow, until it is almost stopped. Time however cannot move faster than the speed of light (the speed of causality). Space moves very differently around things when time slows. What

you can observe changes with how time and space work around an object. Imagine if each of those black dots on a balloon was pushing down on the balloon and creating a dent. The bigger the dent, the bigger the dot. The deeper the dent, the more "dense" the dot. A dent is a distortion in time, that is observed as a distortion in space. The more something resists the balloon at a single point, the more time itself slows down because it is "expanding" slower at that point. If there is no resistance to change, then time flows at its fastest rate, such as in the regions of vast voids between galaxies. When space changes, time changes. They are the "same" thing; they are inherently connected.

The way space moves around objects changes how we see it. The changes in the amount of space between us and a thing, changes our perception of time between two objects. The faster the space increases the more the difference in time between those two things. Things moving away from you slow in time because there is more space between you. If something gets far enough away and is moving fast enough, our observation of it, can slow to almost nothing. We see that time as slowing to almost nothing, and the thing accelerating away from us at faster and faster speeds, doesn't feel themselves slow in time. This perception of slowness is because we are observing more space between two things. The space is not this fixed thing, and our perceptions of a things time is based on how light moves through the altered space around it. It can even slowdown in time, so much as to disappear from our existence. If it didn't disappear, it would have to travel backwards in time for us to observe it and that would just be not cool. Changes in the observed rates of time, preserve locality, and keeps things in reality. The faster you go, the more energy you gain, the more mass you have, the more you resist change, the more distorted the perception of space is around a thing. You can go faster than the speed of light, and if you do, you will enter a vast nothingness of nearly infinite space around you, because you increased space so much that it can no longer be a part of the universe you were in. You can accelerate a thing so much that it becomes like a black hole, possibly.

Time is time wherever you are personally. At any location time is experienced exactly the same. You can't experience time moving slower or faster, in your "frame of reference". If time moved faster or slower, you would experience it exactly the same. Time is how fast things can change, and that is set by the speed of causality. By its very nature, it is a constant, because if it wasn't time wouldn't exist. Everywhere in the universe experiences time differently relative to you, the more distance it has from you. Time is absolutely the same relative to the place it's in. Time is also absolutely set relative to the universe itself. Time actually needs to be observed happening at different rates, so that universe can actually exist. Things have to be separated from each other. Different places need distance between them, to be different. Time is just a sort of distance between things, that lets the universe exist and have different places in it. You need to wait before you can see something happening far away from you. The farther away, the longer you need to wait to even see it.

Space-Time Gravity

Gravity is just a property of space and time. It can both push things apart and pull things in, depending on the situation/geometry.

If you take normal Einstein physics, a black hole has time come to near stop at its event horizon. Inside of its event horizon space expands towards infinity inside it. You can have a black hole with a set and defined surface area on the outside, like a sphere or an empty ball. The spheres outside area doesn't change. Its inside area though can fit a whole galaxy inside it, in a manner of thinking. You can't observe how it moves like a normal thing. Space just goes right through things, and you can't block it with anything.

Space, "moving around", is what creates gravity. Gravity is how space and time move in the universe. Time creates "gravity", by it being slower towards more massive objects, and between two regions of different time rates, they "perceive" the amount of space in the other differently.

Space is form of time, without direction. The curvature of space time is described by how much it dilates time. So, it is time that is curved, and not space exactly. You measure the curvature, by how fast clocks move in a given region compared to another region. If there was more space, then there would be more distance between two things. We can create another kind of distance between things with time, and in how fast time moves between them. That difference in time between two things is expressed by the laws of gravity and inertia. The "curve" is the difference in the rates of time, and those differences create "acceleration" towards slower time. Time is shorter near large dense material objects, and the very distance between things is shorter towards them. Basic visualizations (other than balloons) of this are like memory foam mattress, heavy dense objects make deeper impressions, and things roll towards those objects. Two large massive objects, sort of roll into each other by the bending of mattress space time. The slope in the dent is time, and that dent is a shortening of the "distance" So a straight line follows the curves of time. We can only see time, as things change, yet it functions in much the same way as space in of itself works. Time is like a very special form of space that has direction and can curve.

Equivalence

Equivalence is one of the most important ideas in Einstein's work. The speed of causality causes things to be locally contained. This is called "locality". Space-time arises out this quality and keeps things in their own area, by making space and time, change with how

things move through space. You can't really say what is moving and not moving in space exactly. How things are observed between two objects is changed by their speed, because space and time changed as things move. General relativity came together with a simple beautiful thought. A man falling from a building feels no gravity or acceleration. It is not different than falling through empty space. And if you stand on the ground, or are being pushed up in an elevator, if the force pushing on your feet are the same strength, then these two experiences are exactly equal.

We all know the classic story of Newton under an apple tree, and getting hit by an apple, and this somehow being an insightful moment of how gravity works. The story is missing a key part. He looked up at the moon, and realized what made an apple fall, makes the moon orbit the earth. He created an equivalence between planetary motions, and gravity on earth. No one had done that before. The heavens and the earth were treated as two very separate worlds, with different laws. He then worked out the math/laws to make it possible to explain such large and small things with the same math. Insights in physics are often driven by symmetries. You have to find the right things to make symmetrical though. You find odd pairings of things, and you find symmetry between them, as a way of reducing the universe to more simple laws.

Einstein took Newton's insight to another level. Free falling and gravity are equal. Things just flying in a straight line and gravity are equivalent. The gravity you feel when you stand on the earth, is equivalent to the experience of being pushed up in an elevator, or pushed back into your chair when a jet plane takes off. He wasn't saying they similar, he was saying they were exactly equal. Now making math to explain how that is possible is a lot harder than what Newton had to do. Newton just had to make simple equations to show that movement in the heavens and on earth, were ruled by the same principles.

If you are falling towards the earth, you actually will feel no acceleration. You only feel any acceleration, when something is pushing on you. If you are falling towards the earth, you aren't really accelerating, in a classical way. You won't feel acceleration until you hit the earth.

When you stand on the earth, it is pushing up on your feet. This is the big change you have to make in how you think about gravity. The earth is pushing up on your feet, you are not pushing down on it. The earth in a certain way is expanding out and accelerating you from the ground up. That is how the math works. How even physicists understand that is not all that clear. The fact that gravity is pushing up on us, to make us stick to its surface, and we aren't being pulled into it, is confusing and yet also makes sense. There is no force being acted on you from a distance pulling you with gravity. Gravity doesn't pull you, you feel gravity when it pushes on you, and that should be the important idea to make it feel more real, and realize gravity has no magic power to pull things, and is much more like a ball hitting a ball.

At a deeper level this is making gravitation and inertia equivalent. Which is not such a crazy thing. If you drop 2 balls of very different weights, in a vacuum on the surface of a planet, they will fall at exactly the same rate, and hit the ground at the same time. If you have more inertia, you resist change more, though your experience of gravity is the same as things with less inertia. In a rocket, jetting away in space or standing on the earth, that acceleration pushing on your feet is the same thing. If you are just moving faster, then you have more inertia. A car that is driving faster has more energy and takes more force to slow down, it has more inertia. If you have more material mass, you have more inertia. These are the same thing. Inertia doesn't relate to how fast you move; it relates to how much energy is in a system.

Everything follows straight lines if they have inertia. If something isn't being accelerated, it is following a straight line. If you are going to make math to explain that, then you are going to need to change the definition of a straight line. It is going to be very weird. Curving space and time, made this possible. Einstein just made what things experience as straight, different than what we see, with our immediate senses. They are still absolutely straight in space, they just get curved by time, and thus gravity is a property of time and how it changes.

Inertia is where things keep doing what they are doing. This is basically a straight line. If an object is just sitting and not moving, it is following a straight line in time. Though an object just sitting on the earth is being accelerated up from the earth pushing on it. Gravity pulls and curves things towards it. Objects we throw up, curve back towards the earth and hit the ground again. It is like bending the lines on graph paper to make something curve, rather than making the line curve on a perfect square grid like graph. The curve is not in space; the grid is warped by a hidden dimension of time. The straight lines we see on a graph's grids, can be altered. If you graph time as a dimension like space on a graph, this creates the curves (linear algebra makes this easy to understand). Things are moving in straight lines on the graph paper, the graph paper itself is being warped and changed.

We measure how fast clocks are moving, to measure a gravitational field. We measure time dilation, to measure gravity fields. If you had a stack of clocks reaching from the surface of the earth far into space, then all those clocks would have different rates of time passing on them. We measure space-time curvature, by measuring time. Things fall towards the earth, because it is the shortest route in time. We feel the earth pushing up from under our feet, because of time pushing out (or in) on us.

Newtonian Relativity

Newtonian relativity makes velocity and rest equivalent. You can't define something as moving fast, unless you describe it in relation to something else. Anything you can describe as moving quickly in one direction, you can describe as not moving at all. A ball thrown up in the air can be accelerating up and decelerating down, or you can flip it and say the ball is decelerating up and accelerating down. There is no difference in the math. Negatives and positives can be switched at will, and rest and velocity can be switched at will, all depending on the perspective one takes. The big important thing to realize is that there are more of one thing moving in one way, than another thing. If a ball is moving in a certain direction at a certain speed, we can say it has speed because the earth is so much more than it. Reality becomes what is bigger and more, with speed being more about the things relative to the greatest amount of stuff. The earth is so massive compared to our lives on it, that we can easily lose sight of this relativity between motion and rest.

Special Relativity

Let's say you have a special rocket, and it can always accelerate at 8g non-stop all the time, with endless energy to do so. It is going to be a very special object that is different than the rest of the universe. Compared to the universe it is going to be changing the most compared to it. There is no natural thing in the universe that can do this. Now if you had normal rockets doing this and leaving the earth, you couldn't properly describe one as more proper than the other, if the rocket where to stop accelerating after gaining tremendous speed. To the rocket, the earth would be moving away extremely fast, and in effect be weightless in space. No matter how fast a spaceship goes in space, if it is not accelerating, it is "weight-less" inside, and no different than just floating in empty space

unmoving. The world you came from though will be moving away very quickly. Your future is moving away from where it came from.

Speeds effect how space and time function. With small differences in speed between a rocket and the earth, space-time doesn't change much. This is easy to see when we compare no speed to light speed, and put that difference in speed relative to that.

$$(\text{Speed of the rocket} - \text{Speed of the earth}) / \text{the speed of light} = \text{how much space time alters.}$$

The bigger the speed of the rocket, the closer this number gets to zero. Changes of 100 times the speed of sound, will barely change this ratio relative to such a massive number like the speed of light. As we get numbers approaching 1, we get closer to moving at 100% the speed of light. Think of this as ratio of (a thing) to (space-time). If we look up at the rocket and it going 90% the speed of light, we will see it as having slowed down in time and being crunched up, it flattens in the direction it is moving. It won't stretch out in space-time, it will squish along the direction it is moving and up down relative to it, will stay the same width. As it approaches the speed of light, it flattens towards nothing. This seems really weird to see from earth, to see a thing with a normal shape flatten because it is moving so much faster relative to us, this though makes sense when one considers the way time alters space around that rocket.

From an outside perspective think of it like this. You see 1 meter of space, and the rocket sees 2 meters of space, on each side of it. As the speed increases the rocket sees more and more space, and from the outside, you see the same amount of space, and less and less rocket. It is like a line is stretching in, instead of out. The space stretches

the ship doesn't and we see the same amount of space, and we see the new ratio of space and time. The rocket itself is creating gravity. It is not virtual gravity, it is real gravity. Any acceleration pushing on you like an elevator going up, a rocket launching up, or a rocket accelerating forever, are creating real gravity, described with the same math as the one we use to describe gravity around planets. This equivalence is what creates general relativity.

Trying to go faster than speed of light is just like making black holes. You can think of the black hole event horizon forming into being behind you, as you accelerate to faster and faster speeds approaching the speed of light. "Behind" this 8g rocket is not different gravitationally to "below" you as you stand on the earth, except for the rocket has 8Gs, and the Earth 1G. With stars, the denser the star gets, the deeper its dimple into the universe becomes, and this is the same with how fast any object accelerates to relative to the rest of the universe. Space is forming around the ship, relative to the universe, and eventually all that space will consume that ship, and separate it from the rest of the universe. From the outside, there will be a bending of a small region of space and rocket flattening out to nothing. These are different perspectives of the same thing. That extra space being created is "time". And when that "time kind of space" overwhelms the ship, it enters into a new time frame separate from others.

From the rocket's perspective, the rest of the universe is squishing into nothing. The faster it gets, the more things it is moving away from squish towards smaller and smaller sizes. If it reached light speed, things would squish out of existence for everything but the rocket. If you see this just from a rocket's perspective, the universe begins to shrink to thinner and thinner lengths, the faster you go, and it takes more and more energy for each bit of the universe to get smaller, and more energy to escape from it. For the rocket space around the craft is stretching. It is not harder to go faster, if a

rocket is accelerating towards light speed, the “people” inside will still feel 8Gs of acceleration.

A rocket could accelerate at 8g forever, and experience 8g acceleration forever if it had the energy to do so, and not be affected in its acceleration at all by how close it is to light speed, relative to the rest of the universe. The rocket experiences more space, and simply is moving through more and more space the faster it goes. Space itself is increasing, from the perspective of the outside observer. The reason it gets flatter and flatter to the outside observer is because there is more and more space there. It is seeing an object relative to its distortion of space-time. You are in a sense observing space-time and the object being represented together. The rocket hasn't squished, only your perception of the space around it has increased. If that rocket were in theory to go over light-speed, it wouldn't become infinitely massive or energetic, it would simply leave the universe we live in. There would be more space time around it, than light could escape from, and reach us. It would have to create its own independent space-time from our own. It would move out of any possible influence on us, and cease to exist as a thing we could ever observe. It would cross an event-horizon, like a black hole, and its future would split with ours. Acceleration and speeding away from a thing is altering the very future and what can cause what to do what. It would create so much space between itself and the rest of the universe that light itself couldn't travel across all the space that came into being.

As you approach the speed light everything behind you will start to fade to black and become vastly more distant, with more and “space” to travel through. Only the destination you are speeding towards will be clear. Eventually all the blackness from behind will come around, and slowly shrink the forward view of the universe. At light speed the universe will become black and the last dot of light that you are moving

towards will be closed off with the blackness of space, as you enter into your own universe. You created a new time frame by accelerating. You literally generated new **time**, and it surrounded you. All acceleration is a change in time, and time from a deeper more fundamental understanding of it as a physical thing, that you can describe objectively.

General Relativity and Black Holes

A black hole is like one of those dents in a balloon or a mattress, that goes vertically down really far. That curve downwards is expressed in time dilation. The deeper into the hole the tinier changes in distance can have greater differences in time, which just means more intense acceleration the closer you get. This warping can look like some bottomless pit falling out of space. Visualizing the experience of going through a black hole is not much different than approaching the speed of light for an accelerating object. Approaching an event horizon is approaching a point where the entire universe will cease to exist for anything crossing over an event horizon, at a simple level it is akin to moving into another universe, with different space-time.

To the observer approaching a black hole, there is no steep indent in space-time. The more you are on a sloped space-time, the more space you will observe. There will simply be more space on the side of you away from the event horizon. The event horizon will become more and more the shortest distance to you, and the points away from the black hole behind you will have more and more space growing behind them. You will simply experience more space, which is a representation of the curvature of time. Like any large gravitational object there is more space on the side of you that is opposite the object. Space is "expanding" through you in a sense. Where space is the smallest and

time moving the slowest is pointed straight at the center of gravity of a massive gravitational object like a planet or sun.

Gravity is created by differentials in time on each side of an object, and these differentials in time are expressed in a shifting of the curvature of space. The curve of space is just a bit more space on one side a bit less space on the other. This difference in space between two sides of an object can be measured with two highly accurate clocks. If one clock is pointed towards a black hole and the other pointed away on the other side of an object, the closer they get to the black hole, the greater the difference in time between those two clocks, and this will directly create a distortion in the perception of the amount of space.

Now let's imagine you are on the surface of the Earth, and that the surface of the Earth, wraps up and around you, so that you can see the other side of the planet, directly above you. It would be like being on the inside surface of an empty ball, where everything on the outside would be mirrored on the inside. You would be able to see the entire surface of the Earth by just looking in every direction up and around you. In many ways, this is what it is like to cross over an event horizon. The surface of one world comes to envelope you, and encase you in a new enclosed universe of only itself. The more you approach the event horizon the more space from the black hole will surround and consume you. You can imagine the gradual step by step transition from a positively curved round planet, slowly bending itself into a new curve around you. You are standing on top of a turned over bowl, the bowl slowly flattens, and then eventually the edges of bowl slowly rise up around you. They eventually begin to form a sphere above you, with circle that forms above you, that gets smaller and smaller as the top of the sphere closes of above you like the iris of a camera shutter.

That last little window of a circle that leads out of this negative sphere, is the direction one would have to go to move back into the real universe. Once that window closes, it can never be opened again. This curving is caused by massive amounts of space between you and the rest of the universe, the closer to a black hole you get. You will see the literal event horizon rise up around you as you approach it, and it will eventually surround you completely. In terms of crossing the event horizon, all you experience is just more and more space. There becomes this massive separation between you and the universe that you will see as actually space. That black hole is a near infinitely deep hole and when you are "standing" deep on the surface of the steep hole, you see only more flat space and not deeply curved space. The universe outside of you, is what you see curve.

In a sense, space is expanding through you. As space moves in one direction it is grows in the direction it moves towards. These tiny differences in spaces size as it moves in certain directions, is described by time. As space moves in any direction it expands, and thus the whole universe expands as a product of this spatial expansion that is caused by time itself. The expansion of the universe is a product of time, and time itself is this very expansion of space in the direction it moves. It is a slight and tiny force, that with enough of it adds up into something significant. Nothing can block time itself.

Now it is time to cross that event Horizon.

Crossing an Event Horizon into A New "Universe"

As the last bit of the old universe closes off above you, and you make your final decent into a black hole, you will become aware of the "singularity". When you cross through

the black holes event horizon you will be at the center of a massive sphere, with its walls at large distances from you. The walls are the singularity. Inside a black hole the singularity is not experienced as a singular point, it is seen as the entire inside surface of sphere. The event horizon is the exact center of that sphere. There is no space and no direction you can point that leads out of a black hole. Vast amounts of space exist between you and the old universe, such that space itself is a wall between you and the old, and all you can see is just empty space. In a black hole, all futures and all directions points towards the singularity, thus the in any direction you can spin or orient yourself is the black holes singularity. If you move in any direction, you move closer to the singularity. You can almost imagine the surface of the singularity and the surface of the event horizon as very similar the event horizon rises up around you to form a sphere of space and time, and when it closes the inside of that surface is seen as the singularity itself.

Now let's imagine you orbited the event horizon before crossing into it. Instead of just falling through the event horizon, you accelerated up to 99% the speed of light, with an orbit just above the event horizon of a black hole. This would be the only way to sort of stand still when you cross a black hole's event horizon. Now a magic rocket could be half across an event horizon and half outside of it, in orbit at 99% the speed of light. Rather than be like a ball spinning around a roulette table, the very surface of the roulette wheel would be curving up and closing above the spinning ball. I am not sure the exact geometry of this or how rocket can be half in and half out. The gradation between the event horizon is a smooth surface of a lot space.

Now the event horizon may very well be a point in space and time, where the difference on side to the other side is so different in time, that even at the quantum level of a particle that the differences in time between each side of the particle are so extreme as

to rip that particle in half. If you imagine a 1-km-long-bar of 10-meter-thick steel going towards a black hole, it would certainly stretch into a thin wire way before it reached the event horizon. Difference in time will cause stretching at the most extreme forms of dilation, it would cause noticeable stretching even at the level of an atom. This is unknown to me at the moment, how steep an event horizon is, and the crossing over is more of a visual metaphor, than something a person could actually do.

Crossing the Singularity

The singularity is around you in every direction. It is almost like quantum physics is made real in the large scales of the universe. The expression of a super dense singularity is experienced as a spread-out form in every direction you can see. You are like on the inside of a quantum universe. This surface won't be uniform, and not all directions will be the same. The singularity will have variations across its surface. This is inherent due to the fundamental laws of quantum physics. There is an inherent uncertainty as to the state, momentum and position of anything you observe. Quantum uncertainty is an actual feature of quantum physics and not just a problem with how we observe things.

At the deepest levels of our universe, at its smallest scale, things have randomness. This randomness is one of the great mysteries of all science, social sciences, philosophy and religion. At some level, the universe just flips a coin which direction it is going to go. Up or down, and it is completely random and unpredictable. Tiny variations in randomness can create tiny unique and fleeting structures. If we take empty space, at any point in empty space, anything can appear in that space, with some probability. I exist to some degree in every point in the universe. There is no such thing as an empty box. There is no such thing as zero. If you ask how many apples there are in an empty box, that is perfectly isolated from everything in the universe, the number is never zero. There is

always a certain possibility of an apple being in that box. It is number so small as to be incomprehensible. General relativity can be described as the geometry of space time. It uses the actual shapes and curves of lines in geometry to describe space and time. At the smallest quantum level these sorts of generalized geometries simply can't model uncertainty. We would be looking at quantum uncertainty itself made massive in scale, possibly. Does geometry create quantum uncertainty?

The challenge in understanding what it is to hit the surface of that singularity is to understand the geometry of quantum physics. That surface will have a structure and patterns in it. Those patterns will define what the shape of quantum physics underlying reality really are, rather than just descriptions that work perfectly. Space and time are emergent properties of the geometry of space and time at its fundamental root level. Quantum physics describes how that quantum space time alters systems through time. Matter rests on space and time, and at the tiniest levels it doesn't move like things do at the big. Time is truly a defined and specific structure at this level, that can never be observed, except inside a black hole.

Combining General Relativity and Quantum Physics

Fine Tuning

Physics has a puzzle to solve. It seems the universe can't have its rules if there is just one universe, and there not be a god. Many of the qualities of universe are "fine-tuned". Change one constant of the universe just the tiniest amount and the universe would never exist. With matter, atoms, stars, and planets, there are a lot of "coincidences" that

need to line up to just create the basic universe we live in, with odds of this happening, in ranges of like 1 in 10^{500} . That is a number vastly bigger than all the atoms in universe times the number of atoms in the universe. A common theist would look at that, and say, yep, God created the universe, and their reason would be so wrong. Science was right to question a God, and to find some way this could be possible. How can the universe be fine-tuned to the most utmost level, and it be possible that this was not done by a God creator?

Our own universe is already quite big, it is probably trillions of times larger or more, than the universe we can even see. We will never see nor travel to any of it. A basic idea is that there are many universes like our own super massive one, that sprang into existence at the same time, and that every possible initial configuration of the universe happened at once. You don't need fine tuning if every possible option for that tuning, happened, and we just happen to live in the one that worked. This is the basic idea of a multi-verse, a reason why we need to have more than one universe, with different physical properties than our own. This idea of multiverse is often made into absurd ideas of alternate realities, time travel, and other dimensions. There is nothing magical, nor that amazing about a multiverse, it is actually a bit boring an idea, that is little different than there being more galaxies in our universe, than our own. These other universes spread out from each other by great distances. They can just be really far away, beyond any scale of imagining. It is almost another dimension of the universe would need to create to describe that distance and not our normal 3D space dimensions in our own.

If you could leap far enough in any direction instantaneously, at a certain "distance", our whole universe we live in, would be like a distant star fading away, at a single point of light, and bit farther, absolute darkness and nothing, and bit farther, another universe coming like a small point of light, and we could go enter that universe, and its laws

would be different, though since it is right next door, it would probably be indistinguishable from our own. To actually get to a universe with laws even different than ours would require traveling many whole universes and the distances in between. This is far, and basically so absurd to be nonsense. There is no way to test such a thing. No way to experience any aspect of this, and no possible useful technology to make from it. The more interesting thing we can investigate is how a universe comes into being. Understanding how reality itself forms, and understanding that other universes can exist, can free us to create reality very differently.

Now my idea of "distance" may be very wrong, and very hard to define, because we can never escape our universe, nor even the tiny part of it we are in. the multiverse in this simple explanation, is not what physics believes might be possible. It is just one aspect of the idea.

Unified Field Theory/A Theory of Everything

General Relativity (GR) explains how space and time, are shaped, in such a way to explain gravity. Gravity is able to act on things at a distance with no particle or thing moving between them. Einstein explained this, "spooky action at a distance" by changing how we define the geometry of space and time. Gravity curves space and time, and that curvature changes what a straight line is from our perception of it. There are many ideas in GR that are deeply cherished in physics, and changing any one of them is a big deal. GR is a deterministic model. At its heart, it is still Newtonian (classical).

Gravity is a description of time, and how things point towards the future. Things with more time, experience more time, and thus are slower moving in time. Objects then

orient towards things with more time, and point towards them, and eventually become entangled in that things time frame. The future can be a literal direction a thing is pointed towards and moving towards. This movement and change has to be described with changes in time, rather than just pure motion. At extremes of lots of time and little time, very unique and unintuitive things happen, relative to movement on Earth. We live in the time frame of the earth, and it takes a lot of energy for us to change our future away from it. To travel to another star is to fundamental change one's time frame from their home stars. Seeing gravity as a direction time is pointing, and that its force is driven by changes in geometry that can only be seen with "clocks", is odd. When we go to the tiniest level, the entire way to understand gravity completely changes. It becomes nearly impossible to describe with discrete and indivisible objects.

Quantum Physics, is a way to explain how things work at the tiniest levels, the quantum level. We can't directly observe quantum physics, in the same way we can observe the motion of stars and galaxies. The way matter works at the smallest level is completely contrary to how it works at large scales. Quantum physics is defined by quantum uncertainty. It explains this with probability wave functions. Matter is a set of probabilities understood by the mathematics of waves. The math to explain quantum physics is done with statistics. It works nearly perfectly with empirical observation, and yet defies any intuitive way of understanding it. Things both exist and don't exist, at the same time. The universe is purely random, at the quantum level, according to the math that explains it and the empirical observations made of it. It is non-deterministic. True randomness and not chaos rules. Physics though likes to explain it all in nice ordered deterministic ways, and this creates tremendous conflict among physicists. The current models of physics always strive to push the random non-determinism of quantum physics into determined models, no matter how wild the idea has to be to do it.

These two very contradictory theories of GR and Quantum Mechanics are equally good at describing their scales of the universe. To bring these two ideas into a singular idea requires 2 possible routes, use quantum theory to explain GR, or use GR to explain quantum theory. At a philosophical level there are two possibilities, 1. Build a deterministic model that explains quantum uncertainty, or 2. Build a non-deterministic (probability) model, that can make a deterministic like system. Physics has basically two theories to do this, "Many Worlds" (determinism is right) and the "Multiverse" (non-determinism is right). It is for me, just in my opinion, a difference between a world from everything, a singularity, or a universe from nothing, the ground state.

The Multiverse/Megaverse:

The universe we live in is vastly larger than the one we can see. If you took every atom in the knowable universe and made that into an entire independent, universe, that would still be a very small size. The universe is determined by the initial conditions of the universe, where the laws and the constants of the universe are determined for the life the universe. Currently we describe this with differing big bang theories, in the study of Cosmology. Our universe has many constants, that if changed even the slightest bit, the universe wouldn't exist. The likelihood of this happening by chance is very slim. Stable evolving universes like ours are very rare, with all the possible conditions a universe could have. Most all the universes out there, are just duds, where nothing really happened.

The multiverse is an attempt to explain a fine-tuned universe, with no creator. How we can have this perfectly made and balanced universe, and have no god that created it. In order for this very rare set of initial conditions to have happened, there must have been many universes. All possible initial conditions happened, and we just happen to be in the

one that allowed for life. 10^{500} is a number that get put around for the number of universes that could be in a multiverse. Basically, if you try enough times, crazy rare things can happen. It is not an infinitely big multiverse; its massive size can only be understood in statistics and not in quantity of space to conceive of existing. It is best to imagine the multiverse as colorful images and dream like fantasies of statistics rather than try to "see" the actual universe.

"Parallel" worlds, are not something we can just jump into, any more than travel 1 trillion light years, in 1 second. They are far away. The "Mega-Verse" is a better word for it than the multiverse, because it describes a truly massive universe, with many massive universes. These universes, all exist at the same time, for a basic form of the theory, and are separated by great distances from each other. In a sense our already large universe, which might be trillions of times bigger than we can see, is then multiplied by 10^{500} possible universes. I personally think of this is in a chromatic way. Sort of like a circle rainbow, and our universe is a tiny sliver or even photon in the spectrum of a circular rainbow. And every photon in that rainbow is its own universe, with a slightly different color and wavelength. Photons close to each other would be nearly identical. All the universes "near" us, would probably be indistinguishable from this one, if they all began at the same moment. A chromatic rainbow multi-verse, with color defining each universe unique tuning of its starting properties, makes sense to my imagination. Physics often uses color as analogy to explain quantum theories, so a made-up name like, Quantum Chromatic Cosmology (QCC), wouldn't sound that weird, in physics.

Sequential Multiverse (Time and Space)

Universes could be born out of other universes. In this model, a "multiverse" is separated by lengths in time, and not just distance. You have to travel in time to go into

another universe, rather travel across distances to reach another universe. We can have a multiverse divided by distances in time and/or space. A more developed version of the multiverse also evolves in time, if you go forward in time, you can enter another universe, that arises out of this one. My work in this book will be about how a universe evolves into another universe through time. The multiverse only concerns itself with the initial conditions of the universe. I will explore how to build universes from nothing, that evolve into other universes. We can talk about what happens before the big bang intelligibly, even though, all time we experience begins at that moment, because of how space and time can flip their relationship to each other in white or black holes.

Many Worlds Interpretation (MWI):

A Many-Worlds-Interpretation (MWI) model of the universe can explain a lot and it is a very consistent model. It is a difficult model to fully accept with the existence of free-will. Quantum theory uses a completely probabilistic model to determine quantum states, no quantum state can exactly be known, ever under any conditions, and can only use probability to determine it. It is spinning up with a " $\frac{1}{2}$ " chance, it is spinning down with a " $\frac{1}{2}$ " chance (1 over the square root of 2, if you get basic trig, and use the Pythagorean-theorem). This is the reality of the quantum world.

- Uncertainty principle and Schrodinger's/Dirac's equations.
- We describe particles positions, with probability waves
- We can describe things as being in 2 possible states at the same time.

With different probabilities for those states being actually material real.

Physicists are very divided by this, and think of this probability as more of a math trick, and that we don't understand the real underlying conditions. One solution is to make every possible state that a quantum state can be in, to be contained in its own universe. Every possible state that a quantum particle can occupy as described by a quantum

probability wave, is its own universe. Every particle in the universe, would have a universe for every single state it could be in, for its whole life. 10^{500} particles, time 10^{500} states, with 10^{500} universes, makes for a lot of universes.

We can imagine our universe as like a sheet of paper, and we are just experiencing a line being drawn through that, except that the piece of paper would be like 4-10 dimensions instead of 2 like a flat piece of paper. (take your time to wrap your head around this). if you were to follow a line, that would be one universe. Though you could in theory just draw a line anyway you wanted, and each drawing of a line would be a valid universe. All the different dimensions are just different ways that things can move, the different possibilities. A simple 3-d block just wouldn't be able to contain the possibilities, that quantum states could be in. We are just a pencil line drawn across a 10-dimensional flat piece of paper. The line would be time itself, the direction our universe took through this super universe. The overall 10-dimensional block is just a singular thing, where everything that could ever happen, happened. A sheet of paper (a vector space) has Infinite lines with every possibility the universe could take in its entire life, existing. This also assumes a deterministic universe, with no free will, no freedom, and completely meaningless universe.

The main idea is hard to understand. If there is randomness in quantum physics, the only way to make it determined is to say that all those possibilities something can exist in, exist, and are real, and we just live in a simpler universe where we describe them as probabilities. This is a bit like the allegory of the cave, and there being a higher dimensional world that we only see as simpler dimensional representations of on a cave wall. For MWI, even outside the cave is its own illusion, and a casting on the wall of a higher more real dimensions. Our freedom is an illusion created from a perfect world we

can never exist in or experience directly. This idea is a very powerful one for a creating a strong scientific model that can account for "truth" without probability.

Big Bangs and My Argument on White Hole Cosmology

The universe doesn't have to begin in a singularity, and there is no scientific evidence nor working math for that, in physics now. The big bang theory is not a theory that the universe began in a singularity, all the evidence sort of proved that wrong. What we call the beginning of the universe in the big bang theory, is the exact moment after what is called "inflation". At 10^{-32} seconds is the moment inflation ended, and the moment physics has no idea what happened before it. They assume it was a singularity, and thus use the word "inflation" to describe it. Well if existence didn't begin at the big bang, and the universe we live in came out of another universe, then a single point of super condensed energy is a very unlikely starting state for a universe. The universe beginning from a region of massively expanded space is more likely. This size would be like that of the size of our observable universe at the moment "inflation" "ended". This would of course be an area much larger than we could ever see in theory, because our universe is much larger than what we can ever see, unless we are the exact center of all reality and are the universe. This is so unlikely to the point of being impossible.

Our universe most likely begins in a region of massively expanded empty space, like that described by the size of space caused by cosmic inflation, though many times bigger, depending how much bigger our little universe is compared to what we can see. An interesting fact about the universe is that the net energy in it is zero, there is equal amounts of negative energy, as there is to positive energy. Gravity has negative energy. So, the universe comes from nothing, and nets out in everything, to be nothing. Time then seems to have negative energy.

Chapter 12: Space And Time

Origins of The Universe: Structure of A Big Bang Singularity

I presume the universe begins from nothingness, from purely random forces of probability. The universe at its very root and origin is utterly and completely random without form, meaning, or structure. True nothingness is the "thing" that preexists time, and is the minimal non-material/non-tangible thing, that anything could come from. If probability defines energy, then energy existed in the nothingness before existence, at least the potential for it. It was energy without order, structure, causality, or direction. Energy itself was and is just pure random probabilities in nothingness being expressed. Higher energy things are less likely than lower energy things, and this instability in probability makes them more likely to decay to more probable lower energy states. This correlation can be so high as to be the basis of describing energy itself. Matter is a stable form of low probability fields that don't decay into higher probability low energy fields.

Energy of some form most likely existed before time itself developed, such as how we describe the big bang singularity as a massive ball of energy without time. It was all potential and probable energy and no actual structure, form, causality or time. The singularity of a big bang is itself nearly infinitely small. It is something best described by my 0D models of space, where all of reality is a single non-determined point. The universe comes from a space of nearly infinite smallness. Imagining true nothing nothingness is a hard-mental challenge. Empty space is still something, it can be measured and you can have more or less of it, in mostly exact amounts. True nothingness is like zooming in on any point in the universe and you keep zooming in as far as you can go and never stop zooming. You enter a space of the infinitely small. You

begin to relate to minimal scales the universe forms from, that we can't observe yet, with science.

Space is something. This is truly nothing. Only the possibility of something exists. In nothingness, there is still "something". Timelessness doesn't mean eventlessness. Moments/events come into being from chance, and then die. A singularity of nothingness, can form into anything, and just end. If you imagined our universe having a big bang, then a big crunch, with nothing happening in our universe effecting the next universe, this has a certain timeless quality when you step back and look at all the universes coming and going out of being between big bangs and "big crunches". There is no way to order a sequence of which one came first and which one went after. There is no chain of causality in any form. The random fluctuations in a big bang singularity are equal to the birth and death of a whole universe, without time, so nearly infinite ones can happen and there is no means to define them in sequence. You could just say that all the universes that came and went, all happened at the same time, or in sequence in any order, and it wouldn't matter.

This space where the universe ceases to exist as a thing bound in time, is timeless, and yet is full of energy, possibility and "events". This is a timeless, highly active, nothingness. Just randomness popping in and out of existence, with no real past or future. One event doesn't affect the next event. The random structure has no real form or consistency that can have time, nothing causes something else to happen, nothing can evolve and develop into something else. In theory, anything could pop into existence in this nothingness 0D universe. Our whole universe as it is now, could pop into existence, with nothing happening before it. The more complex the structure, the more unlikely for it to possibly happen in a given moment, from one moment to the next. The big bang could have just formed into the exact universe we see before us, with

us as we are exactly now, with nothing happening before that. The only way to disprove such a reality is probability. It is possible, and it is also the least likely of all possible things to explain our existence. Finding a model that uses the simplest and most likely configurations of the initial universe is a key problem in physics. There are a lot more simpler systems than there are complex systems, and the universe we have most likely evolved from a sort of rare and special simple system.

Maybe trillions upon trillions of “universes” popped out of nothing, and just died in the next moment, one after the other. In a void of nothing, a possibility exists for something to happen. Something happens, and then it ends, the following universe will have no relationship to the one before. There will be no enduring structure. A universe would not have time, nor size, of any meaning in our imagination. Pure nothing, creating meaningless somethings, and back to nothing. Endless moments of near existence, or just nothing. The singularity doesn't need to have all the energy we see in the universe contained in a single point. It just needs the possibility to randomly form into a structure that endures and evolves into the one we have now. A very simple universe much different than our own could have preceded the sort of high complexity universe we see today. Something like the vacuum/space is the most likely thing to pop into random existence out of nothing. The most minimal thing in this universe that can exist and grow, would be the most likely thing to first exist.

Dark Energy, Expanding Space, And Entropy

The vacuum is the least possible thing that exists and is the most likely thing to have been created at random from nothing. The vacuum has the minimum possible energy anything can possess in the universe, and has the least of anything, that a thing can have. Our, normal, very conception of nothing is the vacuum. Nothing is really nothing.

And in nothing random things can happen, because all things are possible. A bunny can just appear from nothing, fully formed, and be the whole of the universe, as a possibility, and this is standard physics. It is also good physics. It is the most likely way to create life, according to probability (Boltzmann Brain). Impossible things don't happen. Very improbable things can happen and do happen. The most direct way for the universe to have a sentient brain, is for it to just randomly appear out of nothing, in a large enough universe that has existed over a long enough time. I called this a "Heisenberg Apple" in an earlier chapter, to describe any object just coming into being from nothingness, and yet in any empty space there is some probability of the possible existence of any possible and real thing emerging in that space, thus making there never be a zero quantity of anything in a set piece of space able to contain possible things. The simplest way to get to absolute intelligence is for it to just appear and not evolve, and this has the lowest statistical possible chance of being the way sentience and a god like brain emerges. Any other solution requires complex ideas, and this part of the book will try to show a way to get there. The vacuum is the most likely thing to appear from empty space because it is the simplest stable structure in the universe, and thus it makes a good basis to propose how to build a sentient being from nothing.

The very first thing was space, the vacuum; Its very first quality is that it got more complex over time. The simplest possible universe, is the most likely universe to randomly pop into existence out of nothingness. Time itself is measured by the expansion of space, by the amount the vacuum has expanded. This is because in order for something to exist it must influence the thing after it, and if empty space is static and doesn't influence itself, it has no time. Entropy itself emerges from the decay of vacuum space into more vacuum space. Universal expansion (dark energy), the properties of the vacuum, time, entropy, are as highly correlated as things can be in modern physics, and thus most likely all the same thing. Time could be an emergent

property of entropy, that itself is the growth of nothingness into something, where one event influences the next event.

Boltzmann Boxes and Boltzmann Brains

This is an idea correlated to Boltzmann's theory of entropy and how the universe began. In a Boltzmann's box, he idealized a simple idea of how the universe began. Imagine a box with a bunch of particles on one side, all bundled up into a singularity. Over time entropy will spread those particles through the box. He defines each particle as able to randomly jump to any location in time, and gain increasing complexity through time. This is like a stove emitting heat to fill a cold room. A single particle can jump across the whole room; it is unlikely but possible. The particles can jump any distance, and with closer distances being more likely. Like in a gas, there is a mixture of cooler and hotter particles all mixed together, and their average temp is what we call the temperature of a gas. There are actually very hot and cold atoms mixed together, and the hotter ones have higher probabilities of jumping larger distances randomly. Particles begin all bundled up, and can randomly move in time. Probability will allow those particles to form into complex structures, because they in theory can just jump into any configuration. One of those structures will form a stability and cause a stable pattern to emerge out of the random chaos. Swirls can move and bounce around in the box, and arise from pure random distribution of particles. Entropy of particles can actually describe a big bang and it forming those particles into the galactic like structures we see today, if certain weird special patterns emerge from randomness. These simple yet impactful patterns that cause new events to happen after them, and lead step by step to more structure in system. A simple pattern, that isn't that unlikely to happen, that can create something after itself and evolve into more and more.

I take basic Boltzmann entropy as it explains cosmology, and add in one more feature. Space is entropic. It isn't just matter rearranging itself, it is also space rearranging itself. It is growing, moving, shifting. In a Boltzmann Box of particles distributing randomly through that box in time, space is also distributing itself. If you imagine a large box and bunch of particles gathered at one end, and then let them go, they will distribute themselves through the box. Space too is distributing itself and "moving". So that space is also expanding, always moving. A box itself expands from space expanding, and not just particles hitting the wall. Space itself may be part of the dispersion of the particles, where tiny amounts of space that can emerge probabilistically in a singularity drive it to disperse through a room. Space is more fundamental than matter, it has energy, it has entropy. It may also be the source of entropy in systems, where we observe matter gaining entropy in time, it is in reality the underlying space that could be driving that change.

Space itself is a quality of that box, and also randomly distributing itself and growing. Boltzmann may have gotten what has entropy wrong in that system, or at least my understanding of his work is wrong. The geometry of vacuum space will become something fundamental to physics if my hypotheses in this chapter are true. I initially saw the structure as big "hexagons" with tiny particles at their loci, all moving and transforming. With later education and looking for better visuals, it seems more similar to "Penrose Tiles". The pattern of space as it relates to entropy and time, is probably a well-defined structure in ancient eastern religious art, somewhere. The math is out there and probably already well worked out, for something seemingly completely unrelated.

In current science the "curvature" of space, creates gravity, and not any sort of particle emitting out from matter. They theorize about a possible particle called a "graviton" that makes gravity possible, though have no way to even explain what it would do, or how to observe it. Space itself might be that "particle" that they call the graviton. Gravity is able to influence things at a distance with nothing exchanging in between. It is not like a ball hitting a ball, and causing something to happen. This "Spooky action at a distance", was a nearly unsolvable puzzle. Newton could not solve it, and directly said that he left this problem to future scientists to solve. Einstein solved this by explaining it as a curvature of space-time altering an object and bringing it towards another object. This then begs the question what is "curvature", what is being "curved", as this all relates to gravity and space-time. Gravity has a high correlation to entropy, universal expansion, time, space, and possibly entanglement. The fact that around matter there is more gravity, is a sort of illusion, that confuses us as to what the real causality might be. The order in which we discover things, is not their order in importance or causality.

Gravity can be seen as a description of how space moves, rather than just curves. The exchange of force is not like being hit by an object with mass. You are hit with a particle that is time itself; it's very movement through things, creates time and the very shape of space in the universe. Space itself is like a super "massive" particle, with no mass that moves through anything. As "time" passes through things, it shifts that thing to one side because space is unequal to the space on the other side due to differences in relative time. Where space is shorter, it moves towards that direction. You can apply this same idea towards time, as time moves slower, matter moves towards slower moving regions of space time. The faster moving space time, that is more expanded pushes it towards the slower moving less dense space region. space pointing away from it towards vast amounts of space.

Space is not curved around a gravitational object. Space doesn't have density. Denser space is larger space, and is measured in size. If the sun was curving space to create gravity, then the distance between the sun and the earth would be increasing. Time is being shifted. If space is entangled and experiences entropy in that entangled structure, then space has a direction that it moves in as it expands. The directionality and flow of space creates what we call curvature, and the more it flows out from something, the more things are attracted towards it. The direction that space moves in is time. Space itself creates time, and is time. The structure of space has a pattern that grows and has entropy, that entropy is like a form of "radiation". The space-time field emits new space as its "particle".

ENTROPY IS QUANTUM GRAVITY

Space-time itself then has to imbed information about the universe in its own structure. At the tiniest levels, space-time can be micro-curved, and those curves can contain information that can influence the universe around it. There are different ways to look at a solution to this and they are all probably roughly equal, eventually. You can create what is called "gravitational waves", and describe them as oscillations at very tiny levels. These waves on larger scales exist, and our scientific instruments to detect them, can only work on the largest possible waves, at this time in history. The waves created in pure space time, can in of themselves have energy, mass, and information imbedded in them. When we consider the entropy of a system, we can also consider that there are micro gravitational waves emitting out from it containing the information of that system. That we observe particles bouncing around, but in reality, the heat we observe coming off of something, also contains its information in space time itself. The way heat moves could be a product of gravity operating on the quantum scale. Entropy is quantum

gravity, and a proper theory of entropy has to explain how it warps time and space at the smallest levels.

Time itself is a kind of “particle” that is not matter. It is better described as a pattern, a replicating system, operating on the tiniest scales of reality. It is itself defined by what the tiniest scale of reality can be, since time is the very “pixels” from which reality is created. This is taken Einstein’s General Relativity, literally and not figuratively like many physicists do. In GR gravity is not a force, and is just how space and time move relative to material objects. Gravity is space-time and inertia according to Einstein’s work. If we want to apply this to the smallest scales, then we just need describe how systems change at the smallest, scales, in time and space, and then say this pattern creates what we see at the largest scales. The specific math of GR doesn’t apply to the small, it applies to the big. It is the general math, ideas and axioms that do apply. Geometry is king.

All the problems found in the study of black-holes and information lost is solved by this shift in thinking, and with the way I described space time in special relativity. We just have to learn to imagine dimensionality and space itself better, so we can see these ideas. All the information in a black-hole is embedded in space-time. That information can cross the event horizon of a black hole, with enough time, because space-time can be “emitted” from a black-hole. The interior of the black-hole and its singularity will contain the structure of space-time at the smallest and most discrete levels on massive scales. A singularity from the inside is massive walls and not a singular point. Seeing a black hole like this is simply a simple literal description of GR.

A Toy Universe

Imagine "a universe", it is singular, and not the one we live in. It had a "big bang" and it will have an end. This universe has a finite life. It is born and it will die. The unique quality of this universe is that the end state of the universe will affect the beginning state of the universe that comes after it. So, this imagined "universe" begins, and this "universe" ends. In sequence, with the conditions of the preceding universe affecting the following universe. It will be the same as the previous one, or adopt a minor change that can add more stability and "longer life" to the universe. In a sequential evolving universe, you need a first universe. Each universe that comes after the one before, will evolve on top of the one before it. The universe arose out of nothing, and due to pure chance and pure probability. Simple order can evolve into larger and more complex systems, that can perpetuate themselves forward.

The whole of a theoretical universe that could evolve and pass its order on, could be as small as say 2 photons. The next universe after it would be 2 photons again, rather than just random. Two entangled photons, spreading out for a trillion years, until that entanglement breaks, and the energy of that resets the universe. The conditions of this next universe would not be purely random; they would arise out the history and end state of that universe of two entangled photons. This universe after many cycles could have a small change that actually works and makes the universe little "better", and it transfers this improvement into the next universe. More entangled photons, would make more universes to grow. If the photons could replicate, then there could be a lot more universe, and lot more opportunities for change.

In a more realistic universe, at one moment, in an empty universe, with no net positive energy, a possibility happened. The smallest possible thing happened. All the random fluctuations created something. This was the smallest possible thing that could exist, and

not a massive singularity. Something more than nothing came into existence. This had happened for nearly infinite times before. On this day, the first day came into existence. It was the tiniest thing, and the tiniest thing that could be and keep existing.

If you do something enough times, some very odd possibilities will happen. This was not a monkey typing the complete works of Shakespeare unlikely. In physics, there is a definitive possibility that the whole universe could just appear as a fully functioning bunny rabbit. It could just appear out of nothing, as a pure possibility. If the universe was created by pure chance in a one-shot deal, with all possible universes being made, then a big bunny rabbit is the most likely universe to come into existence. In physics, they call this a Boltzmann's Brain, I prefer a Big Bunny Rabbit (BBR). The most likely thing to happen is the most absurd one, if all we had was one chance at intelligence in the universe, it would just be a wholly sentient super god brain, that was all things. This though is too unlikely. Something far more likely happened than this, because it was far more likely. The most likely thing that could happen, is that the universe had its very first stable state. It didn't just come and go. It endured a bit. It had some sort of structure. A little bit of vacuum space coming into being, or a BBR, have absolutely different probabilities of having happened. This tiny little state, would have special properties.

ENTANGLEMENT SPACE-TIME GEOMETRY

Quantum entanglement is one of the newest discovery and it is very weird, and it can be the basis and fundamental causality for all creation. Instead of being something we attach on to theories we use to describe the universe, it probably should lie at the foundational root of all theories, and be the primary/first thing we study, even as children. At a quantum level it is structure creating, and has odd properties that are very

difficult to explain when applied to particles. If it was the space beneath particles that was entangled, it might make a theory of entanglement more intuitively accessible, with the right visuals.

The universe before time was endless random events with no connection between the one before it. It was a universe that could be anything at any moment, just flashing into probable configurations. The simple ones would be more common, with complex ones rare. Like a monkey typing out Shakespeare, is nearly impossible. Three words of Shakespeare written at random by a monkey on a typewriter is much more likely, than a whole play. Our universe forming is more like all the poets in history being typed in the order they wrote, by 1 single monkey. It is possible, though the more likely is just a simple sequence that can eventually evolve into the universe now, step by step. The very beginning and end of universes could be as trivial as virtual particles appearing and disappearing in our universe. Three words of Shakespeare would then have to evolve into all the poetry ever written, and this is possible, since we have seen this basic principle happen with life itself and its evolution from near nothingness to demi-god human beings.

The very first thing could have been two entangled pieces of space that had a very specific and unique structure. Nothing entangled to nothing. One piece of nothing went one way, and the other piece of something went the other way.

The entanglement of space can be described mathematically and very simply on a line. You just need to imagine a line with arrows on it, that point in different directions. There are only 2 directions. So, arrows are just like numbers on a number line, that we all learn in basic math in school. For every left, there is a right. If you find a right arrow, you know where a left arrow is exactly. This entanglement can just be directionality. A

dimension is born out of entanglement defining a direction to be opposite of another direction.

- Basic entanglement

<<<<<<<<<<<<<<<>>>>>>>>>>>>>>>>>

- More complex entanglement

<< << << << < << 0 >> >> > >> >> >>

- Impossible entanglement. Is missing one ">" on the right.

<< < << 0 > >> >

Two regions of space entangling are more likely to happen at random, than a stable photon to just appear. This is far simpler, and thus more likely to happen at random. A unique structure can take form, and it can replicate itself. Time is born with the entropy of entangled "space". Entropy is just there being more entanglement, more space, with each passing moment. You can almost just define entropy by space growing, if you give space a simple structure that it emerges from. There is a pattern that can grow and you can determine sequence in it, by the size of its growth. A bigger region is older than a smaller region. The small created the big, and this goes on and on at the minimal rate to make a stable growth. There is no concept of time between each evolution of growth. You can have a sequence and there is no way to describe any amount of time between each small unit of time and growth.

A certain pattern emerged that could replicate itself and grow more space. It could be many possible structures. Many would come into being and most all would fail. You could have many different entangled structures randomly emerge, and if they couldn't grow, they wouldn't exist, they wouldn't endure. By being able to grow, they influence the state of the universe after it, rather than just randomly ceasing to exist and another random disconnected event happening. One unique event, however would cause something new to come into existence, and it would reproduce itself. Space has just had the most basic reproductive system running for a long time, and the bigger it gets, the more it reproduces a really large universe or larger multi/mega-verse. Time could be just the replication of space-time growing, from an actual definable structure and not an amorphous spiritual sense of the word "time" with rainbows and unicorns around it.

This pattern of space-time at its root would have to travel between universes, regardless of what formed in that universe. This pattern can cross over event-horizons. It is the mechanism by which the "genetics" of a universe are transmitted into another universe. Our literal universe we live in right now then could exist as a chain of universes going back in a macro-time of universes. The "genetics" and information of the past universe, embedded in space-time, could then create and define the rules and patterns of how this universe develops. Going back enough universes you would arrive an original universe, that was very simple. All this just depends on the ability of information to be encoded in space-time in things like micro-gravitational-waves and/or the entanglement of space to itself, which can cross event horizons.

The Evolution of Time into Matter

Once you could have a basic universe of time that could grow and replicate itself, it created something even more profound, the ability to create something in time that

could persist and exist even more stable than time itself. Time just grows and isn't all that interesting, nor special. The universe can have a "bias" to like "specialness". This makes it more likely that something new can happen; a unique mutation could occur that would evolve the system to be a more stable, successful, and replicating universe. It may simply evolve into something that evolves faster, and thus exists more in reality. It kept growing, and the bigger it got, the more likely something unique could build on it.

ENTANGLED SPACE

1. there is a basic pattern of entanglement of space, that causes it to grow,
2. this growth is entropic and creates time.
3. this entanglement can create patterns that can mutate into new forms different than time.
4. From these patterns, a specific pattern formed basic matter.
5. Matter arises out of energy, and is made out of only energy.
6. The most basic form of energy in the universe is vacuum energy, the most likely form of energy to randomly come into existence out of nothing. Time creates an ever-increasing supply of formless energy.
7. The more formless energy there is in the universe, the more likely for that energy to form into something with more structure solid and real.
8. Vacuum energy is the most likely state of energy that matter evolved and arose out of to create the material world that we know and are personally made out of.
9. Patterns in the replicating vacuum states, could assemble to fold space and time, inwardly to form matter.
10. Since the vacuum is energy, and matter and energy are equal, it than is possible and probable, that matter arose out of expanding replicating vacuum energy, arranged in a very odd and unique way.

11. The “big bang” created all the matter in the universe, we can see. New matter is not created except in events like the big bang.
12. The “patterns” in Vacuum energy(empty space/entangled space/dark energy) can transmit to and create the following universe. The big bang, that we come from, is probably the culmination of nearly endless big bangs before it, that slowly evolved into the universe we see.

Eventually one random structure, takes on a unique form, after a 100 trillion universes and countless more possible entangled structures pop in and out of existence. The structure of the universe we see, would be built from step by step increases from nothing into a lot of something, which is mostly a lot of a very minimal thing. Simple spatial entanglement can evolve into others forms of spatial entanglement. Patterns and structures in that entanglement can take shape. Different structures will have a rare potential to become something more, and evolve. Space-time must become evolvable into everything we see. Vacuum space doesn't have many properties, like matter does. It can only make more of itself, and if more space exists, things get farther apart.

ENTANGLED SPACE-TIME REPLICATORS

The universe begins with the first replicator. It is built from something like vacuum states. Patterns that survive and “replicate” more, can then survive into the future more and dominate. These structures can randomly evolve to create new structures, that can make them more stable, more able to replicate, more existing, and so on. Time can evolve beyond an expanding field of space and form the basis of new fields. Only consistent things that add stability and structure can survive, so not all fields will endure or even be possible. Entanglement can evolve into more dimensions, such that dimensions themselves are defined by what they entangle and the implications of that

entanglement. The differing forms of matter are just different forms of entanglement. If it is a half spin in one direction it is a half spin another direction, you got a kind of matter. From random probability in nothingness entanglement emerges as a possible structure, one form of entanglement creates a replicating structure that gains entropy, contains information, and in that entropy, creates a structure to time, where one event defines the event after it.

There is a set of information that can change into a new set of information, in a causal relationship. If one set of information evolves into another set of information, then the amount of information in a system will always increase. Causality demands an increase of information, as a system of information evolves into another system of information. The amount of information must always increase, or time would collapse. It needs to be a certain arrangement of information that grows and can change. This is almost an exact description of Entropy in standard physics. You can think of this just like a computer program or computer virus, where if it is replicating or running, then there is time. Entanglement can travel faster than light, because the breaking of entanglement is a breaking of time structure, at some level. It is a breaking of information. That information is embedded in space-time structures, and not photons emitted as heat, or matter.

Entropy and entanglement can create an evolving universe, that builds upon itself through completely random probability. Rare forms of random events, will form stable structures, that can grow and expand. Rare configurations of information can evolve into more complex systems. You can write a set of simple computer code that can evolve into complex computer code, and maintain a consistent stable structure. They will be more stable and exist in time more. This pattern of information grows, replicates, and evolves.

The universe is always expanding, and is always containing variations in its basic underlying pattern across it.

A pre-matter, pure space-time universe would evolve and grow over a very long time. With each death and rebirth. A larger and larger universe would exist. The universe as it gets bigger, makes it more likely for things to happen. More universe, more possibility for things. If you want to make/observe something very improbable happen, then a corresponding size of the universe must relate to that probability. Information can randomly evolve and in a bigger expanding universe, the potential for it to form into something unique, special, and enduring for all time, grows until it becomes a near certainty. Entropy, time, information, entanglement, and even the laws of evolution itself, originate out of one root idea.

The 3 Proposed Axioms

Axiom 1: Evolution of Space-Time

- Axiom 1: A state has time when it can define the states that follow it. States that don't influence the following states, don't have time.
- Abstract version 1: The initial state of a "universe" can be influenced by the preceding "universe".
- Abstract version 2: The initial states of the universe evolve into more stable forms, from the events in the preceding universe into the next one.

This is not a very controversial idea for an axiom if you accept the idea, that universes are born out of universes. This can only be reached as an idea when there are no other ways to explain the universe. An axiom like this, could be developed out in an empirical analysis of vacuum states, and making theoretical models how a stable structure could form and last in it. If one piece of entangled space is going "left", you know its entangled pair is going "right".

In the real world, "virtual particles" forming in the vacuum of space, are just popping in, popping out, with no net energy being made or destroyed. A photon has a probability to appear, it appears sometimes, and then just disappears. It is not stable. It can't build on itself, or exist in time for longer than a single moment. If we can have virtual particles as real things, then we can have odd things happen once in the life a universe, where virtual particles can into something, that just doesn't disappear, in the next moment. Stephen Hawking built an entire career on making virtual particles into stable entities, in a narrow use case. If two entangled virtual particles formed at the event horizon a black hole and one fell through and the other didn't, this would form electromagnetic radiation, that would glow off a black hole, a sort of super minimum heat, that an object could shine off. This "Hawking radiation", is the "coldest"/lowest energy light in the universe, and there is no way to observe it. this light though is a simple theoretical stable structure, that persists in existing, rather than just pops in out of existence like a virtual particle. These two entangled virtual particles take very different paths in time, and become fundamentally divided from each other. One went into a black hole, and the other didn't. This breaking of symmetry creates energy that can be seen as a photon. Light is created from nothing, in Hawking's of model of a black hole. Or in my perspective, light is created from information, found in entropy, that is defined by geometries in space-time, which is "nothing"/empty space, in our real world.

Within those many but small chances, there are these very special rare possibilities, very rare, very small in change, and that build onto the thing that comes before it. In life, mutations that actually produce something useful and completely new are extremely rare. When they do happen, and those mutations don't make a living thing infertile, it can pass those changes on to the next generation, forever. Life on earth, shares mostly the same DNA and mostly all the same biological systems. Even bugs have hearts and blood and the same basic chemistry as us. While we see a great deal of variety of life on earth, it is in reality fairly homogenous, and sharing the majority aspects across different classes of life forms. Tiny variations in the functional forms of life creates all the variety of life we see on earth. Humans and natural watermelons are not that different. *Humans and domesticated watermelons are more similar at a memetic level.* Evolution itself is just a how genetic information moves, and evolves from simple systems to complex systems. All "real" information evolves and grows, and I just take evolutionary information theory, memes, and push it backwards into the origin of universes. We can treat whole universes as an individual "animal", for this argument.

In a theoretical universe, a "trait" gets passed on to the next universe. These traits are part of how a universe evolves and grows, and the mutation itself is creating the next universe, and this is why that mutation can persist into the next universe. Successful mutations are those that survive more, that are "stronger", that "create" more universe, and that add stability for the universe to exist more. In universes giving birth to new universes, stability and being-as-simple-as-possible are the most likely thing to happen, and thus what does happen. The universe which makes more universes, which adds to the growth, and adds to the stability, such that more things become possible, while the universe also doesn't rip itself apart, is the one that will survive. Any universe that is even 1% less productive, will over time, fade into nothing compared to more successful universe.

The universe moves in two directions, outwards with space and time as it expands, and inwards towards its initial ground state. Entropy adds complexity to the system, and arises out of replication and growth of stable systems of information on top of stable systems, that creates an expansion of entangled space-time structures. Things go to their ground state, and lowest possible energy state, whenever they can, that is the most probably and the simplest way to go. It is a driving law of our universe. Going towards simplicity, going towards the ground state, creates an orientation towards what is possible and not possible. Probability contained in the initial ground state of the universe, still influences the universe. Every single point in all of existence is just a replication of that initial singular point all of time and existence comes out of. It contains nearly infinite dimensions and can be any number. It goes with more likely things, more simple things, and only possible things. The ground state is everywhere.

Axiom 2: The Consistency of Time

Axiom of Time: The speed of causality is consistent across all possible universes

Space should be viewed like a particle, in some aspects. It can pass through anything. It has an orientation and a directionality in its movement. The directionality of space's movement, is where time itself emerges from. Space passing through things, creates a directionality in time, an experience in time. So as space expands out from a planet or sun, it slows time opposite the direction it flows towards. This dilation causes gravitational attractive forces

- Time is space with a direction
- That direction is created from the entropy of entanglement

- Entangled space has only direction as a property

Time has a structured and ordered sequence across all universes, because that is the requirement for time to even exist, if it didn't there wouldn't be causality and without causality, there would be no time. You can never personally experience any other speed of time, different than the one you occupy. All experience time as *absolute* relative to themselves in the same way. If you were placed in different regions of space time with time traveling at different rates, you would experience time exactly the same. It is impossible to directly experience time as anything different than your own experience of it, regardless of any concepts of consciousness. Clocks don't feel time dilation. A second is second, in your own time frame. You can never experience time move at a different rate, you can't experience time in slow or fast motion, as anything more than your brain processing things differently. Even if between every second whole universes were born and died, a second would just pass.

Imagine we had a massive multiverse, the only thing that would preserve the stability and separation of universes is the inability to cross over into the other. There needs to be an inherent separation of universes. If the speed of causality wasn't constant, there couldn't be one universe after the other. The previous universe could just jump into the next if separated by time, or jump over to the next, if separated by distance. Or in our own universe, all the light of every star we could possibly see could just bombard us at once.

You can measure the time of the universe by its expansion, in absolute terms. A person by a black hole's event horizon, will experience a slower rate of time, than a person a few kilometers away. Certain galaxies distances from each other is an absolute clock for the universe, no matter where you are observing them, so long as they are truly distant

from you. This distance between galaxies will be a "clock" for the universe. The person near a black hole event horizon, where time travels slower, will see this happen faster. There can be a measure of "simultaneous" events determinable by the relationship each observer has of this separation. Though they will observe each galaxy differently. A person near an event horizon seeing the universe passing by faster than any other possible place in the universe, would also be bombarded with all the light from a large amount of time, in a short time. Where one might even feel the heat of a distant star, like the way we feel the sun.

The consistency of time is the consistency of the universe as a real thing. We may all see time differently and yet can all understand each other's different time frames in absolute terms. If you have 2 different time frames viewing the same thing, the sort of information you would need to transform your observation of an event, into the perceived observation of another's event, would describe your theory of time. Observer A sees one thing and Observer B sees another. This can be described a set of information for each observer. The information will look different. The sort of rule or set of information can I use to transform one set of information into other set of information, observing the same this, is a theory of time. If there is no rule nor way to do this, then time is not consistent, and is based solely on how it is observed in that moment, and with no relation to anything else. This is just my very poor way of understanding Einstein's conflict with Niels Bohr about reality and "God not playing dice with the universe".

If time is just the size of the universe, then that size can be consistent, even across universes. This would be possible if universes are created from the space-time of a previous universe. And if information from a previous universe could be contained in the space time of a previous universe, then universes could create universes and pass on information between them, to create one from the other. Time being consistent across

all universes can be translated into a statement about all of reality being one mega-verse definable by one consistent set of information/"number". The big bang can't be a wholly strict separation from other universes in a possible multi-verse, nor the soul beginning of time.

Axiom 3: The Probability of the Possible and Impossible

Axiom 2: The only things that can exist are what are possible to exist.

My Probability Rule: The least likely things to happen are the most likely things to happen, because there are a lot more unlikely things, then there are likely things.

Determinism arises out a fundamental feature of probability, that which is possible, and that which is impossible. Space time must be defined by probability and quantum phenomena, and not some clockwork universe model of stiff materialism. A deterministic model has practical use in its math and ease of use. In some ways determinism is merely seeing the absolute universe beginning from its end. What is possible, and most probable is a very narrow and very fine path, that leads to the universe we see before us, and the amazing "created" universe we see. Physics and logic may love to believe the universe is all determined, and yet we as humans know from the act of thinking that the universe is not determined. Reality is clearly non-deterministic. On its largest scales it is very deterministic, and on its smaller scales more random. Human thought is far more connected to the smallest scales and how quantum information is processed, rather than how stars move across the heavens or cannonballs fire from a cannon.

-The universe begins in pure probability. For something to exist is an act of probability. For this probability to be known beforehand, would eliminate the probability, and thus eliminate its existence.-

Complexity can be built out of chain of very small and very unlikely things, stacking on top of each-other. A simple stable structure could form, then over time a rare event can form another tiny change that adds more stability and structure. This can go on and on, until the universe we live, comes to exist. Lots of small unlikely things aligning over enough time, rather than just one huge creation of everything in one moment, is the most likely way thing came into existence.

We can think of biology's survival of the fittest applying to the tiniest aspects of the universes laws, and they evolved into the ones we have. The evolution of life and all its laws and logic, can themselves be emergent properties of how the universe and its laws evolved from the tiniest details in step by step fashion with the tiniest of mutations, where only a few out of near countless amounts would ever even work. Life is a natural expression of the creation of the universe, from its very start.

A step towards more stability is very unlikely and very rare. When it does happen, it becomes part of the "identity" of existence, and the next universe will develop with that new very unlikely state, and not have to "learn" it again out of pure random chance.

If there was a moment where all possible starting states of the universe all began at once, there wouldn't be many that lasted. And our universe would probably look like most of those that did last. Though this is an event too unlikely to happen and not necessary to create the sheer complexity we see. We don't need 10^{500} universes popping into existence at once. We only need $10^{200-300}$ popping up randomly to get to one single configuration that can evolve into the more complex one we have now. This difference in odds is so large that only simple evolving systems would be likely possible to create the universe, by 200-300 sigma (0's). Evolution from simple to complex, step by step, with lots of trial and error, in sequence, is far more likely. I doubt in all of creation, in all universes that ever existed, fully formed sentient beings formed out of nothing, even if it is possible, because the mega-verse isn't old enough/big enough.

The universe had to evolve from state to state, and have almost no direction or purpose. It just became a bit more stable, and all times absolutely consistent with what is possible, within probability and the minimal basic states of the universe. Complexity had to develop from simple to more complex, step by step, it couldn't just leap, it was very unlikely. The most likely thing is just a very small change, not BBRs appearing from nothing.

What is possible and not possible, is probably the most important idea in all of creation.

This simple idea seems to be a simple truism, it is however giving directionality to creation and time. It is a basic axiom, and a pretty minimal one at that. It is for me the basis of what people see as god in creation. In evolutionary biology, there are not that many configurations of life, that can actually live and reproduce something possibly better than it. This creates a great constraint on what is possible, and the creation of

things that look designed, but are in reality created out of pure probability. It is very unlikely that sentient space faring alien life, will be that much different from us as human beings, simply because there might only be "one" route to even develop to that stage of life.

The universe had to develop with the most unlikely steps, and not highly likely ones. This is more probable. Likely things are very unlikely to happen. A chain of very unlikely events can be the most likely thing if they form into a chain that adds increased stability, and that are in a sense a very rare super unlikely thing, that which is possible. If you have 900 $1/1000$ chances, and one $1/10$ chance. It is more likely that you will experience a $1/1000$ thing than a $1/10$ thing. These more unlikely things are more common and thus the simpler and the more likely things. More common and unlikely things are simpler, than rare more likely things. Small and frequent jumps of energy can end up at a large energy state. Small energy bursts are much more likely to happen, and thus the more likely cause of the universe. Lower energy events are more likely than high energy events. Energy itself might just be a measure of contained probabilities. Low energy things are more likely than high energy things. A sentient mind forming from nothing is a very high energy event, that is the simplest way to get there, and very unlikely. A low energy event is far more common and simple, where a simple arrangement of information is the most likely starting point of a universe, by a very large statistical amount.

CHAPTER SUMMARY

Quantum reality most likely created space and time, because things develop from smaller to bigger, in general, with evolution. Time emerged from space-time geometry in

the very small, in nothingness. Absolutely nothing has qualities, and features, that are independent of time and space. Very few qualities can exist independent of time.

- GR is defined by Space time geometry
- Quantum Physics is defined by probability

Quantum Gravity is the geometry of space-time, that can influence the state changes of quantum systems in time and create space time itself as a smooth structure outside of the quantum scale. Quantum gravity drives things to their ground state, and evolves systems, it is not an attractive or repulsive force at the quantum scale, because the force we call gravity is created by massive amounts of new space-time structures being formed. Space and time are physical entities that are created in some ways like matter. Time is a physical property of actual space, that is defined by the creation of new space and new information. The universe grows very little at the small scales, and even at stellar scales. Atoms give off light, but aren't themselves light. The motions of subatomic particles themselves are defined by how space and time move, rather than cause space and time to move. The odd properties of quantum phenomena could possibly be described in space-time geometry. That Geometry is what we call "gravity", and its primary qualities are not about gravitational attraction.

Space-time is structured at the tiniest levels in a "replicating" pattern, that always expands. This expansion is time itself. Entropy comes from the expansion and replication of space-time arranged in a specific pattern. With each replication, it alters the state of things and it expands. There is a state becoming another state, and we can infer some form of causality between them, that links them together in a sequence. This causality is time. Causality doesn't require exact cause and effect, only probable, because that is a simpler system to evolve from nothing, than a defined and perfect cause and effect,

which in itself could just be seen as a single thing. Time needs probability to be time, since all determined things could be defined as a single state from a higher dimensional perspective.

The universe can't stop expanding, and time can't stop moving forward. The evolution of quantum systems arises from how matter interacts from this quantum space-time geometry. It is like hills emerging and pushing things into new states. Shifts in probabilities create shifts in energy. There is an up and there is a down. Expansion creates energy, and this creates real shifts in how quantum systems evolve. Every state in the universe changes moment to moment. Time never freezes. If time could stop, it would never start again. Time by its very nature, causes constant causality between 2 states, so that no 2 things can ever be equal in time, and that the following system is always bigger than the previous system. No atom can be the same between any 2 moments in time. Things in space possibly could be equal, if they exist in the same time.

Randomness creates all the mutations in life that evolve into the complex and very useful structures, that life needs to survive and replicate. Life needs prolonged periods of stability to spread and grow. The more a species spreads-and-grows stably through time, the more likely it is to create mutations that will actually benefit that species. There is a tension between a need for randomness and emerging stability in evolving system. The universe needs similar conditions to evolve in.

Only possible things can exist, and this has an underlying power to it, that can't be readily expressed. It appears to arise out of randomness yet has a determined quality to it, there is a duality to it, that may lie at the heart of all real dualities. The randomness

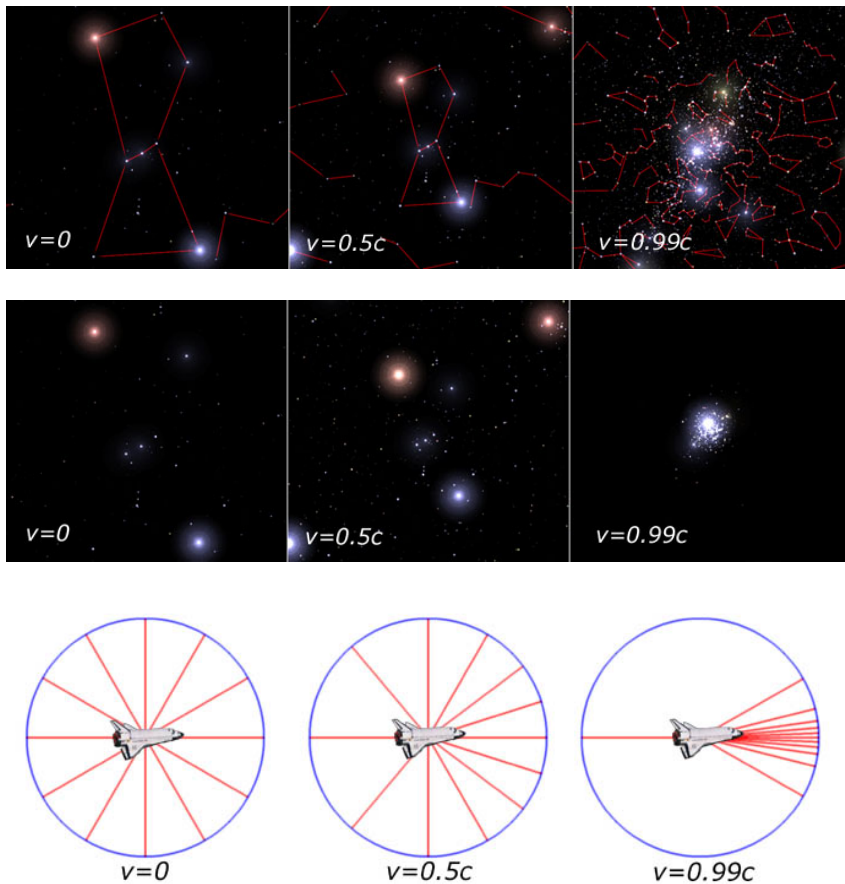
in the ground state contains the possibility to evolve into a determined system through an evolution in time, where there is a very narrow range of means to reach such an end such that the path is so unique, as to be deterministic, yet inherently unknowable. The universe always evolving into more, leaves room for an improvement upon itself at each step, and this can go on without end. Information can always increase, and always evolve. Evolution and entropy are two very interacting qualities of information, and all things that exist in the real universe. I can't yet explain their exact distinction.

Chapter 13: Quantum Gravity and Evolving Universes

As things approach the speeds of causality, they begin to create new universes or go into others. This is not a new universe with anything in it, or of any moral value. It is just a space that is separate from our own, and can never be reached. An event horizon is a boundary between universes, that is not wholly separated from one universe to the next. In simple geometry, they are whole new universes, and totally separate. In the deeper geometry of time, they are connected. If you stood on the surface of the earth, you had gravity increase and increase, then eventually you would see the very earth curve up and around you. What is curved in one direction, can be curved in another direction, when time dilates enough.

As you fall through an event horizon, you would see the surface of the event horizon surround you slowly. It would consume you in so much new space, that the old universe would disappear. As you emerge inside a black hole, the singularity is spread out like a spherical surface all around you. If a singularity was "red", you would see red at a singular distance in every direction you looked. The event horizon you just crossed would look like and is, pure empty space. You cross over into the dead center of a sphere; the event horizon is the very origin and center of that sphere. Things crossing the event horizon, from the outside universe, would enter one by one from that singular point at the center, and it would probably emit energy almost like a sun. Matter would be flinging out it in all directions, seemingly coming from nothing. It could be thought of like a white hole at the center of a massive spherical space, that keeps making the black hole bigger, the more stuff you add in.

<http://math.ucr.edu/home/baez/physics/Relativity/SR/Spaceship/spaceship.html>



From the outside as you enter, all of the universe you could possibly see all around you, will narrow towards a single point. You are surrounded in the space of the event horizon, and still able to enter the universe you came from. It will look like the whole universe of all the stars you knew is flying away in the distance, like a rocket flying away from its home planet and it is getting smaller and smaller. If you tried to escape, even going near the speed of light, that point would only slowly get closer, and it would take a long time to escape. Space is literally coming into being around you, so you are moving through massive amounts of space. This new space surrounding, is just seen as slowed down time, to those on the outside. They will see you go slower in time, flatten away to nothing, and not see a sea of space consume you. These two perspectives are exactly equal, when you allow for time and space to invert themselves.

A black hole could be seen as leaping into zero-dimensional space, rather than anything outside of our universe. You are basically zooming in and in until there is no more zooming possible. The new space is created out of this zooming into space, caused by the dilation of time. As you slow in time, you sink into the well of nothingness, that is the zero-dimension. The event horizon doesn't create exactly another universe, it transitions things into the domain of quantum reality. In Zeno's paradox, the process of taking shorter and shorter steps alters the very time it takes to travel those steps, and alters the very structure of time as to be different. In a sense, you never take a shorter step, and always take the same length step, and are zooming deeper into reality with each step.

On a 2D graph of Zeno's paradox, as you approach the wall on the x axis, the line curves on the y axis towards an infinite value. What we see as reality, in a given spot, is always as a person standing like a tangent on that line, with flat space. The tangent line itself becomes the x axis of any person, and there is no correct x-axis in the entire universe. Different reference frames basically say that what we think of as the x axis is relative, and not the actual measurements on that axis are relative. We all have our own x-axis, they are all tilted differently. Space-time dilations and relativity are described with this simple explanation. When the tangent line goes vertical, a new universe is created, with an inversion of space and time, with an inversion of x and y. The Y axis keeps all these different versions of x consistent, by changing how time moves. In positions closer to the wall, time is slower, time is more, and for those farther from the wall time is more flat and normal. The closer to the wall you get, the more you leap into a new reality and into a new universe. You literally go into new galaxies by how fast you accelerate a spaceship, and that trip is described in this math. Not there is any wall to actually

approach. The wall itself is just relative to the rest of the universe you come from. What we call a wall or the x-axis, are just dependent on how similar they are to everything else. Earth has a very clear wall, and x-axis, that is very similar to nearly everything we will ever experience as humans.

A "ROPE" UNIVERSE: A VISUAL ANALOGY FOR RELATIVITY VISUALIZATIONS.

To translate between different time frames, no matter how extreme is basically fairly simple in math, and in visual metaphor. It is just the Pythagorean theorem of $A^2+B^2=C^2$, with space and time as legs of a triangle, and not only space like in traditional geometry. You are making triangles out of the different x and y axis lengths, relative to each other. Standing on a planet in its gravity, or speeding a rocket, create the same changes in time, with the same basic math. Let's say you have 2 ropes, and they are held apart by two people, where one rope is stretched tight and the other is loose. We can actually add as many ropes as you want, with different tensions between them. This tension will create different lengths for each rope. If this is space time, the tension on the ropes is time, and this tension changes the length of ropes between the two people, while never actually changing their real distance from each other. We always see space the same, and only measure the tension to be different between two different time frames.

If you are standing on the earth, you can imagine the rope just dipping a little. If you are "standing" on the Sun, the rope will dip even more. If you are in a rocket speeding fast enough away from everything else, you can dip really deeply, and have a very loose rope. Near a black hole's event horizon this rope keeps dipping more and more and

more. That dip is time, and yet we will always see ourselves on a tight and taught rope. To translate between perspectives, is just about comparing rope length. The longer your rope, the more space you see around you, though to the universe outside your rope, your rope isn't longer, your time is slower. If there is a rocket speeding at 99% the speed of light across from your frame, you will see a sort of pre-black-hole zipping by you. The space ship will be tiny compared to this distortion in space, and you probably won't be able to see the actual ship. That "pre-black-hole" will be moving at 99% the speed of light towards whatever it is moving towards. You are seeing a bubble of space contained in a small area, moving at a tremendous speed. All that contained space, make in anything in that bubble have distorted perceptions of scale. It is being seen through a lens, and you can describe the image exactly with the basic math of lenses. The length of rope describes the curve in this lens.

The space ship inside, will see the whole universe around them as a narrow circle in front of them, with mostly vast empty space consuming them from behind. The dip in the rope is consuming them, just like near an event horizon. Time is the very dark space around them. They can even look at another spaceship going 99% the speed of light, zipping along, and observe them, as slowed in time, and like this black hole like object zipping across everything. They both can see each other as slowed in time, contained in large time bubbles, that shrink the perspective of things to flatten out towards nothing. To help with your own intuitive understanding, just think about the ropes. The speeding away ship has a different rope length, that creates a different dip. Each rope sees itself as tight, and sees every other rope has being the same length, with different observed speeds of time. Each rope can become its own universe, if it changes enough compared to the rest. Relativity just describes how ropes see each other, and doesn't say any one of them is correct, though the whole universe we live is sort of its own rope, and changes from that can be seen as a kind of absolute time.

Every real point in the universe, can be its own universe, and has its own unique event horizon. Time is about the relationship between all these finite points that make up the real structure of reality. Ropes describe the potential event horizon of each potential universe. If two ropes are so far apart, with one being tight and another having a dip going to infinity, they will split from each other. A rope just by itself, will slowly get longer and longer and bend more and more. The very nature of rope is that it itself grows and fractures into more ropes. An expanding universe creates more universes. Our own observable universe will split off from itself, into many universes, many ropes, in the future. Our observable universe is its own event horizon, that we can't cross, from this little point we all come from, on this planet Earth. Galaxies are constantly crossing this event horizon, until one day, there won't be any other galaxies in our observable universe except our own. If you get confused by expanding universes, just think of the rope getting longer in it of itself, the tension in time, causes new things to pop into existence.

An Evolving Multiverse

Every tiny region in the universe can eventually expand out so far, that they can be contained in their own universe larger than the one we see. Now it would take one cubic centimeter of space 14 billion years to double in size, and take many more doublings until it reached a size capable of changing into a universe. The rope itself is always growing no matter what we do to it. time is always moving forward. We can warp our perceptions of time, and flip space and time by how we move it. In the end, the rope of time, and its tension has its own geometry that leads it to grow and not just have different depth of dips in it from different lengths of rope being held at the same distance.

You could describe the field matter particle of a space-time field as universes themselves.

Regions of space expanded away from each other, can splinter off into separate times and nearly separate universes. The universe itself is a replicating entity, that's replication at quantum levels creates gravity and gradual expansion, and when done over large periods of time on huge scales, replicates whole universes itself. The way we would sequence and organize all those universes beyond our universe, if they existed, would take at least 1 dimension, if our universe just came from a universe before it. You could make a sequence of universes, its own dimension, and this extra dimension would exist at the most expanded scales of the universe and define the geometry between totally different universes. The future contains in our universe another dimension of our universe, that has emerging from it new universes.

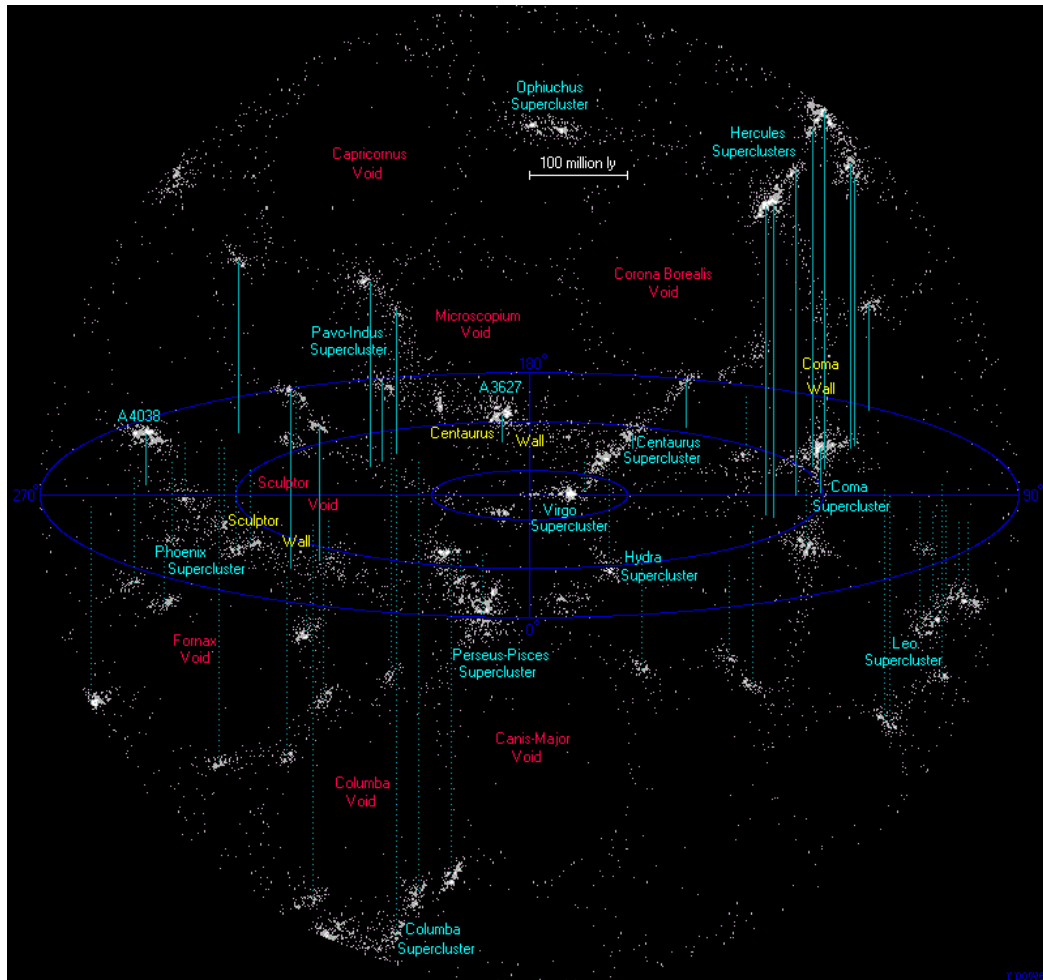
If we go back to the beginnings of creation of time emerging from nothing, it would take tremendous growth of a basic universe before it ever even evolved into any sort of complex universe we see today. Nothingness expanding, had to grow quite large, and create a vast number of independent universes. The multiverse itself had to grow to 10^{200} universes before even one of them has a realistic chance of creating a random event, that leads to the creation of the evolving material reality we see. There doesn't need to be infinite universes, because we don't need to account for all possibilities. We just need enough possibilities, that our realities beginning, becomes a more likely

possibility, to have at least happened once, in one tiny region of empty space in vast nothingness.

The multiverse will emerge in our own universe and in things we can directly observe, if we look for them. If universes are born in universes, then we should be able to see the beginnings of these new universes forming in our own right now. By the predictions in this work, these will most likely arise in areas with the most space and the least matter, around them. A universe that comes from nothing, would be most likely born in a space of the most nothing. If space time emits universes as a "particle" than only in areas that have the most space-time energy, would there contain the energy capable of making a universe "particle". This would be the pre-formation of white holes, before they "explode" into an independent universe. You would look for white holes and an emerging multiverse, where there is the most nothingness and the most amount of spatial expansion, and not for high energy matter and particles. These regions exist, and behave exactly as predicted, in vast supervoids. (*note: I had no idea they did exist when coming up with this idea*)

This would be as if another universe was intersecting and entering into our universe from the multiverse. You could describe the time dimension spatially, and imagine an existing separate multiverse universe, intersecting and merging into our own. Since we live in time, we would observe this as a universe emerging into our universe from nothing. Its expansion and growth in size would correlate to the line segment of a 2D circle passing through a 1D line, and just showing two points expanding apart from each other. The edge of the circle would touch the line, and it would create a bigger and bigger intersection as time passed, and the 2 points would get wider apart. Since we live in a 4D universe, a sort of solid cylinder, we would see an actual sphere emerge as the

intersection, as a reduction in dimensionality into our 3D spatial perceptions. It would keep growing in size, until a critical point, and became its own contained universe.



A General Relativity Interpretation of the “Many Worlds Interpretation”

-Unto to each his own a universe to be a master of with absolute dominion. -

Each and every point is expanding in of itself. If you took the immediate space right in front of you, and divided it up into a few sections, eventually at a certain point in the future, all those space will grow large enough to be their own universe, and grow so separately from each other to have no causal linkage between them. Space expands flatly in time, because the expansion happens at small levels, and is distributed. It is not a singular expansion of a singular frame of time. Each and every place in the universe has its own unique place in space and time. Time moves differently in every place in all of existence. Each one of these unique frames of reference is its own universe that grows and expands. Universes are born out of universes. The universes born in one universe had their infancy in ones before it. A tiny space sitting in front of you may have to cross over more than one universes existence, before it can grow large enough and independent enough to create its own white hole event. Space builds up in size before becoming a white hole and exploding into a new universe. The universe we live in is a nursery to many new universes. Gravity arises out of tension and differences between these different frames. The geometry of space time is lines of tension between many expanding frames of reference interacting.

You can look at the stars in galaxies as pre-event horizon debris that will eventually fall into the black hole at the center of each galaxy. Galaxies themselves can be seen as just large black holes and everything around them, as just the future substance that will make them up. Our planet and star exist in the time line of a black hole, and will eventually be consumed by one. In some ways, we exist in the event horizon of a black hole, if we can't escape from this galaxy into another galaxy. The gravity of this galaxy, makes any matter escaping it, very unlikely. Light can escape for now, and possibly massive technological machines created solely to escape this galaxy can get out. We are not yet contained within a perfect event horizon where only space and time can escape.

We are in a probabilistic event horizon of black-hole at the center of our galaxy, with a 99.99999% chance or more of solid matter never escaping it.

ON WHITE HOLES

Vast voids make up the universe we live in. We can observe, approach, and enter into these vast voids in the universe, that will one day become new universes. The cosmic voids of the universe are most likely pre-formed white holes, gradually growing in size before than rupture in a white hole forming event. A perpetual expanding universe almost requires that regions of space will split off from each other and have no causal connection between them. When these voids and distances become so great, they possible can create truly new universes. When they form into real white holes they will have event horizons that nothing in an old universe can cross over, without leaping back in time to a point before the white hole existed. All futures point away from a white hole, just as all futures point towards a black hole.

A white hole can emerge from a gradual process of ever increasing vacuum space. We can approach an emerging white hole before it becomes a real white hole, by going towards large voids in the universe, where space has expanded the most. Dark energy is the only possible source of new energy to create a new universe from within our universe. This amount of energy can grow to nearly any size, and accounts for 70% or more of the energy in our little universe. In a time far into the future, this amount will approach 100%. Eventually our galaxy will not have visible neighboring galaxies. In certain regions of space without any galaxies, the space between galaxies, dark energy and the vacuum will be 100% of everything in possible existence. This is a "singularity" of massive empty darkness. Time expands outward, and pushes things apart. Galaxies are flinging apart from each other at faster than the speed of light, the farther you go

out from our universe. The amount of super small expansion, in tiny pieces of spaces, across a long enough distance of space, can enough growth together, that the points at each end expand from each other faster than light.

A white hole is where all possibly futures point *out* from it. The event horizon of a white hole is the past itself. All events we experience must point away from it, and all time moves away from it, and thus there is no means to see it, unless we look far out into the past of distant places in the universe, and see how time emerged in their locations. The farther we look out, the farther we see back into the time of another place, and not our own. We can only assume that things were the same in the distant past of other parts of the universe, as they are in ours. Nothing can cross the event horizon of white hole without functionally traveling backwards in time. The faster you push towards a white hole, the more you would just stand still in time, and see nothing change around you. Instead of seeing the universe pass by you, you would look out at the universe and see it slow to nothing, and become more and distant and black. You would enter a vast sea of nothingness. A real white hole's event horizon can never be approached, since in each moment, you got closer to it, you would also be getting farther away from it. Our universe is a white hole, and we exist in its domain, and it controls all our perceptions of time.

TIME TRAVEL

A time machine that can travel back in time, with you in it, is just as possible as a machine to transport into a cartoon movie world. There is no past to travel to, and thus any attempt to do so, is no different than traveling to a fantasy world. The past is not real and concrete. The only thing that exists is reality and what we can see. To travel back in time is form of cognitive delusion in how our minds create fantasies about the

past. We believe the past actually happened and is real, and this delusion helps keep us alive. Our memories of the past don't even need to be connected to actual reality.

Memory only serves a purpose in giving us greater dominion over things, so we can survive. The idea of things being real and solid, is how a mind driven to believe its own survival is everything, would make a mind conceive time and space. There are many different ways we as humans conceive the past, present and future, and in shockingly levels of diversity. Almost no one can imagine a different conception of time, and then the one they have, and the sense of reality it brings them.

The past does literally exist, and it is a material thing existing in our world right now. Our own past is however something we can never observe. We can only observe the past of other places distant from us. When I see a car crash right in front of me, I am actually seeing the past experience of a time from a certain distance from me. It is nearly instantaneous, because we are so close, and yet it is still the past I am seeing. Extending out from our planet are shells of time. If you wanted to "travel backwards in time" 100 years, you would need to go 100 lights in every single direction around the earth, and record all the information contained in space at that moment. It would be 200 light years across and have a surface area of immense proportions. This layer of space-time is our past from 100 years ago. It is the literal past, with all the information about that past contained within it. it experiences no sense of time, and moves instantaneously through the universe, relative to itself. The speed it moves is the speed of light, and this is absolute.

The future then would be contained in something smaller than an atom, and inside the very fabric of nothingness. It emerges out of space in each moment from nothingness. The future is a literal thing that exists at the tiniest scales, and grows into our present, and expands out to infinity in our future. Time is a physical and real thing, and not a

metaphysical concept. We can never catch up with our own past. If you go 99.9999% the speed of light, the past will always be a bit ahead of you, and always get a bit farther away in each and every moment. If you freeze matter to its coldest possible temperature, time still moves in it and emerges from it. There is an inherent "heat" to the universe constantly expressing itself. Time is heat emerging from nothingness. There is always more universe, there is always more heat, there is always more time. The future arises out entropy and thermodynamics, and the past can be described by an understanding and arrangement of this sort of information after it emerges.

If we made a superpower that could time travel, using real physics, it would be the power to travel into the future. A time travel power is always shown as slowing time around a person, and they can move faster than anyone and everyone around them. They move so quickly they can leap through time. They can catch up with time that escaped from them. The degree of this power would be how close they could get to a mean of light moving away from them, or if they could actually see the very light, they just emitted, by going faster than light. You could think of traveling into the past, as being able to move so fast, that you could go from where you are standing, to another point, and see yourself standing in that old location, and then leap to the location you are in at the moment.

Time paradoxes of traveling into the past, would arise from jumping a certain distance faster than light, and being able to observe what happened before, and observe your very own self. Instead of fragmenting and creating multiple time lines, you would fragment whatever moved back in time and create multiple versions of itself, and often occupying the very same positions in time and space. If you leaped out 1000 light years, and sent a message, it would still only arrive after you left. If you leapt 1000 light years, observed what happened, and then jumped back, you would still be in the world you

came from. It is this fantasy idea of the past being a real thing, and not understanding the reality of what the past is, that creates absurd fantasy ideas of time travel.

Traveling to the future is certainly possible, and would make for a very interesting time traveling super power. A real master of time, would be able to slow time, towards near stillness. If I slowed myself in time, I would leap forward in time, and be able to attract objects towards me with gravity. Since this would destroy me, creating so much gravity from myself, I would have to project time slowing around me to different locations. It could create the power to crush any object by freezing it in time such that gravity inside of it would crush it down to its tiniest size, and even attract things around it towards it. You basically would gain the power to control gravity, if you could leap forward in time. A black hole itself can be seen as a massive time machine that leaps into the most distant future.

A time machine that we can build, based on ways we know we can change time, would just be a rocket that accelerates up to near light speed. It could travel great distances, and arrive in a very short time relative to itself. The rocket can't arrive anywhere faster than light speed, from the earth's perspective. Light itself travels instantly across the universe, relative to itself, since time is frozen in light, and experiences not traveling of time. If you go fast enough, your sense of time slows, and your actual distances covered can be great. You can travel into the future with a fast rocket, as your time machine. Going faster in a rocket is the only known way to control the flow of time, by slowing it down relative to the rest of the universe.

Inflation:

The science and math of the current "big bang theory" can only go as far back as cosmic inflation, when the universe was very large, and not back to a single point of energy that many theorize the universe begins in. When light first came into being, its heat level was consistent across the entire observed universe. Inflation was invented as a means to explain how all of the observable early universe, could be so consistent, and there could be a singularity that the universe begins from. This rapid expansion evens out the energy, and can help explain the physical and thermal structure of the universe we could see at the earliest moments light existed. "The Big Bang" as it stands now, is actually the moment after inflation happened, and what happened before it exists in this realm of near unknowable magic. To solve this problem, in a new way, we just need a way this whole huge region could have a simultaneous event happen across it, other than a matter singularity. There isn't even math to describe a singularity, because singularities as super dense singular points, don't exist.

Instead of a densely packed singularity of matter that created the universe, it could be a massive space time "singularity" formed out of vacuum energy. A mass of space, may form such an immense amount of energy, and expand at such a rate, that it creates a white hole. That event, that moment, inverts space into time, and turns all events from that white hole to be pointed away from it, and makes all possible futures as beings caused by it. We can never directly observe that event or look into the past universe, because all of time points away from it, and towards our future. This is the literal past, and traveling there takes time travel. A white hole would emerge from the gradual buildup of vacuum space, until it reached a critical size and "mass" that would cause it to break from the space time of the universe it emerged from. The universe can emerge from another universe, as the most probable and easy way to explain how the universe works. We then need to properly define what space and time are in order to understand how this is possible.

Entanglements Are Faster Than Light

Entanglements of matter can interact at faster than the speed of light. This is a great mystery in physics. Its solution is also very simple, if one looks at the basic logic and facts. Entanglement itself is what creates time, and thus the breaking and altering of entanglements transcends time and causality, because entanglement itself is what creates causality in the universe. Entanglement is the basic structure of the universe that makes it possible to define one moment as causing another moment. The very structure of space contains within it a defined pattern, that allows time to emerge from it. Space entangles to create time itself, and so all entanglement are properties of space-time. Space-time entanglements can interact at faster than the speed of light, because they are the structure that creates time itself. Matter can't cause anything to happen faster than time itself. Time itself can cause time to happen in whatever structure time is arranged in. There is no other possible explanation for how entanglement can move faster than light, other than that entanglement is the structure of space-time itself, or a fundamental property of what creates it. Entropy, probability, and uncertainty all play important roles too.

The only "thing" in a white hole, is nothingness. "Nothing" can cause "nothing" to happen at any speed it wants, and create an entirely independent and new space time. A vast region of highly entangled empty space could simultaneously interact with itself, without violating the speed of light, because it would be only space interacting with space, and that is what creates time and not a cause of one thing on to the next. It creates causality itself and not causes on specific things. "Breaking" time itself, doesn't violate time, nor creates any paradoxes. A white hole emerges from a universe where time has spread the universe out so far, that it breaks into separate pieces with independent space times.

A big bang is the event of a white hole forming into an independent universe. All matter is created at "big bangs", and no matter from a previous universe can survive into another one, because new universes emerge from "nothingness", from vast regions of empty space. Something in the nature of vast entangled regions of pure space-time, there exists a means for this to collapse and create a singular energetic event. The maxim that no matter or energy can be created or destroyed after the big bang, can still be true, so long as you don't include vacuum energy. If that energy could then convert into something in else, in a singular grand moment, then you could explain how the energy of a new universe can be created from inside an old universe. The white hole itself creates all energetic matter in a universe in a single moment.

Space-time is time, and to understand time you need to understand how it changes space. Nothing can "happen" before a big bang, because "nothingness" itself created all big bangs, and time itself curved from one universe into a wholly new and separate universe. Empty space has underlying patterns, and these patterns create what we see as time and causality. This structure has odd rules that seem impossible to discern, because people associate with them, the interaction of matter we experience in our everyday lives. When space-time moves and interacts it defies all easy understanding, and creates entirely new realities. This interaction has a concrete and real structure in the quantum world, and it is described in how space entangles.

This structure of entangled space, is too fine to see and can't be observed with cameras or light sensing devices. Time is measured with clocks, and this very subtle kinds of time at the quantum level would take a level of time measurement beyond what exists in my time now. Just now in physics we have observed gravitational waves. These are massive waves in space-time, that have no particle or matter, carrying massive amounts of

energy and information across the universe. Information/energy can be imbedded in gravitational waves. We could think of the fine structure of the universe with entangled space as very fine gravitational waves, with information and energy embedded in them. This underlying structure if true, would solve most all the problems current physics has with black holes. Energy/information can be lost in black holes, and the only thing that can escape a black hole is space-time itself. If space time can have information imbedded in it, as part of its intrinsic nature, then it changes the foundations of modern physics.

The implications of such idea radically change all possible views of how the universe was created, will end, and on the nature of God. It is almost like the universe has ESP, and it can broadcast information with nothing observed in between. This magic like quality of entanglement that can instantly connect things light years apart, is bound in how certain classes information is transmitted through space and time itself, with no matter. This information can't cause things to happen, and yet it influences all of reality. I would guess this pattern is highly correlated to crystals, and how life forms. It causes certain structures and patterns to be more likely to emerge than others. It orders things at a very fine scale, and sometimes those orderings lead to replicating systems, like crystals and life. The very nature of life might be embedded in this structure, and thus could be the central driving force of our human reality.

The Day Everything Changed: Anthropic Principle

Sentient life is a well know thing, that we are fairly certain exists and thus must be accountable for, in the very laws that govern how the universe exists and comes to be *(I oddly find this the most controversial statement in my book).*

One-day sentient life evolved on this world we live. It is much rarer than non-sentient life. Most planets and places where life evolves will probably be destroyed before sentient life evolves. The window of time in the universe where life can evolve and the window of time in solar system where life can evolve, are usually not long enough for complex life forms to evolve. Our own planet had life for billions of years, before it even formed into basic animals during the Cambrian explosion. Life on earth was mostly single celled organisms for 3-4 billion years. While life could be abundant in the universe, it will rarely ever get beyond a few celled organisms. Planets have short windows where life is possible. Sentient life is even more rare, because it needs even more specific conditions to evolve.

Sequential Evolving Universes

Expanded Axiom: The end state of one universe can influence the creation of the next universe

- *Universes evolve towards increased stability that increases the likelihood of more replicating universes surviving and creating more replicating universes. Any increase in stability, no matter how small, will "evolve" the universe by the tiniest amount.*
- *A Universe is alive if it can replicate and evolve*

If universes come from universes, then stability and evolution would be selected features for the survival of new universes. Any living universe, such as ours, has a stable end. That stability pre-existed life, and is the foundation from which all things could evolve, the universe included. In physics, they found that our universe won't collapse back onto itself, and have gravity pull it all back together. It also won't start expanding so fast,

that even matter is ripped apart by the acceleration expansion of the universe. It expands at this precise and exact level, where stability is maximized. This is such a fine distinction between the universe ripping apart, or crunch back together, that even tiny changes, would lead to this stability falling apart. *The real universe is consistent with this theory, because we actually exist.* The state of the previous universe will replicate into the following universe, and “mutations” are possible.

Stable long living universes are more likely to create more universes, than short lived ones, because they grow longer and create more vacuum energy to create new universes. The easiest way to imagine a stable expanding universe creating more universes, is if universes themselves are born out of massively expanded regions of space. The longer the universe expands, the more likely there will be massive regions of space that are large and energetic enough to create a new universe. The universe evolves in a direction of stability, so a universe that would grow into one like ours and then experience a big crunch or a big rip, would be impossible, and doesn't exist in any universe or possible condition of the universe that life exists in. A living universe goes towards a stable end of sorts that allows it to pass on its “knowledge”, to the next universe.

At a certain point in the evolution of living universes its rules and initial conditions become very stable, and evolve only at the tiniest levels. It will evolve towards a near perfect rate of growth and stability at the macro-level, that fosters the most replicated universes possible coming into existence. If the universe mutates and becomes more stable, it passes on that increased stability to the next universe. This creates a trend that expands down into far more nuanced things, that select for a life bearing universe. The conditions that make a universe maximally capable of replicating itself, lead to a

universe able to create life on the micro level using the same evolutionary processes of the macro. The forces that drive universal evolution emerge from quantum features, and become more optimal at micro levels, when there are longer macro time windows for life to randomly come into existence. The fine tuning “knobs” for the physical galaxy would be much bigger, and the fine tuning “knobs” of life, much smaller.

The First Living Universe: Narrow Windows of Time for Life's Development

- ***The tiniest differences in the initial states of the universe effect the chances of life in the universe and its development.***
- **The universe can select for more life bearing universes, if life or the conditions for life bearing influences the survivability and replication of universes.**

Life exists in our universe. At a minimum, it happened in one universe, at one exact moment, in one exact place, and only there. The conditions for life to exist are the rarest of all conditions. In a grand multiverse, most all possible universes would have no life, and no chance of life ever existing. There are just so few ways we can arrange the constants of the universe such that they produce life. And even in a universe with rules and conditions where life could come into being, it is extremely unlikely that life would even come into existence. You could have trillions upon trillions of universes with all the same conditions, and only one even forms the most basic form of life possible.

The window of time where life is possible, is very narrow, in any universe. Stars must form, and those stars have to explode over and over, until they stabilize and produce ever more complex and dense forms of new matter. A simple universe of all hydrogen, that never even forms into stars, would never form life. Most universes other than our own, would resemble stages of our early universe, except where that stage goes on for much longer, and the universe ends before forming into something more complex like our current universe. A universe may end sooner, collapse, or even expand so fast it rips itself apart in a short time.

We can see looking back in the history of our universe and find the earliest stages of galactic development, where life could possibly evolve in. We can also calculate how long the universe will have until it is inhospitable to life. The longer this window, the more life there will be in a universe, probabilistically. The universe may seek out a maximization of this window of time, that life can arise out of, and prosper in. Enough diversity for things to evolve in, and enough stability that things can survive when they do form.. With longer periods of time, and more stable galaxies and more stable star systems, life has more likelihood to come into existence. Life on earth needed so many specific conditions to evolve, and those conditions are rare, though computable. Eventually over enough universes life becomes more probably in each successive universe. Increasing stability and increasing complexity are fundamental laws of the universe. Life itself may increase the likelihood or more survivable universes that prosper more, such that the more life there is, the more chances that quality life bearing universes will exist and dominate over non-life bearing universes. Iff life itself can affect the birth of universes.

- https://en.wikipedia.org/wiki/Drake_equation
- https://en.wikipedia.org/wiki/Timeline_of_the_formation_of_the_Universe
- https://en.wikipedia.org/wiki/Extraterrestrial_life
- https://en.wikipedia.org/wiki/Fermi_paradox
 - https://en.wikipedia.org/wiki/Modal_logic

- https://en.wikipedia.org/wiki/Freeman_Dyson

More Stable and More Complex, Universes Are More Suitable for Life.

A universe with life is a more stable universe. The qualities that drove the evolution of the universe, may also drive the evolution of life, and to such a degree, that life itself comes to drive the evolution of the Universe. The first universe with life couldn't be much different than our own universes starting conditions. However as more life existed it creates conditions to make starting conditions for universes more beneficial to life. What is possible and not possible is a very narrow range of things, and this becomes especially so with living things. Life expands on the possible, and never becomes the impossible. These probabilities that evolve life directly correspond to the evolution of random quantum systems, to some degree. Life is a quantum based entity, where the irregularities of quantum features that created the universe, define black holes, and defines the properties of matter, act to create the fundamental features of life. Life may also have a great proclivity to exist, because quantum features drive it to exist, as an expression of possible outcomes, as an expression of itself manifested more clearly.

Ever increasing stable complexity, makes life more possible to exist.

The Universe appears to have a trend of becoming more life bearing as a function of the same principles that formed it out of nothing. This is not a big leap of thinking, for a basic theory of how the universe works. It is simple logic, and going for the simplest most direct explanation to account for life in the universe, the actual state of the

universe we see, the laws of physics as we know them, and the simple logical outcomes of the thought experiments that develop modern physics.

A Grand Note: Avoiding Metaphysics

A one in infinity chance, is zero

If the multi-universe is just a massive number of possible universes and not infinite amount of them, then the constraints on that infinity negate all possible quantum configurations existing. If it isn't infinite, then not all possible universes came into being, then this introduces a degree of randomness and probability into our universe coming into being. There then is some amount non-determinism at some level into quantum physics. The question then becomes how small a multiverse needs to be made for it to be possible to create our universe, and to hunt for every single way to constrain that number, to the smallest amount. If we don't, then we deal with an idealized concept of infinite universes, that creates zero possibilities for existence. Probability is the only way to create a universe, because absolute deterministic models have zero possibilities of existing, due to simple math $1/(\text{infinity})$, would be our place in the universe. The more we limit the size of the multiverse, the more probable we make our existence, up to a certain point.

The geometry of space time at a quantum level demands a universe consistent with it, things that are more consistent with these properties, will exist more in the universe. The very universe and all its complexity are expressions of the nothingness that created the universe. This nothing gained evolution, it gained time, and it evolves towards its own self. These patterns of time and space are real, and they imprint themselves on the

universe, and life emerges in response to these patterns, and sentient lives very mental structures must reflect them, or not be able to work. Life itself down to all our very thoughts must be consistent with the geometry of space time, down to the tiniest scales, or it would not be consistent, possible or real. Only the possible can exist. As time emerges it annihilates that which can't be, and favors that which can be. This is a sort of conceptual friction that annihilates inconsistency in the universe. As a system arises out of randomness, and grows to nearly infinite, yet not infinite, scales, it must be constrained by reality and what is possible. These constraints are deep, everlasting and precise.

Consider this: The math developed by Einstein to explain General Relativity is the underlying math that was used to model human consciousness and form the basic programming language of modern AI and deep learning. The more we evolve an understanding of this geometry, the more we reveal the consciousness of humanity, and possibly the purpose of the Universe. To all physicists studying the stars, you may want to give all your time, thought, and dedication to revealing the power of AI, and creating a new age of technology, as the greatest act of service to science, truth, and humanity possible. If this seems a simple coincidence to your mind, then you are a fool. A theory of space-time is a theory of the information embedded in it. The truth of this is unavoidable and absolute. Its consequences when integrated into our technology and cultural vision of ourselves, will transform everything.

Chapter 14: A Guide to Practical Magic

LIVING LINEAR EVOLVING UNIVERSE

A living universe can give birth to another living universe. If universes can be sequential, then they can be created by living sentient beings, as the very product and purpose of life itself. Life is an expression of the quantum functions that created the universe from random nothingness, and evolve towards one definitive possible end.

Our universe is most likely created by another universe that existed before ours. It had nearly or exactly identical starting conditions as our own. It is possible that a civilization of aliens nearly identical to us, in a preceding universe, created our universe. This has happened or will happen, because we are alive and have minds connected to reality. The conscious domestication of universes can perpetuate itself and make life more likely to be possible in each successive universe. Life has the potential to spread across all the knowable universe. We may only reach a billion galaxies, and occupy every solar system in them, with human life filling every place. We may create more human beings than there are atoms on the surface of the earth. Humanity has the potential to be a universal species spreading across the entire travelable universe.

The technology to expand out to the whole universe is mostly right here in front of us, already. All it needs is to be developed more, made more efficient, cheaper, and smaller. We don't need humans to travel on spaceships to create a universal empire. We can use the most basic forms of technology to build it from things right here on the earth now. Antimatter, warp drives, warp gates, interdimensional travel, cold fusion, are not needed to get this done. Space travel is just a fantasy. Zero-Cargo-Space-Ships, Neo-Androids,

and an interstellar high-latency-internet, are the most likely, fastest, and cheapest way to colonize the universe. With the right development, the fastest, most humane, and cheapest way to send humanity to the stars is to broadcast humanity in data. In the end, our DNA is just data, and the upbringing children receive from parents, is just data. All that makes a person human is just data, that imprints onto a unique and special mind born from a mother's body. Each mind is unique and special, and the data to create us and train us can be broadcast.

If we can spread to the galaxy and spread to the universe, we can alter the entirety of the universe into a living machine supporting massive ecosystems of life that fulfill us and expresses us as people. The whole of our selves can shape not just a planet, but all of reality. We transform the earth and domesticate it to human needs. We transform reality to support our lives with technology, and any technology that doesn't match with the reality of human identity doesn't survive. Evolution extends itself out into a creature that can transform reality to match its own inner reality, to better serve its own survival. Our dreams and inner visions have to conform to reality eventually, and do. When those dreams become reality, it is the creation of something more human, and humanity fulfilling. Our galaxy will be alive, and moving to the will of human intelligence and thought.

At the smallest level of the universe, true and absolute randomness and probability governs absolutely. These random possibilities created from the smallest thing, the entire ordered, structured, and seemingly designed universe, that we know. If we are living in a universe than could spawn new universes, then sort of "consciousness" could itself emerge in time, in the nature of what is possible and not possible. That consistency and randomness in nature can be a universal driving force to reality. This is an argument that "God" is in the laws of physics, in a far deeper and precise way.

Extremely rare star-faring alien life won't be that different from us. In the broad strokes of culture and how consciousness works there will have to be deep and fundamental similarities. There just isn't that many ways to make a sentient mind with consciousness. It needs culture, it needs ethics, it needs collective groups with selfish individuals, and so on. Technology itself has to develop in sequence and all star-faring life will go through the same gradual development, and face equivalent social challenges, that must be overcome by adaptations and development of culture interconnected with technology. Life and sentience is rather linear and has a very limited range of possible ways to evolve into something more than it was before, and become something that can create a universe. It has to conform to reality.

There is something beneath the objectively knowable universe that is real, that our minds can connect into. We have to feel, in order to be sentient. We have to know how the very rules can play out in its variations, from the very insides of our beings. Sentience is created from random evolution that forms towards innate structures universal to all existence. The universe survives and evolves, people survive and evolve, and the underlying system seems to have a fundamental and intrinsic connection. Things that aren't inconsistent with reality are less likely to survive. Over vast scales of time, even tiny differences in probability lead toward near absolute outcomes. The easiest and most likely steps, that can stack on top of the one before an evolve, are a very rare and narrow set of things, that can gain a near deterministic quality from its apparent randomness. The very consistency of human life with all the tiniest features of reality formed from step by step evolution, was guided by random chance and the constrained by what is possible to exist. That range of possibility is narrow, and over time becomes singular.

The Domestication of the Galaxy

Mankind can colonize the known galaxy, with every solar system domesticated to human will. 100 billion star systems with billions to trillions of human lives inhabiting each one, can be done in just a few million years. The number of people in the galaxy will eventually exceed 10^{21} people. Our own planet earth may eventually hold over a trillion of our kind, and nature itself will be even more bountiful and diverse. In the scale of all that humanity, all that possibility, unique minds of great value will arise. Humanity itself in all its form, will be the very wealth of the universe. Technology will release the inner potential of all people, and be able to transform that into concrete reality. People will be able to work, by just being themselves the most they can be. Computer intelligence will surround all life on a scale near unimaginable in our current terms. Each person wielding the power of the combined computing power of the Earth as it is today.

That task of colonizing space, is a very hard one, and makes it very unlikely we will ever get to another star system, much less another galaxy, because of the inherent conflict between humans. Technology can get us to the stars, and how well we integrate that technology successfully into our lives will determine its success. The most likely outcome is a rebellion against technology itself, as it consumes and destroys our very selves and identities. We will always be in permanent conflict, and consumed by constant moral and ethical failure. It is far too easy and more likely to transform people into something fundamentally not human, than more human, with technology. We will make constant mistakes; if we don't take risks, we face an absolute end, with nothing. Culture will not be a singular homogenous entity ruled by a centralized authority, nor will people ever truly embrace individual responsibility and freedom. The path to the future is narrow, brutal, and absolute, we can all die at any time. Humans are nasty and will be for all

time. We may never embrace ourselves, and the fundamentally value of having the most possible individual lives living in freedom, as the true wealth of the universe.

If we go to the stars to be all the evil things in our hearts that we dream we can be, then what goes to the stars will never develop, grow, evolve, and endure through time itself. The creation of a perfect life and a perfect human will never be achieved. Being human and living will always be nasty and difficult. Anything too perfect, and too civilized is just a reflection of one person's own idealized hatred and disgust towards life and their selves.

We live and we grow, life pushes forward. We do not yield; we do not surrender. We endure, we survive, we create our very destinies from the nothingness of the universe, from the very laws of all creation that created all space, matter, stars, galaxies, planets, life, people, cultures, and technology. These things are bound in and grow out of the same unifying ideas and are absolute. We may give up, die, and become nothing but star dust and background radiation. There is a chance to be something more. We could create a better universe. One civilization though can do this, may do this, and will succeed. Life doesn't just die and give up in the face of a universe fading away into nothing.

A GRAND THOUGHT EXPERIMENT ON GALACTIC TRAVEL

- **We will have the power to create new universes. This may have happened before, and it may not have happened before. We can make it happen.**
- **Can the mind of a single person define the probability of success?**

We can't broadcast ourselves into another Galaxy like we can in theory between closely neighboring star systems. To cross over into another galaxy is a completely different problem. We may need to send people to other galaxies, this will take thousands of years. These will need to be civilization ships, world ships. They will have to contain all of humanity and who we are, embedded in them. This journey, beyond the rim, will not be like the one taken to cross our own universe. It will require a far deeper commitment of our humanity. Great ships will set out from our galaxy to other galaxies. Most of these missions won't even know about the other missions, because the galaxy is too big to readily communicate across. Whole star systems will be transformed into machines to launch mankind towards other galaxies. We will have to go ourselves, in our flesh bodies, and journey as a people to reach a new galaxy.

There will be only one path that will move us forward, and we can never know it, until we try the countless multitudes. Certain individuals will rise above all others, and contain within them the ideas and the will, to create a better world than the one they live in now. We can be better, in each and every moment, with our thoughts and mindfulness. Through generation after generation. We can find in certain people ideas that will drive us forward to better and more human lives. The more people, the more possibilities of life improving and reality altering ideas emerging from nothingness into our lives. Humanity itself is the treasure of life and being human. These ideas and people will be the cargo of this ship.

THE THOUGHT EXPERIMENT:

- What sort of people could travel thousands of years between galaxies, to form new galactic civilizations, spread life to an entire galaxy, and remain purely and truly human through all of it?

- What singular emotions, ideas, vision, and drive would they need?
- What sort of individual would rise to the top of such a culture? How would they rise up? What power would they properly have?
- If things went horribly bad on the mission, and most everyone died, what would it take to continue that mission?
- What sort of ranges of change and variation should this culture have? Should they always be the same? Is being a singular unchanging culture even human?
 - Would they need to be able to become or at least contain any culture that has ever existed in humanity?
- What cultures on our world now should be on that ship? What races?
- Can more than one ship go to the same galaxy? Should different ships even talk?
- Could this trip be done by one person?
- Does the journey even need to be taken if there is a process that can create a people noble and great enough to do this?
- Why go?

WE WILL LIVE AS HUMAN BEINGS

We will live as human beings,

Sleep in beds,

eat food, breath air,

dance,

sing,

take drugs,

have sex,

argue with our parents,

look for love,

feel alone,

be bored,

be boring,

Do stupid things

Betray our humanity

Break the law

Deny who we really are to even ourselves

Find new things in ourselves we never knew possible

Surrender who we are to another in pure respect...and love

Betray the ones who trust us and love us

We will be human

We will find a way to make money from anything

Be communist revolutionaries

Soldiers for lost causes

Dreamers

Realists

Christians

Chinese Mandarins

Red skinned, yellow skinned, pink skinned, olive skinned

Broken

Lost

Dying

Dead

Remembered

Scorned

Re-enacted in the movie of our lives, by future generations

Pedophiles?

Gays?

Dwarfs?

Mental retards?

Autistics?

Killers?

Genocidal Warlords of the Cosmos, seeking immortal life, and absolute power over all creation to screw over the bastard who did us wrong and make a reality born out of their spite, where we annihilate our enemies from even existing as a concept?

Human

Lovers

Friends

Joshua?

Magical Thinking

This section crosses the line into fantasy, and is not purely science, nor even possible. Where that exact line is, I don't know. Takes this section serious. Separate the fantasy from science with your own judgement. Science Fiction and Fantasy, are one genre, delving into different things. You might think of this as the science in fantasy, or the root archetypal humanity in science fiction.

WILD IDEA 1: EVOLUTIONARY RECORDS

The study of the formation of our own Universe, can be a study of the evolution of sequential universes themselves. Possibly the only way to model and understand certain observations of our universe, will require such a complex evolutionary system to model and understand them, and not be explainable with one elegant formula. The very history of every possible state the universe actually had, is recorded in the history of this universe, possibly. I theorized that the universe had to grow to massive scale with no matter in it, before matter even evolved. Matter itself would have to evolve in sequence universe to universe. When each new universe forms, it can have tiny mutations that slowly evolve the system. The universe in sense lasts longer, and longer, and becomes more and more complex, with every condition before it happening in a chain of sequences building on it.

The stages of universal development in our universe, may reflect in some way how things evolved through trillions up trillions of former universes, that evolved in sequence, into the one we live in now. Possibly, we can observe the earliest forms of the evolution of the universe in the patterns of things we can see in the present. The farther

we can look out to the light from regions of space vastly far away from us, and where the causality of the origins of the universe is still hitting us, we can gain insights, beyond what we have now. We will have to understand evolution as a universal concept that drives all of reality. Complex systems don't just appear from nothing.

We can see in the stages of how we grow into a living being, the very history of how life developed on earth. All very unlikely, but much more likely than the universe developing and coming into a more stable and evolving form. The conditions of life evolved before the first life came into existence. The chemical cycles of ATP that give life/DNA energy, had to exist before life existed, and the conditions for that chemical cycle had to evolve and come into being. The very first life on earth was just strands of RNA floating around. They were possibly trapped in sulfur bubbles for cell walls, because life needed a cell wall to exist. DNA/RNA had to evolve to produce its own cell wall. Life evolved from the tiniest things, and all those stages still exist in some form in the world we live in. Different forms of primitive life evolved, and came together in symbiosis to create modern cells.

WILD POSTULATE 1: THE FUTURE HAS HAPPENED IN THE PAST

Wild postulate 1: Our future states will be conducive to a sentient stare faring species to influence the whole of creation, in a sequential step by step evolution of a galactic civilization.

Longer lasting and more stable universes are selected to exist more. Increased complexity adds increased survivability only with the rarest of possibilities. These possibilities are so rare as to be almost deterministic like. Such that through pure

random chance, the same basic result could arise in nearly all situations, given enough time for that system to evolve. Determinism in a system, can arise out of pure random probability, and lead towards a near certain end, in a vast enough Space-Time. Life itself is part of this increased stability. Sentient life may play a great role in the nature of the universe and hard science.

Probability defines the evolution of the universe, and impossible things don't come into existence. The structure and systems we see, that look designed, and are born from randomness, and have those structures for a reason. Across nearly all possible highly stable universes, the differences between them would be minimal. The range of possible things, that can add onto themselves, improve themselves, and propagate, are very small. But there exists a probability for that to exist, and the more probable ones are much more likely to exist. There are not weird universes out there operating on completely different rule than our own. The slightest difference in survivability would lead to extinction of any poorly forming system.

If you take two starships and send them on a journey, with even the tiniest degree of difference in the orientation, the distance between them as time approaches infinity will be infinity. If one is going in the right direction, and the other is not, and the distance of the destination approaches near infinite sums, then any difference is absolute on universal scales. 0 and 1 become absolutely determined observations of any random probabilistic system, given enough time. That direction becomes a kind of destiny that feeds into itself and favors all systems that go towards it.

The universe evolves into something more stable. The universe seeks stability, and the more stable it becomes, the more it exists. And so on and so on. For instance, at this

moment, physics has found that the universe won't have a big crunch or a big rip. It won't expand so fast that it rips apart, or expand so slow that it collapses, it has this near perfect sweet spot of acceleration, that creates maximal stability. From my way of seeing the universe, that would be the most likely outcome, since we live in a highly evolved universe that evolved from another universe nearly identical to this one, and thus should have in the future very high degrees of stability. The future of the universe could be assumed to be hospitable to certain aspects of life influencing its creation of new universes.

This would be a wild though interesting postulate to explore. The very structure of our reality should conform to the possibility of us being able to make universes. We can make predictions on this, and they will work out in time.

There is always a step forward, and there is not an absolute inherent dead end. We can always be better than before. To survive is a constant pursuit of improvement. That which improves itself survives more than what doesn't. To evolve evolution itself and take conscious active control of our evolution is an evolutionary leap into sentience, that by its nature will lead to ever increasing and radical evolutions towards survival and mass replication. Our species has the potential to domesticate galaxies, and create a universe.

The path to colonizing all space, has already been taken, in the past, and thus can be taken. The tech stack is possible, because it is most likely that our universe was created by creatures like us from a preceding universe. There is a way to develop advanced technology, be a galactic super "empire", and not destroy ourselves, because it is mostly likely that our universe was created by such a species. They themselves evolved most likely the same as we have evolved up until now. The purpose of creating a universe is

to maximize our existence and potential to create universes, towards some grand goal, of ever expanding stability and survivability. We are their survival, and in a cosmic scale one species. To be human can transcend even our DNA, and life on Earth. All-star faring civilizations are human.

Our future has happened in the past. The means to develop all our tech in a perfect technological tech stack, that leads to an idealized future, is a solved problem. The subjective journey of that path will be brutal, chaotic, and fundamentally driven by free humans making mistakes. Do not mistake this idea as a call for centralized power believing they are god and know what is right, because all centralization of power leads to absolute death.

There are basically just 2 kinds of views of creation, in its absolute terms. Pure white light, or pure darkness. Pure light super energy, a singularity, a set determined universe, clear laws, a chaotic but ordered destiny (the future is known), or, an absolute darkness, a universe born out of nothing, pure probability. randomness, and an unknowable future. Different people feel in them, different destinies and visions of being and meaning.

WILD IDEA 2: GOD IN SPACE-TIME

In many aspects, my conception of "God" builds out of an idea of extreme evolution. The possible things that can exist, function, add value, reproduce, and evolve is so small as to be almost singular. Life on different planets across all universe and planets is going to have serious similarities. Culture, ethics, love itself, will have universal characteristics.

The possible and the not possible, what can add a tiny bit of change and make something more survivable, is a process of gradual change towards an order different than the one before. The universe gains form, the universe creates life, most of the time it does nothing of the sort. We exist in this narrow band of reality where things do work and come together through gradual step by step evolution. These structures of cognition and sentience are bound by the very things that shape space-time and the origins of matter. The birth of universes, galaxies, stars, planets and life bearing ecosystems constrain what sentience can be and what drives it. It is all meaningless, yet meaningful. Magicless and yet pure magic.

Love is born out of replication, and continual improvement towards the future in successive generations. Only things that reproduce, evolve, and improve can exist, and at basic level, that is the foundational roots for love itself coming into being. Love may very well have created the universe we live in, and could create the different possible universes that can arise out of this one. It isn't outside the realm of the possible and may be the only probable means that creates the universe. If we came out of the universes messy soup of possibilities, it would seem that what we are imbedded in the very laws of the universe.

Our very identities are not just random code, operating on deterministic destinies like clockwork machines unbound. We aren't robots with no free will; we are not billiard balls bouncing on a pool table for a few trillion years, with the exact end knowable from the exact measurements of the beginning. The laws of the universe don't create a clockwork universe. A God conceived as the laws of a clockwork universe doesn't exist. Love however may exist in the very form of the laws of the universe. Love guides a universe in chaos towards the future and survival. We can become anything, be anything, and

have absolute freedom in everything, down to all the rules of reality. Love survives, and it alone can sacrifice for and build the future.

Our very feelings give rise to our consciousness, and this consciousness gives rise to our very models of reality and science. Those models actually work, and they were born from human feelings, where love can reign over all.

Love seems a useful force, and idea to grow and develop. Love requires freedom. Freedom is based in randomness. Pure random chance is where our choices are born out of. In a reality with no randomness, there is no morality, no love, no meaning, no freedom, no purpose. The danger though lies in projecting one's own simple conception of love onto the universe, science, technology, life and culture. This is how most problems arise now in all these fields. People have mostly un-reflected upon root ideas of love and goodness, that they spend a life defending in some way. Your idea of love is probably totally insane and wrong. It is all too human.

- How would love shape a people traveling a thousand years on a ship between galaxies?
- What sort of humanity and person would arise out of that crucible of absolute creation, and desolation?
- Could such a culture, create a universe?
- Could a single person contain the moral truth and love, to create the whole of human reality that we see?

A single person, can be a point of fundamental leadership and inspiration towards the future. God can be represented as the orientation in time, we must go as, in the flesh of a human being. Newton, Aristotle, Christ, Steve Jobs can be seen gods and demi-gods that directed man's development. Living sentient beings could have created state of the end universe that built the following universe. The pursuit of science near the end of the universe is to conceive an initial configuration of the universe that will generate a better universe, in the following one. And possibly influence its development after the "big bang". Or to find a way to simply replicate and expand humanity more readily into a following universe as a means of metaphysical reproduction. The idea we are living in a simulation is not what it would appear to be to the theorist speculating about it. It would be a search for God, and thus the simulation itself, to the those that made it, would have more moral value than even their own reality. The people in it, would be the very modeling of God, to broadcast into another universe.

Broadcasting from the old universe into the next one, while respecting the speed of light, is possible, if you receive a signal from far enough away, and that signal was broadcast for a very long time in the preceding universe. This paradigm may extend to even the creation of a new universe. From planet to planet, we develop a greater morality, from galaxy to galaxy another level of character, and in seeding life into another universe we rise to something totally different. We can replicate through data transmission between planets, and possibly eventually between galaxies, and then universes. If we look far enough away from us, and with the right tools, we will be able to see the beginning of the universe, the information from a very distant point in space and time. If there was something existing before the big bang, as seems very likely to be true, we may be able to observe something of the universe before our own, if it was transmitted to us.

We in theory will be able to observe information that travelled to us from another universe. If it was sent by a nearly god like species, then that information could very well influence and configure a receiver. If they knew how life could evolve, they could make it a signal that living things could receive. This is my best attempt to explain people getting messages from "god" that could be true, and how life from the very beginning could be influenced. Revelation, sentience, and our feelings of the divine, could have a divine form, mixed in with the 99.9999% self-delusion. This is not "aliens are in my head telling me what to do!". Humans have magical thinking, and it seems fundamental to being human. This is also how one has to solve the information paradox of a black hole, and what can transmit out from one. Space.

Wild Idea 3: Memories of The Past and Future

Creatures precognition may be more than analysis of the present and past. It may be possible to send signals back from the future, and not create a paradox. In that we can only know if we received a message from the future after that future event has passed. The sort of "information" that can be known in the past from information from the future, may be only recognized if it has already been experienced before. The future that happened in a past universes civilization nearly identical to our own can be seen as the future of our universe, such that we will recognize and dream of forms from that past universe as the ideal possible future of our own.

Memory in some forms, works like a computer hash. A hash takes a large file of data, and creates a short unique code for that file, if one letter is changed, the hash radically changes. We don't remember the past. We remember compressed encodings of the past, that when presented with those memories, we can recognize them to be correct or not. We don't remember the past. We can recognize what was the past and is not the

past, by these compressed encodings. When we see in front of us the past, that we have an encoding for, we can recognize it and sequence it temporal hashes.

Déjà vu is experiencing something that falsely triggers this feeling of a past memory. You recognize something that can't be memory as something you remember. The underlying memory that is falsely corresponding, doesn't even have to be similar.

- A hash of a picture is remembered.
- Every unique image will have a unique hash.
- If presented with that unique image you have a hash for, you will remember that image as a correct memory.
- You can have false memories and false identifications of having seen something before, that you have not, based on hashes not being 100% unique.

The information broadcast through time is these hashes, and can't be known unless they match an actual experienced event. Maybe we can know if we have received a message or not. The very existence of messages implies a future to exist in, and thus a reason to have hope. Knowing you can survive is of great use to survival. Human destiny, our sense of purpose, and the frame work, that we construct our identity around, are bound in a sense of time, that projects into and from the future. Or at least at some level, people experience life with an illusion like this to guide them. Information could be broadcast from the past from a very distant and long dead civilization. While we also could make that same technology and broadcast to our past. No information could be broadcast back, like "buy this stock". Life is probability and knowing the outcome of a probable thing, destroys its existence.

Emotions arise out of the way human memory is formed and experienced, as an embodiment of time. Evolution and Survival is the subjective emotional experience of time as flesh and blood. Our emotions arise from a will to live and reproduce and to experience our existence and reproduction as meaningful. Reproduction must always be a set of emotions favored over individual survival, while not favoring group survival. If this sense of survival can be tuned to the right frequency, then it can amplify the potential for a star-faring civilization to exist and prosper. An animal could naturally evolve such a receiver, if ones that did survived and prospered more. The message itself could influence the configuration of even crystals, or certain rare events, where life comes into being initially. At a simple level, it seems possible, that something like gravitational-waves could influence the configuration of quantum systems towards certain configurations, and thus transmit self-replicating information, which could then evolve into sentient life. Love is an interesting emotion, that drives us to be better.

The Dark Night Of The Soul

Within us as humans, we can look into this place in us, with absolute darkness, absolute nothingness, and find meaning. People have two senses of the divine, absolute nothingness and absolute light and everything. They are equivalent.

The amount of light you see depends on how much "god" you put into the absolute nothing of the universe from which all things came into existence out of, and how far into the future your sense of reality is dependent on. Teleological thinking orients towards light, and deductive thinking towards darkness. This dualistic interplay of divinity as absolute nothingness and absolute being, are very real cognitive experiences in many "spiritual" human beings, between the two essences of human nature.

I experienced absolute darkness and fell through absolute nothingness for eternity. I found the tiniest and most distant star in that darkness to orient, after I died. I can't universalize my experience to all people. If the universe was like a ying-yang symbol, I would be fixated on the white dot in the blackness. I found in the nothingness a guiding direction. A point of light in nothingness, to orient creation towards. I found myself, a direction to guide me, from a voice both inside and outside me, at once. Eastern dualism, as compared to western dualism, lies more at the direct human experience of reality. We can also conceive creation in it. It is not good and evil. Different souls will see different things, and both be true.

There is a structure of to our consciousness that correlates to the very structure of the universe at its very origins. We couldn't even be sentient, unless we evolved a mind that connected to this. Possible things are things connected to the very laws of creation at the beginning of the universe, by the laws of probability and what is even possible to exist. Random evolution can create sentience, by creating random combinations, until one combination hit on the one that correlated to what can actually work and not work. This can be a gradual process, until you hit sentience, an actual awareness of that initial state of the universe. We as humans are collectively sentient, and not individually fully aware. It takes a large grouping of humans to contain all that "knowledge" contained within nothingness, in beautiful tapestry of different cognitive structures.

Our minds seem connected to more than just random survival. We have a sense of time, a sense of destiny. We have built real things from this feeling. We built civilizations and grand achievements in technology. These things are real, and we made them from the

feelings inside of us. Children have looked out at the world, and felt the laws of nature; through a life of thinking and study, they at rare and special times create the most world changing ideas. Newton and Einstein's true formative thoughts, that shaped all of their successes, were born out of their younger childhood minds.

The ability to truly hold onto your identity and childhood dreams, creates a will capable of changing the world.